Topology Course Notes

Fall 2013 to Spring 2015

Gary Gruenhage Mark Guest

2019

Auburn University

© 2019 by Gary Gruenhage & Mark Guest. This work may be downloaded for personal use. Publishing it elsewhere or selling it is strictly forbidden.

The 350 theorems herein are theorems stated by Gruenhage in IBL style graduate topology courses over a two year period; most topics covered the first year are standard, while more specialized topics in set-theoretic topology were covered during the second year. The organization of these theorems, together with hints and proofs, into this book is due to Guest. A large percentage of the proofs are due to Guest, with other proofs due to other students in the classes.

Contents

1 The Basics **1**

1.1 Some Set Theory We Should Know 1

1.1.1 Cardinality & Cardinal Numbers 1

1.1.2 Ordinal Numbers 2

1.1.3 Transfinite Induction 4

1.1.4 Axiom of Choice & Equivalents 6

1.2 Topological Spaces 6

1.2.1 Closed Sets 7

1.2.2 Interiors & Boundaries 8

1.2.3 Bases 9

1.2.4 Subspaces 10

1.2.5 Continuous Functions 10

1.2.6 Homeomorphisms 12

1.2.7 Convergent Sequences 12

1.3 Some Types of Spaces 12

1.3.1 Metric Spaces 12

1.3.2 Product Spaces 13

1.3.3 Quotient Spaces 15

2 Topological Properties **17**

2.1 Separation Axioms 17

2.1.1 T_0, T_1, and T_2 17

2.1.2 T_3, $T_{3.5}$, and T_4 17

2.1.3 Urysohn's Lemma 18

2.1.4 Completely Normal 19

2.1.5 Perfectly Normal 19

2.2 Covering Properties & Countability Axioms 20

2.2.1 Compact and Lindelöf 20

2.2.2 Filters & Tychonoff's Theorem 22

2.2.3 Countably Compact 23
2.2.4 First-Countable 24
2.2.5 Separable & Second-Countable 24
2.2.6 Sequentially Compact 26
2.2.7 Locally Compact & Baire Spaces 26
2.3 Connectedness 28
2.3.1 Connected & Path Connected 28
2.3.2 Components & Path Components 29
2.3.3 Locally Connected & Locally Path Connected . . 30
2.3.4 Quasicomponents 30
2.3.5 Some Theorems 31
2.4 Paracompactness 32
2.4.1 Point-Finite & Locally Finite Collections 32
2.4.2 Point-Finite & Locally Finite Refinements 32
2.4.3 Partitions of Unity 33
2.5 Modifications of Some Properties 33
2.5.1 The Countable Chain Condition 33
2.5.2 Collectionwise Normal 34
2.5.3 Monotonically Normal 35

3 More About Metrizable Spaces 37
3.1 Metrization Theorems 37
3.1.1 Urysohn's Metrization Theorem 37
3.1.2 Metrizable Spaces Are Paracompact 38
3.1.3 Nagata-Smirnov Metrization Theorem 39
3.1.4 G_δ-Diagonals 39
3.1.5 Networks 40
3.1.6 Perfect Images 40
3.2 Completeness 41
3.2.1 Complete Metrics 41
3.2.2 $C^*(X)$ 42
3.2.3 Isometries 43

4 More About Linearly Ordered Spaces 45
4.1 The Cantor Set 45
4.1.1 The Construction 45

4.1.2 A Space-Filling Curve 46
4.2 Characterizing Some Ordered Spaces 46
4.2.1 $\mathbb{R}$ & $[0,1]$ 46
4.2.2 $\mathbb{Q}$ 47
4.2.3 Compact Spaces 47
4.3 The Ordinal ω_1 48
4.3.1 Closed & Unbounded Subsets 48
4.3.2 Stationary Subsets 48
4.4 Characterizing Paracompact Ordered Spaces 49
4.4.1 Cofinality of Limit Ordinals 50
4.4.2 Club & Stationary Subsets of Cardinals 50
4.4.3 The Characterization 51
4.5 Hereditarily Lindelöf but not Separable 51
4.5.1 *ccc* Linearly Ordered Spaces 51
4.5.2 Suslin Lines 52

5 Some Set-Theoretic Topology 53
5.1 Trees 53
5.1.1 The Cantor Tree 53
5.1.2 The Irrationals 54
5.1.3 Aronszajn and Suslin Trees 55
5.2 Martin's Axiom 56
5.2.1 The Statement 56
5.2.2 Working with $\mathsf{MA}(\kappa)$ 57
5.2.3 First Category Subsets of $\mathbb{R}$ 59
5.3 Small Cardinals 59
5.3.1 Unbounded and Dominating Families in ω^ω . . . 59
5.3.2 Subsets of the Irrationals 60
5.4 Normal vs. Collectionwise Normal 61
5.4.1 Tangent Disc Spaces 61
5.4.2 Q-Sets 62
5.5 Countably Compact vs. Pseudocompact 63
5.5.1 Almost Disjoint Families 63
5.5.2 Mrówka Spaces 63
5.6 Hereditarily Separable vs. Hereditarily Lindelöf 64
5.6.1 Some History 64

5.6.2 The Kunen Line 64

6 The Stone-Čech Compactification 67

6.1 Hausdorff Compactifications 67
6.1.1 One-Point Compactifications 67
6.1.2 Compactification by Adding Points 68
6.1.3 Compactification by Embedding in $[0,1]^\kappa$ 68
6.1.4 The Stone-Čech Compactification 69
6.1.5 Four More Results 70
6.2 The Topology of $\beta\omega$ 71
6.2.1 ω^* is not separable 73
6.2.2 A strange property of $\beta\omega$ 73
6.2.3 Closed subsets of $\beta\omega$ 74
6.2.4 P-points 74
6.2.5 ω^* is not homogeneous 75
6.2.6 Products of countably compact spaces 75
6.2.7 Products of sequentially compact spaces 76
6.2.8 Characterizing βX 77
6.2.9 Continua 79

7 Topological Games 81

7.1 Completeness & the Baire Property 81
7.1.1 Complete metric spaces 81
7.1.2 Čech complete 82
7.1.3 The Baire property 83
7.2 The Choquet game 85
7.2.1 Strategies 85
7.2.2 Choquet Spaces 86
7.2.3 Products of Baire Spaces 87
7.3 W-Spaces 88
7.3.1 The W-Game 88
7.3.2 w-spaces 90
7.3.3 w but not W 91

8 Hints 93

9 Proofs 101

1 The Basics

1.1 Some Set Theory We Should Know

1.1.1 Cardinality & Cardinal Numbers

Two sets A and B are said to have the same *cardinality*, and we write $|A| = |B|$, if there exists a bijection $f\colon A \to B$. We also say $|A| \leqslant |B|$ if there exists an injection $f\colon A \to B$. The *Schröder-Bernstein Theorem* says: If there are one-to-one maps $f\colon A \to B$ and $g\colon B \to A$, then $|A| = |B|$.

A set is called *countable* if it is either finite or has the same cardinality as the set $\mathbb{N}$ of positive integers. To specify countable but not finite, we say *countably infinite*.

Theorem 1.

(a) *A countable union of countable sets is countable.*

(b) *If $A_1, A_2, \ldots, A_n$ are countable, so is $A_1 \times \cdots \times A_n$.*

(c) *If A is countable, so is the set of all finite subsets of A, as well as the set of all finite sequences of elements of A.*

(d) *The set $\mathbb{Q}$ of all rational numbers is countable.*

Theorem 2. *The following sets have the same cardinality as the set $\mathbb{R}$ of real numbers:*

(a) *the set $\mathcal{P}(\mathbb{N})$ of all subsets of the natural numbers $\mathbb{N}$;*

(b) *the set of all functions $f\colon \mathbb{N} \to \{0, 1\}$;*

(c) *the set of all infinite sequences of 0's and 1's;*

(d) *the set of all infinite sequences of real numbers.*

The cardinality of $\mathbb{N}$ (and any countable infinite set) is denoted by $\aleph_0$. $\aleph_1$ denotes the next infinite cardinal, $\aleph_2$ the next, etc. *Cantor's Continuum Hypothesis* (CH) says that the cardinality of $\mathbb{R}$ is $\aleph_1$; it turns out that CH is undecidable from the usual axioms (ZFC) of set theory. The next theorem shows that for any cardinal, there is a bigger one.

★ **Theorem 3.** *For any set X, we have $|X| < |\mathcal{P}(X)|$, where $\mathcal{P}(X)$ is the set of all subsets of X.*

If the set X has cardinality κ, then the cardinal 2^κ is defined to be the cardinality of the set $\mathcal{P}(X)$—which has the same cardinality as the set of all functions from X into $\{0, 1\}$, by the same argument that gets Theorem 2(b). So $2^{\aleph_0}$ is the cardinality of $\mathbb{R}$, and CH can be stated as $2^{\aleph_0} = \aleph_1$.

1.1.2 Ordinal Numbers

A *relation* R on a set X is a subset of $X \times X$; we write xRy to mean $\langle x, y\rangle \in R$. A *(strict) partial order*[1] on X is a relation $<$ satisfying, for every $x, y, z \in X$:

(1) $x < y$ and $y < x$ are not both true, so $<$ is *antireflexive*; and

(2) $x < y$ and $y < z$ implies $x < z$, so $<$ is *transitive*.

If $<$ is a relation, we may write $x \leqslant y$ to mean "$x < y$ or $x = y$." We say that $<$ is a *linear order* if it satisfies

(3) the *trichotomy property*: for any $x, y \in X$, exactly one of $x < y$, $y < x$, or $x = y$ holds.

And finally, a linear order $<$ is a *well-order* if

(4) every non-empty subset of X has a $<$-least element.

[1] A *non-strict partial order* on X is a relation $\leqslant$ on X which is *reflexive* (so $x \leqslant x$ for all x), *antisymmetric* (so $x \leqslant y$ and $y \leqslant x$ if and only if $x = y$), and transitive. We will just say that a set is *partially ordered* and let context indicate if the relation is strict or not.

If the well-order is understood, we may simply say a least element.

Two linearly ordered sets $\langle A, <_A \rangle$ and $\langle B, <_B \rangle$ are *order-isomorphic* if there is an *order-preserving* bijection $h\colon A \to B$; that is, $x <_A y$ if and only if $h(x) <_B h(y)$.

We will assume the following:

Theorem 4. *There is a class* **Ord** *and a linear ordering* $<$ *on* **Ord** *with the following properties:*

(a) *Every nonempty subset of* **Ord** *has a least element;*

(b) *Given any well-ordered set* $\langle X, \prec \rangle$*, there exists a unique* $\alpha \in \mathbf{Ord}$ *such that* $\langle X, \prec \rangle$ *is order-isomorphic to* $\langle P_\alpha, < \rangle$*, where* P_α *is the set* $\{\beta \in \mathbf{Ord} : \beta < \alpha\}$ *of predecessors of* α*.*

Elements of **Ord** are called *ordinals*. Intuitively, an ordinal number denotes position in a well-ordered sequence. The least ordinal is denoted by 0, the next by 1, the next by 2, etc. The least ordinal with infinitely many predecessors, that is, the least ordinal greater than all the finite ordinals $0, 1, 2, \ldots$, is denoted by ω. Then comes $\omega + 1, \omega + 2, \ldots$. The least ordinal bigger than $\omega + n$ for every $n < \omega$ is denoted by $\omega \cdot 2$. One can similarly define $\omega \cdot 3, \omega \cdot 4, \ldots$. The least ordinal bigger than $\omega \cdot n$ for all $n < \omega$ is denoted by $\omega \cdot \omega = \omega^2$. Etc.

Note that all ordinals mentioned in the above paragraph have only countably many predecessors. Such ordinals are called *countable ordinals*. The least ordinal with uncountably many predecessors, that is, the least ordinal greater than all of the countable ordinals, is denoted by ω_1. A better definition of $\aleph_1$ than the one given previously is that $\aleph_1$ denotes the cardinality of the set of predecessors of ω_1. $\aleph_2, \aleph_3, \ldots$ may be defined similarly. See below where $\aleph_\alpha$ is defined by transfinite induction for any ordinal α. Denote by **Card** the class of all cardinals $\aleph_0, \aleph_1, \aleph_2, \ldots$.

The position in modern set theory, and the position we shall take advantage of, is that an ordinal *is* the set of its predecessors. Thus $0 = \varnothing$, $1 = \{0\}$, $2 = \{0, 1\}$, $\ldots$, $\omega = \{0, 1, 2, \ldots\}$, $\omega + 1 = \omega \cup \{\omega\}$, etc. Moreover, for $\alpha, \beta \in \mathbf{Ord}$ we have $\alpha < \beta$ if and only if $\alpha \in \beta$. This viewpoint gives us the following, though our limited definitions may make it difficult to prove without the hint.

★ **Theorem 5.** *If A is a set of ordinals, then $\sup A = \bigcup A$ is the least ordinal greater than every member of A.*

Corollary 6. *If C is any countable subset of ω_1, then there is a countable ordinal α such that $\alpha \geqslant \beta$ for all $\beta \in C$.*

The following notation will be used throughout. If A and B are sets, α is an ordinal, and κ is a cardinal, then

(1) $[A]^{\kappa} = \{X \subset A : |X| = \kappa\}$;

(2) $[A]^{<\kappa} = \bigcup \{[A]^{\lambda} : \lambda < \kappa\}$;

(3) $[A]^{\leqslant\kappa} = [A]^{<\kappa} \cup [A]^{\kappa}$;

(4) B^A is the set of all functions from A to B, so A^{α} is the set of all functions from α to A (which is also the set of all sequences of length α in A);

(5) $A^{<\alpha} = \bigcup \{A^{\beta} : \beta < \alpha\}$; and

(6) $A^{\leqslant\alpha} = A^{<\alpha} \cup A^{\alpha}$.

1.1.3 Transfinite Induction

Theorem 7, Principle of Transfinite Induction. *Let κ be an ordinal. Suppose we have for each $\alpha < \kappa$ a statement S_{α} such that:*

(1) *S_0 is true;*

(2) *If $\beta < \kappa$ and S_{α} is true for every $\alpha < \beta$, then S_{β} is true.*

Then S_{γ} is true for every $\gamma < \kappa$.

★ **Theorem 8, Principle of Definition by Transfinite Induction.** *Let κ be an ordinal, B a set, and $\mathcal{F}$ the set of all functions (including the empty function) into B whose domain is the set $\{\beta \in \mathbf{Ord} : \beta < \alpha\}$ for some $\alpha < \kappa$. Let $R\colon \mathcal{F} \to B$. Then there is a unique function G with domain $\{\alpha \in \mathbf{Ord} : \alpha < \kappa\}$ such that $G(\alpha) = R(G \restriction \{\beta \in \mathbf{Ord} : \beta < \alpha\})$ for all $\alpha < \kappa$.*

Interpret this as follows: Let $f \in \mathcal{F}$. Then f is a function whose domain is the predecessors of some ordinal α. Think of f as a possibility for what you have done in the induction prior to α. Then R is a "recipe" for what you then do at step α: you do $R(f)$. The conclusion of the principle says that there is a unique function G whose domain is the predecessors of κ (which is the ordinal you are inducting on; $\kappa = \omega$ in ordinary induction) such that $G(\alpha)$ is always what the recipe tells you to do at step α, given $G(\beta)$ for all $\beta < \alpha$.

What do you do at the first step in the induction? Well, you haven't done anything yet, so you do $R(\varnothing)$. And note that

$$G(0) = R(G \restriction \{\beta \in \mathbf{Ord} : \beta < 0\}) = R(G \restriction \varnothing) = R(\varnothing).$$

So given R, $G(0)$ is uniquely defined! What next? Note that we want $G(1) = R(G \restriction \{\beta : \beta < 1\}) = R(G \restriction \{0\})$, which we see now is uniquely defined since $G(0)$ is uniquely defined. And so on.

The Transfinite Induction Principle and the Principle of Definition by Transfinite Induction also hold if in their statements we replace κ by the class **Ord** of all ordinals.

Now the Principle of Definition by Transfinite Induction justifies defining ω_α and $\aleph_\alpha$ for any ordinal α as follows: $\aleph_0$ is as defined previously, and $\omega_0 = \omega$. Suppose α is an ordinal and ω_β and $\aleph_\beta$ have been defined for every $\beta < \alpha$.

(1) If α has an immediate predecessor γ (so $\alpha = \gamma + 1$), then let ω_α be the least ordinal with more than $\aleph_\gamma$-many predecessors, and let $\aleph_\alpha$ denote the cardinality of ω_α.

(2) If α has no immediate predecessor, in which case α is called a *limit ordinal*, then let ω_α be the least ordinal greater than ω_β for every $\beta < \alpha$. Again, let $\aleph_\alpha$ denote that cardinality of the set of predecessors of ω_α.

Exercise 9. *For $\alpha < \omega_1$, define collections $\mathcal{B}_\alpha$ of subsets of a set X as follows: $\mathcal{B}_0$ is an arbitrary collection of subsets of X. If $\mathcal{B}_\alpha$ has been defined, then let $\mathcal{B}_{\alpha+1} = \mathcal{B}_\alpha \cup \{X \setminus A : A \in \mathcal{B}_\alpha\} \cup \{\bigcap \mathcal{A} : \mathcal{A} \in [\mathcal{B}_\alpha]^{\leqslant \aleph_0}\}$. Finally, if α is a limit ordinal let $\mathcal{B}_\alpha = \bigcup_{\beta<\alpha} \mathcal{B}_\beta$.*

Let $\mathcal{B} = \bigcup_{\alpha<\omega_1} \mathcal{B}_\alpha$. Prove that $\mathcal{B}$ is closed under complements, countable intersections, and countable unions.

If X is a topological space and $\mathcal{B}_0$ is the collection of all open sets of X, then $\mathcal{B}$ is precisely the collection of all *Borel* subsets of X. A standard definition of this collection is that it is the smallest σ-algebra containing the open sets, where a collection $\mathcal{B}$ of subsets of a set X is a *σ-algebra* if it is closed under complements, countable intersections, and countable unions.

1.1.4 Axiom of Choice & Equivalents

The Axiom of Choice (AC) says: given any collection $\mathcal{A}$ of nonempty sets, there is a function $f\colon \mathcal{A} \to \bigcup \mathcal{A}$ such that $f(A) \in A$ for every $A \in \mathcal{A}$; such an f is called a *choice function* for $\mathcal{A}$.

Zorn's Lemma (ZL) works as follows. Suppose $\langle P, < \rangle$ is a partially ordered set with the following property: If L is any subset of P linearly ordered by $<$, then there is $p \in P$ such that $\ell \leqslant p$ for any $\ell \in L$. Then there is an element q in P such that no member of P is strictly greater than q. Colloquially: if every *chain* in the partial order $\langle P, < \rangle$ is bounded above in P, then P has a *maximal element*.

Theorem 10. *The following are equivalent:*

(a) *The Axiom of Choice;*

(b) *Zorn's Lemma;*

(c) *Every set can be well-ordered.*

1.2 Topological Spaces

A *topological space* is an ordered pair $\langle X, \mathcal{T} \rangle$, where X is a set and $\mathcal{T}$ is a collection of subsets of X such that

(1) $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$;

(2) $U \cap V \in \mathcal{T}$ whenever $U, V \in \mathcal{T}$; and

(3) $\bigcup \mathcal{U} \in \mathcal{T}$ whenever $\mathcal{U} \subset \mathcal{T}$.

$\mathcal{T}$ is called a *topology* on the set X; elements of $\mathcal{T}$ are called *open sets*. To put it in other words, (2) says that pairwise (and hence finite) intersections of open sets are open, and (3) says that any union of open sets is open.

We will often write X *is a topological space* or just X *is a space* to mean that X is a set with some fixed but unspecified topology; if we say U *is open in* X, we mean that U is a member of whatever topology on X we happen to be working with.

1.2.1 Closed Sets

If X is a space, and $A \subset X$, then A is *closed* if and only if $X \setminus A$ is open.

Lemma 11, De Morgan's Laws. *If X is any set and $\mathcal{A}$ is any collection of subsets of X, then*

(a) $X \setminus \bigcup \mathcal{A} = \bigcap \{X \setminus A : A \in \mathcal{A}\}$ *and*

(b) $X \setminus \bigcap \mathcal{A} = \bigcup \{X \setminus A : A \in \mathcal{A}\}$.

Corollary 12. *In any space X, the sets $\varnothing$ and X are closed, the union of finitely many closed sets is closed, and the intersection of any number of closed sets is closed.*

Let X be a space and $A \subset X$. The *closure* of A, denoted $\overline{A}$ or $\operatorname{cl} A$, is the intersection of all closed sets containing A; note $\overline{A}$ is closed by Corollary 12, and A is closed if and only if $\overline{A} = A$. We can think of the closure of A as being the "smallest" closed set containing A.

If $x \in X$ and every open set containing x contains a point of A other than x, we say that x is a *limit point* of A; the set of limit points of A is denoted by A^d or $\lim A$.

Theorem 13. *For any subset A of a space X we have $\overline{A} = A \cup A^d$.*

Theorem 14. *If A and B are subsets of a space X, then*

(a) $\overline{\varnothing} = \varnothing$,

(b) $A \subset B \to \overline{A} \subset \overline{B}$,

(c) $\overline{\overline{A}} = \overline{A}$, *and*

(d) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Theorem 15. *If* $\{A_\alpha : \alpha \in \Lambda\}$ *is any collection of subsets of a space* X*, then*

(a) $\overline{\bigcap_{\alpha\in\Lambda} A_\alpha} \subset \bigcap_{\alpha\in\Lambda} \overline{A}_\alpha$ *and*

(b) $\overline{\bigcup_{\alpha\in\Lambda} A_\alpha} \supset \bigcup_{\alpha\in\Lambda} \overline{A}_\alpha$

1.2.2 Interiors & Boundaries

The *interior* of A, denoted A° or $\operatorname{int} A$, is the union of all open sets contained in A; we can think of the interior of A as the "largest" open set contained in A. If $x \in A^\circ$ we say that A is a *neighborhood* of x, so $\overline{A}$ consists of those points whose every neighborhood meets A.

Theorem 16. *For any subset* A *of a space* X*:*

(a) $A^\circ = X \setminus \overline{X \setminus A}$ *and*

(b) $\overline{A} = X \setminus (X \setminus A)^\circ$.

The *boundary* of A, denoted ∂A or $\operatorname{bd} A$, consists of those points whose every neighborhood meets both A and $X \setminus A$.

Theorem 17. *For any subset* A *of a space* X*:*

(a) $\partial\partial A \subset \partial A$*;*

(b) $\partial A = \overline{A} \cap \overline{X \setminus A} = \overline{A} \setminus A^\circ$*, whence* ∂A *is always closed;*

(c) A *is closed if and only if* $\partial A \subset A$*;*

(d) $\partial A \cap A^\circ = \varnothing$*, and* A *is open if and only if* $\partial A \cap A = \varnothing$*;*

(e) A *is closed-and-open if and only if* $\partial A = \varnothing$

The *isolated points* of A are those $x \in A$ having a neighborhood U for which $A \cap U = \{x\}$; if every point of A is isolated we say A is a *discrete* subset of X, and if every point of X is isolated then X has the *discrete topology.*

Theorem 18. *Let A be a subset of a space X, and let A' consist of the isolated points of A. Then*

(a) $A^d \cap A' = \varnothing$,

(b) $\overline{A} = A^d \cup A'$, *and*

(c) *the isolated points of $\overline{A}$ are contained in A'.*

1.2.3 Bases

Let $\langle X, \mathcal{T} \rangle$ be a topological space, and let $\mathcal{B}$ be a subcollection of $\mathcal{T}$. We say $\mathcal{B}$ is a *base*, and refer to its members as *basic* open sets, for the topological space $\langle X, \mathcal{T} \rangle$ if and only if every nonempty member of $\mathcal{T}$ is a union of members of $\mathcal{B}$.

A subcollection $\mathcal{S}$ of $\mathcal{T}$ is a *subbase* for $\langle X, \mathcal{T} \rangle$ if the set of finite intersections in $\mathcal{S}$ is a base for $\langle X, \mathcal{T} \rangle$.

Theorem 19. *Let X be a set and let $\mathcal{B}$ be a collection of subsets of X. Then there is a (unique) topology $\mathcal{T}$ on X such that $\mathcal{B}$ is a base for $\langle X, \mathcal{T} \rangle$ if and only if*

(a) *Each $x \in X$ is in some member of $\mathcal{B}$;*

(b) *Whenever $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, there exists $B \in \mathcal{B}$ with $x \in B \subset B_1 \cap B_2$.*

Collections $\mathcal{B}_1$ and $\mathcal{B}_2$ of subsets of a set X are *equivalent bases* if and only if there is a topology $\mathcal{T}$ on X such that $\mathcal{B}_1$ and $\mathcal{B}_2$ are both bases for $\langle X, \mathcal{T} \rangle$.

Theorem 20. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, let $\mathcal{B}_1$ be a base for $\langle X, \mathcal{T} \rangle$, and let $\mathcal{B}_2$ be a collection of subsets of X. If*

(1) $x \in B_1 \in \mathcal{B}_1 \to \exists B_2 \in \mathcal{B}_2 (x \in B_2 \subset B_1)$, *and*

(2) $x \in B_2 \in \mathcal{B}_2 \to \exists B_1 \in \mathcal{B}_1 (x \in B_1 \subset B_2)$,

then $\mathcal{B}_1$ and $\mathcal{B}_2$ are equivalent bases.

Theorem 21. *Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for the topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a set X. Then $\mathcal{T}_2$ is finer than $\mathcal{T}_1$ if and only if every member of $\mathcal{B}_1$ is a union of members of $\mathcal{B}_2$.*

1.2.4 Subspaces

A *subspace* of a topological space $\langle X, \mathcal{T} \rangle$ is a pair $\langle A, \mathcal{T}_A \rangle$, where $A \subset X$ and $\mathcal{T}_A = \{U \cap A : U \in \mathcal{T}\}$; $\mathcal{T}_A$ is called the *subspace topology on A*, the *relative topology* on A, or the *topology on A induced by $\mathcal{T}$*.

Theorem 22. *Let A be a subspace of a space X.*

(a) *H is closed in A if and only if $H = A \cap K$ for some K closed in X.*

(b) *If U is open in A and A is open in X, then U is open is X.*

(c) *If K is closed in A and A is closed in X, then K is closed in X.*

Theorem 23. *If $\langle X, \mathcal{T} \rangle$ is a topological space, $\mathcal{B}$ a base for $\mathcal{T}$, and $A \subset X$, then the collection $\{B \cap A : B \in \mathcal{B}\}$ is a base for $\langle A, \mathcal{T}_A \rangle$.*

Theorem 24. *Let A be a subspace of a space X, and let $H \subset A$. Then $\mathrm{cl}_A H = \mathrm{cl}_X H \cap A$, where $\mathrm{cl}_A H$ indicates we are taking the closure of H in the subspace A.*

1.2.5 Continuous Functions

The following notation will help us differentiate between applying a function to a point which is a set and applying a function to each point of a set: if $f\colon X \to Y$, $A \subset X$, and $B \subset Y$, then $f[A] = \{f(a) : a \in A\}$ and $f^{-1}[B] = \{x \in X : f(x) \in B\}$.

If $f\colon X \to Y$, where X and Y are spaces, and $x_0 \in X$, we say f *is continuous at* x_0 if, given any open set V in Y with $f(x_0) \in V$, there is

an open set U in X with $x_0 \in U$ and $f[U] \subset V$. Also, f *is continuous* if f is continuous at every $x_0 \in X$.

Theorem 25. *If $f\colon X \to Y$, where X and Y are spaces, then the following are equivalent:*

(a) *f is continuous;*

(b) *$f^{-1}[V]$ is open in X whenever V is subbasic open in Y;*

(c) *$f^{-1}[V]$ is open in X whenever V is basic open in Y;*

(d) *$f^{-1}[V]$ is open in X whenever V is open in Y;*

(e) *$f^{-1}[K]$ is closed in X whenever K is closed in Y; and*

(f) *whenever $A \subset X$ and $x \in \overline{A}$, then $f(x) \in \overline{f[A]}$ (in other words, $f[\overline{A}] \subset \overline{f[A]}$).*

Theorem 26. *Let A be a subspace of X.*

(a) *The inclusion map $j\colon A \to X$ is continuous.*

(b) *If $f\colon X \to Y$ is continuous, then the restriction of f to A, denoted $f \restriction A$, is a continuous function from A to Y.*

Theorem 27. *Constant functions are continuous, and the composition of continuous functions is continuous.*

Theorem 28.

(a) *Let $f\colon X \to Y$, where $X = \bigcup_{\alpha\in\Lambda} U_\alpha$ with each U_α open. If $f \restriction U_\alpha$ is continuous for each $\alpha \in \Lambda$, then f is continuous.*

(b) *Let $X = H \cup K$ with H and K closed, and let $f\colon H \to Y$ and $g\colon K \to Y$ be continuous. If $f \restriction (H \cap K) = g \restriction (H \cap K)$, then $h\colon X \to Y$ defined so that $h \restriction H = f$ and $h \restriction K = g$ is continuous.*

1.2.6 Homeomorphisms

A function $f\colon X \to Y$ is *closed* (resp., *open*) if the image of every closed set (resp., open set) in X is closed (resp., open) in Y.

A function $h\colon X \to Y$ is a *homeomorphism* if h is one-to-one, onto, and both h and h^{-1} are continuous. If such a homeomorphism exists then the spaces X and Y are said to be *homeomorphic*, denoted by $X \cong Y$.

Theorem 29. *If $h\colon X \to Y$ is one-to-one and onto, then the following are equivalent:*

(a) *h is a homeomorphism;*

(b) *h is open and continuous; and*

(c) *h is closed and continuous.*

Exercise 30. *Find spaces X and Y and a mapping $f\colon X \to Y$ which is one-to-one, onto, and continuous, but not a homeomorphism.*

1.2.7 Convergent Sequences

Let X be a topological space, $x \in X$, and $\langle x_n \rangle_{n\in\mathbb{N}}$ a sequence of points in X. We say $\langle x_n \rangle$ *converges to* x, and write $\langle x_n \rangle \to x$, if every (open) neighborhood of x contains x_n for sufficiently large $n \in \mathbb{N}$; that is, for any (open) neighborhood U of x, there is $k \in \mathbb{N}$ such that $x_n \in U$ for all $n \geqslant k$. Alternatively, we may say that each neighborhood of x contains a *tail* of the sequence.

Theorem 31. *If $f\colon X \to Y$ is continuous and $\langle x_n \rangle_{n\in\mathbb{N}} \to x$ in X, then $\langle f(x_n) \rangle \to f(x)$ in Y.*

1.3 Some Types of Spaces

1.3.1 Metric Spaces

Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is called a *metric* on X if, for each $x, y, z \in X$, d satisfies:

(1) $d(x, y) \geqslant 0$ and $d(x, y) = 0$ if and only if $x = y$;

(2) $d(x, y) = d(y, x)$; and

(3) $d(x, z) \leqslant d(x, y) + d(y, z)$.

Conditions (2) and (3) say that d is *symmetric* and satisfies the *triangle inequality*, respectively.

A *metric space* is a pair $\langle X, d\rangle$, where X is a set and d is a metric on X. If $\varepsilon > 0$, then $B_d(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}$ is the *ε-ball about x (with respect to d)*; if it is understood what metric we are talking about, we may omit the subscript d and write $B(x, \varepsilon)$.

Lemma 32. *Let $\langle X, d\rangle$ be a metric space, and let $y \in B_d(x, \varepsilon)$. Then there exists $\delta > 0$ such that $B_d(y, \delta) \subset B_d(x, \varepsilon)$.*

Lemma 33. *Let $\langle X, d\rangle$ be a metric space. Then the collection of all ε-balls is a base for a topology on X.*

Let $\langle X, d\rangle$ be a metric space; the topology on X given in Lemma 33 is called the *metric topology (generated by the metric d)*. If d and d' are two metrics on the same set X which generate the same topology, then d and d' are called *equivalent metrics*. A topological space $\langle X, \mathcal{T}\rangle$ is said to be *metrizable* if there is a metric on X which generates $\mathcal{T}$.

Corollary 34. *Let $\langle X, d\rangle$ be a metric space. Then a subset U of X is open in the metric topology if and only if, for every $x \in U$, there is $\varepsilon > 0$ such that $B_d(x, \varepsilon) \subset U$.*

Theorem 35. *Let $\langle X, d\rangle$ be a metric space, and let $Y \subset X$. Let d_Y be the metric d restricted to $Y \times Y$. Then $\langle Y, d_Y\rangle$ is a metric space, and the metric topology on Y is the same as the subspace topology with respect to the metric topology on X.*

1.3.2 Product Spaces

Let $\{\langle X_\alpha, \mathcal{T}_\alpha\rangle : \alpha \in \Lambda\}$ be a collection of spaces; put $X = \prod_{\alpha\in\Lambda} X_\alpha$. A point of X can be thought of as a sequence $\vec{x} = \langle x_\alpha\rangle_{\alpha\in\Lambda}$, where

$x_\alpha \in X_\alpha$ for all α, or as a function $f\colon \Lambda \to \bigcup_{\alpha\in\Lambda} X_\alpha$ with $f(\alpha) \in X_\alpha$ for all α.

Denote by $\mathcal{B}$ be the collection of all sets of the form $\prod_{\alpha\in\Lambda} U_\alpha$, where $U_\alpha \in \mathcal{T}_\alpha$ for all α, and $U_\alpha = X_\alpha$ for cofinitely[2] many α. Then $\mathcal{B}$ is a base for a topology on X; this topology is called the *product topology*, or the *Tychonoff (product) topology*, and X with this topology is called the *product space* of $\{\langle X_\alpha, \mathcal{T}_\alpha\rangle : \alpha \in \Lambda\}$. Each X_α is called a *coordinate space* of X.

If F is a finite subset of Λ, we may write a typical basic open set in the product space as $B = \prod_{\alpha\in F} U_\alpha \times \prod_{\alpha\in\Lambda\setminus F} X_\alpha$, where U_α is open in X_α. The finite set F is sometimes called the *support* of the basic open set B.

Observe: if for each α, $\mathcal{B}_\alpha$ is a base for X_α, then the collection of all sets of the form $\prod_{\alpha\in\Lambda} B_\alpha$, where $B_\alpha = X_\alpha$ for cofinitely many α and $B_\alpha \in \mathcal{B}_\alpha$ if $B_\alpha \neq X_\alpha$, is an equivalent base for the product topology.

For each $\alpha \in \Lambda$, the function $\pi_\alpha\colon X \to X_\alpha$ defined by $\vec{x} \mapsto x_\alpha$ is called the *projection map onto* X_α. Note that if $U_\alpha \subset X_\alpha$, then $\pi_\alpha^{-1}[U_\alpha] = \prod_{\beta\in\Lambda} V_\beta$, where $V_\alpha = U_\alpha$ and $V_\beta = X_\beta$ if $\beta \neq \alpha$. Thus a typical basic open set in the product topology can be denoted by $\bigcap_{i\leqslant n} \pi_{\alpha_i}^{-1}[U_{\alpha_i}]$, where $\alpha_i \in \Lambda$ and U_{α_i} is open in X_{α_i} for each $i \leqslant n$. So, the collection

$$\{\pi_\alpha^{-1}[U_\alpha] : \alpha \in \Lambda \wedge U_\alpha \in \mathcal{T}_\alpha\}$$

is a subbase for the product topology.

Theorem 36. *Projection maps are always continuous and open.*

Theorem 37. *Let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be a product space, and let $\vec{x} \in X$. Fix $\gamma \in \Lambda$. Then the "cross-section" $\{\vec{y} \in X : \forall\alpha \neq \gamma(y_\alpha = x_\alpha)\}$ is homeomorphic to X_γ.*

Theorem 38. *Let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be a product space, and let $A_\alpha \subset X_\alpha$ for all $\alpha \in \Lambda$. Then $\overline{\prod_{\alpha\in\Lambda} A_\alpha} = \prod_{\alpha\in\Lambda} \overline{A}_\alpha$.*

[2] Or, all but finitely.

Lemma 39. *Let $\langle X, d\rangle$ be a metric space and let $M > 0$. Define d' by $d'(x, y) = M$ if $d(x, y) \geqslant M$ and $d'(x, y) = d(x, y)$ otherwise. Then d' is an equivalent metric on X.*

Theorem 40. *Let $\{\langle X_n, d_n\rangle : n \in \mathbb{N}\}$ be metric spaces such that, for each $n \in \mathbb{N}$, $d_n(x, y) \leqslant 1/2^n$ for every $x, y \in X_n$. For $\vec{x}, \vec{y} \in X = \prod_{n\in\mathbb{N}} X_n$, define $d(\vec{x}, \vec{y}) = \max\{d_n(x_n, y_n) : n \in \mathbb{N}\}$. Then d is a metric on X and the topology generated by d is the same as the product topology.*

Corollary 41. *A countable product of metrizable spaces is metrizable.*

1.3.3 Quotient Spaces

Let X and Y be spaces, and $q\colon X \to Y$ a surjective map. Then q is called a *quotient map* provided U is open in Y if and only if $q^{-1}[U]$ is open in X. Note that quotient maps are continuous.

Theorem 42. *Open continuous surjections and closed continuous surjections are quotient maps.*

Theorem 43. *Let $q\colon X \to Y$ be a quotient map, and $f\colon Y \to Z$ arbitrary. Then f is continuous if and only if $f \circ q\colon X \to Z$ is continuous.*

Let $q\colon X \to Y$ be a quotient map. A subset U of X is called *saturated* if $U = q^{-1}[q[U]]$; equivalently, U meets a *fiber* $q^{-1}(y)$ if and only if U contains it.

Theorem 44. *Let $q\colon X \to Y$ be a quotient map. Then the collection of open sets of Y is equal to $\{q[U] : U$ is a saturated open subset of $X\}$.*

Let $\langle X, \mathcal{T}\rangle$ be a space, Y a set, and $f\colon X \to Y$ a surjection. Define $\mathcal{T}_f = \{U \subset Y : f^{-1}[U] \in \mathcal{T}\}$. Then $\mathcal{T}_f$ is easily seen to be a topology on Y, under which f is a quotient map. $\mathcal{T}_f$ is called the *quotient topology induced by f*.

Let X be a space, and let $\mathcal{D}$ be a partition of X (that is, $\mathcal{D}$ is a pairwise-disjoint cover of X). Define $p\colon X \to \mathcal{D}$ so that $p(x) = D$ if

and only if $x \in D$. Then $\mathcal{D}$ with the quotient topology induced by p is called a *quotient space* of X. We say that $\mathcal{D}$ is the space obtained from X by identifying points $x, y \in X$ if they are in the same member of $\mathcal{D}$. If $\mathcal{D}$ has only one element A with more than one point, the resulting quotient space is also denoted by X/A.

Let X be a space and let $\sim$ be an equivalence relation on X. Then $X/\sim$ denotes the quotient space of X where $\mathcal{D}$ is the collection of $\sim$-equivalence classes.

For example, let $X = [0,1] \times [0,1]$.

(1) If we identify $\langle x, 0\rangle$ with $\langle x, 1\rangle$ for each $x \in [0,1]$, the resulting quotient space is a cylinder.

(2) If we also identify $\langle 0, y\rangle$ with $\langle 1, y\rangle$ for each y, the resulting quotient space is a torus.

(3) If we identify as in (1), and also identify $\langle 0, y\rangle$ with $\langle 1, 1-y\rangle$, the result is a *Klein bottle*.

(4) If we identify $\langle x, 0\rangle$ with $\langle 1-x, 1\rangle$, the resulting quotient space is a *Möbius strip*.

(5) If we identify as in (4), and also identify $\langle 0, y\rangle$ with $\langle 1, 1-y\rangle$, the result is a *real projective plane*. We also get a projective plane if we identify antipodal points of (the surface of) a sphere.

2 Topological Properties

2.1 Separation Axioms

2.1.1 T_0, T_1, and T_2

A space X is a T_0-*space* if whenever $x, y \in X$ with $x \neq y$, there is an open set containing one of these points but not the other. If there is always an open set containing x and missing y, then X is a T_1-*space*. If there are always disjoint open sets containing x and y, then X is a T_2-*space* or a *Hausdorff space*.

Theorem 45. *A space X is a T_1-space if and only if every point of X is a closed set.*

2.1.2 T_3, $T_{3.5}$, and T_4

A space X is a T_3-*space*, or *regular space*, if X is a T_1-space, and whenever $x \in X$ and H is a closed set not containing x, then there are disjoint open sets containing x and H, respectively.

Theorem 46. *A T_1-space X is regular if and only if*

(a) *whenever $x \in X$ and H is a closed subset of X not containing x, there exists an open set U containing x whose closure misses H; or,*

(b) *whenever x is in an open V, there is an open set U containing x such that $\overline{U} \subset V$.*

A space X is a $T_{3.5}$-*space*, or *completely regular space*, if X is a T_1-space, and whenever $x \in X$ and H is a closed set not containing x, there

is a continuous function $f\colon X \to [0,1]$ with $f(x) = 0$ and $f(y) = 1$ for all $y \in H$.

X is a T_4-*space*, or *normal space*, if X is a T_1-space and any two disjoint closed sets are contained in disjoint open sets.

Theorem 47. *A T_1-space X is normal if and only if*

(a) *whenever H and K are disjoint closed subsets of X, there exists an open set U such that $U \supset H$ and $\overline{U} \cap K = \varnothing$; or,*

(b) *whenever a closed set H is contained in an open set V, there is an open set U containing H such that $\overline{U} \subset V$.*

Some authors don't include T_1 in the definitions of regular or normal, and define T_3 to be T_1 plus regular, and T_4 to be T_1 plus normal. Still others do as in the previous sentence but with T_3 and regular swapped, and T_4 and normal swapped, in which case T_3 and T_4 don't imply T_1!

Theorem 48. $T_4 \to T_3 \to T_2 \to T_1 \to T_0$ *and* $T_{3.5} \to T_3$.

Theorem 49. *Metrizable spaces are normal.*

Theorem 50. *Let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be a product space. If each X_α is a T_i-space, where $i = 0, 1, 2, 3$ or 3.5, then so is X.*

2.1.3 Urysohn's Lemma

★ **Theorem 51.** *Suppose H and K are disjoint closed subsets of a normal space X. Then one can assign to each rational number r with $0 < r < 1$ an open set U_r such that*

(a) $H \subset U_r$ *and* $\overline{U}_r \cap K = \varnothing$;

(b) $r < r' \to \overline{U}_r \subset U_{r'}$.

Theorem 52, Urysohn's Lemma. *The following are equivalent for a T_1-space X:*

(a) *X is normal;*

(b) *Whenever H and K are disjoint closed subsets of X, there exists a continuous function $f\colon X \to [0,1]$ with $f(x) = 0$ for all $x \in H$ and $f(x) = 1$ for all $x \in K$.*

Corollary 53. *$T_4 \to T_{3.5}$; that is, every normal space is completely regular.*

Theorem 54, Tietze Extension Theorem. *Suppose H is a closed subset of a normal space X. Then any continuous function $f\colon H \to \mathbb{R}$ can be extended to a continuous function $\hat{f}\colon X \to \mathbb{R}$. Furthermore, the range of $\hat{f}$ can be made to be contained in the convex hull of the range of f.*

2.1.4 Completely Normal

Two subsets A and B of a space X are *separated* if $\overline{A} \cap B = \varnothing$ and $\overline{B} \cap A = \varnothing$; that is, no point of any one of the sets is in the closure of the other set. A T_1-space X is said to be *completely normal* if given any two separated sets A and B, there are disjoint open sets containing A and B, respectively.

Theorem 55. *A space X is completely normal if and only if every subspace of X is normal.*

Corollary 56. *Metrizable spaces are completely normal.*

If P is a property of topological spaces, we say that a space X is *hereditarily P* if and only if every subspace of X has property P. For example, "completely normal" is equivalent to "hereditarily normal" (and the property is sometimes called that).

2.1.5 Perfectly Normal

Lemma 57. *Suppose A and B are subsets of a space X, and that there are open sets $U_1, U_2, \ldots$ and $V_1, V_2, \ldots$ such that:*

(a) *$A \subset \bigcup_{n\in\mathbb{N}} U_n$ and $B \subset \bigcup_{n\in\mathbb{N}} V_n$; and*

(b) *for each* $n \in \mathbb{N}$, $\overline{U}_n \cap B = \varnothing = \overline{V}_n \cap A$.

Then there are disjoint open sets U *and* V *containing* A *and* B*, respectively.*

X is said to be *perfectly normal* if X is normal and every closed set is G_δ (a countable intersection of open sets).

Theorem 58. *Metrizable spaces are perfectly normal.*

Theorem 59, the Vedenissoff Theorem. *The following are equivalent for a* T_1*-space* X*:*

(a) X *is perfectly normal;*

(b) *For any closed subset* H *of* X*, there is a collection* $\{U_n : n \in \mathbb{N}\}$ *of open sets such that* $H = \bigcap_{n\in\mathbb{N}} U_n = \bigcap_{n\in\mathbb{N}} \overline{U}_n$*;*

(c) *For any closed subset* H *of* X*, there exists a continuous function* $f\colon X \to [0,1]$ *such that* $H = f^{-1}(0)$*;*

(d) *Whenever* H *and* K *are disjoint closed subsets of* X*, there exists a continuous function* $f\colon X \to [0,1]$ *such that* $H = f^{-1}(0)$ *and* $K = f^{-1}(1)$*.*

Theorem 60. *Every subspace of a perfectly normal space is perfectly normal, hence perfectly normal spaces are completely normal.*

2.2 Covering Properties & Countability Axioms

2.2.1 Compact and Lindelöf

A collection $\mathcal{U}$ of subsets of a space X is said to be a *cover* of X if every point of X is in some member of $\mathcal{U}$, so $X = \bigcup \mathcal{U}$. If $\mathcal{U}$ is a cover of X, then a subcollection $\mathcal{V}$ of $\mathcal{U}$ is a *subcover* if $\mathcal{V}$ is also a cover. A cover $\mathcal{U}$ is an *open cover* if every member of $\mathcal{U}$ is open.

A space X is *compact* (resp., *Lindelöf*) if every open cover of X has a finite (resp., countable) subcover.

Theorem 61.

(a) *Every continuous image of a compact (resp., Lindelöf) space is compact (resp., Lindelöf).*

(b) *Every closed subset of a compact (resp., Lindelöf) space is compact (resp., Lindelöf).*

Theorem 62. *Every compact subspace of a Hausdorff space is closed.*

Exercise 63. *Let* $\alpha \in \mathbf{Ord}$. *Then the set* $[0, \alpha] = \{\beta \in \mathbf{Ord} : 0 \leqslant \beta \leqslant \alpha\}$ *with the order topology (using the usual ordering on the ordinals) is a compact Hausdorff space.*

Note that $[0, \alpha] = \alpha + 1$ in our view of the ordinals.

Theorem 64. *Suppose* $f\colon X \to Y$ *is continuous, one-to-one, and onto. If* X *is compact and* Y *is Hausdorff, then* f *is a homeomorphism.*

Exercise 65. *Let* $X = [0, 2\pi]$, *and let* $A = \{0, 2\pi\}$. *Show that* X/A *is homeomorphic to the unit circle*

$$S^1 = \{\langle x, y\rangle \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Similarly, the space obtained from the unit disk

$$B^2 = \{\langle x, y\rangle \in \mathbb{R}^2 : x^2 + y^2 \leqslant 1\}$$

by identifying its boundary S^1 *to a single point is homeomorphic to the 2-sphere*

$$S^2 = \{\langle x, y, z\rangle \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}.$$

Theorem 66. *Suppose* $f\colon X \to \mathbb{R}$ *is continuous, where* X *is a compact space. Then there exists* $x_0 \in X$ *such that* $f(x_0) \geqslant f(x)$ *for every* $x \in X$.

Theorem 67.

(a) *Every compact Hausdorff space is regular.*

(b) *Every compact Hausdorff space is normal.*

★ **Theorem 68.** *Every regular Lindelöf space is normal.*

★ **Example 69.** *The Sorgenfrey line $\mathbb{S}$ is a regular Lindelöf space, and is completely normal, but $\mathbb{S}^2$ is neither Lindelöf nor normal.*

Theorem 70. *Let X be a compact metric space with metric d. If $\mathcal{U}$ is an open cover of X, then there exists $\lambda > 0$ such that every ball of radius λ is contained in some member of $\mathcal{U}$.*

Sometimes the conclusion of the previous theorem is stated in the following way: "every set of diameter less than λ is contained in some member of $\mathcal{U}$." The number λ is called a "Lebesgue number" of the open cover.

Lemma 71, the Tube Lemma. *Let X be a space, and let Y be compact. Fix $x_0 \in X$. Then for any collection $\mathcal{U}$ of open sets in $X \times Y$ with $\{x_0\} \times Y \subset \bigcup \mathcal{U}$, there is a finite subcollection $\mathcal{V}$ of $\mathcal{U}$ and an open neighborhood W of x_0 such that $W \times Y \subset \bigcup \mathcal{V}$.*

Theorem 72. *If X and Y are compact, so is $X \times Y$.*

2.2.2 Filters & Tychonoff's Theorem

A collection $\mathcal{F}$ of subsets of a set X is said to have the *finite intersection property* if the intersection of any finite subcollection of $\mathcal{F}$ is nonempty.

Theorem 73. *A space X is compact if and only if for every collection $\mathcal{F}$ of closed sets with the finite intersection property, $\bigcap \mathcal{F} \neq \varnothing$.*

A collection $\mathcal{F}$ of subsets of a set X is called a *filter on X* if:

(1) $F_1 \cap F_2 \in \mathcal{F}$ whenever $F_1, F_2 \in \mathcal{F}$;

(2) $G \in \mathcal{F}$ whenever $X \supset G \supset F$ for some $F \in \mathcal{F}$;

(3) $\varnothing \notin \mathcal{F}$.

In other words, a filter is a collection of nonempty sets which is closed under supersets and finite intersections. A filter $\mathcal{F}$ on X is called an *ultrafilter* if $\mathcal{F}$ is not properly contained in any other filter on X.

Theorem 74. *Let $\mathcal{F}$ be a filter on X. Then the following are equivalent:*

(a) *$\mathcal{F}$ is an ultrafilter;*

(b) *If $G \subset X$ and $G \cap F \neq \varnothing$ for every $F \in \mathcal{F}$, then $G \in \mathcal{F}$;*

(c) *For every $G \subset X$, either $G \in \mathcal{F}$ or $X \setminus G \in \mathcal{F}$.*

Theorem 75. *Suppose $\mathcal{F}$ is a collection of subsets of X having the finite intersection property. Then $\mathcal{F}$ is contained in some maximal collection $\mathcal{G}$ of subsets of X with the finite intersection property; furthermore, any such $\mathcal{G}$ is an ultrafilter on X.*

Let X be a space, $p \in X$, and $\mathcal{F}$ a filter on X. We say that $\mathcal{F}$ *clusters at* p if $p \in \overline{F}$ for every $F \in \mathcal{F}$, and we say $\mathcal{F}$ *converges to* p if every neighborhood of p contains a member of $\mathcal{F}$ (equivalently, every neighborhood of p is in $\mathcal{F}$).

Theorem 76. *The following are equivalent for a space X:*

(a) *X is compact;*

(b) *Every filter on X clusters at some point;*

(c) *Every ultrafilter on X converges to some point.*

Theorem 77, Tychonoff's Theorem. *Any product of compact spaces is compact.*

2.2.3 Countably Compact

X is *countably compact* if every countable open cover of X has a finite subcover.

Theorem 78. *A space X is compact if and only if X is both countably compact and Lindelöf.*

Theorem 79.

(a) *If X is countably compact, then every infinite subset of X has a limit point in X.*

(b) *If X is a T_1-space, then X is countably compact if and only if every infinite subset of X has a limit point in X.*

2.2.4 First-Countable

A *local base* at a point x of a space X is a collection $\mathcal{B}_x$ of open neighborhoods of x such that, whenever $x \in U$ where U is open, there is some $B \in \mathcal{B}_x$ with $x \in B \subset U$. A space X is *first-countable* if every point of X has a countable local base.

If $\{U_n : n \in \mathbb{N}\}$ is a local base at x and we put $V_n = \bigcap \{U_i : i \leqslant n\}$, then $\langle V_n \rangle_{n\in\mathbb{N}}$ is another countable local base at x which is decreasing (in the sense that $i < j$ implies $V_i \supset V_j$). So a point has a countable local base if and only if it has a countable decreasing local base.

Theorem 80. *Let X be a first-countable space. A point x is in the closure of a subset A of X if and only if there is a sequence $\langle a_n \rangle_{n\in\mathbb{N}}$ in A such that $\langle a_n \rangle \to x$.*

Theorem 81. *If X is first-countable, then $f\colon X \to Y$ is continuous if and only if $\langle f(x_n) \rangle \to f(x)$ in Y whenever $\langle x_n \rangle_{n\in\mathbb{N}} \to x$ in X.*

Theorem 82. *Every metric space is first-countable.*

Exercise 83. *Let $X = \mathbb{N} \times [0,1]$ with its topology as a subspace of the plane. Note that each $\{n\} \times [0,1]$ is open and closed in X; we may say X is a* topological sum *of countably many copies of $[0,1]$. Let $A = \mathbb{N} \times \{0\}$. Describe the topology of X/A (that is, what do the neighborhoods of points of X/A look like?) Is X/A metrizable?*

2.2.5 Separable & Second-Countable

A subset D of a topological space X is *dense* in X if $X = \overline{D}$; equivalently, every non-empty open set contains a point of D. X is *separable* if it

has a countable dense subset. A space X is *second-countable* if it has a countable base.

Theorem 84. *Every subspace of a second-countable space is second-countable, separable, and Lindelöf.*

Theorem 85. *The following are equivalent for a metrizable space X:*

(a) *X is second-countable,*

(b) *X is Lindelöf,*

(c) *X is separable,*

(d) *X is hereditarily separable, and*

(e) *X is hereditarily Lindelöf.*

Theorem 86. *If X_n is a first-countable (resp., second-countable) space for each $n \in \mathbb{N}$, then $X = \prod_{n\in\mathbb{N}} X_n$ is first-countable (resp., second-countable).*

Example 87. *Let 2 denote a two-point discrete space (with elements 0 and 1, say). Let Λ be any uncountable index set. Then 2^Λ is neither first- nor second-countable.*

Theorem 88. *If X_n is separable for each $n \in \mathbb{N}$, then $\prod_{n\in\mathbb{N}} X_n$ is separable.*

Let A be a subset of a space X and let $x \in X$. If every neighborhood of x contains uncountably many points of A, then we call x a *complete accumulation point* of A.

Theorem 89.

(a) *Every infinite subset of a compact space X has a limit point in X.*

(b) *Every uncountable subset A of a Lindelöf space X has a complete accumulation point in X.*

Corollary 90. *For every uncountable subset A of a separable metric space, there is a complete accumulation point a of A such that $a \in A$.*

2.2.6 Sequentially Compact

A space X is *sequentially compact* if every infinite sequence in X has an infinite convergent subsequence.

Theorem 91.

(a) *Sequentially compact spaces are countably compact.*

(b) *If X is first-countable, then X is countably compact if and only if X is sequentially compact.*

Theorem 92. *The following are equivalent for a separable metric space X:*

(a) *X is compact;*

(b) *X is countably compact;*

(c) *X is sequentially compact.*

The previous theorem is in fact true for *any* metric space, separable or not, but the proof is more difficult.

Exercise 93. *Give the set ω_1 of all countable ordinals the order topology (using the usual order on the ordinals). Then ω_1 with this topology is sequentially compact (and so also countably compact) but not compact.*

2.2.7 Locally Compact & Baire Spaces

A space X is *locally compact* if every point has a compact neighborhood; that is, for each $x \in X$, there is a compact subset N of X such that $x \in N^\circ$.

Theorem 94. *Let $\langle X, \mathcal{T} \rangle$ be a locally compact non-compact Hausdorff space. Then there is a compact Hausdorff space $\langle X^*, \mathcal{T}^* \rangle$ satisfying:*

(a) $X^* = X \cup \{\infty\}$, *where* $\infty \notin X$;

(b) $\mathcal{T}_X^* = \mathcal{T}$ *(that is, the relative topology on* X *as a subspace of* X^* *is the same as* $\mathcal{T}$*); and*

(c) U *is an open neighborhood of* ∞ *in* X^* *if and only if* $X^* \setminus U$ *is a compact subset of* X.

The space $\langle X^*, \mathcal{T}^* \rangle$ in the previous theorem is called the *one-point compactification* of the locally compact Hausdorff space $\langle X, \mathcal{T} \rangle$.

Theorem 95. *Every locally compact Hausdorff space is completely regular.*

Lemma 96. *Let* X *be a locally compact Hausdorff space. Then the collection of compact neighborhoods of a point* x *in* X *is a local neighborhood base at* x *(that is, every open set containing* x *contains a compact neighborhood of* x*).*

Lemma 97. *Suppose* $\langle H_n \rangle_{n \in \mathbb{N}}$ *is a decreasing sequence of non-empty closed sets in a space* X*, and* H_1 *is (countably) compact. Then* $\bigcap_{n \in \mathbb{N}} H_n$ *is non-empty.*

Theorem 98. *If* $\{G_n : n \in \mathbb{N}\}$ *are dense open subsets of a locally compact Hausdorff space* X*, then* $\bigcap_{n \in \mathbb{N}} G_n$ *is dense.*

A space X with the property of Theorem 98 is called a *Baire space*. So all locally compact Hausdorff spaces, in particular $\mathbb{R}^n$ for all $n \in \mathbb{N}$, are Baire spaces.

Lemma 99. *Every open subspace of a Baire space is Baire.*

Theorem 100. *Suppose* X *is a Baire space, and* $X = \bigcup_{n \in \mathbb{N}} X_n$*. Then for some* $n \in \mathbb{N}$, $\operatorname{int}(\operatorname{cl} X_n) \neq \varnothing$*; that is, some* X_n *is dense in some open subset of* X*. Moreover,* $\bigcup_{n \in \mathbb{N}} \operatorname{int}(\operatorname{cl} X_n)$ *is dense in* X.

2.3 Connectedness

2.3.1 Connected & Path Connected

A space X is *connected* if and only if X cannot be written as the union of two disjoint nonempty open sets; equivalently, X has no proper nonempty subset which is both open and closed.

Theorem 101. *Let $\langle X, \mathcal{T} \rangle$ be a topological space, and let $C \subset X$. Then the following are equivalent:*

(a) *$\langle C, \mathcal{T}_C \rangle$ is connected;*

(b) *C is not the union of two nonempty separated sets in X.*

It follows from Theorem 101 (by taking $C = X$) that a space X is connected if and only if X is not the union of two nonempty separated sets. In fact, this is often taken as the definition of connected.

Theorem 102. *Let X be a space, $A \subset X$, and suppose $A \subset B \subset \overline{A}$. If A is connected, so is B. In particular, the closure of a connected set is connected.*

Theorem 103. *A subspace X of the real line $\mathbb{R}$ is connected if and only if X is convex (that is, $a < b \in X$ implies $[a, b] \subset X$).*

It is easily seen that a subset of $\mathbb{R}$ is convex if and only if it is an interval (infinite intervals such as (a, ∞) or $(-\infty, b)$ allowed).

A space X is *path connected* if and only if for any two points a, b in X, there is a continuous function $f\colon [0, 1] \to X$ such that $f(0) = a$ and $f(1) = b$. The function f is called a *path* from a to b.

Theorem 104. *The continuous image of a connected (resp., path connected) space is connected (resp., path connected).*

Theorem 105. *Suppose $\{A_\alpha : \alpha \in \Lambda\}$ is a collection of connected (resp., path connected) subspaces of a space X, and for any $\alpha, \beta \in \Lambda$, $A_\alpha \cap A_\beta \neq \varnothing$. Then $\bigcup_{\alpha \in \Lambda} A_\alpha$ is connected (resp., path connected).*

Theorem 106.

(a) *If X and Y are connected spaces, so is $X \times Y$.*

(b) *Any product of connected spaces is connected.*

Theorem 107. *Every path connected space is connected.*

Theorem 108. *Let $X = \prod_{\alpha \in \Lambda} X_\alpha$ be a product space, and let Y be any space. A function $f : Y \to X$ is continuous if and only if $\pi_\alpha \circ f : Y \to X_\alpha$ is continuous for each $\alpha \in \Lambda$.*

Theorem 109. *Any product of path connected spaces is path connected.*

Lemma 110. *Let X and Y be topological spaces, and let $f : X \to Y$ be continuous. Let $G = \{\langle x, f(x)\rangle : x \in X\} \subset X \times Y$ be the "graph" of f. Then G is homeomorphic to X.*

Example 111. *The topologist's sine curve*

$$\{\langle x, \sin 1/x\rangle : 0 < x \leqslant 1\} \cup (\{0\} \times [-1, 1]) \subset \mathbb{R}^2$$

is connected but not path connected.

Note that the topologist's sine curve is also a closed and bounded subset of the plane, hence compact. A compact, connected metric space is often called a *continuum*. Continuum theory is an important subfield of general topology.

2.3.2 Components & Path Components

Let X be a space, and $x \in X$. The *component* $C(x)$ of x is the union of all connected subsets of X which contain x. Similarly, the *path component* $P(x)$ of x is the union of all path connected subsets of X which contain x.

Theorem 112.

(a) *The component (resp., path component) of a point x is the largest connected (resp., path connected) subset of X which contains x.*

(b) *The collection of components (resp., path components) of a space X forms a partition of X (that is, they are pairwise disjoint and their union is X) into connected (resp., path connected) subsets of X.*

(c) *Each component of X is closed in X.*

Unlike components, path components need not be closed: it is easy to see that the topologist's sine curve has a path component which is not closed.

2.3.3 Locally Connected & Locally Path Connected

A space X is *locally connected* (resp., *locally path connected*) if whenever $x \in U$ where U is open in X, there is a connected (resp., path connected) subset N of U such that $x \in N^\circ$.

The topologist's sine curve is an example of a connected space which is not locally connected.

Lemma 113. *The components (resp., path components) of a locally connected (resp., locally path connected) space are open.*

Theorem 114. *A space X is locally connected (resp., locally path connected) if and only if X has a base consisting of connected (resp., path connected) subsets.*

Theorem 115. *If X is locally path connected, then the components and path components of X are the same. In particular, a connected space which is locally path connected is path connected.*

2.3.4 Quasicomponents

Let X be a space and $x \in X$. The *quasicomponent* $Q(x)$ of x is the intersection of all closed-and-open, or *clopen*, sets containing x.

Theorem 116. *For any space X and $x \in X$, the component $C(x)$ of x is contained in the quasicomponent $Q(x)$ of x.*

Lemma 117. *Let X be a compact space, let $\mathcal{H}$ be a collection of closed subsets of X, and let $H = \bigcap \mathcal{H}$. If $H \subset U$, where U is open, then there is a finite subcollection $\mathcal{H}'$ of $\mathcal{H}$ such that $H \subset \bigcap \mathcal{H}' \subset U$.*

Theorem 118. *If X is a compact Hausdorff space, then the components of X are the same as the quasicomponents of X.*

Let X be the following subspace of the plane:

$$\{\langle 0,0\rangle, \langle 0,1\rangle\} \cup \bigcup_{n\in\mathbb{N}} (\{1/n\} \times [0,1]).$$

Then the component in X of the point $\langle 0,0\rangle$ is itself, but the quasicomponent of $\langle 0,0\rangle$ is $\{\langle 0,0\rangle, \langle 0,1\rangle\}$.

2.3.5 Some Theorems

Theorem 119. *If $\langle K_n\rangle_{n\in\mathbb{N}}$ is a decreasing sequence of compact, connected subsets of a Hausdorff space X, then $K = \bigcap_{n\in\mathbb{N}} K_n$ is compact and connected.*

Theorem 120, "To the boundary" Theorem. *Let H be a proper closed subspace of a compact connected Hausdorff space X. Then any component of the subspace H meets the boundary of H.*

Lemma 121. *Let X be a compact connected Hausdorff space. Suppose $\mathcal{K} = \{K_n : n \in \mathbb{N}\}$ is a collection of disjoint closed sets covering X, at least two of which are nonempty. Then for each $k \in \mathbb{N}$, there is a compact connected subspace C_k of X which meets at least two members of $\mathcal{K}$.*

Theorem 122. *Let X be a compact connected Hausdorff space. Then every partition $\mathcal{K}$ of X into disjoint closed sets is either uncountable or contains only one element.*

2.4 Paracompactness

2.4.1 Point-Finite & Locally Finite Collections

A collection $\mathcal{U}$ of subsets of a space X is said to be *point-finite* (resp., *locally finite*) if, for each $x \in X$, the set $\{U \in \mathcal{U} : x \in U\}$ is finite (resp., there is a neighborhood O_x of x such that the set $\{U \in \mathcal{U} : O_x \cap U \neq \varnothing\}$ is finite).

A collection $\mathcal{U}$ of subsets of X is called *closure-preserving* if, for any $\mathcal{U}' \subset \mathcal{U}$, we have $\bigcup\{\overline{U} : U \in \mathcal{U}'\} = \overline{\bigcup\{U : U \in \mathcal{U}'\}}$.

Lemma 123. *Suppose $\mathcal{U}$ is a locally finite collection of subsets of X. Then:*

(a) *$\mathcal{U}$ is closure-preserving;*

(b) *If X is compact, then $\mathcal{U}$ is finite.*

Exercise 124.

(a) *Find a locally finite cover of the real line $\mathbb{R}$ by open intervals.*

(b) *Find a point-finite cover of $\mathbb{R}$ which is not locally finite.*

(c) *Find a point-finite open cover of $[0, 1]$ which is not closure-preserving.*

(d) *Find an infinite collection of subsets of $[0, 1]$ which is closure-preserving.*

2.4.2 Point-Finite & Locally Finite Refinements

Let $\mathcal{U}$ be a cover of a space X. A cover $\mathcal{V}$ of X is called a *refinement* of $\mathcal{U}$ if for every $V \in \mathcal{V}$, there is some $U \in \mathcal{U}$ with $V \subset U$. A space X is said to be *paracompact* (resp., *metacompact*) if every open cover $\mathcal{U}$ of X has a locally finite (resp., point-finite) open refinement $\mathcal{V}$.

Theorem 125.

(a) *Every paracompact T_2-space is regular.*

(b) *Every paracompact T_2-space is normal.*

Theorem 126. *Every regular Lindelöf space is paracompact.*

Theorem 127. *Let X be a T_1-space. Then X is compact if and only if X is paracompact and countably compact.*

2.4.3 Partitions of Unity

If ϕ is a function from X into $[0,1]$, then the *support* of ϕ, denoted $\operatorname{supp}\phi$, is defined to be the closure of the set $\{x \in X : \phi(x) \neq 0\}$. Now let $\{U_\alpha : \alpha \in J\}$ be an indexed open cover of X. We say that an indexed family $\{\phi_\alpha : \alpha \in J\}$ of functions from X into $[0,1]$ is a *partition of unity on X dominated by,* or *subordinated to,* $\{U_\alpha : \alpha \in J\}$ if:

(1) $\operatorname{supp}\phi_\alpha \subset U_\alpha$ for all $\alpha \in J$;

(2) the collection $\{\operatorname{supp}\phi_\alpha : \alpha \in J\}$ is locally finite; and

(3) $\sum_{\alpha \in J} \phi_\alpha(x) = 1$ for all $x \in X$.

Theorem 128. *Let X be a paracompact Hausdorff space. Then every indexed open cover $\{U_\alpha : \alpha \in J\}$ has a partition of unity subordinated to it.*

A partition of unity can sometimes be used to patch together functions defined locally on a space to get a function defined on the whole space.

2.5 Modifications of Some Properties

2.5.1 The Countable Chain Condition

A space X is said to have the *countable chain condition* (*ccc*) if every pairwise-disjoint collection of open subsets of X is countable.

Theorem 129. *Every separable space has the ccc.*

Theorem 130. $[0,1]^{[0,1]}$ *is separable.*

Let $\mathfrak{c} = |[0,1]|$. By a similar argument, any product of $\mathfrak{c}$ (or fewer) separable spaces is separable. Let $\mathfrak{c}^+$ denote the least cardinal greater than $\mathfrak{c}$.

Theorem 131. $[0,1]^{\mathfrak{c}^+}$ *is not separable.*

★ **Theorem 132, Δ-system Lemma.** *Let $\mathcal{F}$ be an uncountable collection of finite sets. Then there is an uncountable subcollection $\mathcal{G}$ of $\mathcal{F}$ and a set R such that $G_1 \cap G_2 = R$ for any distinct $G_1, G_2 \in \mathcal{G}$.*

In Theorem 132, the collection $\mathcal{G}$ is called a Δ-*system* and the set R is called the *root* of the Δ-system.

Theorem 133. *Let $\{X_\alpha : \alpha < \kappa\}$ be a collection of ccc spaces. Then $\prod_{\alpha<\kappa} X_\alpha$ has the ccc if and only if every finite subproduct has the ccc, where a finite subproduct is a product of the form $\prod_{\alpha\in F} X_\alpha$ for some finite subset F of κ.*

Corollary 134. $[0,1]^{\mathfrak{c}^+}$ *has the ccc but is not separable.*

2.5.2 Collectionwise Normal

A collection $\mathcal{H}$ of subsets of a space X is said to be *discrete* in X if every point of X has a neighborhood meeting at most one member of $\mathcal{H}$. A T_1-space X is said to be *collectionwise normal* if, given any discrete collection $\mathcal{H}$ of closed sets, there is a pairwise-disjoint collection $\{U_H : H \in \mathcal{H}\}$ of open sets with $H \subset U_H$ for every $H \in \mathcal{H}$.

For example, let $\mathbb{S}$ be the Sorgenfrey line and let $\mathcal{H} = \{\langle x, -x\rangle : x \in \mathbb{S}\}$. Then $\mathcal{H}$ is a discrete collection of closed subsets of $\mathbb{S}^2$. Note that there is no pairwise-disjoint collection of open sets separating the members of $\mathcal{H}$. So $\mathbb{S}^2$ is not collectionwise normal. (We already know it's not even normal, but not collectionwise normal is easier to see.)

Lemma 135. *If $\mathcal{H}$ is a discrete collection of closed sets, then $\mathcal{H}$ is pairwise-disjoint, and $\bigcup \mathcal{H}'$ is closed for every subcollection $\mathcal{H}'$ of $\mathcal{H}$.*

Theorem 136. *Every paracompact T_2-space is collectionwise normal.*

Corollary 137. *Every metrizable space is collectionwise normal.*

★ **Example 138, Bing's G.** *Let A be an uncountable set, and let $\mathcal{P}(A)$ be the set of all subsets of A. For each $\alpha \in A$, define $e_\alpha \colon \mathcal{P}(A) \to \{0,1\}$ by $e_\alpha(B) = 1$ if $\alpha \in B$ and $e_\alpha(B) = 0$ if $\alpha \notin B$. Let $E = \{e_\alpha : \alpha \in A\}$. Note that E can be considered to be a subset of $2^{\mathcal{P}(A)}$.*

Let X be the set $2^{\mathcal{P}(A)}$ with the topology defined by declaring every point of $X \setminus E$ to be isolated, while each e_α has its usual product neighborhoods in $2^{\mathcal{P}(A)}$. Then X is normal but not collectionwise normal.

2.5.3 Monotonically Normal

A space X is said to be *monotonically normal* if to each pair $\langle H, K\rangle$ of disjoint closed sets, one can assign an open set $U(H,K)$ such that

(a) $H \subset U(H,K) \subset \overline{U(H,K)} \subset X \setminus K$;

(b) If $H \subset H'$ and $K \supset K'$, then $U(H,K) \subset U(H',K')$.

An operator $U(H,K)$ satisfying the above conditions is called a *monotone normality operator* for X.

Theorem 139. *Metrizable spaces are monotonically normal.*

Lemma 140. *If X is monotonically normal, then there is a monotone normality operator $U(H,K)$ for X satisfying $U(H,K) \cap U(K,H) = \varnothing$ for any pair $\langle H, K\rangle$ of disjoint closed sets.*

Theorem 141. *Monotonically normal spaces are collectionwise normal.*

Theorem 142. *The following are equivalent for a T_1-space X:*

(a) *X is monotonically normal;*

(b) *To each $x \in X$ and open neighborhood U of x, one can assign an open neighborhood U_x of x satisfying:*

$$U_x \cap V_y \neq \varnothing \to x \in V \text{ or } y \in U;$$

(c) *Same as* 142(b), *but with the neighborhoods* U *restricted to members of a given base* $\mathcal{B}$.

Theorem 143. *Every subspace of a monotonically normal* T_1*-space is monotonically normal.*

Theorem 144. *Every linearly ordered space is monotonically normal.*

M.E. Rudin proved in 2001 that the class of compact Hausdorff monotonically normal spaces is exactly the class of continuous images of compact linearly ordered spaces. The reverse direction is relatively easy: it's not hard to show that the closed image of a monotonically normal space is monotonically normal. But the forward direction is a deep and very difficult result.

3 More About Metrizable Spaces

3.1 Metrization Theorems

3.1.1 Urysohn's Metrization Theorem

Let $\mathcal{F}$ be a family of continuous functions from a space X to the unit interval $[0,1]$. We say $\mathcal{F}$ *separates points* if, given $x_1 \neq x_2 \in X$, there exists $f \in \mathcal{F}$ such that $f(x_1) \neq f(x_2)$. We say $\mathcal{F}$ *separates points from closed sets* if, given $x \in X$ and any closed set H with $x \notin H$, there exists $f \in \mathcal{F}$ with $f(x) \notin \overline{f[H]}$. The *evaluation function determined by* $\mathcal{F}$ is the function $e_{\mathcal{F}} \colon X \to [0,1]^{\mathcal{F}}$ defined by $x \mapsto \langle f(x) \rangle_{f \in \mathcal{F}}$.

Theorem 145. *Let $\mathcal{F}$ be a collection of continuous functions from X into $[0,1]$. Then:*

(a) *$e_{\mathcal{F}}$ is continuous;*

(b) *If $\mathcal{F}$ separates points, then $e_{\mathcal{F}}$ is one-to-one;*

(c) *If X is a T_1-space and $\mathcal{F}$ separates points from closed sets, then $e_{\mathcal{F}} \colon X \to e_{\mathcal{F}}[X]$ is a homeomorphic embedding of X into $[0,1]^{\mathcal{F}}$.*

Theorem 146, Urysohn's Metrization Theorem. *The following are equivalent for a space X:*

(a) *X is a separable metrizable space;*

(b) *X is a regular second-countable space;*

(c) *X is homeomorphic to a subspace of the Hilbert cube $[0,1]^{\mathbb{N}}$.*

Theorem 147. *A space X is completely regular if and only if X is homeomorphic to a subspace of $[0,1]^\kappa$ for some cardinal κ.*

3.1.2 Metrizable Spaces Are Paracompact

★ **Theorem 148.** *The following are equivalent for a regular space X:*

(a) *X is paracompact;*

(b) *Every open cover of X has a locally finite refinement;*

(c) *Every open cover of X has a locally finite closed refinement.*

★ **Theorem 149.** *Let X be a regular space. If every open cover of X has an open refinement $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$, where each $\mathcal{V}_n$ is locally finite, then X is paracompact.*

A collection that is the union of countably many collections with property P may be called a σ-P collection. So, $\mathcal{V}$ in Theorem 149 is *σ-locally finite*.

Lemma 150. *Let X be metrizable with metric d. Let $\mathcal{U} = \{U_\alpha : \alpha < \kappa\}$ be an open cover of X indexed by the ordinal κ. For each $\alpha < \kappa$ and $n \in \mathbb{N}$, let*

$$H_{\alpha,n} = \{x \in U_\alpha \setminus \bigcup_{\beta<\alpha} U_\beta : d(x, X \setminus U_\alpha) \geqslant 1/n\},$$

where $d(x,Y) = \sup\{d(x,y) : y \in Y\}$, and let

$$V_{\alpha,n} = \bigcup_{x\in H_{\alpha,n}} B_d(x, 1/4n).$$

Then for each n, $\mathcal{V}_n = \{V_{\alpha,n} : \alpha < \kappa\}$ is locally finite, and $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$ is a refinement of $\mathcal{U}$.

In fact, in Lemma 150 each point of X has a neighborhood meeting at most one member of $\mathcal{V}_n$. Note the use of AC in the form of the well-ordering principle in Corollary 151.

Corollary 151. *Every metrizable space is paracompact.*

It is immediate from Corollary 151 and Theorem 127 that countably compact metrizable spaces are compact. Thus Theorem 92—that compact, countably compact, and sequentially compact are equivalent in metric spaces—becomes easy.

3.1.3 Nagata-Smirnov Metrization Theorem

Lemma 152. *If X is regular and has a σ-locally finite base, then X is perfectly normal.*

Lemma 153. *Suppose X is regular and has a base $\mathcal{B} = \bigcup\{\mathcal{B}_n : n \in \mathbb{N}\}$, where each $\mathcal{B}_n$ is locally finite. For each $n \in \mathbb{N}$ and each $B \in \mathcal{B}_n$, let $f_{n,B} \colon X \to [0, 1/n]$ be continuous such that $f_{n,B}(x) = 0$ if and only if $x \notin B$, and let*

$$\mathcal{F} = \{f_{n,B} : n \in \mathbb{N} \wedge B \in \mathcal{B}_n\}.$$

Now define $d \colon X \times X \to \mathbb{R}$ by $d(x, y) = \sup\{|f(x) - f(y)| : f \in \mathcal{F}\}$. Then d is a metric on X which generates the topology of X.

Theorem 154, The Nagata-Smirnov Metrization Theorem. *A space X is metrizable if and only if X is regular and has a σ-locally finite base.*

Theorem 155. *If a paracompact T_2-space is locally metrizable—that is, if every point has a metrizable neighborhood—then it is metrizable.*

3.1.4 G_δ-Diagonals

★ **Lemma 156.** *Let X be a compact Hausdorff space and suppose there is a countable collection $\mathcal{U}$ of open sets of X such that, whenever $x \neq y \in X$, there is some $U \in \mathcal{U}$ with $x \in U$ and $y \notin \overline{U}$. Then X is metrizable.*

A space X is said to have a *G_δ-diagonal* if the diagonal, $\Delta = \{\langle x, x\rangle \in X^2 : x \in X\}$, is a G_δ set in X^2 (that is, there are open subsets U_n, $n < \omega$, of X^2 such that $\Delta = \bigcap_{n<\omega} U_n$).

Theorem 157. *A compact Hausdorff space X is metrizable if and only if X has a G_δ diagonal.*

3.1.5 Networks

A collection $\mathcal{N}$ of subsets of a space X is a *network* for X if whenever $x \in U$, where U is open in X, there is some $N \in \mathcal{N}$ with $x \in N \subset U$.

Theorem 158. *If X has a countable network, then so does every continuous image of X.*

★ **Theorem 159.** *The following are equivalent for a T_1-space X:*

(a) *X has a countable network;*

(b) *X is the continuous image of a separable metric space.*

Because of Theorem 159, spaces having a countable network are sometimes called *cosmic* spaces.

Theorem 160. *If X is regular and has a countable network, then X has a G_δ-diagonal.*

Theorem 161. *If X is a compact metrizable space, then so is every Hausdorff continuous image of X.*

3.1.6 Perfect Images

Theorem 162. *Let $f\colon X \to Y$ be a continuous surjection. Then the following are equivalent:*

(a) *f is closed;*

(b) *Whenever $y \in Y$ and U is an open set in X containing $f^{-1}(y)$, there is an open set V in Y containing y with $f^{-1}[V] \subset U$;*

(c) *For each open set U in X, the set $f^*[U] = \{y \in Y : f^{-1}(y) \subset U\}$ is open in Y.*

A continuous surjection $f\colon X \to Y$ is said to be *perfect* if f is closed and $f^{-1}(y)$ is compact for each $y \in Y$.

Theorem 163. *The perfect image of a separable metrizable space is separable metrizable.*

Indeed the perfect image of any metrizable space—separable or not—is metrizable, but the argument is more complicated.

3.2 Completeness

3.2.1 Complete Metrics

Let X be a metric space with metric d. A sequence $\langle x_n \rangle_{\mathbb{N}}$ is *d-Cauchy* if for each $\varepsilon > 0$ there is $k \in \mathbb{N}$ such that $d(x_i, x_j) \leqslant \varepsilon$ whenever $i, j \geqslant k$. The metric d is *complete* if every d-Cauchy sequence converges. A metrizable space X is *completely metrizable* if there is a complete metric d for X which generates the topology on X.

The usual metric on $\mathbb{R}^n$ is complete. The usual metric on the space of irrationals is not complete, but the irrationals are completely metrizable (this fact will follow from subsequent results).

Theorem 164. *The following are equivalent for a metric d:*

(a) *d is complete;*

(b) *Whenever $A_1, A_2, \ldots$ is a decreasing sequence of nonempty closed sets with $\operatorname{diam} \langle A_n \rangle \to 0$, then $\bigcap_{n \in \mathbb{N}} A_n \neq \varnothing$, where $\operatorname{diam} A = \sup \{d(x, y) : x, y \in A\}$.*

Corollary 165. *If X is compact and metrizable, then every metric which generates the topology is complete.*

Theorem 166. *A completely metrizable space is a Baire space; that is, the intersection of countably many dense open sets is always dense.*

Theorem 167. *If d is a complete metric for a space X and $A \subset X$, then $d \restriction A \times A$ is a complete metric for the subspace A if and only if A is closed in X.*

Theorem 168. *A countable product of completely metrizable spaces is completely metrizable.*

Lemma 169.

(a) *Let U be an open subset of a metrizable space X with metric d. Then the map defined by $x \mapsto \langle x, 1/d(x, X \setminus U)\rangle$ is a homeomorphism of U onto a closed subset of $X \times \mathbb{R}$.*

(b) *Let G be a G_δ-subset of a metrizable space X. Then G is homeomorphic to a closed subset of $X \times \mathbb{R}^{\mathbb{N}}$.*

Corollary 170. *A subspace Y of a completely metrizable space X is completely metrizable if and only if Y is G_δ in X.*

Lemma 171. *If a locally compact space Y is a dense subspace of a Hausdorff space X, then Y is open in X.*

Corollary 172. *Every locally compact metrizable space is completely metrizable.*

3.2.2 $C^*(X)$

Given a space X, define $C^*(X)$ to be the set of all bounded continuous real-valued functions on X. For $f, g \in C^*(X)$, define

$$d(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

Theorem 173. *The function $d\colon C^*(X) \times C^*(X) \to \mathbb{R}$ defined above is a complete metric on $C^*(X)$.*

This doesn't work for $C(X)$, the set of all (not necessarily bounded) continuous real-valued functions on X, because in this case the "metric" d as defined for $C^*(X)$ wouldn't necessarily be defined for all $f, g \in$

$C(X)$—the sup could be infinite. However, note that it does work if X is compact, for then $C(X)$ and $C^*(X)$ are the same.

$C^*(X)$ is a vector space over $\mathbb{R}$ under the usual operations of addition of functions and scalar multiplication. Also $C^*(X)$ has a "norm" $\|\cdot\|$ defined by

$$\|f\| = \sup_{x \in X} |f(x)|.$$

That is, $\|\cdot\|$ satisfies:

(1) $\|f\| \geqslant 0$ and $\|f\| = 0$ if and only if $f = 0$;

(2) $\|f + g\| \leqslant \|f\| + \|g\|$;

(3) $\|cf\| = |c| \cdot \|f\|$ for all $c \in \mathbb{R}$.

So, $C^*(X)$ is what is called a *normed linear space*.

Given a norm, one can define a metric d by declaring $d(f, g) = \|f - g\|$. For the norm $\|f\| = \sup_{x \in X} |f(x)|$ on $C^*(X)$, the associated metric is precisely the metric on $C^*(X)$ defined earlier.

A *Banach space* is a complete normed linear space; that is, complete in the metric defined by the norm. So by Theorem 173, $C^*(X)$ is a Banach space.

3.2.3 Isometries

Let $\langle X, d\rangle$ and $\langle X', d'\rangle$ be metric spaces. A mapping $j\colon X \to X'$ is called an *isometry* if for each $x, y \in X$, $d'(j(x), j(y)) = d(x, y)$; that is, j preserves distances.

Theorem 174. *Let $\langle X, d\rangle$ be a metric space and let $a \in X$; for each $x \in X$, let $f_x \in C^*(X)$ be defined by $f_x(z) = d(z, x) - d(z, a)$. Then the map $j\colon X \to C^*(X)$ defined by $j(x) = f_x$ is an isometry under the usual distance on $C^*(X)$.*

If the metric d in the above theorem were bounded, defining $f_x(z) = d(z, x)$ would suffice.

Corollary 175. *Every metric space $\langle X, d\rangle$ is isometrically embeddable in a complete metric space.*

★ **Theorem 176.** *For every metric space $\langle X, d\rangle$ there is a unique (up to isometry) complete metric space $\langle \tilde{X}, \tilde{d}\rangle$ with X a dense subspace of $\tilde{X}$ and $\tilde{d} \restriction X \times X = d$.*

The metric space $\langle \tilde{X}, \tilde{d}\rangle$ given by Theorem 176 is called the *completion* of $\langle X, d\rangle$. It can also be defined (by a method analogous to a classical way of defining the reals from the rationals) by "adding a point" to X for each Cauchy sequence in X which does not converge in X, calling two Cauchy sequences $\langle x_n\rangle_{\mathbb{N}}, \langle y_n\rangle_{\mathbb{N}}$ equivalent if $d(x_n, y_n) \to 0$. Each "point" of $\tilde{X} \setminus X$, then, is an equivalence class of Cauchy sequences in X. The distance $\tilde{d}$ between two equivalence class with representatives $\langle x_n\rangle$ and $\langle y_n\rangle$ is defined by $\tilde{d}(\langle x_n\rangle, \langle y_n\rangle) = \lim_{n\to\infty} d(x_n, y_n)$.

4 More About Linearly Ordered Spaces

4.1 The Cantor Set

Recall that we may represent a finite sequence of length $n < \omega$ in a set X as a function $\sigma\colon n \to X$, so that $\sigma = \langle \sigma(0), \ldots, \sigma(n-1) \rangle = \langle \sigma_0, \ldots, \sigma_{n-1} \rangle$. If $x \in X$, then $\tau = \sigma^\frown x$ is a new sequence $\tau\colon n+1 \to X$ such that $\tau \restriction n = \sigma$ and $\tau(n) = x$; that is, we have appended x to the end of our sequence σ.

Also recall that a function is a set of ordered pairs, so $|\sigma|$ gives the length of σ.

4.1.1 The Construction

For each finite sequence σ of 0's and 1's (including the empty sequence $\varnothing$), we define a closed subinterval I_σ of $[0,1]$ as follows. Start by setting $I_\varnothing = [0,1]$. Then if I_σ has been defined, let $I_{\sigma^\frown 0}$ and $I_{\sigma^\frown 1}$ be the left and right thirds, respectively, of I_σ. Thus $I_0 = [0, 1/3]$, $I_1 = [2/3, 1]$, $I_{00} = [0, 1/9]$, $I_{01} = [2/9, 1/3]$, etc. For each $n < \omega$, let $C_n = \bigcup \{I_\sigma : |\sigma| = n\}$. So $C_0 = [0,1]$, $C_1 = [0,1/3] \cup [2/3, 1]$, etc. Finally, $\mathbb{C} = \bigcap_{n<\omega} C_n$ is the *Cantor set*.

Theorem 177. *The Cantor set $\mathbb{C}$ as defined above is (as a subspace of the real line $\mathbb{R}$) compact, metrizable, has no isolated points, and has a countable base of open and closed sets. It is an uncountable closed subset of $\mathbb{R}$ with empty interior in $\mathbb{R}$.*

A space which has a base of open and closed sets is sometimes called *zero-dimensional*, or more precisely, is said to have *small inductive*

dimension zero. There are several concepts of dimension.

★ **Theorem 178.** *Suppose X is a compact Hausdorff metrizable space with no isolated points, and has a countable base of clopen sets. Then X is homeomorphic to the Cantor set.*

Corollary 179. *The following spaces are homeomorphic to the Cantor set $\mathbb{C}$:*

(a) $\{0,1\}^{\mathbb{N}}$;

(b) *Any product space $\prod_{n\in\mathbb{N}} X_n$, where each X_n is a finite discrete space with at least two points;*

(c) $\mathbb{C}\times F$ *for any nonempty finite discrete space F, so the topological sum of a finite number of copies of $\mathbb{C}$ is homeomorphic to $\mathbb{C}$;*

(d) $\mathbb{C}\times\mathbb{C}$;

(e) $\mathbb{C}^{\mathbb{N}}$.

4.1.2 A Space-Filling Curve

Theorem 180. *The function $f\colon 2^{\mathbb{N}}\to[0,1]$ defined by $\vec{x}\mapsto\sum_{n=1}^{\infty} x_n/2^n$ is continuous and onto.*

★ **Theorem 181.** *There is a continuous function from $[0,1]$ onto $[0,1]^2$ — or even onto the Hilbert cube $[0,1]^{\mathbb{N}}$.*

4.2 Characterizing Some Ordered Spaces

4.2.1 $\mathbb{R}$ & $[0,1]$

Theorem 182. *Suppose $\langle X,<\rangle$ and $\langle Y,\prec\rangle$ are countable linearly ordered sets with no first or last point, and both are densely ordered (that is, between any two points there is another point). Then there is an order-preserving bijection $f\colon X\to Y$.*

Theorem 183. *A linearly ordered space X is connected if and only if*

(a) *X is densely ordered, and*

(b) *every bounded above subset of X has a least upper bound.*

Theorem 184. *Suppose X is a connected linearly ordered space with no first or last point, and is separable. Then X is order-isomorphic to the real line $\mathbb{R}$ with the usual order.*

Corollary 185. *Suppose X is a separable compact connected linearly ordered space. Then X is homeomorphic to the unit interval $[0, 1]$.*

4.2.2 $\mathbb{Q}$

Lemma 186. *If X is completely regular and $|X| < |\mathbb{R}|$, then X has a base of clopen sets.*

★ **Lemma 187.** *Suppose κ is an infinite cardinal. If κ is the least cardinal of a base for a space X, then for every base $\mathcal{B}$, there is a base $\mathcal{C} \subset \mathcal{B}$ such that $|C| = \kappa$.*

The least cardinal of a base for a space X is called the *weight* of X and is denoted by $w(X)$.

Theorem 188. *Suppose X is countable, regular, first-countable, and has no isolated points. Then X is homeomorphic to the rationals $\mathbb{Q}$.*

Corollary 189. *The following are homeomorphic to the space $\mathbb{Q}$ of rationals.*

(a) *$\mathbb{Q}^n$ for every positive integer n;*

(b) *Any countable dense subset of $\mathbb{R}^n$ for any $n \in \mathbb{N}$, or of the Cantor set $\mathbb{C}$;*

(c) *$\mathbb{Q}$ with the right half-open interval topology.*

4.2.3 Compact Spaces

Theorem 190. *A linearly ordered space X is compact if and only if every subset of X has a least upper bound and a greatest lower bound.*

★ **Theorem 191.** *Every linearly ordered space X is a dense subset of a compact linearly ordered space $\hat{X}$.*

4.3 The Ordinal ω_1

We now discuss paracompactness of ordered spaces, with the eventual goal of the characterization theorem of the next section. The following result is a corollary of a couple of results from previous chapters.

Theorem 192. *The space ω_1 of countable ordinals is not paracompact.*

4.3.1 Closed & Unbounded Subsets

Theorem 193.

(a) *Let C and D be closed (in the order topology) and unbounded subsets of ω_1. Then $C \cap D$ is unbounded.*

(b) *Let C_n, $n < \omega$, be club—closed and unbounded—in ω_1. Then $\bigcap_{n<\omega} C_n$ is club.*

★ **Theorem 194.** *ω_1 is not perfectly normal.*

Theorem 195. *Let C be club in ω_1. Then C is homeomorphic to ω_1.*

Theorem 196. *If $f\colon \omega_1 \to \omega_1$, then $C = \{\alpha : \forall\beta < \alpha(f(\beta) < \alpha)\}$ is club.*

4.3.2 Stationary Subsets

A subset S of ω_1 is said to be *stationary* in ω_1 if $S \cap C \neq \varnothing$ for every closed unbounded set C.

Theorem 197, Pressing Down Lemma. *Let S be a stationary set in ω_1. If for each ordinal α in S, $\alpha > 0$, we choose an ordinal $\beta_\alpha < \alpha$, then there is some $\beta < \omega_1$ such that $\beta = \beta_\alpha$ for uncountably many $\alpha \in S$. (Such a map $\alpha \mapsto \beta_\alpha$ is said to be "pressing down.")*

In fact, there is a $\beta < \omega_1$ such that $f^{-1}(\beta)$ is not just uncountable but stationary. It is possible to use this stronger version of the pressing down lemma to show that there is a collection of ω_1-many disjoint stationary sets. We can, however, show the following.

Proposition A. There are disjoint stationary sets.

Theorem 198. *Let S be a stationary subset of ω_1. Then S is not paracompact.*

Theorem 199. *Let C be a closed subset of a linearly ordered space X. Then $X \setminus C$ is the union of a disjoint collection of convex open sets.*

Theorem 200. *The following are equivalent for a subspace S of ω_1:*

(a) *S is metrizable,*

(b) *S is paracompact, and*

(c) *S is non-stationary.*

★ **Example 201.** *The space $\omega_1 \times (\omega_1 + 1)$ is not normal. Hence, the product of a normal space and a compact Hausdorff space need not be normal.*

4.4 Characterizing Paracompact Ordered Spaces

It will be convenient to define an ordinal κ to be a *cardinal* if there is no function from an ordinal $\alpha < \kappa$ onto κ. With this notation, ω and ω_1 denote the least infinite cardinal and the least uncountably infinite cardinal, respectively (as well as the least ordinals with infinitely many and uncountably many predecessors, respectively). A cardinal κ is called a *successor cardinal* if there is a cardinal $\lambda < \kappa$ such that κ is the least cardinal greater than λ. In this case, κ is often denoted by λ^+. A cardinal κ which is not a successor cardinal is called a *limit cardinal*. For example $\omega_1, \omega_2, \ldots$ are successor cardinals while ω and $\omega_\omega = \sup\{\omega_n : n < \omega\}$ are limit cardinals.

4.4.1 Cofinality of Limit Ordinals

Let λ be a limit ordinal. We define the *cofinality* of λ, denoted $\operatorname{cf}\lambda$, to be the least cardinal κ such that there is a function $f : \kappa \to \lambda$ with $\{f(\alpha) : \alpha < \kappa\}$ unbounded in λ. For example, if λ is any countable limit ordinal, then $\operatorname{cf}\lambda = \omega$. Similarly, $\operatorname{cf}\omega_\omega = \omega$.

An infinite cardinal κ is said to be *regular* if $\operatorname{cf}\kappa = \kappa$. Otherwise, κ is called *singular*. Clearly, ω_ω is singular.

Proposition B. ω_1 is a regular cardinal.

★ **Lemma 202.**

(a) *If κ is any infinite cardinal, then the union of $\leqslant \kappa$-many sets, each of cardinality $\leqslant \kappa$, has cardinality $\leqslant \kappa$;*

(b) *if κ is a successor cardinal, then κ is regular;*

(c) *For any limit ordinal λ, $\operatorname{cf}\lambda$ is a regular cardinal.*

Theorem 203. *The following are equivalent for an infinite cardinal κ:*

(a) *κ is regular;*

(b) *For any $A \subset \kappa$, if $|A| < \kappa$, then $\sup A < \kappa$;*

(c) *The union of $< \kappa$-many sets, each with cardinality $< \kappa$, has cardinality $< \kappa$.*

4.4.2 Club & Stationary Subsets of Cardinals

Closed unbounded and stationary sets are defined for any uncountable regular cardinal κ in the same way they were defined for ω_1, and the analogues of Theorem 193, Theorem 195, and the Pressing Down Lemma hold by similar arguments.

Theorem 204. *Let κ be an uncountable regular cardinal.*

(a) *If $\mathcal{C}$ is a collection of $< \kappa$-many club subsets of κ, then $\bigcap \mathcal{C}$ is club;*

(b) *If $f\colon \kappa \to \kappa$, then $C = \{\alpha < \kappa : \forall \beta < \alpha (f(\beta) < \alpha)\}$ is club;*

(c) *If $S \subset \kappa$ is stationary, and to each $\alpha \in S$ with $\alpha > 0$ we choose $\beta_\alpha < \alpha$, then there is some $\beta < \kappa$ such that $\beta_\alpha = \beta$ for κ-many $\alpha \in S$.*

Theorem 205. *Let S be a stationary subset of a regular cardinal κ. Then S is not paracompact.*

4.4.3 The Characterization

Let $\mathcal{U}$ be a collection of sets and let $U, V \in \mathcal{U}$. A *finite linked chain in $\mathcal{U}$ from U to V* is a sequence $U_1, U_2, \ldots, U_n$ of members of $\mathcal{U}$ such that $U_1 = U$, $U_n = V$, and $U_i \cap U_{i+1} \neq \varnothing$ for any $i = 1, 2, \ldots, n-1$. $\mathcal{U}$ is said to be *connected* if there is a finite linked chain in $\mathcal{U}$ between any two members of $\mathcal{U}$.

The next result has nothing to do with ordered spaces, but is good to know.

Theorem 206. *A space X is connected if and only if every cover of X by nonempty sets is connected.*

★ **Theorem 207.** *A linearly ordered space X is paracompact if and only if X does not contain a closed subspace homeomorphic to a stationary subset of a regular uncountable cardinal.*

4.5 Hereditarily Lindelöf but not Separable

4.5.1 ccc Linearly Ordered Spaces

Lemma 208. *The following are equivalent for a space X:*

(a) *X is hereditarily Lindelöf;*

(b) *Every open subspace of X is Lindelöf;*

(c) *For any collection $\mathcal{U}$ of open subsets of X, there is a countable $\mathcal{V} \subset \mathcal{U}$ such that $\bigcup \mathcal{V} = \bigcup \mathcal{U}$;*

(d) *There is no subset* $\{x_\alpha : \alpha < \omega_1\}$ *of* X *with the property that, for each* $\alpha < \omega_1$, $x_\alpha \notin \overline{\{x_\beta : \beta > \alpha\}}$.

A set $\{x_\alpha : \alpha < \omega_1\}$ satisfying the condition in Lemma 208(d) is said to be *right-separated in type* ω_1. *Left-separated in type* ω_1 is defined analogously, and the analogous result is that a space is hereditarily separable if and only if it does not contain a subspace which is left-separated in type ω_1.

★ **Theorem 209.** *Supose* X *is a ccc linearly ordered space. Then* X *is hereditarily Lindelöf.*

4.5.2 Suslin Lines

A linearly ordered space X is a *Suslin line* if X is *ccc*, connected, has no first or last point, and is not homeomorphic to the real line. By Theorem 184, a Suslin line cannot be separable.

★ **Theorem 210.** *If there is a ccc nonseparable linearly ordered space, then:*

(a) *there is one which is densely ordered and such that no nonempty open interval is separable;*

(b) *there is a Suslin line.*

Sometimes a Suslin line is defined to be a nonseparable *ccc* linearly ordered space. By Theorem 210, this is equivalent to our definition (in the sense that one exists if and only if the other exists).

5 Some Set-Theoretic Topology

5.1 Trees

A partially ordered set $\langle T, \leqslant \rangle$ is a *tree* if the set $P_t = \{s \in T : s < t\}$ is well-ordered by $\leqslant$ for all $t \in T$. The elements of T are called the *nodes* of T. A maximal linearly ordered subset of T is called a *branch* of T.

Recall that any well-ordered set is order-isomorphic to (the predecessors of) an ordinal. If $\langle T, \leqslant \rangle$ is a tree, then for any ordinal α, the set

$$L_\alpha = \{t \in T : P_t \text{ is isomorphic to } \alpha\}$$

is called the αth *level* of T. The least α such that L_α is empty is called the *height* of T.

5.1.1 The Cantor Tree

Let T be the set of all finite sequences of 0's and 1's (including the empty sequence). We can equivalently describe T as the set of all functions σ from some natural number n into 2, where here we think of n as the set $\{0, 1, \ldots, n-1\}$ and 2 as the set $\{0, 1\}$. (Or, σ is a sequence in 2 of length n.) If $\sigma, \tau \in T$, define $\sigma < \tau$ if and only if σ is an *initial segment* of τ; for example, $110 < 1100$, $01 < 0111$, etc. Then $\langle T, \leqslant \rangle$ is called the *Cantor tree*. The set $2^{<\omega}$, ordered by extension, is another way to describe the Cantor tree.

Note that the branches of the Cantor tree can be identified with 2^ω, the set of all functions $f \colon \omega \to 2$. Given $f \in 2^\omega$, $\{f \restriction n : n < \omega\}$ is a branch, and given a branch b, $\bigcup b$ is a function from ω to 2. Note $b = \{\bigcup b \restriction n : n < \omega\}$.

The Cantor tree $\langle T, \leqslant \rangle$ is completely described (up to isomorphism) as follows:

(1) The least level L_0 of T has exactly one node;

(2) Each node σ has exactly two (immediate) successors; and

(3) T has height ω.

Theorem 211. *Recall our construction of the Cantor set $\mathbb{C}$ in Section 4.1. If $\mathcal{B} = \{I_\sigma \cap \mathbb{C} : \sigma \in 2^{<\omega}\}$, then $\mathcal{B}$ is a countable base of clopen sets in $\mathbb{C}$ such that $\langle \mathcal{B}, \supset \rangle$ is isomorphic to the Cantor tree.*

Theorem 212. *A space X is homeomorphic to the Cantor set $\mathbb{C}$ if and only if X has a base $\mathcal{B}$ consisting of clopen sets such that $\langle \mathcal{B}, \supset \rangle$ is isomorphic to the Cantor tree and the intersection of each branch of this tree is a single point.*

★ **Theorem 213.** *Let $\langle T, \leqslant \rangle$ be the Cantor tree. Let X be the set of all branches of T. Note that if b is a branch of T, then $\bigcup b \in 2^\omega$. Define a linear order $\prec$ on X as follows: if $b, b' \in X$ and n is the least integer such that $\bigcup b(n) \neq \bigcup b'(n)$, then $b \prec b'$ if and only if $\bigcup b(n) = 0$ and $\bigcup b'(n) = 1$. Then X with the topology induced by this order is homeomorphic to $\mathbb{C}$.*

★ **Theorem 214.** *The following are continuous images of $\mathbb{C}$:*

(a) *the unit interval $[0, 1]$;*

(b) *the Hilbert cube $[0, 1]^\omega$;*

(c) *any closed subset of $\mathbb{C}$; and*

(d) *any compact metric space.*

5.1.2 The Irrationals

For each $n < \omega$, let ω^n denote the set of all functions from n into ω, and let $\omega^{<\omega} = \bigcup_{n<\omega} \omega^n$. If $\sigma, \tau \in \omega^{<\omega}$, let $\sigma < \tau$ if and only if σ is an initial segment of τ. Then $\langle \omega^{<\omega}, < \rangle$ is a tree of height ω with one node at the least level (the empty sequence) and such that every node has a countable infinite number of immediate successors.

Exercise 215. *Let $\mathbb{P}$ denote the irrationals (as a subspace of $\mathbb{R}$). Show that $\mathbb{P}$ has a base $\mathcal{B}$ of clopen sets such that $\langle \mathcal{B}, \supset \rangle$ is a tree isomorphic to $\langle \omega^{<\omega}, \leqslant \rangle$, and such that the intersection of each branch is a singleton.*

A space X is said to be *nowhere-locally-compact* if no point of X has a compact neighborhood.

Theorem 216. *The following are equivalent for a space X:*

(a) *X is homeomorphic to the space of irrationals (as a subspace of the real line with the usual Euclidean topology);*

(b) *X has a base $\mathcal{B}$ consisting of clopen sets such that $\langle \mathcal{B}, \supset \rangle$ is isomorphic to the tree $\langle \omega^{<\omega}, \leqslant \rangle$, and the intersection of each branch of this tree is a single point; and*

(c) *X is a nowhere-locally-compact complete separable metric space and has a countable base of clopen sets.*

Corollary 217. *Let $\mathbb{P}$ denote the space of irrationals. Then $\mathbb{P}$ is homeomorphic to $\mathbb{P}^2$, $\mathbb{P}^\omega$, and ω^ω (where ω is given the discrete topology).*

A subset A of the real line $\mathbb{R}$ is said to be *analytic* if there is a continuous surjection $f\colon \mathbb{P} \to A$. It is known that every Borel set is analytic, every analytic set is Lebesgue measurable, and that there are analytic sets that are not Borel (a highly nontrivial result!) and measurable sets that are not analytic. Analytic sets play a major role in the field of "descriptive set theory."

5.1.3 Aronszajn and Suslin Trees

Theorem 218. *If T is a tree of height ω, and every level of T is finite, then T has an infinite branch.*

An *Aronszajn tree* is a tree of height ω_1 such that every branch and every level is countable. A *Suslin tree* is a tree of height ω_1 such that every branch and every antichain is countable. Note that every Suslin tree is Aronszajn.

Theorem 219. *There is an Aronszajn tree.*

Theorem 219 shows that the natural analogue of Theorem 218 for trees of height ω_1 is false: every level can be countable yet there is no branch going all the way to the top of the tree. If, however, you require every level to be finite, then there is such a branch.

★ **Theorem 220.** *If there is a Suslin line, then there is a Suslin tree.*

★ **Theorem 221.** *If there is a Suslin tree, then there is a Suslin line.*

★ **Theorem 222.** *If S is a Suslin line, then S^2 does not have the ccc.*

★ **Theorem 223.** *If there is a Suslin line, there is one such that no non-degenerate interval is separable. Such a Suslin line is the union of ω_1-many nowhere-dense sets.*

5.2 Martin's Axiom

Like the Continuum Hypothesis (CH), Martin's Axiom (MA) is an axiom of set theory that is known to be consistent with and independent of ZFC. Martin's Axiom has had many applications in certain parts of general topology and real analysis. It's most powerful when conjuncted with the *negation* of the Continuum Hypothesis. Informally, assuming $\mathsf{MA} + \neg\mathsf{CH}$ one can bump up to ω_1 or higher some results that are true in ZFC for ω. For example, in ZFC you know how to prove that the real line is not the union of countably many nowhere dense sets. Assuming $\mathsf{MA} + \neg\mathsf{CH}$, $\mathbb{R}$ is not the union of ω_1-many nowhere-dense sets. We'll also see that $\mathsf{MA} + \neg\mathsf{CH}$ kills Suslin lines.

5.2.1 The Statement

Let $\langle P, \leqslant \rangle$ be a partially ordered set. We say $D \subset P$ is *dense* in P if for any $p \in P$ there is $q \leqslant p$ with $q \in D$. A subset G of P is called a *filter* in P if

(a) For each $p, q \in G$, there is $r \in G$ with $r \leqslant p$ and $r \leqslant q$;

(b) For each $p \in G$, if $p \leqslant q$ then $q \in G$.

For example, let X be any set and let $\mathcal{P}(X) \setminus \{\varnothing\}$ be the collection of all nonempty subsets of X. For $p, q \in \mathcal{P}(X) \setminus \{\varnothing\}$, define $p \leqslant q$ if and only if $p \subset q$. Then G is a filter on $\mathcal{P}(X) \setminus \{\varnothing\}$ if and only if G is a filter of subsets of X in the usual sense.

We say that two elements p, q of P are *comparable* if $p \leqslant q$ or $q \leqslant p$, *compatible* (abbreviated $p \not\perp q$) if there is $r \in P$ with $r \leqslant p$ and $r \leqslant q$, and *incompatible* (abbreviated $p \perp q$) if they are not compatible. In the above example of $\mathcal{P}(X) \setminus \{\varnothing\}$, p and q are comparable if and only if one is a subset of the other, compatible if and only if they have nonempty intersection, and thus incompatible if and only if they are disjoint.

An *antichain* in P is a subset A of P such that every two elements of A are incompatible. We say that $\langle P, \leqslant \rangle$ has the *ccc* if every antichain is countable, or equivalently, every uncountable subset of P has a pair of compatible elements.

Note that $\mathcal{P}(X) \setminus \{\varnothing\}$ will not have the *ccc* if X is uncountable. But now let X be a topological space, let $\mathcal{O}(X)$ be the collection of all nonempty open sets, and define $p \leqslant q$ if and only if $p \subset q$. Then the partial order on $\mathcal{O}(X)$ has the *ccc* if and only if the space X has the *ccc*. Note that a subset $\mathcal{D}$ of $\mathcal{O}(X)$ is dense in this partial order if and only if $\mathcal{D}$ is a collection of nonempty open sets such that every nonempty open set contains a member of $\mathcal{D}$. (Such a collection $\mathcal{D}$ is sometimes called a *π-base* for the space X.)

Finally, we now define Martin's Axiom (MA): Let κ be a cardinal. $\mathsf{MA}(\kappa)$ is the following statement: Whenever $\langle P, \leqslant \rangle$ is a *ccc* partial order, and $\mathcal{D}$ is a family of $\leqslant \kappa$-many dense sets, there is a filter G in P such that $G \cap D \neq \varnothing$ for every $D \in \mathcal{D}$.

MA is the statement that $\mathsf{MA}(\kappa)$ holds for every $\kappa < 2^{\omega}$.

5.2.2 Working with MA(κ)

Theorem 224. $\mathsf{MA}(\omega)$ *is true, hence* $\mathsf{CH} \to \mathsf{MA}$.

★ **Theorem 225.** *Assume* $\mathsf{MA}(\kappa)$. *Let X be a compact Hausdorff ccc space. If $\mathcal{U}$ is a collection of $\leqslant \kappa$-many dense open sets, then $\bigcap \mathcal{U} \neq \varnothing$.*

Corollary 226. $\mathsf{MA}(2^\omega)$ *is false.*

Corollary 227. *Assume* $\mathsf{MA}(\omega_1)$. *Then there are no Suslin lines.*

The topological statement in Theorem 225 is actually equivalent to $\mathsf{MA}(\kappa)$. So Martin's Axiom is equivalent to the statement that no compact Hausdorff *ccc* space is the union of fewer than 2^ω-many nowhere-dense sets. In particular, MA implies that the real line is not the union of few than 2^ω-many nowhere-dense sets. One can also show that MA implies the real line is not the union of fewer than 2^ω-many Lebesgue measure zero sets.

★ **Lemma 228.** *Assume* $\mathsf{MA}(\omega_1)$. *Suppose* X *is ccc and* $\{U_\alpha : \alpha < \omega_1\}$ *is a collection of nonempty open subsets of* X. *Then there is an uncountable subset* A *of* ω_1 *such that* $\{U_\alpha : \alpha \in A\}$ *has the finite intersection property.*

A space X is said to have *property* K if every uncountable collection $\mathcal{U}$ of nonempty open sets contains an uncountable subcollection $\mathcal{V}$ such that $V_1 \cap V_2 \neq \varnothing$ for every $V_1, V_2 \in \mathcal{V}$.

Corollary 229. *Assume* $\mathsf{MA}(\omega_1)$. *Then every ccc space has property* K.

Theorem 230. *Assume* $\mathsf{MA}(\omega_1)$. *If* X *and* Y *are ccc, so is* $X \times Y$. *Hence any product of ccc spaces is ccc.*

Note that the above theorem also yields Corollary 227.

Lemma 231.

(a) *A regular Lindelöf space* X *is hereditarily Lindelöf if and only if* X *is perfectly normal.*

(b) *If* X *is compact Hausdorff, then* X *is first-countable if and only if every point of* X *is a* G_δ*-set. Consequently, compact Hausdorff perfectly normal spaces are first-countable.*

★ **Theorem 232.** *Assume* $\mathsf{MA}(\omega_1)$. *If* X *is a compact Hausdorff hereditarily Lindelöf space, then* X *is hereditarily separable.*

★ **Lemma 233.** *Assume* MA(κ). *Let* $\{U_\alpha : \alpha < \kappa\}$ *be a collection of dense open subsets of the real line* $\mathbb{R}$. *Then there is a dense* G_δ*-set* G *such that* $G \subset \bigcap_{\alpha<\kappa} U_\alpha$.

5.2.3 First Category Subsets of $\mathbb{R}$

A subset X of $\mathbb{R}$ is said to be of *first category*—some texts call sets of first category *meagre*, and their complements *comeagre*—in $\mathbb{R}$ if X is contained in the union of countably many nowhere-dense subsets of $\mathbb{R}$. Since the closure of a nowhere-dense sets is nowhere-dense, it is equivalent to say that X is first category if and only if the complement of X contains a dense G_δ-set.

Theorem 234. *Assume* MA. *Then the union of* $< 2^\omega$*-many first category subsets of* $\mathbb{R}$ *is first category. In particular, any subset of* $\mathbb{R}$ *of cardinality* $< 2^\omega$ *is first category.*

It is also true that assuming MA, the union of $< 2^\omega$-many Lebesgue measure zero subsets of $\mathbb{R}$ has measure zero, and in particular, any subset of $\mathbb{R}$ of cardinality $< 2^\omega$ has measure zero.

5.3 Small Cardinals

5.3.1 Unbounded and Dominating Families in ω^ω

Let $f, g \in \omega^\omega$. Define $f <^* g$ if and only if there is $k < \omega$ such that $f(n) < g(n)$ for all $n \geqslant k$. Call a subset $\mathcal{F}$ of ω^ω

(1) $<^*$*-unbounded* if there is no $g \in \omega^\omega$ such that $f <^* g$ for all $f \in \mathcal{F}$, and

(2) $<^*$*-dominating* if for every $g \in \omega^\omega$ there is $f \in \mathcal{F}$ such that $g <^* f$.

For convenience we may say a family is simply *unbounded* or *dominating*.

Define $\mathfrak{b}$ (resp., $\mathfrak{d}$) to be the least cardinal of a $<^*$-unbounded (resp., $<^*$-dominating) family in ω^ω.

Theorem 235. *$\omega_1 \leqslant \mathfrak{b} \leqslant \mathfrak{d} \leqslant \mathfrak{c}$, so* CH $\to \mathfrak{b} = \mathfrak{d} = \mathfrak{c}$.

Theorem 236. MA $\to \mathfrak{b} = \mathfrak{d} = \mathfrak{c}$.

Theorem 237. *There exists a collection $\{f_\alpha : \alpha < \mathfrak{b}\} \subset \omega^\omega$ which is unbounded and $<^*$-increasing.*

Since every cofinal subset of an unbounded $<^*$-increasing family is also unbounded, it is easy to see that $\mathfrak{b}$ is a regular cardinal; $\mathfrak{d}$ need not be.

Theorem 238. *If $\mathfrak{b} = \mathfrak{d}$, then there exists $\{f_\alpha : \alpha < \mathfrak{d}\} \subset \omega^\omega$ which is dominating and $<^*$-increasing.*

A dominating $<^*$-increasing family in ω^ω may be called a *scale*.

5.3.2 Subsets of the Irrationals

The cardinal $\mathfrak{d}$ has important connections to the structure of compact subsets of the irrationals. Recall $\mathbb{P} \cong \omega^\omega$. Call a collection $\mathcal{A}$ of compact subsets of ω^ω *dominating* if every compact subset of ω^ω is contained in some $A \in \mathcal{A}$.

★ **Theorem 239.** *Let $\mathfrak{C}$ be the collection of compact subsets of ω^ω, $\mathcal{A}$ the subcollections of $\mathfrak{C}$ which cover ω^ω, and $\mathcal{B}$ the subcollections of $\mathfrak{C}$ which are dominating. Then $\mathfrak{d} = \min\{|A| : A \in \mathcal{A}\} = \min\{|B| : B \in \mathcal{B}\}$.*

Next we have a topological characterization involving $\mathfrak{b}$. Call a subset A of the irrationals (think in the real line now) *concentrated about the rationals* $\mathbb{Q}$ if for any open $U \supset \mathbb{Q}$, the set $A \setminus U$ is countable.

Proposition C. The set of all irrationals is not concentrated about $\mathbb{Q}$.

Proposition D. Define a subset $A \subset \mathbb{P}$ to be $(<\kappa)$-concentrated about $\mathbb{Q}$ if $|A \cap C| < \kappa$ for every $C \subset \mathbb{P}$ closed in the reals. Then

$$\mathfrak{b} = \min\{\kappa > \omega : \exists A \in [\mathbb{P}]^\kappa (A \text{ is } (<\kappa)\text{-concentrated about } \mathbb{Q})\}.$$

Theorem 240. *$\mathfrak{b} = \omega_1$ if and only if there is an uncountable subset A of the irrationals concentrated about $\mathbb{Q}$.*

Theorem 240 takes added interest due to the following result.

Example 241. *Let A be an uncountable subset of the irrationals concentrated about $\mathbb{Q}$. Let X be the space $A \cup \mathbb{Q}$, where points of A are isolated and points of $\mathbb{Q}$ have their usual Euclidean neighborhoods (in the subspace $A \cup \mathbb{Q}$ of $\mathbb{R}$). Then X is a regular Lindelöf space but its product with the irrationals is not Lindelöf.*

So $\mathfrak{b} = \omega_1$, and in particular CH, implies that there is a regular Lindelöf space whose product with the irrationals is not Lindelöf. But the following question of E.A. Michael has remained open for over 40 years: Is there in ZFC a regular Lindelöf space whose product with the irrationals is not Lindelöf?

A recent result on this problem says you do not have to worry about "regular;" any example, even not T_0, implies that there is a regular one.

5.4 Normal vs. Collectionwise Normal

5.4.1 Tangent Disc Spaces

We begin by defining a modification of the Tangent Disc Space. Let A be a subset of the real line $\mathbb{R}$. Let $X(A)$ be the space whose set is

$$(A \times \{0\}) \cup \{\langle q, r\rangle : q, r \in \mathbb{Q} \wedge r > 0\}.$$

That is, $X(A)$ consists of the points on the x-axis corresponding to $a \in A$, together will all points in the upper half-plane with rational coordinates.

Let the points in the upper half-plane be isolated, and for each $a \in A$ and $n > 0$, let

$$D(a, n) = \{\langle a, 0\rangle\} \cup \{\langle q, r\rangle \in \mathbb{Q}^2 : \sqrt{(q-a)^2 + (r - 1/n)^2} < 1/n\}$$

be a basic neighborhood of $\langle a, 0\rangle$;that is, $D(a, n)$ consists of the point $\langle a, 0\rangle$ together with all points with rational coordinated in the interior of a disk of radius $1/n$ tangent to the x-axis at $\langle a, 0\rangle$.

Note that $A \times \{0\}$ is closed in $X(A)$, and discrete as a subspace. It follows that the collection of singletons $\{\{\langle a, 0\rangle\} : a \in A\}$ is a discrete collection of closed sets, and hence the union of any subcollection, that is, $B \times \{0\}$ for any $B \subset A$, is closed in $X(A)$.

★ **Theorem 242.** *Let $A \subset \mathbb{R}$, and let $X(A)$ be as defined above. Then:*

(a) *If $A = \mathbb{R}$, then $X(A)$ is not normal;*

(b) *If A is uncountable, then $X(A)$ is not collectionwise normal;*

(c) *If every subset of A is a G_δ-set in A, then $X(A)$ is normal.*

5.4.2 Q-Sets

An uncountable subset A of $\mathbb{R}$ whose every subset is G_δ in A is called a *Q-set.*

★ **Theorem 243.** *If $2^\omega < 2^{\omega_1}$, then there are no Q-sets.*

★ **Theorem 244.** *Let κ be an uncountable cardinal, and assume* $\mathsf{MA}(\kappa)$. *Then every subset of $\mathbb{R}$ of cardinality κ is a Q-set.*

Corollary 245. *Assume* $\mathsf{MA}(\omega_1)$. *Then there is a Q-set, and hence there is a normal first-countable separable non-collectionwise normal space.*

Corollary 246. *Assume* MA. *Then $2^\kappa = 2^\omega$ for every infinite cardinal $\kappa < 2^\omega$.*

Bob Heath showed that if there is a normal first-countable separable non-collectionwise normal space, then there is a Q-set. So, by Theorem 242(c), the existence of such a space is equivalent to the existence of a Q-set.

Lemma 247, Jones' Lemma. *Let X be a separable space, and let D be a closed discrete subset of X. If $2^{|D|} > 2^\omega$, then X is not normal.*

Since $2^{2^\omega} > 2^\omega$, Lemma 247 gives another way to show Theorem 242(a), the square of the Sorgenfrey line is not normal, and the like.

★ **Corollary 248.** *If $2^\omega < 2^{\omega_1}$, then every normal separable space is collectionwise normal.*

5.5 Countably Compact vs. Pseudocompact

5.5.1 Almost Disjoint Families

A collection $\mathcal{A}$ of infinite subsets of ω is said to be *almost-disjoint* if the intersection of any two distinct members of $\mathcal{A}$ is finite. By a standard Zorn's Lemma argument, every almost-disjoint family $\mathcal{A}$ is contained is a maximal almost-disjoint family $\mathcal{A}'$.

Theorem 249. *There is an almost-disjoint family of subsets of ω of cardinality 2^ω.*

Proposition E. Any finite partition of ω into infinite sets is a maximal almost-disjoint family.

Lemma 250. *No countably infinite almost-disjoint family of subsets of ω is maximal.*

Theorem 251. *Assume* MA. *Then every infinite maximal almost-disjoint family of subsets of ω has cardinality 2^ω.*

5.5.2 Mrówka Spaces

Let $\mathcal{A}$ be an almost-disjoint family of subsets of ω. Define a space $\psi(\mathcal{A})$ as follows. The underlying set for $\psi(\mathcal{A})$ is $\omega \cup \{x_A : A \in \mathcal{A}\}$, where $\{x_A : A \in \mathcal{A}\}$ is a set of distinct points not in ω. Define the topology by declaring the points of ω to be isolated, and the nth member of a local base at x_A to be $b(x_A, n) = \{x_A\} \cup (A \setminus n)$.

Theorem 252. *$\psi(\mathcal{A})$ is locally compact Hausdorff, and $\{x_A : A \in \mathcal{A}\}$ is a closed discrete subset of $\psi(\mathcal{A})$. Thus, if $\mathcal{A}$ is infinite then $\psi(\mathcal{A})$ is not countably compact.*

A space X is said to be *pseudocompact* if every continuous $f\colon X \to \mathbb{R}$ has a bounded range.

Theorem 253. *Every countably compact space is pseudocompact.*

Theorem 254. *If $\mathcal{A}$ is an infinite maximal almost-disjoint family of subsets of ω, then $\psi(\mathcal{A})$ is pseudocompact (but not countably compact).*

5.6 Hereditarily Separable vs. Hereditarily Lindelöf

5.6.1 Some History

An *S-space* is a regular hereditarily separable space which is not hereditarily Lindelöf; an *L-space* is a regular hereditarily Lindelöf space which is not hereditarily separable. By Theorem 232, assuming $\mathsf{MA}(\omega_1)$ there are no compact L-spaces. $\mathsf{MA}(\omega_1)$ also implies there are no compact S-spaces. On the other hand, a Suslin line (compactified by adding a first and last point) is a compact L-space. In this section, we show the Continuum Hypothesis can be used to construct a compact S-space.

The results in these notes related to S- and L-spaces were proven by the late 1970's. In the early 1980's, S. Todorcevic proved that under the Proper Forcing Axiom (PFA)—an axiom stronger than $\mathsf{MA}(\omega_1)$—there are no S-spaces. Whether or not there are L-spaces in ZFC remained unsettled until 2006 when J. Moore surprised everyone by constructing an L-space without assuming any special axioms of set theory.

5.6.2 The Kunen Line

Throughout this section, $2^\omega = \omega_1$ is assumed. Let $\mathbb{R} = \{x_\alpha : \alpha < \omega_1\}$, where $\mathbb{Q} = \{x_n : n < \omega\}$.

Lemma 255. *Let $\mathcal{A} = \{C \subset \mathbb{R} : |C| = \omega, |\overline{C}| > \omega\}$. It is possible to enumerate $\mathcal{A}$ as $\{A_\alpha : \omega \leqslant \alpha < \omega_1\}$ such that $\forall \alpha \geqslant \omega (x_\beta \in A_\alpha \to \beta < \alpha)$.*

Let $X = \mathbb{R}$, and for each $\alpha \leqslant \omega_1$ let $X_\alpha = \{x_\beta : \beta < \alpha\}$. Let $\mathcal{A}$ be

indexed as in Lemma 255. We inductively define a topology $\mathcal{T}_\alpha$ on X_α. Let $\mathcal{T}_\omega$ be the discrete topology on X_ω. Note that $A_\omega \subset X_\omega$. Choose a sequence $n_0 < n_1 < \cdots$ such that $\langle x_{n_i} \rangle \to x_\omega$ in the Euclidean topology and such that $x_{n_i} \in A_\omega$ if $x_\omega \in \overline{A}_\omega$. Let

$$\mathcal{B}_\omega = \{\{x_{n_i}\}_{i \geqslant k} \cup \{x_\omega\} : k \in \omega\},$$

and let $\mathcal{T}_{\omega+1}$ be the topology on $X_{\omega+1}$ generated by $\mathcal{T}_\omega \cup \mathcal{B}_\omega$.

Note that $\mathcal{T}_{\omega+1}$ is locally compact Hausdorff, has a countable base of clopen sets, and every subset of $X_{\omega+1}$ which is open in the Euclidean topology of $X_{\omega+1}$ is open in $\mathcal{T}_{\omega+1}$ (so $\mathcal{T}_{\omega+1}$ is finer than the Euclidean topology). Also, $x_\omega \in \overline{A}_\omega$ in $\mathcal{T}_{\omega+1}$ if this is true in $\mathbb{R}$.

★ **Lemma 256.** *Suppose $\alpha < \omega_1$ and for all $\beta < \alpha$ we have defined a topology $\mathcal{T}_\beta$ on X_β satisfying:*

(a) *$\langle X_\beta, \mathcal{T}_\beta \rangle$ is a locally compact Hausdorff space with a countable base of clopen sets, and $\mathcal{T}_\beta$ is finer than the Euclidean topology on X_β;*

(b) *$\gamma < \beta \to \mathcal{T}_\gamma \subset \mathcal{T}_\beta$ and $\mathcal{T}_\gamma = \mathcal{T}_\beta \restriction X_\gamma = \{U \cap X_\gamma : U \in \mathcal{T}_\beta\}$;*

(c) *If $\gamma < \delta < \beta$ and x_δ is in the Euclidean closure of A_γ, then x_δ is in $\mathrm{cl}_{\mathcal{T}_{\delta+1}} A_\gamma$.*

Then there is a topology $\mathcal{T}_\alpha$ on X_α satisfying 256(a), 256(b), *and* 256(c)*; hence this defines $\mathcal{T}_\alpha$ for all $\omega \leqslant \alpha < \omega_1$.*

Theorem 257. *Let X_α and $\mathcal{T}_\alpha$, $\alpha < \omega_1$, be as in* Lemma 256. *Let $\mathcal{T}_{\omega_1}$ be the topology on $X = \mathbb{R}$ generated by $\bigcup_{\omega \leqslant \alpha < \omega_1} \mathcal{T}_\alpha$. Then $\langle \mathbb{R}, \mathcal{T}_{\omega_1} \rangle$ is locally compact Hausdorff, hereditarily separable, and non-Lindelöf, hence its one-point compactification is a compact S-space.*

6 The Stone-Čech Compactification

6.1 Hausdorff Compactifications

A *compactification* of a completely regular space X is a compact T_2-space αX and a homeomorphic embedding $\alpha\colon X \to \alpha X$ such that $\alpha[X]$ is dense in αX.

To explain the separation assumptions in the above definition: we will only be concerned with compactifications which are Hausdorff. Note that only completely regular spaces can have a T_2-compactification. Hence in this section, unless otherwise stated, all spaces are assumed to be completely regular.

A few of the concepts and results below are from previous chapters but are included here for completeness.

6.1.1 One-Point Compactifications

Let X be a locally compact non-compact Hausdorff space. The *one-point compactification* ωX of X is the space $X \cup \{\infty\}$, where points of X have their usual neighborhoods, and a neighborhood of ∞ has the form $\{\infty\} \cup X \setminus K$ where K is a compact subset of X. In the framework of the previous definition, the map $\omega\colon X \to \omega X$ is the identity map on X.

Of course, X is open in its one-point compactification. We'll see that this is the case for any compactification of a locally compact X.

Lemma 258. *If X is a dense subset of a compact T_2-space Z, and X is locally compact, then X is open in Z.*

Corollary 259. *If αX is a compactification of a locally compact space X, then $\alpha[X]$ is open in αX.*

If αX and γX are compactifications of X, we say $\alpha X \geqslant \gamma X$ if and only if there exists a continuous function $f\colon \alpha X \to \gamma X$ such that $f \circ \alpha = \gamma$. If such f exists which is a homeomorphism, we say that αX and γX are *equivalent* compactifications, and write $\alpha X \approx \gamma X$.

Theorem 260. *If X is locally compact, ωX is the one-point compactification, and αX is any compactification, then $\alpha X \geqslant \omega X$.*

So ωX is the "smallest" compactification of X, as it intuitively ought to be.

6.1.2 Compactification by Adding Points

We often think of obtaining a compactification αX of X by adding some points to X to make it compact; that is, X is a dense subset of αX (instead of merely homeomorphic to one) and the map $\alpha\colon X \to \alpha X$ is the identity on X. The next result says we don't lose anything by thinking of compactifications in this way.

★ **Theorem 261.** *If αX is any compactification of X, then there is an equivalent compactification γX such that $X \subset \gamma X$ and the map $\gamma\colon X \to \gamma X$ is the identity map.*

Lemma 262. *Let f and g be continuous mappings from a space X into a Hausdorff space Y. If f and g agree on a dense subset of X, then $f = g$.*

Theorem 263. *$\alpha X \approx \gamma X$ if and only if $\alpha X \geqslant \gamma X$ and $\gamma X \geqslant \alpha X$.*

6.1.3 Compactification by Embedding in $[0,1]^\kappa$

Let $\mathcal{F}$ be a family of continuous functions from a space X to the unit interval $[0,1]$. We say $\mathcal{F}$ *separates points* if, given $x_1 \neq x_2 \in X$, there exists $f \in \mathcal{F}$ such that $f(x_1) \neq f(x_2)$. We say $\mathcal{F}$ *separates points from*

closed sets if, given $x \in X$ and any closed set H with $x \notin H$, there exists $f \in \mathcal{F}$ with $f(x) \notin \overline{f[H]}$. The *evaluation function determined by* $\mathcal{F}$ is the function $e_{\mathcal{F}} \colon X \to [0,1]^{\mathcal{F}}$ defined by $x \mapsto \langle f(x) \rangle_{f \in \mathcal{F}}$.

Theorem 264. *Let $\mathcal{F}$ be a collection of continuous functions from X into the unit interval. Then:*

(a) *$e_{\mathcal{F}}$ is continuous;*

(b) *If $\mathcal{F}$ separates points, then $e_{\mathcal{F}}$ is one-to-one;*

(c) *If X is a T_1-space and $\mathcal{F}$ separates points from closed sets, then $e_{\mathcal{F}} \colon X \to e_{\mathcal{F}}[X]$ is a homeomorphic embedding of X into $[0,1]^{\mathcal{F}}$.*

Corollary 265. *A space X is completely regular if and only if X is homeomorphic to a subspace of $[0,1]^{\kappa}$ for some cardinal κ.*

Corollary 266. *Any completely regular space X has a Hausdorff compactification.*

Let $\mathcal{C}(X,[0,1])$ be the collection of all continuous functions from a completely regular space X into the unit interval $[0,1]$, and let $\mathcal{F}$ be any subfamily of $\mathcal{C}(X,[0,1])$ which separates points from closed sets. Let $e_{\mathcal{F}}$ be the embedding of X into $[0,1]^{\mathcal{F}}$ defined above, and let $e_{\mathcal{F}}X = \overline{e_{\mathcal{F}}[X]}$, where the closure is taken in $[0,1]^{\mathcal{F}}$. It follows from Theorem 264(c) that $e_{\mathcal{F}}X$ is a compactification of X.

Theorem 267. *If αX is any compactification of X, there exists a subfamily $\mathcal{F}$ of $\mathcal{C}(X,[0,1])$ for which $e_{\mathcal{F}}X \approx \alpha X$.*

6.1.4 The Stone-Čech Compactification

The *Stone-Čech compactification* of a completely regular space X is defined to be $e_{\mathcal{F}}X$, where $\mathcal{F} = \mathcal{C}(X,[0,1])$. The Stone-Čech compactification of X is denoted by βX, and the embedding $e_{\mathcal{C}(X,[0,1])}$ is denoted by β_X, or just β if X is understood.

Theorem 268. *Suppose $h \colon X \to Y$ is a continuous map from X into Y (both $T_{3.5}$). Then there exists a continuous $H \colon [0,1]^{\mathcal{C}(X,[0,1])} \to$*

$[0,1]^{\mathcal{C}(Y,[0,1])}$ *such that* $H \circ \beta_X = \beta_Y \circ h$. *In particular,* H *restricted to* βX *maps* βX *into* βY.

Theorem 269. *For every compact* T_2*-space* Y *and each continuous map* $f\colon X \to Y$, *there exists a continuous* $F\colon \beta X \to Y$ *such that* $f = F \circ \beta_X$.

If, as often is done, we think of X as a subset of βX, and dispense with the map $\beta_X : X \to \beta X$, then Theorem 269 says that any continuous map from X to a compact space can be extended to βX.

Corollary 270. *If* αX *is any compactification of* X, *then* $\beta X \geqslant \alpha X$. *Also, any compactification of* X *which satisfies the condition of* Theorem 269 *is equivalent to* βX.

Theorem 271. $[0, \omega_1]$ *is the only (up to equivalence) Hausdorff compactification of the space* $[0, \omega_1)$ *of countable ordinals. Hence* $\beta[0, \omega_1) = [0, \omega_1]$.

Theorem 272. *Let* ω *be the discrete space of natural numbers. Then* $\beta\omega$ *maps continuously onto any separable compact Hausdorff space.*

Corollary 273. $|\beta\omega| \geqslant 2^{\mathfrak{c}}$.

Lemma 274. *If* X *is a separable* T_2*-space, then* $|X| \leqslant 2^{\mathfrak{c}}$.

Corollary 275. $|\beta\omega| = 2^{\mathfrak{c}}$.

★ **Theorem 276.** *Let* Z *be a compact Hausdorff space, and let* X *be a dense subspace of* Z *(so* Z *is a compactification of* X*). If every continuous* $f\colon X \to \mathbb{I}$ *extends some continuous* $f^*\colon Z \to \mathbb{I}$, *then* $Z \approx \beta X$.

6.1.5 Four More Results

In the next 4 results assume, as we may, that $X \subset \beta X$ and that the mapping $\beta_X\colon X \to \beta X$ is the identity on X.

Theorem 277. *If* X *is normal, then any two disjoint closed subsets of* X *have disjoint closures in* βX.

Theorem 278. *Let X be a normal space, and let $H \subset X$ be closed. Then $\mathrm{cl}_{\beta X} H \approx \beta H$.*

Corollary 279. *$\beta\omega$ is homeomorphic to a subspace of $\beta[0,1)$; also of $\beta\mathbb{R}$.*

Theorem 280. *$\beta[0,1) \setminus [0,1)$ is connected, and $\beta\mathbb{R} \setminus \mathbb{R}$ has exactly two components.*

6.2 The Topology of $\beta\omega$

Now we will investigate the space $\beta\omega$. We have already seen that it is surprisingly large. Next we'll see that this space can be conveniently described as the space of all ultrafilters on ω with a certain topology.

Recall, a collection $\mathcal{F}$ of subsets of a set X is called a *filter* on X if:

(1) Whenever $F_1, F_2, \ldots, F_n \in \mathcal{F}$, then $\bigcap_{i=1}^n F_i \in \mathcal{F}$;

(2) Whenever $F \in \mathcal{F}$ and $F \subset G \subset X$, then $G \in \mathcal{F}$;

(3) $\varnothing \notin \mathcal{F}$.

In other words, a filter on X is a collection of nonempty subsets of X which is closed under supersets and finite intersections.

Also, a filter $\mathcal{F}$ on X is called an *ultrafilter* if $\mathcal{F}$ is not properly contained in any other filter on X. A trivial example of an ultrafilter on X is the collection of all subsets of X containing a fixed element x_0 in X. Ultrafilters like this are called *fixed* ultrafilters. A filter $\mathcal{F}$ is said to be *free* if $\bigcap \mathcal{F} = \varnothing$. The Axiom of Choice is needed to show the existence of free ultrafilter's. A standard Zorn's Lemma argument (or a nearly trivial Tukey's Lemma argument) shows that any filter on X is contained in an ultrafilter, so applying this to any free filter, for example, to the filter of co-finite subsets of an infinite set X, gets you a free ultrafilter on X.

Theorem 281. *Let $\mathcal{F}$ be a filter on X. Then the following are equivalent:*

(a) *$\mathcal{F}$ is an ultrafilter;*

(b) *If $G \subset X$ and $G \cap F \neq \varnothing$ for every $F \in \mathcal{F}$, then $G \in \mathcal{F}$;*

(c) *For every $G \subset X$, either $G \in \mathcal{F}$ or $X \setminus G \in \mathcal{F}$.*

Theorem 282. *Let $p \in \beta\omega \setminus \omega$, and let $\mathcal{F}_p = \{U \cap \omega : p \in U^\circ\}$. Then:*

(a) *$\mathcal{F}_p$ is a free ultrafilter on ω;*

(b) *$p \neq q \to \mathcal{F}_p \neq \mathcal{F}_q$;*

(c) *If $\mathcal{F}$ is a free ultrafilter on ω, then there is a unique $p \in \beta\omega \setminus \omega$ with $\mathcal{F}_p = \mathcal{F}$.*

Theorem 283.

(a) *If $A \subset \omega$, then $\mathrm{cl}_{\beta\omega} A = A \cup \{p \in \beta\omega \setminus \omega : A \in \mathcal{F}_p\}$;*

(b) *For each $A \subset \omega$, $\mathrm{cl}_{\beta\omega} A$ is clopen in $\beta\omega$;*

(c) *The collection $\{\mathrm{cl}_{\beta\omega} A : A \subset \omega\}$ forms a base for $\beta\omega$.*

Now define a space as follows. The set for the space is

$$\omega \cup \{\mathcal{F} : \mathcal{F} \text{ is a free ultrafilter on } \omega\}.$$

It is easy to check that, for any $A, B \subset \omega$, we have $\overline{A} \cap \overline{B} = \overline{A \cap B}$. It follows that $\{\overline{A} : A \subset \omega\}$ is a base for a topology on the space.

Theorem 284. *The space as defined above is homeomorphic to $\beta\omega$.*

We now explore more topological properties of $\beta\omega$. By compactness, every infinite subset of ω has a limit point in $\beta\omega$, and said limit point must of course be in $\beta\omega \setminus \omega$, but we'll see we do not have sequential convergence:

Theorem 285. *No nontrivial—that is, not eventually constant—sequence in ω converges to a point of $\beta\omega$.*

Of course, the "interesting" points of $\beta\omega$ are the points of the remainder $\beta\omega \setminus \omega$, which is often denoted by ω^*. Note that ω^*, being closed in $\beta\omega$, is also a compact space. For each infinite $A \subset \omega$, let $A^* = \mathrm{cl}_{\beta\omega} A \cap \omega^*$.

Then by Theorem 283, the collection $\{A^* : A \subset \omega \wedge |A| = \aleph_0\}$ is a clopen base for ω^*.

Lemma 286. *Let $A, B \subset \omega$. Then:*

(a) $A^* \cap B^* = (A \cap B)^*$*;*

(b) $A^* \cap B^* = \varnothing$ *if and only if $A \cap B$ is finite;*

(c) $A^* \subset B^*$ *if and only if $A \setminus B$ is finite.*

6.2.1 ω^* is not separable

For $A, B \subset \omega$, we say that:

(a) A and B are *almost disjoint* if $A \cap B$ is finite;

(b) A is *almost included* in B (denoted $A \subset^* B$) if $A \setminus B$ is finite.

Of course, any disjoint family of subsets of ω is countable, but as we saw last chapter...

Theorem 287. *There is a pairwise almost disjoint family $\mathcal{A}$ of subsets of ω of cardinality $\mathfrak{c}$.*

Corollary 288. *There is a family of $\mathfrak{c}$-many disjoint open subsets of ω^*; in particular, ω^* is not separable.*

6.2.2 A strange property of $\beta\omega$

Now we want to see a very strange property of $\beta\omega$: it contains no nontrivial convergent sequences at all. First, recall that a subset D of a space X is *relatively discrete* if the subspace topology on D is the discrete topology. Equivalently, D being relatively discrete means that no point of D is a limit point of D; this should be compared with the concept of D being *closed discrete*, which means no point of the whole space X is a limit point of D.

Lemma 289. *If D is a countably infinite relatively discrete subset of a regular space, then there is a family of disjoint open sets $\{U_d : d \in D\}$ with $d \in U_d$ for each $d \in D$.*

Now our "no convergent sequences" in $\beta\omega$ result follows quite easily from:

Theorem 290. *If $D \subset \beta\omega$ is countably infinite and relatively discrete, then $\overline{D} \approx \beta D$ (which of course is homeomorphic to $\beta\omega$).*

Corollary 291. *$\beta\omega$ contains no nontrivial convergent sequence.*

Lemma 292. *Every infinite Hausdorff space contains an infinite relatively discrete subset.*

6.2.3 Closed subsets of $\beta\omega$

The following says that closed subsets of $\beta\omega$ are either small (finite) or very big (cardinality $2^{\mathfrak{c}}$).

Theorem 293. *Every infinite closed subset of $\beta\omega$ (or of ω^*) contains a copy of $\beta\omega$.*

Lemma 294. *If $A_0, A_1, A_2, \ldots$ are infinite subsets of ω, and $A_{n+1} \subset^* A_n$ for all $n < \omega$, then there is an infinite $A \subset \omega$ such that $A \subset^* A_n$ for all $n < \omega$.*

Such a set A as in Lemma 294 is sometimes called a *pseudo-intersection* of the A_n.

Theorem 295. *Every non-empty G_δ set in ω^* has non-empty interior.*

6.2.4 P-points

A point p in a space X is called a *P-point* if every G_δ set containing p is a neighborhood of p (equivalently, the intersection of countably many neighborhoods of p is again a neighborhood of p).

★ **Theorem 296.** *Assume* CH. *Then ω^* has a P-point.*

Lemma 297. *Every infinite compact T_2-space has a non-P-point.*

6.2.5 ω^* is not homogeneous

A space X is said to be *homogeneous* if for any $x, y \in X$, there is a homeomorphism $h\colon X \to X$ such that $h(x) = y$.

Corollary 298. *Assume CH. Then ω^* is not homogeneous.*

With a significantly more difficult argument, in the late 1960's, Z. Frolik proved that ω^* is not homogeneous without using any special (that is, beyond ZFC) axioms of set theory.

As we've seen, ω^* is the space of free ultrafilters on ω with a certain topology. When one first encounters the notion of free ultrafilters, they may seem rather elusive since you can't define one without the help of the Axiom of Choice. So Corollary 298 may be a little surprising, because it shows there are some free ultrafilters that really are different from other free ultrafilters.

6.2.6 Products of countably compact spaces

Let X be a space and let A be an infinite subset of X. Call a point p an *infinite accumulation point* of A if every neighborhood of p contains infinitely many points of A.

Theorem 299. *A topological space X is countably compact if and only if every infinite subset of X has an infinite accumulation point.*

In a T_1-space, a point p is a limit point of a set A if and only if it is an infinite accumulation point of A. So we have the following corollary.

Corollary 300. *If X is a T_1-space, then X is countably compact if and only if X does not contain an infinite subset with no limit point (which would be an infinite closed discrete set).*

Lemma 301. *Suppose $X, Y \subset \beta\omega$ and $X \cap Y = \omega$. Then $X \times Y$ is not countably compact.*

Theorem 302.

(a) *There is a subset X of ω^* such that X and $\omega^* \setminus X$ meet every infinite closed subset of ω^*;*

(b) *If X is as in* 302(a)*, then $\omega \cup X$ is countably compact.*

Corollary 303. *There are countably compact spaces $X, Y \subset \beta\omega$ such that $X \cap Y = \omega$ and hence $X \times Y$ is not countably compact.*

6.2.7 Products of sequentially compact spaces

In contrast to countable compactness, sequential compactness is countably productive.

★ **Theorem 304.** *If X_n, $n < \omega$, are sequentially compact then so is $\prod_{n<\omega} X_n$.*

But not every product of sequentially compact spaces is sequentially compact.

★ **Theorem 305.** $[0,1]^{[0,1]}$ *is not sequentially compact.*

Let $A \subsetneq^* B$ denote that $A \subset^* B$ but $A \neq^* B$; that is, $A \setminus B$ is finite but $B \setminus A$ is infinite. Let α be an ordinal. A sequence A_β, $\beta < \alpha$, of infinite subsets of ω is called a (decreasing) *tower* of length α if $\beta < \gamma < \alpha$ implies $A_\gamma \subsetneq^* A_\beta$. If an infinite set A has the property that $A \subset^* A_\beta$ for every $\beta < \alpha$, we call A a *pseudo-intersection* of the tower $\{A_\beta : \beta < \alpha\}$.

Given any decreasing tower, if it has an infinite pseudo-intersection we can extend it. If we inductively keep extending until we can extend no more, we arrive at a decreasing tower which does have an infinite pseudo-intersection. We define $\mathfrak{t}$ to the least cardinal κ such that there is a tower of length κ with no infinite pseudo-intersection.

Obviously, $\mathfrak{t} \leqslant \mathfrak{c}$. It follows from Lemma 294 that $\mathfrak{t} \geqslant \omega_1$.

★ **Theorem 306.** $\mathfrak{t} \leqslant \mathfrak{b}$.

Theorem 307. *Assume* MA*. Then* $\mathfrak{t} = \mathfrak{c}$.

The next theorem is a strengthening of Theorem 304.

Theorem 308. *A product of fewer than* $\mathfrak{t}$*-many sequentially compact spaces is sequentially compact, and the product of* $\mathfrak{t}$*-many sequentially compact spaces is countably compact.*

In particular, since $\mathfrak{t} \geqslant \omega_1$, the product of ω_1-many sequentially compact spaces is always countably compact. The following related problem is a well-known unsolved problem (the "Scarborough-Stone problem"):

Question. Must the product of any family of regular sequentially compact spaces be countably compact?

The regular assumption is in there because there is known to be a collection of Hausdorff sequentially compact spaces whose product is not countably compact. In 1984, E.K. van Douwen proved that assuming $\mathfrak{b} = \mathfrak{c}$ (in particular, CH or MA) that there is a family of $2^{\mathfrak{c}}$-many sequentially compact regular spaces whose product is not countably compact (that $2^{\mathfrak{c}} = |\beta\omega|$ is not a coincidence: he used $\beta\omega$ to assist him in the construction).

It is still not known if one can construct without assuming any special axioms of set theory a family of regular sequentially compact spaces whose product is not countably compact.

6.2.8 Characterizing βX

Now we want to get a characterization of βX in general, in the same spirit as out characterization of $\beta\omega$ in terms of ultrafilters. Roughly speaking, the characterization is obtained by replacing ultrafilters on ω with ultrafilters of zero-sets in X. We'll apply our characterization to obtain an interesting property of $\beta[0,\infty) \setminus [0,\infty)$ (which by Corollary 259 is compact and by Theorem 280 is connected).

A subset F of a space X is called a *zero-set* in X if $F = g^{-1}(0)$ for some continuous $g\colon X \to \mathbb{R}$. It is equivalent to say, instead of $F = g^{-1}(0)$, that $F = g^{-1}(C)$ for some closed $C \subset \mathbb{R}$. The complement of a zero-set is called a *cozero-set*.

Lemma 309.

(a) *Each point of a completely regular space has a neighborhood base of zero-sets and an open neighborhood base of cozero-sets.*

(b) *Every zero-set F in X is closed and a regular G_δ set; that is, there are open sets U_n, $n < \omega$, such that $F = \bigcap_{n<\omega} U_n = \bigcap_{n<\omega} \overline{U}_n$.*

(c) *In a normal space, the zero-sets are precisely the G_δ sets; so in a perfectly normal space, every closed set is a zero-set.*

In the following results, when relevant, we always assume $X \subset \beta X$.

Lemma 310.

(a) *If F_1, F_2 are zero-sets, so is $F_1 \cup F_2$.*

(b) *If F_n is a zero-set for each $n < \omega$, then $F = \bigcap_{n<\omega} F_n$ is a zero-set.*

(c) *If F is a zero-set in X, and $Y \subset X$, then $F \cap Y$ is a zero-set in Y.*

(d) *If F_1 and F_2 are disjoint zero-sets, then there exists a continuous $f\colon X \to [0,1]$ with $F_1 = f^{-1}(0)$ and $F_2 = f^{-1}(1)$.*

(e) *Disjoint zero-sets in X have disjoint closures in βX.*

A *z-filter* in X is a collection $\mathcal{F}$ of zero-sets satisfying:

(1) $\varnothing \notin \mathcal{F}$;

(2) $F_1, F_2 \in \mathcal{F} \to F_1 \cap F_2 \in \mathcal{F}$;

(3) If $F \in \mathcal{F}$ and G is a zero-set containing F, then $G \in \mathcal{F}$.

A maximal z-filter is called a *z-ultrafilter*. It is straightforward to show that a z-filter $\mathcal{F}$ is a z-ultrafilter if and only if $\mathcal{F}$ has the following property: For each zero-set Z, if $Z \cap F \neq \varnothing$ for each $F \in \mathcal{F}$, then $Z \in \mathcal{F}$.

Theorem 311. *Let X be a completely regular space, and for each $p \in \beta X \setminus X$ let*

$$\mathcal{F}_p = \{F \subset X : F \text{ is a zero-set in } X \text{ and } p \in \mathrm{cl}_{\beta X} F\}.$$

Then:

(a) *$\mathcal{F}_p$ is a z-ultrafilter in X.*

(b) *For every zero-set F in X, $\mathrm{cl}_{\beta X} F = F \cup \{p \in \beta X \setminus X : F \in \mathcal{F}_p\}$.*

(c) *The collection $\{\beta X \setminus \mathrm{cl}_{\beta X} F : F \text{ is a zero-set in } X\}$ is a base for the topology of βX.*

(d) *For each cozero-set U in X, let*

$$\mathrm{Ex}\, U = U \cup \{p \in \beta X \setminus X : \exists F \in \mathcal{F}_p (F \subset U)\}.$$

Then $\mathrm{Ex}\, U = \beta X \setminus \mathrm{cl}_{\beta X}(X \setminus U)$, and $\{\mathrm{Ex}\, U : U \subset X,\ U \text{ cozero}\}$ forms a base for βX.

(e) *For a cozero subset U of X, $\mathrm{Ex}\, U$ is the largest open subset of βX whose intersection with X equals U.*

6.2.9 Continua

A *continuum* is a compact connected Hausdorff space, though many topologists include the adjective "metrizable" in the definition. A continuum is *indecomposable* if it cannot be written as the union of two proper subcontinua. The Knaster continuum, or buckethandle continuum, constructed from the Cantor set by joining certain pairs of points with semicircles, is an example of an indecomposable continuum in the plane.

★ **Theorem 312.** *A continuum X is indecomposable if and only if every proper subcontinuum is nowhere-dense in X.*

Let $\mathbb{H}$ be the half-line $[0, \infty)$. By Theorem 280, $\mathbb{H}^* = \beta\mathbb{H} \setminus \mathbb{H}$ is a continuum. Our goal now is to prove it is indecomposable.

Lemma 313. *Let O be open in $\mathbb{H}$ such that $0 \notin O$ and $\mathbb{H} \setminus \overline{O}$ is unbounded. Then there are $a_n, b_n \in \mathbb{H}$ with $a_0 < b_0 < a_1 < b_1 < a_2 < \dots$ such that $O \subset \bigcup_{n<\omega}(a_n, b_n)$.*

Lemma 314. *Let K be a proper subcontinuum of $\mathbb{H}^*$. Then there are $a_n, b_n \in \mathbb{H}$ with $a_n < b_n < a_{n+1}$ for all n such that:*

(a) *If $O = \bigcup_{n<\omega}(a_n, b_n)$, then $K \subset \operatorname{Ex} O$;*

(b) *If $u = \{A \subset \omega : K \subset \operatorname{cl}_{\beta H}(\bigcup_{n\in A}[a_n, b_n])\}$, then u is an ultrafilter on ω.*

Proposition F. If $p \in \operatorname{Ex} O \setminus X$, then there exists $F \in \mathcal{F}_p$ where $F \subseteq O$. Since $F \in \mathcal{F}_p$, $p \in \operatorname{cl}_{\beta X} F \subset \operatorname{cl}_{\beta X} O$.

Theorem 315. *$\mathbb{H}^*$ is indecomposable.*

7 Topological Games

7.1 Completeness & the Baire Property

In this section we will introduce Čech completeness, which for metrizable spaces is equivalent to completely metrizable. It makes sense to study this after our chapter on compactifications, because the definition of Čech complete is in terms of the Stone-Čech compactification. This study will also lead naturally into the topic of topological games.

In this section, when we talk about a compactification αX of X we will assume that $X \subset \alpha X$ and that the map $\alpha\colon X \to \alpha X$ is the identity on X.

7.1.1 Complete metric spaces

First, recall that a metric space $\langle X, d\rangle$ is *complete* if every d-Cauchy sequence converges. A space $\langle X, \mathcal{T}\rangle$ is *completely metrizable* if X has a complete metric compatible with $\mathcal{T}$. We have also proved:

(1) The following are equivalent for a complete metric d on a space X:

  (a) d is complete;

  (b) If $A_0, A_1, \ldots$ is a decreasing sequence of closed sets and $\operatorname{diam} A_n \to 0$, then $\bigcap_{n<\omega} A_n \neq \varnothing$.

(2) If $\langle X, d\rangle$ is complete and $Y \subset X$, then $\langle Y, d \restriction Y^2\rangle$ is complete if and only if Y is closed in X, but Y is completely metrizable if and only if Y is a G_δ set in X.

(3) Any metric space $\langle X, d\rangle$ has a completion; that is, a unique (up to isometry) complete metric space $\langle \tilde{X}, \tilde{d}\rangle$ such that X is dense in $\tilde{X}$ and $\tilde{d} \restriction X \times X = d$.

(4) Every completely metrizable space is a Baire space; that is, the intersection of countably many dense open sets is dense.

For example, since the subset $\mathbb{P}$ of irrationals in the real line $\mathbb{R}$ is not closed in $\mathbb{R}$, the Euclidean metric on $\mathbb{P}$ is not a complete metric; however, $\mathbb{P}$ is completely metrizable since $\mathbb{R}$ is and $\mathbb{P}$ is G_δ in $\mathbb{R}$. One can obtain a complete metric on $\mathbb{P}$ using $\mathbb{P}\cong\omega^\omega$ and for $f \neq g$ in ω^ω defining $d(f,g) = 1/2^n$ if and only if n is least such that $f(n) \neq g(n)$.

On the other hand, the space $\mathbb{Q}$ of rationals is not G_δ in $\mathbb{R}$ so $\mathbb{Q}$ is not completely metrizable and thus there is not a complete metric on $\mathbb{Q}$ which generates the usual topology of $\mathbb{Q}$.

7.1.2 Čech complete

A completely regular space X is *Čech complete* if X is a G_δ set in some compactification αX of X (equivalently, $\alpha X \setminus X$ is σ-compact).

Note that every locally compact completely regular space is Čech complete, since locally compact spaces are open in any compactification. The irrationals are G_δ in the one point compactification of the reals, so they are Čech complete. We will soon see that in fact any completely metrizable space is Čech complete.

Lemma 316. *If αX and γX are compactifications of X, and $f\colon \alpha X \to \gamma X$ is a continuous function with $f \restriction X = \mathrm{id}_X$, then $f^{-1}[\gamma X \setminus X] = \alpha X \setminus X$.*

Theorem 317. *The following are equivalent for a completely regular space X:*

(a) *X is Čech complete;*

(b) *X is a G_δ set in βX;*

(c) *X is a G_δ set in every compactification of X;*

(d) *X is a G_δ set in every completely regular space in which it is densely embedded.*

Theorem 318. *The following are equivalent for a completely regular space X:*

(a) *X is Čech complete;*

(b) *There is a sequence $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \ldots$ of open covers of X such that: whenever $\mathcal{F}$ is a filter of closed sets such that for each $n < \omega$ there is some $F_n \in \mathcal{F}$ with $F_n \subset U$ for some $U \in \mathcal{U}_n$, then $\bigcap \mathcal{F} \neq \varnothing$.*

Exercise 319. *True or False:*

(a) *The Sorgenfrey line is Čech complete.*

(b) *The tangent disk space is Čech complete.*

Theorem 320. *The following are equivalent for a metrizable space X:*

(a) *X is Čech complete;*

(b) *X is completely metrizable.*

Corollary 321. *Let X be completely metrizable, and $A \subset X$. Then A is completely metrizable if and only if A is G_δ in X.*

Theorem 322.

(a) *If X is a closed or G_δ subspace of a Čech complete space Y, then X is Čech complete.*

(b) *If X_n is Čech complete for each $n < \omega$, then $\prod_{n<\omega} X_n$ is Čech complete.*

★ **Example 323.** *ω^{ω_1} is not Čech complete.*

7.1.3 The Baire property

Theorem 324. *Every Čech complete space is a Baire space.*

There is no decreasing sequence version (as there is for complete metric spaces) of Theorem 318(b) (by this we mean replacing the filter $\mathcal{F}$ with a decreasing sequence of closed sets $\langle F_n \rangle_{n<\omega}$) that is equivalent to Čech completeness. To see this, not that if there were, then any

completely regular countably compact space would be Čech complete (since in a countably compact space, every decreasing sequence of closed sets has non-empty intersection). But the countably compact space we encountered in Theorem 302 is not Čech complete:

Example 325. *Let X be a subspace of ω^* such that X and $\omega^* \setminus X$ meet every infinite closed subset of ω^*. Then X is complete regular and countably compact but not Čech complete.*

Regular countably compact spaces are, however, Baire:

Theorem 326. *Every regular countably compact space is a Baire space.*

There are separable metrizable spaces that are Baire but not Čech complete; equivalently, not completely metrizable. A subset B of the real line is *Bernstein* if both B and $\mathbb{R} \setminus B$ meet every uncountable closed subset of $\mathbb{R}$.

Theorem 327. *Let B be a Bernstein subset of the real line. Then B is a Baire space but is not completely metrizable.*

Theorem 328.

(a) *If X is an open subspace, or a dense G_δ subspace, of a Baire space, then X is Baire.*

(b) *If X has a dense Baire subspace, then X is Baire.*

★ **Example 329.** *There is a subspace X of the real line which is Baire but has a closed (and hence G_δ too) non-Baire subspace.*

★ **Theorem 330.** *If X is Baire and Y is Baire and second-countable, then $X \times Y$ is Baire.*

Theorem 331. *If X is Baire, and Y is regular and countably compact or Čech complete, then $X \times Y$ is Baire.*

7.2 The Choquet game

Let X be a space. Define a two-person infinite game on X, called the *Choquet game*, or the *Banach-Mazur game*, and denoted $\mathrm{Ch}(X)$, as follows. There are two players, Empty (E) and Nonempty (NE). In the first round, E chooses a nonempty open set U_0, and NE then chooses a nonempty open subset V_0 of U_0. In the nth round, E chooses a nonempty open subset U_n of V_{n-1}, and NE responds with a nonempty open $V_n \subset U_n$. There is a round for every $n < \omega$.

A play of the game produces a decreasing sequence

$$U_0 \supset V_0 \supset U_1 \supset V_1 \supset U_2 \supset \dots$$

of nonempty open sets. We say NE *wins* the game if $\bigcap_{n<\omega} U_n \neq \varnothing$ (note that $\bigcap_{n<\omega} U_n = \bigcap_{n<\omega} V_n$, since the sequence is decreasing). Otherwise E wins; that is, E wins if $\bigcap_{n<\omega} U_n = \varnothing$.

7.2.1 Strategies

A *strategy* for player E is a function σ from finite decreasing sequences of nonempty open sets (including the empty sequence) of the form

$$\langle U_0, V_0, U_1, V_1, \dots, U_n, V_n \rangle$$

to the collection of all nonempty open sets such that, for each $n < \omega$,

$$\sigma(\langle U_0, V_0, U_1, V_1, \dots, U_n, V_n \rangle) \subset V_n.$$

Such a strategy σ is called a *winning strategy* for E if whenever

$$U_0, V_0, U_1, V_1, U_2, \dots$$

is an infinite decreasing sequence of open sets such that $U_0 = \sigma(\varnothing)$ and

$$U_{n+1} = \sigma(\langle U_0, V_0, U_1, V_1, \dots, U_n, V_n \rangle)$$

for each $n < \omega$, then $\bigcap_{n<\omega} U_n = \varnothing$. A strategy and winning strategy from NE is defined analogously.

Obviously, E and NE cannot both have a winning strategy. But it is possible that neither does; in such a case we say the game is *undetermined*.

When considering so-called "perfect information" winning strategies as above, we may consider a winning strategy for E or NE to be a function σ defined on finite sequences of the opponents moves only, and such that if $\sigma(U_0, \ldots, U_n)$ is the player's move given the opponent has played $U_0, \ldots, U_n$, then that player is guaranteed to win. This can simplify notation for arguments involving such strategies.

Theorem 332. *A space X is a Baire space if and only if E has no winning strategy in the game* $\mathrm{Ch}(X)$.

Theorem 333. *If X is either Čech complete, or regular and countably compact, then NE has a winning strategy in* $\mathrm{Ch}(X)$.

Theorem 334. *Let B be a Bernstein subset of the real line. Then* $\mathrm{Ch}(B)$ *is undetermined.*

It is sometimes useful to know that one may assume a player is choosing from a certain base $\mathcal{B}$ for X. Let $\mathrm{Ch}(X, \mathcal{B})$ denote the game in which both players always choose a member of $\mathcal{B}$. The next theorem implies that the game $\mathrm{Ch}(X, \mathcal{B})$ is equivalent to the original game $\mathrm{Ch}(X)$ in the sense that a player has a winning strategy in one game if and only if they have one in the other.

Theorem 335. *Let $\mathcal{B}$ and $\mathcal{C}$ be bases for a space X. Then E (resp., NE) has a winning strategy in* $\mathrm{Ch}(X, \mathcal{B})$ *if and only if E (resp., NE) has a winning strategy in* $\mathrm{Ch}(X, \mathcal{C})$.

7.2.2 Choquet Spaces

A space X is said to be *Choquet* if NE has a winning strategy in the game $\mathrm{Ch}(X)$; clearly, Choquet spaces are Baire.

When the game is called the Banach-Mazur game, the players E and NE are denoted β and α, respectively (and yes, β goes first!), and the property "Choquet" as defined here is called *weakly α-favorable*.

α-favorable means NE has a winning strategy depending only on E's last move (sometimes called a *stationary* winning strategy, or a *tactic*).

Theorem 336. *Any product of Choquet spaces is Choquet.*

Corollary 337. *Any product of Čech complete spaces is Choquet, hence Baire.*

Note that by Theorem 322 and Example 323, while a countable product of Čech complete spaces is always Čech complete, an uncountable product need not be Čech complete. But the above corollary says the product will still be Baire; in particular, ω^{ω_1} is Baire but not Čech complete, and any product of real lines is Baire.

7.2.3 Products of Baire Spaces

It's time we saw an example of Baire spaces X and Y whose product is not Baire. By various results above, neither X nor Y can be Čech complete, regular countably compact, or second-countable Baire. In particular, if we want metrizable such X and Y, then neither factor can be separable. But we will presently see that there are metrizable Baire spaces X and Y of weight ω_1 such that $X \times Y$ is not Baire. The example is due to Fleissner and Kunen; examples were previously known, but their example is simpler, and a nice application of stationary sets.

First let's fix some notation. The spaces are going to be subspaces of ω_1^ω, where the set ω_1 of countable ordinals is given the discrete topology. So points in ω_1^ω are functions from ω to ω_1, or equivalently, infinite sequences (in order type ω) of countable ordinals.

Let $\omega_1^{<\omega}$ denote the set of all finite sequences of countable ordinals. If $\sigma \in \omega_1^{<\omega}$, let $[\sigma] = \{f \in \omega_1^\omega : \sigma \subset f\}$. Note that $\sigma \subset f$ simply means that the infinite sequence f extends the finite sequence σ. It is easy to check that $\mathcal{B} = \{[\sigma] : \sigma \in \omega_1^{<\omega}\}$ is a base for ω_1^ω. Also, for each $f \in \omega_1^\omega$, the collection $\{[f \restriction n] : n < \omega\}$ is a local base at the point f.

For $f \in \omega_1^\omega$, let $f^* = \sup\{f(n) : n < \omega\}$. Of course, f^* is a countable ordinal. For $A \subset \omega_1$, let $A^* = \{f \in \omega_1^\omega : f^* \in A\}$.

★ **Lemma 338.**

(a) *If $A \subset \omega_1$ is uncountable, then A^* is dense in ω_1^ω.*

(b) *If A and B are uncountable disjoint subsets of ω_1, then $A^* \times B^*$ is not Baire.*

By the previous lemma, if we can find two disjoint uncountable subsets A and B of ω_1 such that A^* and B^* are Baire, we'll have our example. It turns out that disjoint stationary sets (we saw last chapter that they exist) will do the job. First a technical lemma to be used as a tool in the proof of the theorem.

Lemma 339. *Let U be dense open in ω_1^ω. For each $\sigma \in \omega_1^{<\omega}$, choose an extension σ_U of σ such that $[\sigma_U] \subset U$. Then*

$$C_U = \{\delta \in \omega_1 : \forall \sigma \in \delta^{<\omega}(\sigma_U \in \delta^{<\omega})\}$$

is club in ω_1.

Theorem 340. *Let A be a stationary subset of ω_1. Then A^* is Baire.*

Corollary 341. *There are metrizable Baire spaces X and Y such that $X \times Y$ is not Baire.*

7.3 W-Spaces

7.3.1 The W-Game

Consider the following topological game, denoted $G_{O,P}(X, x)$, played in a space X at a point $x \in X$: In round n, Player O (for "Open") plays an open neighborhood U_n of x, and Player P (for "Point") selects a point $x_n \in U_n$. Player O *wins* the game if $\langle x_n \rangle \to x$; otherwise P wins. Let us call this game the *W-game* (on X, at x) for short. If O has a winning strategy in the W-game at every $x \in X$, then X is called a *W-space*.

A space X is said to be a *Fréchet space* if whenever $x \in \overline{A}$, there are $x_n \in A$ with $\langle x_n \rangle \to x$.

Theorem 342. *Every first-countable space is a W-space and every W-space is a Fréchet space.*

The one-point compactification of an uncountable discrete space is an example of a W-space which is not first-countable: all O needs to do is to make P choose a different point each time.

Here's an example of a Fréchet space which is not a W-space (in fact, P has a winning strategy). It is called the *sequential fan*, and is denoted by S_ω. It can be described as the quotient space of the topological sum of countably many convergent sequences obtained by identifying the limit points of the sequences to a single point ∞. This makes ∞ the only non-isolated point. The space can also be described as the set $(\omega \times \omega) \cup \{\infty\}$, where points of $\omega \times \omega$ are isolated, and for each function $f\colon \omega \to \omega$, a basic neighborhood $U(f)$ of ∞ is defined by $U(f) = \{\infty\} \cup \{\langle m, n\rangle \in \omega \times \omega : n \geqslant f(m)\}$.

Example 343. *The sequential fan S_ω is Fréchet, but P has a winning strategy in the W-game at ∞.*

Theorem 344. *Every countable W-space is first-countable, as is every regular separable W-space.*

Theorem 345.

(a) *Every subspace of a W-space is a W-space; in particular, every countable subspace of a W-space is first-countable;*

(b) *The product of countably many W-spaces is a W-space.*

Something more general than Theorem 345(b) above holds. For each $\alpha < \kappa$ pick a point p_α in a space X_α. Then the set

$$\Sigma_{\alpha<\kappa} X_\alpha = \{\vec{x} \in \Pi_{\alpha<\kappa} X_\alpha : |\{\alpha : x_\alpha \neq p_\alpha\}| \leqslant \omega\},$$

with the topology as a subspace of the usual product topology, is a called a Σ-*product* of the X_α's (with *base point* $\vec{p} = \langle p_\alpha \rangle_{\alpha<\kappa}$).

Theorem 346. *Let $\{X_\alpha : \alpha < \kappa\}$ be a collection of W-spaces. Then $\Sigma_{\alpha<\kappa} X_\alpha$ is also a W-space.*

7.3.2 w-spaces

If the point picker does not have a winning strategy in the W-game at x in a space X, then x is called a *w-point* and X is a *w-space* if every point is a w-point. So, if X is a w-space which is not a W-space, then at some point in X the game is undetermined.

Though weaker than W, w-spaces still have strong convergence properties. A space X is said to be *strongly Fréchet*, or *countably bi-sequential*, if whenever $A_0, A_1, \ldots$ is a decreasing sequence of subsets of X with $x \in \overline{A}_n$ for all $n < \omega$, then there are $x_n \in A_n$ with $\langle x_n \rangle \to x$. A space X is said to have *countable tightness* if whenever $x \in \overline{A}$, then there is a countable subset B of A with $x \in \overline{B}$.

Some of the implications in the next theorem are trivial or already done; they are listed for the sake of comparison and future reference. By 347(c)$\leftrightarrow$347(d) below, the w-space property is equivalent to a slightly more general form of strongly Fréchet.

★ **Theorem 347.** *Each statement below implies the next, and* 347(d) *implies* 347(c).

(a) *X is a W-space;*

(b) *X has countable tightness and every countable subspace is first-countable;*

(c) *Whenever $x \in \overline{A}_n$ for every $n < \omega$, then there are $x_n \in A_n$ with $x_n \to x$;*

(d) *X is a w-space;*

(e) *X is strongly Fréchet;*

(f) *X is Fréchet;*

(g) *X has countable tightness.*

We saw that the sequential fan S_ω is Fréchet; it is easily seen not to be strongly Fréchet. If p is a free ultrafilter on ω, then the subspace $\omega \cup \{p\}$ of $\beta\omega$ has countable tightness (trivially), but is not Fréchet.

Finding a w-space that is not a W-space is harder; indeed, Gruenhage left it as an open question in his original paper on W-spaces. The following example is due to Istvan Juhasz. If T is a tree, the *interval topology* on T is the topology with basis sets of the form $(s,t] = \{x \in T : s < x \leqslant t\}$. Since the predecessors of any element of T is well-ordered, this is a locally compact topology, and so has a one-point compactification $T \cup \{\infty\}$. Let $t^{\downarrow} = \{s \in T : s \leqslant t\}$. It is easy to see that $K \subset T$ is compact if and only if $K \subset \bigcup_{t \in F} t^{\downarrow}$ for some finite $F \subset T$. Note that a countably infinite subset S of T converges to ∞ if and only if $t^{\downarrow} \cap S$ is finite for each $t \in T$. Examples of such S are antichains and cofinal subsets of some branch.

Example 348. *Let T be an Aronszajn tree with the interval topology, and $T \cup \{\infty\}$ its one-point compactification. Then $T \cup \{\infty\}$ is not a W-space, but it has countable tightness and every countable subset is first-countable (hence it is a w-space).*

7.3.3 w but not W

Example 348 leaves open the question of whether or not there could be a *countable* w-space which is not W. It turns out there are such examples, one such being obtained from something called a "Hausdorff gap." A *Hausdorff gap* is a pair $\langle \langle a_\alpha \rangle_{\omega_1}, \langle b_\alpha \rangle_{\omega_1} \rangle$ of ω_1-sequences of subsets of ω satisfying:

(1) $a_\alpha \subset^* \alpha_\beta \subset^* b_\beta \subset^* b_\alpha$ for each $\alpha < \beta < \omega_1$;

(2) There is no $c \subset \omega$ such that $a_\alpha \subset^* c \subset^* b_\alpha$ for each $\alpha < \omega_1$.

So the sets a_α on the "left side" of the gap are $\subset^*$-increasing, the right side is $\subset^*$-decreasing, every set on the right contains mod-finite every set on the left, and there is no set in between. To get a feel for this, try the following exercise.

Theorem 349. *Let $\langle a_n \rangle$ and $\langle b_n \rangle$ be sequences of subsets of ω with $a_m \subset^* a_n \subset^* b_n \subset^* b_m$ for each $m < n < \omega$. Then there is a subset c of ω such that $a_n \subset^* c \subset^* b_n$ for each $n < \omega$.*

The existence of a Hausdorff gap, sometimes called an (ω_1, ω_1^*)-gap,

in ZFC (which was proven by Hausdorff himself!) might be surprising in view of Theorem 307, the proof of which shows that every decreasing tower of length ω_1 has an infinite pseudo-intersection assuming $\mathsf{MA}(\omega_1)$. So its existence can be thought of as illustrating an absolute property of the cardinal ω_1, which can't be negated by MA or anything else.

Here's why we are interested now in Hausdorff gaps. Given such, let $\mathcal{I} = \{x \subset \omega : x \cap a_\alpha \text{ is finite for every } \alpha < \omega_1\}$. Note that $\mathcal{I}$ is closed under finite unions, and that besides finite sets, for any α the set $\omega \setminus b_\alpha$ is in $\mathcal{I}$.

Now let $X = \omega \cup \{\infty\}$, where the points of ω are isolated, and a neighborhood of ∞ has the form $X \setminus I$, where $I \in \mathcal{I}$. There are three easy obervations about X:

(1) every a_α converges to ∞;

(2) $\{\infty\} \cup b_\alpha$ is open for each α; and

(3) if $\infty \in \overline{A}$, then A must meet some a_α in an infinite set.

Example 350. *The space $X = \omega \cup \{\infty\}$ defined above is a w-space but not a W-space.*

8 Hints

Hint for Theorem 3
Given $f\colon X \to \mathcal{P}(X)$, consider the set $Y = \{x \in X : x \notin f(x)\}$.

Hint for Theorem 5
A transitive subset of **Ord** is a member of **Ord**. (A set X is called *transitive* if $Y \subset X$ whenever $Y \in X$.)

Hint for Theorem 8
Note that $G(0)$ and $G(1)$ are uniquely defined, given R. Let $\gamma < \kappa$. If there is a function g with domain δ, where $\gamma < \delta$, such that for any $\alpha < \delta$, $g(\alpha) = R(g \restriction \alpha)$, then define $G(\gamma)$ to be $g(\gamma)$. Prove by transfinite induction that G is uniquely defined for every $\alpha < \kappa$ and satisfies the condition $G(\alpha) = R(G \restriction \alpha)$ for any $\alpha < \kappa$.

Hint for Theorem 51
Well-order $\mathbb{Q} \cap [0, 1]$, handling the endpoints carefully, and induct.

Hint for Example 69
Consider $Y = \{\langle x, -x\rangle : x \in \mathbb{S}\}$ as a subspace of $\mathbb{S}^2$.

Hint for Theorem 68
Recall Lemma 57.

Hint for Theorem 132
Without loss of generality, every member of $\mathcal{F}$ has the same cardinality k. Induct on k.

Hint for Example 138

For non-collectionwise normal, show that $\{\{e_\alpha\} : \alpha \in A\}$ is a discrete collection of singleton sets and use the fact that the usual product topology on $2^{\mathcal{P}(A)}$ is *ccc*.

Hint for Theorem 148

For (c $\to$ a): Let $\mathcal{U}$ be an open cover of X. By assumption, there is a locally finite closed refinement $\mathcal{A}$ of $\mathcal{U}$. For each $A \in \mathcal{A}$, choose $U_A \in \mathcal{U}$ with $A \subset U_A$. Since $\mathcal{A}$ is locally finite, the collection $\mathcal{V}$ of all open sets which meet only finitely many elements of $\mathcal{A}$ is an open cover of X. Let $\mathcal{B}$ be a locally finite closed refinement of $\mathcal{V}$. For each $A \in \mathcal{A}$, let $W_A = X \setminus \bigcup\{B \in \mathcal{B} : B \cap A = \varnothing\}$. Finally, let $O_A = W_A \cap U_A$. Show that $\mathcal{O} = \{O_A : A \in \mathcal{A}\}$ is a locally finite open refinement of $\mathcal{U}$.

Hint for Theorem 149

Use Theorem 148(b).

Hint for Lemma 156

$\mathcal{B} = \{X \setminus \overline{\bigcup \mathcal{V}} : \mathcal{V} \in [\mathcal{U}]^{<\omega}\}$, where $[\mathcal{U}]^{<\omega}$ is the collection of all finite subsets of $\mathcal{U}$, is a countable base for X.

Hint for Theorem 159

For (a $\to$ b): If $\mathcal{N}$ is a countable network for X, show that the topology on X obtained by taking $\mathcal{N} \cup \{X \setminus N : N \in \mathcal{N}\}$ as a subbase is a separable metrizable topology finer than the original topology on X.

Hint for Theorem 176

It follows readily from Corollary 175 that such an $\langle \tilde{X}, \tilde{d} \rangle$ exists (consider the closure of the image of X under the isometric embedding). You need to show uniqueness.

Hint for Theorem 178

Construct what is often called a "Cantor tree" of clopen sets, as follows. Let $B_1, B_2, \ldots$ enumerate a countable base of nonempty clopen sets. Divide $X = X_\varnothing$ into two nonempty clopen pieces X_0 and X_1 such that,

unless B_1 is the whole space, then B_1 is one of the pieces. Then divide each of X_0 and X_1 into two pieces, say X_{00}, X_{01}, X_{10}, and X_{11} such that each piece is either contained in B_2 or completely misses B_2. And so on. Show that given any infinite sequence $\vec{e} = e_1 e_2 e_3 \ldots$ of 0's and 1's, the intersection of the "branch" $X_\varnothing, X_{e_1}, X_{e_1 e_2}, \ldots$ is a single point p, and the nodes of this branch, that is, $\{X_{\vec{e}\restriction n} : n \in \mathbb{N}\}$, form a base at p.

Hint for Theorem 181

Get such a function whose domain is $\mathbb{C}$ and then extend linearly to $[0, 1]$.

Hint for Lemma 187

For every infinite cardinal κ, the set of all finite subsets of κ also has cardinality κ.

Hint for Theorem 191

Given X, call a subset A of X *left-closed* if A is closed, and $a \in A$ and $b < a$ implies $b \in A$. For example, for each $x \in X$, the set $\{a \in X : a \leqslant x\}$ is left-closed. (Other left-closed sets are often called "gaps.") Let $\hat{X}$ be all left-closed sets, ordered by $\subset$. Include the empty set in $\hat{X}$ if and only if X has no least element.

Hint for Theorem 194

Show that the set H of all limit ordinals is a closed set, and any open superset of H contains all but countably many points of the space.

Hint for Example 201

Let $H = \{\langle \alpha, \alpha \rangle : \alpha < \omega_1\}$ and $K = [0, \omega_1) \times \{\omega_1\}$. Show H and K are disjoint closed sets which can't be separated.

Hint for Lemma 202

Note that 202(a) is essentially equivalent to $\kappa = |\kappa \times \kappa|$.

Hint for Theorem 207

For the "if" direction, first show that it suffices to construct a locally finite open refinement of $\mathcal{U}$ on $\bigcup \mathcal{U}$, where $\mathcal{U}$ is a connected collection of open intervals.

Hint for Theorem 209

Since open subspaces of *ccc* linearly ordered space are also *ccc* linearly ordered, it suffices to show X is Lindelöf. Let $\mathcal{U}$ be an open cover of X. Define $x \sim y$ if and only if $[x, y]$ is covered by some countable subcollection of $\mathcal{U}$. Show that each equivalence class E is open and is covered by some countable subcollection of $\mathcal{U}$.

Hint for Theorem 210

For 210(a), define $x \sim y$ if and only if the interval from x to y is separable. Note that equivalence classes are convex. Show that the set of equivalence classes with the natural order satisfies the desired conditions.

Hint for Theorem 213

Show that 2^ω with the topology induced by the lexicographic order is the same as the usual Tychonoff product topology.

Hint for Theorem 214

For 214(a), define $f\colon 2^\omega \to [0, 1]$ so that $f(\vec{x}) = \sum_{n<\omega} x_n/2^{n+1}$, where x_n is the nth-coordinate of $\vec{x}$. For 214(c), if K is a closed subset of $\mathbb{C}$ note that $K \times \mathbb{C}$ is homeomorphic to $\mathbb{C}$.

Hint for Theorem 220

Let X be a Suslin line. Define closed subsets C_α, $\alpha < \omega_1$, of X as follows. Let $C_0 = \varnothing$. If α is a limit ordinal and C_β has been defined for all $\beta < \alpha$, let $C_\alpha = \overline{\bigcup_{\beta<\alpha} C_\beta}$. If $\alpha = \gamma + 1$ and C_γ has been defined, let L_γ be the convex components of $X \setminus C_\gamma$. For each $I \in L_\gamma$, choose a countable sequence of points converging to each endpoint of I. Let C'_γ be the collection of these chosen points for all $I \in L_\gamma$. Then let $C_\alpha = \overline{C_\gamma \cup C'_\gamma}$.

Now let L_α be the convex components of $X \setminus C_\alpha$, and let $T = \bigcup_{\alpha<\omega_1} L_\alpha$. Show that T ordered by $\supset$ is a Suslin tree.

Hint for Theorem 221
Let T be a Suslin tree, and let $\prec$ be an arbitrary linear order on T. Let X be the set of all branches of T. For any branch b, let $b(\alpha)$ be the member of b in level α of the tree. Given $b_1, b_2 \in X$, let α be minimal such that $b_1(\alpha) \neq b_2(\alpha)$, and then define $b_1 < b_2$ if and only if $b_1(\alpha) \prec b_2(\alpha)$. Show that X with this order is *ccc* and nonseparable.

Hint for Theorem 222
Let $T = \bigcup_{\alpha<\omega_1} L_\alpha$ be as in the hint for Theorem 220. For each α, choose $I_\alpha \in L_\alpha$. There are disjoint $I^0_\alpha, I^1_\alpha \in L_{\alpha+1}$ contained in I_α. Show that $\{I^0_\alpha \times I^1_\alpha : \alpha < \omega_1\}$ is pairwise-disjoint.

Hint for Theorem 223
For the second part, study the hint for Theorem 220.

Hint for Theorem 225
Use the partial order $\mathcal{O}(X)$ defined on page 57.

Hint for Lemma 228
Show there exists $\alpha_0 < \omega_1$ such that $\overline{\bigcup_{\beta>\alpha} U_\beta} = \overline{\bigcup_{\beta>\alpha_0} U_\beta}$ for all $\alpha > \alpha_0$. Let $P = \{O \in \mathcal{O}(X) : O \subset \bigcup_{\beta>\alpha_0} U_\beta\}$ and apply MA, where $\mathcal{O}(X)$ is the partial order defined on page 57.

Hint for Theorem 232
Suppose not; there are points x_α, $\alpha < \omega_1$, with $x_\alpha \notin \overline{\{x_\beta : \beta < \alpha\}}$. Let $Y = \overline{\{x_\alpha : \alpha < \omega_1\}}$. It follows from first-countability that every point of Y is in $\overline{\{x_\beta : \beta < \alpha\}}$ for some $\alpha < \omega_1$. Use compactness of Y and Lemma 228 applied to Y to get a contradiction.

Hint for Lemma 233
Let $\mathcal{B}$ be a countable base for $\mathbb{R}$. Let P be all pairs of the form $\langle \vec{C}, F\rangle$, where $\vec{C}$ is a finite sequence $\langle C_0, \ldots, C_n\rangle$ of members of $\mathcal{B}$ and F is a finite subset of κ. Define a partial order $\leqslant$ on P so that $\langle \vec{C}', F'\rangle \leqslant \langle \vec{C}, F\rangle$ if and only if

(1) $\vec{C}'$ extends $\vec{C}$,

(2) $F' \supset F$, and

(3) for each $i \in \operatorname{dom} \vec{C}' \setminus \operatorname{dom} \vec{C}$ we have $C_i \subset \bigcap_{\alpha \in F} U_\alpha$.

For each $B \in \mathcal{B}$, $k < \omega$, and $\alpha < \kappa$ put

$$D_{B,k} = \{\langle \vec{C}, F\rangle \in P : \exists i > k(C_i \subset B)\}$$

and $E_\alpha = \{\langle \vec{C}, F\rangle \in P : \alpha \in F\}$. Now what can we do with $\mathsf{MA}(\kappa)$?

Hint for Theorem 239

Note that if $f \in \omega^\omega$, then the set $K_f = \{g \in \omega^\omega : \forall n[g(n) \leqslant f(n)]\}$ is compact.

Hint for Theorem 242

For 242(c) recall when working with Bing's G—Example 138—we noted that to prove normality for a space consisting of a closed discrete set plus isolated points, we need only show that any subset of the closed discrete set, and its complement in the closed discrete set, can be put into disjoint open sets.

Hint for Theorem 243

How many G_δ sets can there be in a space with a countable base?

Hint for Theorem 244

Let $A \subset \mathbb{R}$ have cardinality κ, and let $B \subset A$. Use $\mathsf{MA}(\kappa)$ to show that there is a sequence $\langle C_0, C_1, \ldots\rangle$ of basic open sets of A such that every $x \in B$ is in infinitely many C_i's, while every $y \in A \setminus B$ is in only finitely many C_i's. Then note that $B = \bigcap_{n<\omega}(\bigcup_{i>n} C_i)$.

Hint for Corollary 248

In a normal space, any countable discrete collection of closed sets can be separated by disjoint open sets.

Hint for Lemma 256
For limit α, simply let $\mathcal{T}_\alpha$ be the topology generated by $\bigcup_{\beta<\alpha} \mathcal{T}_\beta$. If α is a successor, say $\alpha = \beta + 1$, then $X_\alpha = X_\beta \cup \{x_\beta\}$, and the task is to define a neighborhood base $\mathcal{B}_\beta$ at x_β so that the topology generated by $\mathcal{T}_\beta \cup \mathcal{B}_\beta$ satisfies 256(a), 256(b), and 256(c).

Hint for Theorem 261
Let $\gamma X = X \cup (\alpha X \setminus \alpha[X])$. Define a natural one-to-one and onto function $h\colon \gamma X \to \alpha X$ and use h to "pull back" the topology on αX to define a topology γX.

Hint for Theorem 276
Let $f = \mathrm{id}_X\colon X \to Z$ and apply Theorem 269; show that the resulting $F\colon \beta X \to Z$ is a homeomorphism.

Hint for Theorem 296
Construct an ω_1-sequence $\langle A_\alpha \rangle_{\omega_1}$ of infinite subsets of ω such that $\alpha < \beta \to A_\beta \subset^* A_\alpha$ and $u = \{B \subset \omega : \exists \alpha < \omega_1 (A_\alpha \subset^* B)\}$ is an ultrafilter.

Hint for Theorem 304
Apply Lemma 294.

Hint for Theorem 305
Show that $\mathbb{I}^{\mathbb{I}}$ has a closed subspace which is not sequentially compact.

Hint for Theorem 306
For an infinite subset A of ω, let f_A denote its increasing enumeration. Observe that if $B \subsetneq^* A$, then $f_A <^* f_B$.

Hint for Theorem 312
Make use of Theorem 120.

Hint for Example 323
Show that ω^{ω_1} is not G_δ in $(\omega + 1)^{\omega_1}$.

Hint for Example 329

Look for an X that is Baire because it contains most of $\mathbb{R}$, but has a closed subspace homeomorphic to $\mathbb{Q}$.

Hint for Theorem 330

Note that if O is dense open in $X \times Y$, and B is open in Y, then $\pi_X[O \cap (X \times B)]$ is dense open in X.

Hint for Lemma 338

For 338(b), let U_i be the collection of all $\langle f, g\rangle \in \omega^{\omega_1} \times \omega^{\omega_1}$ such that $\min\{f^*, g^*\} > \max\{f(i), g(i)\}$. Then U_i is dense open in $\omega^{\omega_1} \times \omega^{\omega_1}$ and $\bigcap_{i\in\omega} U_i \cap (A^* \times B^*) = \varnothing$.

Hint for Theorem 347

For (c $\to$ d), observe that 347(c) implies Fréchet. Suppose σ is a strategy for P in the W-game at $p \in X$, where X satisfies 347(c). We need to show that σ can be beat. Let

$$P_\varnothing = \{\sigma(U) : U \text{ open}, p \in U\}.$$

Then $p \in \overline{P}_\varnothing$, so there is a countable subset $A_\varnothing$ of $P_\varnothing$ with $p \in \overline{A}_\varnothing$ ($A_\varnothing$ could be taken to be a convergent sequence, but this doesn't matter for this argument). Then for each $y_0 \in A_\varnothing$, let U_{y_0} be such that $y_0 = \sigma(U_{y_0})$ and let

$$P_{y_0} = \{\sigma(U_{y_0}, U) : U \text{ open}, p \in U\}.$$

Let A_{y_0} be a countable subset of P_{y_0} with $p \in \overline{A}_{y_0}$. Now for each $y_1 \in A_{y_0}$, let $U_{y_0y_1}$ be such that $y_1 = \sigma(U_{y_0}, U_{y_0y_1})$ and let

$$P_{y_0y_1} = \{\sigma(U_{y_0}, U_{y_0y_1}, U) : U \text{ open}, p \in U\}.$$

And so on. Apply 347(c) to the resulting $A_{y_0y_1\ldots y_n}$'s.

9 Proofs

Proof of Theorem 1
Let p_n denote the nth prime number; so $p_1 = 2$, $p_2 = 3$, etc.

(a) Note $\{p_n^k : n, k \in \mathbb{N}\} = \bigcup_{n\in\mathbb{N}} \{p_n^k : k \in \mathbb{N}\}$. Certainly the right-hand side is a countable union of countable sets. The left-hand side is countable by the map defined as follows: $1 \mapsto p_1^1$, $2 \mapsto p_1^2$, $3 \mapsto p_2^1$, $4 \mapsto p_1^3$, $5 \mapsto p_2^2$, $6 \mapsto p_3^1$, $7 \mapsto p_1^4$, $8 \mapsto p_2^3$, $9 \mapsto p_3^2$, etc.

(b) $A \times B = \bigcup \{\{a\} \times B : a \in A\}$, and

$$\langle\langle a_1, \ldots, a_{n-1}\rangle, a_n\rangle \mapsto \langle a_1, \ldots, a_n\rangle$$

is a bijection between $(A_1 \times \cdots \times A_{n-1}) \times A_n$ and $A_1 \times \cdots \times A_n$.

(c) We may assume $A = \{p_n^k : n, k \in \mathbb{N}\}$; there is a one-to-one correspondence between the finite subsets of A and $\mathbb{N}$ given by $\varnothing \mapsto 1$ and $\{p_{n_1}^{k_1}, \ldots, p_{n_m}^{k_m}\} \mapsto p_{n_1}^{k_1} \cdots p_{n_m}^{k_m}$.

The map $\langle a_1, \ldots, a_n\rangle \mapsto \{\langle a_1, 1\rangle, \ldots, \langle a_n, n\rangle\}$ gives an injection from the set of finite sequences in A into $A \times \mathbb{N}$.

(d) Certainly $|\mathbb{N}| \leqslant |\mathbb{Q}|$ and $|\mathbb{Q}^+| \leqslant |\mathbb{N} \times \mathbb{N}|$, where $\mathbb{Q}^+$ are the positive rational numbers. ■

Proof of Theorem 2
(a) Certainly $r \mapsto \{q < r : q \in \mathbb{Q}\}$ witnesses $|\mathbb{R}| \leqslant |\mathcal{P}(\mathbb{Q})| = |\mathcal{P}(\mathbb{N})|$. For each $X \in \mathcal{P}(\mathbb{N})$, let $X(n) = 2$ if $n \in X$ and $X(n) = 0$ otherwise; then $X \mapsto \sum_{n=1}^{\infty} X(n)/3^n$ is an injection from $\mathcal{P}(\mathbb{N})$ to $\mathbb{R}$.

(b) The map $f \mapsto f^{-1}(1)$ gives a bijection between the functions from $\mathbb{N}$ to $\{0, 1\}$ and $\mathcal{P}(\mathbb{N})$.

(c) An infinite sequence of 0's and 1's is a function from $\mathbb{N}$ to $\{0,1\}$.

(d) Let F be a bijection between $\mathbb{R}$ and the set of functions from $\mathbb{N}$ to $\{0,1\}$. Then

$$\begin{aligned}&\langle r_1, r_2, \ldots\rangle \mapsto \\ &F^{-1}(\langle F(r_1)(1), F(r_1)(2), F(r_2)(1), F(r_1)(3), F(r_2)(2), F(r_3)(1), \ldots\rangle)\end{aligned}$$

is a bijection between $\mathbb{R}$ and the infinite sequences in $\mathbb{R}$. ■

Proof of Theorem 3
Let $f\colon X \to \mathcal{P}(X)$ and put $Y = \{x \in X : x \notin f(x)\}$; then

$$x \in Y \to x \notin f(x) \to f(x) \neq Y$$

and

$$x \notin Y \to x \in f(x) \to f(x) \neq Y,$$

so $Y \notin \operatorname{ran} f$. ■

Proof of Theorem 4
See any book about ZFC, such as Kunen's *Set Theory*. ■

Proof of Theorem 5
Let $A \subset \mathbf{Ord}$ and put $a = \bigcup A$. Certainly $a \subset \mathbf{Ord}$, so we need to show a is transitive. Let $Y \in a$; there exists $\alpha \in A$ such that $Y < \alpha$. If $\beta < Y$ we necessarily have $\beta < \alpha$, so any predecessor of Y is a predecessor of α. This yields $Y \subset \alpha$ (whence ordinals are transitive sets). But $\alpha \subset a$ by construction, so we must have $Y \subset a$. Thus a is transitive and an ordinal by the hint.

That a is an upper bound of A is immediate. For any $\beta < a$ there is $\alpha \in A$ with $\beta < \alpha$, so no predecessor of a is an upper bound of A. Thus, a is the least upper bound of A. ■

Proof of Corollary 6
Observe $\alpha = \bigcup C$ is an ordinal with countably many predecessors. ■

Proof of Theorem 7
Suppose on the contrary that the set A of those $\alpha < \kappa$ for which S_α is false is nonempty; let $\beta = \min A$. By 7(1), $0 < \beta$. Since β is minimal, we must have that S_α is true for each $\alpha < \beta$. Then S_β is true by 7(2) so $\beta \notin A$, a contradiction. ■

Proof of Theorem 8

Proof of Exercise 9
Given $A \in \mathcal{B}$, there exists $\alpha < \omega_1$ with $A \in \mathcal{B}_\alpha$. Then $X \setminus A \in \mathcal{B}_{\alpha+1}$, so $\mathcal{B}$ is closed under complements.

Let $\mathcal{A} = \{A_n : n < \omega\} \subset \mathcal{B}$; by Corollary 6 there exists $\alpha < \omega_1$ such that $A_n \in \mathcal{B}_\alpha$ for each $n < \omega$. By construction, $\bigcap \mathcal{A} \in \mathcal{B}_{\alpha+1}$ so $\mathcal{B}$ is closed under countable intersections. Note that we may write $\bigcup \mathcal{A} = X \setminus \bigcap_{n<\omega}(X \setminus A_n)$, so $\mathcal{B}$ is closed under countable unions as well. ■

Proof of Theorem 10
(a → b) Let $\langle P, <\rangle$ be a partially ordered set with no maximal element and let $f\colon \mathcal{P}(P) \setminus \{\varnothing\} \to P$ be a choice function. For each $a \in P$ put $P_a = \{x \in P : a < x\}$ and observe each P_a must be nonempty.

Fix $a_0 = f(P)$. Suppose a_α has been defined for all $\alpha < \beta$. If $\beta = \alpha+1$, put $a_\beta = f(P_{a_\alpha})$. If β is a limit ordinal and $\bigcap_{\alpha<\beta} P_{a_\alpha} \neq \varnothing$, define $a_\beta = f(\bigcap_{\alpha<\beta} P_{a_\alpha})$ and continue.

Note that this process must terminate; that is, there exists a limit ordinal γ such that $\bigcap_{\alpha<\gamma} P_{a_\alpha} = \varnothing$. It follows that $\langle a_\alpha \rangle_{\alpha<\gamma}$ is a chain in $\langle P, <\rangle$ which has no upper bound in P.

(b → c) Let X be a (nonempty) set and let P be the set of all pairs $\langle A, \prec_A\rangle$ such that $A \subset X$ and $\prec_A$ is a well-order on A. Define a partial order $<$ on P so that $\langle A, \prec_A\rangle < \langle A', \prec_{A'}\rangle$ if and only if $\langle A', \prec_{A'}\rangle$ extends $\langle A, \prec_A\rangle$; that is, $\langle A, \prec_A\rangle$ is a proper initial segment of $\langle A', \prec_{A'}\rangle$. It is easy to check that $\langle P, <\rangle$ is a partially ordered set whose every chain has an upper bound in P.

By ZL there is a maximal element $\langle M, \prec_M\rangle$ in $\langle P, <\rangle$. It must be the case that $M = X$, else we could extend $\langle M, \prec_M\rangle$. Hence, $\prec_M$ is a

well-order on X.

(c → a) Let $\mathcal{A}$ be a collection of nonempty sets; well-order each $A \in \mathcal{A}$. Then $A \mapsto \min A$ defines a choice function on $\mathcal{A}$. ■

Proof of Lemma 11
Certainly

$$\begin{aligned} x \in X \setminus \bigcup \mathcal{A} &\leftrightarrow x \notin \bigcup \mathcal{A} \leftrightarrow \forall A \in \mathcal{A}(x \notin A) \\ &\leftrightarrow \forall A \in \mathcal{A}(x \in X \setminus A) \leftrightarrow x \in \bigcap \{X \setminus A : A \in \mathcal{A}\} \end{aligned}$$

and

$$\begin{aligned} x \in X \setminus \bigcap \mathcal{A} &\leftrightarrow x \notin \bigcap \mathcal{A} \leftrightarrow \exists A \in \mathcal{A}(x \notin A) \\ &\leftrightarrow \exists A \in \mathcal{A}(x \in X \setminus A) \leftrightarrow x \in \bigcup \{X \setminus A : A \in \mathcal{A}\}. \end{aligned}$$ ■

Proof of Corollary 12
By definition we need to show the complement of each is open, using Lemma 11 where necessary. Certainly $X \setminus \varnothing = X$ and $X \setminus X = \varnothing$ are open. If $F_1, \ldots, F_n$ are closed in X, then

$$X \setminus (F_1 \cup \cdots \cup F_n) = (X \setminus F_1) \cap \cdots \cap (X \setminus F_n)$$

is open as a finite intersection of open sets. If $\mathcal{F}$ is a collection of closed subsets of X, then

$$X \setminus \bigcap \mathcal{F} = \bigcup \{X \setminus F : F \in \mathcal{F}\}$$

is open as a union of open sets. ■

Proof of Theorem 13
Note that any open set containing x but missing A corresponds to a closed set containing A but missing x, and every open set containing a point of $A \cup A^d$ meets A. ■

Proof of Theorem 14
(a) We know $\varnothing$ is closed.

(b) Certainly $\overline{B}$ is a closed set containing A.

(c) Note that $\overline{A}$ is already closed.

(d) Note that $A \cup B \subset \overline{A} \cup \overline{B}$ and $A, B \subset \overline{A \cup B}$; use 14(b). ■

Proof of Theorem 15
Note $\bigcap \{A_\alpha : \alpha \in \Lambda\} \subset \overline{A}_\beta$ and $\overline{\bigcup \{A_\alpha : \alpha \in \Lambda\}} \supset A_\beta$ for each $\beta \in \Lambda$.■

Proof of Theorem 16
(a) Note $\overline{X \setminus A} \subset K \subset X \setminus A^\circ$ implies $A^\circ \subset X \setminus K \subset X \setminus \overline{X \setminus A}$, so there is no closed set strictly between $\overline{X \setminus A}$ and $X \setminus A^\circ$.

(b) By 16(a), $X \setminus (X \setminus A)^\circ = X \setminus [X \setminus \overline{X \setminus (X \setminus A)}] = \overline{X \setminus (X \setminus A)} = \overline{A}$.
■

Proof of Theorem 17
(a) Let U be an open neighborhood of $x \in \partial\partial A$. Then U is an open neighborhood of each point of $U \cap \partial A$, which must be nonempty. It follows that U meets A and $X \setminus A$, so $x \in \partial A$.

(b) Note $x \in \overline{A}$ if and only if every neighborhood of x meets A, $x \in \overline{X \setminus A}$ if and only if every neighborhood of x meets $X \setminus A$, and $x \notin A^\circ$ if no neighborhood of x lies entirely inside A.

(c) A is closed if and only if every point outside of A has a neighborhood missing A, if and only if every point outside of A is not in the boundary of A.

(d) Observe that A° is a neighborhood of each of its points missing $X \setminus A$, so no point in the interior of A is a point in the boundary of A.
If A is open then $A = A^\circ$, so $\partial A \cap A = \partial A \cap A^\circ = \varnothing$. If $\partial A \cap A = \varnothing$ then every point of A has a neighborhood lying entirely inside A, so A is open.

(e) By 17(c) and 17(d), A is closed-and-open if and only if ∂A is a

subset of A which misses A. ■

Proof of Theorem 18

(a) By definition, an isolated point x of A has a neighborhood which does not meet A at a point other than x.

(b) Note that $x \in A \backslash A^d$ if and only if x has a neighborhood U such that $A \cap U = \{x\}$. By Theorem 13, $\overline{A} = A \cup A^d = (A \setminus A^d) \cup A^d = A' \cup A^d$.

(c) If x is an isolated point of $\overline{A}$, there is a neighborhood U of x such that $\overline{A} \cap U = \{x\}$. But this implies $A \cap U \subset \{x\}$, so x is not a limit point of A. By 18(a) and 18(b), we must have $x \in A'$. ■

Proof of Theorem 19

$\rightarrow$ X, being open, may be written as a union of members of $\mathcal{B}$, so each point of X is in some member of $\mathcal{B}$.

The intersection of two members of $\mathcal{B}$ is open, so it may be written as a (possibly empty) union of members of $\mathcal{B}$; points in such an intersection are necessarily in some member of $\mathcal{B}$.

$\leftarrow$ Put $\mathcal{T} = \{\bigcup \mathcal{U} : \mathcal{U} \subset \mathcal{B}\}$; certainly $\mathcal{T}$ is closed under arbitrary unions and $\varnothing, X \in \mathcal{T}$. Fix $U, V \in \mathcal{T}$. If $x \in U \cap V$, there exist $U_x, V_x \in \mathcal{B}$ such that $x \in U_x \subset U$ and $x \in V_x \subset V$. By assumption there exists $B \in \mathcal{B}$ such that $x \in B \subset U_x \cap V_x$, so $U \cap V \in \mathcal{T}$. Thus $\mathcal{T}$ is a topology on X for which $\mathcal{B}$ is a base.

Suppose $\mathcal{T}'$ is another topology on X generated by $\mathcal{B}$. If $U \in \mathcal{T}'$, then $U = \bigcup \mathcal{U}$ for some $\mathcal{U} \subset \mathcal{B}$. But this implies $U \in \mathcal{T}$, so $\mathcal{T}' \subset \mathcal{T}$. By the same argument $\mathcal{T} \subset \mathcal{T}'$. ■

Proof of Theorem 20

Certainly every point of X is contained in some member of $\mathcal{B}_2$. Take $B_1, B_2 \in \mathcal{B}_2$ with $x \in B_1 \cap B_2$. Then there exist $B_1', B_2' \in \mathcal{B}_1$ such that $B_1' \subset B_1$, $B_2' \subset B_2$, and $x \in B_1' \cap B_2'$. Since $\mathcal{B}_1$ is a base, let $B' \in \mathcal{B}$ such that $x \in B' \subset B_1' \cap B_2'$. Then there exists $B \in \mathcal{B}_2$ such that $x \in B \subset B' \subset B_1' \cap B_2' \subset B_1 \cap B_2$. By Theorem 19, $\mathcal{B}_2$ is a base for a topology $\mathcal{T}'$ on X.

Observe that for each $U \in \mathcal{B}_1$ there exists $\mathcal{B}_U \subset \mathcal{B}_2$ such that $U = \bigcup \mathcal{B}_U$. So if $\mathcal{U} \subset \mathcal{B}_1$ we have $\bigcup \mathcal{U} = \bigcup \{\mathcal{B}_U : U \in \mathcal{U}\}$, implying $\mathcal{T} \subset \mathcal{T}'$. A similar argument shows $\mathcal{T}' \subset \mathcal{T}$, whence $\mathcal{B}_1$ and $\mathcal{B}_2$ are equivalent bases. ■

Proof of Theorem 21

If $\mathcal{T}_1 \subset \mathcal{T}_2$ and $B \in B_1$, then B is open in $\mathcal{T}_2$ and may be written as a union of members of $\mathcal{B}_2$.

Now suppose every member of $\mathcal{B}_1$ is a union of members of $\mathcal{B}_2$; fix $U \in \mathcal{T}_1$. There is a subcollection $\mathcal{U}$ of $\mathcal{B}_1$ such that $U = \bigcup \mathcal{U}$, and for each $B \in \mathcal{U}$ there is a subcollection $\mathcal{U}_B$ of $\mathcal{B}_2$ such that $B = \bigcup \mathcal{U}_B$. Then $U = \bigcup \{\bigcup \mathcal{U}_B : B \in \mathcal{U}\}$, so U is open in $\mathcal{T}_2$. ■

Proof of Theorem 22

(a) If H is closed in A, then $A \setminus H = U \cap A$ for some U open in X. This means $(X \setminus U) \cap A = H$, with $X \setminus U$ closed in X.

If K is closed in X and $H = A \cap K$, then $(X \setminus K) \cap A = A \setminus H$ is open in A.

(b) There exists U' open in X such that $U = U' \cap A$. Then U is a finite intersection of open subsets of X, so U is open in X.

(c) There exists U' open in X such that $A \setminus K = U' \cap A$. Certainly $X \setminus A$ is open in X, so $X \setminus (U' \cup (X \setminus A)) = K$ is closed in X. ■

Proof of Theorem 23

Let U' be open in $A \subset X$. Then $U' = U \cap A$ for some U open in X and there exists a subcollection $\mathcal{B}'$ of $\mathcal{B}$ such that $U = \bigcup \mathcal{B}'$. It follows that $U' = U \cap A = (\bigcup \mathcal{B}') \cap A = \bigcup \{B \cap A : B \in \mathcal{B}'\}$. ■

Proof of Theorem 24

$\subset$ Note $\operatorname{cl}_X H \cap A$ is closed in A and contains H.

$\supset$ Let $x \in A \setminus \operatorname{cl}_A H$. There exists U open in X such that $x \in U \cap A$ and $(U \cap A) \cap H = U \cap H = \varnothing$. This implies $x \notin \operatorname{cl}_X H$. ■

Proof of Theorem 25
(a → b) We may write $f^{-1}[V] = \bigcup\{U_x : x \in f^{-1}[V]\}$, where U_x is an open neighborhood of x with $f[U_x] \subset V$.

(b → c) Observe $f^{-1}[V_1 \cap V_2] = f^{-1}[V_1] \cap f^{-1}[V_2]$.

(c → d) Observe $f^{-1}[\bigcup_\Lambda V_\alpha] = \bigcup_\Lambda f^{-1}[U_\alpha]$.

(d → e) Observe $f^{-1}[K] = X \setminus f^{-1}[Y \setminus K]$.

(e → f) Observe that $f^{-1}[\overline{f[A]}]$ is closed containing A.

(f → a) Fix $x \in X$ and V open in Y with $f(x) \in V$; put $A = f^{-1}[Y \setminus V]$. Then $f[\overline{A}] \subset \overline{f[f^{-1}[Y \setminus V]]} \subset Y \setminus V$, so $x \in X \setminus \overline{A}$ and $f[X \setminus \overline{A}] \subset V$. ■

Proof of Theorem 26
(a) Given U open in X we have $j^{-1}[U] = \{x \in A : x \in U\} = A \cap U$, which is open in A by definition.

(b) Given $V \subset Y$ we have $(f \restriction A)^{-1}[V] = A \cap f^{-1}[V]$. ■

Proof of Theorem 27
If $f\colon X \to Y$ is constant then $f^{-1}[V] \in \{\varnothing, X\}$, so preimages (of open sets) are open.

If $f\colon X \to Y$ and $g\colon Y \to Z$, then $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$. ■

Proof of Theorem 28
(a) Observe $f^{-1}[V] = \bigcup_\Lambda (f^{-1}[V] \cap U_\alpha) = \bigcup_\Lambda (f \restriction U_\alpha)^{-1}[V]$. By Theorem 22(b), $(f \restriction U_\alpha)^{-1}[V]$ is open in X for each $\alpha \in \Lambda$ whenever V is open in Y.

(b) Observe $h^{-1}[A] = f^{-1}[A] \cup g^{-1}[A]$. By Theorem 22(c), $f^{-1}[A]$ and $g^{-1}[A]$ are closed in X whenever A is closed in Y. ■

Proof of Theorem 29

(a → c) If K is closed in X, then $h[K] = (h^{-1})^{-1}[K]$ is closed in Y.

(c → b) If U is open in X, then $h[U] = Y \setminus h[X \setminus U]$ is open in Y.

(b → a) If U is open in X, then $(h^{-1})^{-1}[U] = h[U]$ is open in Y. ■

Solution of Exercise 30

Let X have more than one point; let $\mathcal{T}_1$ and $\mathcal{T}_2$ be the discrete and indiscrete topologies on X, respectively. Since $\mathcal{T}_2 \subset \mathcal{T}_1$, the identity map $j\colon \langle X, \mathcal{T}_1\rangle \to \langle X, \mathcal{T}_2\rangle$ is a continuous bijection. That j^{-1} is not continuous follows from $\mathcal{T}_1 \not\subset \mathcal{T}_2$. ■

Proof of Theorem 31

Let $V \subset Y$ be open containing $f(x)$. Then $f^{-1}[V]$ is an open neighborhood of x, so it contains a tail of $\langle x_n\rangle$. It follows that V contains the corresponding tail of $\langle f(x_n)\rangle$, so $\langle f(x_n)\rangle \to f(x)$. ■

Proof of Lemma 32

Put $\delta = \varepsilon - d(x, y)$ and let $z \in B(y, \delta)$. Then

$$d(z, x) \leqslant d(z, y) + d(y, x) < \delta + d(y, x) = \varepsilon - d(x, y) + d(y, x) = \varepsilon,$$

so $z \in B(x, \varepsilon)$. ■

Proof of Lemma 33

Certainly each $x \in X$ is in some member of $\mathcal{B}$.

Let $B_1, B_2 \in \mathcal{B}$ and suppose $x \in B_1 \cap B_2$. By Lemma 32, there exist $\delta_1, \delta_2 > 0$ such that $B(x, \delta_1) \subset B_1$ and $B(x, \delta_2) \subset B_2$; define $\delta = \min\{\delta_1, \delta_2\}$. Then $B(x, \delta) \subset B_1 \cap B_2$ and $B(x, \delta) \in \mathcal{B}$, so $\mathcal{B}$ is a base by Theorem 19. ■

Proof of Corollary 34

By Lemma 33, any open set may be written as a union of ε-balls and any union of ε-balls is open. ■

Proof of Theorem 35

Let $x, y, z \in Y$. Certainly

(1) $d_Y(x, y) = d(x, y) \geqslant 0 \wedge d_Y(x, y) = d(x, y) = 0 \leftrightarrow x = y,$

(2) $d_Y(x,y) = d(x,y) = d(y,x) = d_Y(y,x)$, and

(3) $d_Y(x,z) = d(x,z) \leqslant d(x,y) + d(y,z) = d_Y(x,y) + d_Y(y,z)$,

so d_Y is a metric on Y.

Let $\mathcal{B}_1 = \{d(x,\varepsilon) \cap Y : x \in X, \varepsilon > 0\}$ be a base for the subspace topology on Y, and let $\mathcal{B}_2 = \{d_Y(y,\varepsilon) : y \in Y, \varepsilon > 0\}$. Certainly $\mathcal{B}_2 \subset \mathcal{B}_1$. Let $x \in X$ and $\varepsilon > 0$ such that $B_d(x,\varepsilon) \cap Y \neq \varnothing$ and let $y \in B_d(x,\varepsilon) \cap Y$. By Lemma 32, there exists $\delta > 0$ such that $B_d(y,\delta) \subset B_d(x,\varepsilon)$. Then $B_{d_Y}(y,\delta) = B_d(y,\delta) \cap Y \subset B_d(x,\varepsilon) \cap Y$, so $\mathcal{B}_1$ and $\mathcal{B}_2$ are equivalent bases by Theorem 20. By Lemma 33 $\mathcal{B}_2$ is a base for the metric topology on $\langle Y, d_Y \rangle$, so the subspace and metric topologies must be the same. ■

Proof of Theorem 36

Let $X = \prod_\Lambda X_\alpha$; fix $\beta \in \Lambda$ and consider the projection map π_β. If $U_\beta \subset X_\beta$ is open then $\pi_\beta^{-1}[U_\beta] = U_\beta \times \prod_{\Lambda \setminus \{\beta\}} X_\alpha$ is a basic open set of X, so π_β is continuous.

Now let $U \subset X$ be open and take $x_\beta \in \pi_\beta[U]$; fix $\vec{x} \in U$ such that $\pi_\beta(\vec{x}) = x_\beta$. There is a basic open set $B = \prod_\Lambda U_\alpha$ such that $\vec{x} \in B \subset U$. It follows that $x_\beta \in \pi_\beta[B] = U_\beta \subset \pi_\beta[U]$, so π_β is open. ■

Proof of Theorem 37

Set $Y = \{\vec{y} \in X : \forall \alpha \neq \gamma (y_\alpha = x_\alpha)\} = \prod_\Lambda Y_\alpha$, where $Y_\alpha = \{x_\alpha\}$ for $\alpha \neq \gamma$ and $Y_\gamma = X_\gamma$. By Theorem 36, the projection map $\pi_\gamma \colon Y \to Y_\gamma$ is open and continuous. Clearly π_γ is one-to-one and onto, thus a homeomorphism by Theorem 29. ■

Proof of Theorem 38

Let $\vec{x} = \langle x_\alpha \rangle_\Lambda \in X$, $A = \prod_\Lambda A_\alpha$, $L = \overline{A}$, and $R = \prod_\Lambda \overline{A}_\alpha$.

If $\vec{x} \notin R$, there exists $\beta \in \Lambda$ such that $x_\beta \notin \overline{A}_\beta$; let $U_\beta \subset X_\beta$ be open containing x_β and missing A_β. Then $\pi_\beta^{-1}[U_\beta]$ misses A, so $\vec{x} \notin L$.

If $\vec{x} \notin L$, there is a basic open neighborhood $U = \prod_\Lambda U_\alpha$ of $\vec{x}$ missing A; take $\beta \in \Lambda$ such that U_β misses A_β. Then $x_\beta \notin \overline{A}_\beta$, so $\vec{x} \notin R$. ■

Proof of Lemma 39

Fix $x, y, z \in X$.

(1) That $d'(x,y) \geqslant 0$ and $d'(x,y) = 0$ if and only if $x = y$ is immediate from the definition of d' and the fact that d has the same

properties.

(2) If $d(x,y) \geqslant M$ then $d(y,x) \geqslant M$ as well, so $d'(x,y) = M$ and $d'(y,x) = M$. If $d(x,y) < M$ then $d(y,x) < M$ as well, so $d'(x,y) = d(x,y) = d(y,x) = d'(y,x)$.

(3) First note that $d'(x,y) + d'(y,z)$ is always the minimum of

$$D = \{d(x,y) + d(y,z), M + d(y,z), d(x,y) + M, 2M\}.$$

If $M \leqslant d(x,z) \leqslant d(x,y) + d(y,z)$, then

$$d'(x,z) = M \leqslant \min D = d'(x,y) + d'(y,z).$$

If $d(x,z) < M$ we have

$$d'(x,z) = d(x,z) \leqslant \min D = d'(x,y) + d'(y,z),$$

since $d(x,z) \leqslant d(x,y) + d(y,z)$.

It follows that d' is a metric.

Fix $x \in X$ and $\varepsilon > 0$. With $\delta = \min\{\varepsilon/2, M\}$, observe

$$B_d(x, \delta/2) \subset B_{d'}(x,\delta) = B_d(x,\delta) \subset B_d(x,\varepsilon).$$

Then a union of d-balls may be written as a union of d'-balls and vice versa, so d and d' must generate the same topology. ■

Proof of Theorem 40

Fix $\vec{x}, \vec{y}, \vec{z} \in X$.

(1) Certainly $d(\vec{x}, \vec{y}) \geqslant 0$ since $d_n(x_n, y_n) \geqslant 0$ for all $n \in \mathbb{N}$. Also, $d(\vec{x}, \vec{y}) = 0$ if and only if $d_n(x_n, y_n) = 0$ for all $n \in \mathbb{N}$ if and only if $x_n = y_n$ for all $n \in \mathbb{N}$ if and only if $\vec{x} = \vec{y}$.

(2) Since each d_n is symmetric,

$$\begin{aligned} d(\vec{x}, \vec{y}) &= \max\{d_n(x_n, y_n) : n \in \mathbb{N}\} \\ &= \max\{d_n(y_n, x_n) : n \in \mathbb{N}\} \\ &= d(\vec{x}, \vec{y}). \end{aligned}$$

(3) Each d_n satisfies the triangle inequality, so

$$\begin{aligned} d(\vec{x}, \vec{z}) &= \max\{d_n(x_n, z_n) : n \in \mathbb{N}\} \\ &\leqslant \max\{d_n(x_n, y_n) + d_n(y_n, z_n) : n \in \mathbb{N}\} \\ &\leqslant \max\{d_n(x_n, y_n) : n \in \mathbb{N}\} + \max\{d_n(y_n, z_n) : n \in \mathbb{N}\} \\ &= d(\vec{x}, \vec{y}) + d(\vec{y}, \vec{z}). \end{aligned}$$

It follows that d is a metric.

Let $B = \prod_{\mathbb{N}} U_n$ be basic open with support F; fix $\vec{x} = \langle x_n \rangle_{\mathbb{N}} \in B$. For each $i \in F$, let $\varepsilon_i > 0$ such that $B_{d_i}(x_i, \varepsilon_i) \subset U_i$. Put $\varepsilon = \min\{\varepsilon_i : i \in F\}$ and take $\vec{y} = \langle y_n \rangle_{\mathbb{N}} \in B_d(\vec{x}, \varepsilon)$. For each $i \in F$ we have

$$d_i(x_i, y_i) \leqslant d(\vec{x}, \vec{y}) < \varepsilon \leqslant \varepsilon_i,$$

implying $y_i \in B_{d_i}(x_i, \varepsilon_i) \subset U_i$. For each $i \in \mathbb{N} \setminus F$ we have $U_i = X_i$, which of course contains y_i. It follows that $\vec{y} \in B$.

Now, fix $\vec{x} \in X$ and $\varepsilon > 0$. Let $n \in \mathbb{N}$ such that $1/2^n < \varepsilon$ and put

$$B = \prod_{i=1}^{n-1} B_{d_i}(x_i, \varepsilon) \times \prod_{i=n}^{\infty} X_i.$$

Certainly $\vec{x} \in B$; take $\vec{y} \in B$. For $1 \leqslant i < n$ we have $d_i(x_i, y_i) < \varepsilon$, and for all $i \geqslant n$ we have $d_i(x_i, y_i) \leqslant 1/2^i \leqslant 1/2^n < \varepsilon$. Thus

$$d(\vec{x}, \vec{y}) = \max\{d_i(x_i, y_i) : i \in \mathbb{N}\} < \varepsilon,$$

so $\vec{y} \in B(\vec{x}, \varepsilon)$.

By Theorem 20, the topology generated by d is equivalent to the product topology on X. ■

Proof of Corollary 41

Use Lemma 39 to get the bounded metrics necessary to apply Theorem 40. ■

Proof of Theorem 42

Let $q\colon X \to Y$ be a continuous surjection. By definition $q^{-1}[V]$ is open in X whenever V is open in Y. Take $V \subset Y$ such that $q^{-1}[V]$ is open in X.

If q is open, then $q[q^{-1}[V]] = V$ is open in Y. If q is closed, then $q[X \setminus q^{-1}[V]] = Y \setminus q[q^{-1}[V]] = Y \setminus V$ is closed in Y. In either case, V being open makes q a quotient map. ■

Proof of Theorem 43
$\rightarrow$ Quotient maps are continuous, so this follows from Theorem 27.

$\leftarrow$ Let $W \subset Z$ be open; observe $(f \circ q)^{-1}[W] = q^{-1}[f^{-1}[W]]$ is open in X. Since q is a quotient map we have $f^{-1}[W]$ open in Y. ■

Proof of Theorem 44
If V is open in Y we have $q^{-1}[q[q^{-1}[V]]] = q^{-1}[V]$, so $q^{-1}[V]$ is an open saturated subset of X. If $U \subset X$ is open and saturated then $q^{-1}[q[U]] = U$ is open in X, so $q[U]$ is open in Y by definition. ■

Proof of Theorem 45
$\rightarrow$ Fix $x \in X$; each $y \in X \setminus \{x\}$ has a neighborhood missing x, so $X \setminus \{x\}$ is open.

$\leftarrow$ Given $x \neq y \in X$, $X \setminus \{y\}$ is open containing x but not y. ■

Proof of Theorem 46

Proof of Theorem 47

Proof of Theorem 48
That $T_4 \rightarrow T_3 \rightarrow T_2 \rightarrow T_1 \rightarrow T_0$ is straightforward.

Let X be a $T_{31/2}$-space, let $x \in X$, and let $H \subset X$ be closed not containing x. Then there exists a continuous $f\colon X \rightarrow [0,1]$ such that $f(x) = 0$ and $f[H] = \{1\}$. Certainly $f^{-1}[[0,1/3)]$ and $f^{-1}[(2/3,1]]$ are disjoint open sets containing x and H, respectively. ■

Proof of Theorem 49
Let $\langle X, d\rangle$ be a metric space and let A and B be disjoint closed subsets

of X. For each $a \in A$ choose $\varepsilon_a > 0$ such that $B(a, \varepsilon_a) \cap B = \varnothing$. Similarly, for each $b \in B$ choose $\varepsilon_b > 0$ such that $B(b, \varepsilon_b) \cap A = \varnothing$. Put $U = \bigcup_{a \in A} B(a, \varepsilon_a/2)$ and $V = \bigcup_{b \in B} B(b, \varepsilon_b/2)$; certainly U and V are open sets containing A and B, respectively.

Suppose there exists $z \in U \cap V$. Fix $a \in A$ and $b \in B$ such that $z \in B(a, \varepsilon_a/2) \cap B(b, \varepsilon_b/2)$. Without loss of generality we may assume $\varepsilon_a \leqslant \varepsilon_b$, whence $d(a, b) \leqslant d(a, z) + d(z, b) < \varepsilon_a/2 + \varepsilon_b/2 \leqslant \varepsilon_b$. It follows that $a \in B(b, \varepsilon_b)$, a contradiction. ■

Proof of Theorem 50

Suppose X_α is a T_0-space for each $\alpha \in \Lambda$. Let $\vec{x} \neq \vec{y} \in X$. Then there exists $\alpha \in \Lambda$ such that $x_\alpha \neq y_\alpha$, so there exists some U_α open in X_α containing one of x_α, y_α but not the other. It follows that $\pi_\alpha^{-1}(U_\alpha)$ is open in X and contains one of $\vec{x}, \vec{y}$ but not the other. The argument for T_1 and T_2-spaces is similar.

Now suppose X_α is T_3 for each $\alpha \in \Lambda$. Let $\vec{x} \in U$, with $U \subset X$ open. There is a basic open set $\prod_{\alpha \in F} U_\alpha \times \prod_{\alpha \in \Lambda \setminus F} X_\alpha \subset U$ containing $\vec{x}$, where $F \subset \Lambda$ is finite and each U_α is open in X_α. For each $\alpha \in F$, take V_α open X_α such that $\overline{V}_\alpha \subset U_\alpha$. Put $V = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$. Certainly $\vec{x} \in V$ and

$$\overline{V} = \prod_{\alpha \in F} \overline{V}_\alpha \times \prod_{\alpha \in \Lambda \setminus F} X_\alpha \subset \prod_{\alpha \in F} U_\alpha \times \prod_{\alpha \in \Lambda \setminus F} X_\alpha \subset U.$$

Hence, X is T_3.

Finally suppose X_α is completely regular for each $\alpha \in \Lambda$. Let $A \subset X$ be closed and take $\vec{b} \in X \setminus A$. Then there exists a basic open set $B = \bigcap_{i=1}^n \pi_{\alpha_i}^{-1}(U_{\alpha_i})$ such that $\vec{b} \in B$ and $B \cap A = \varnothing$. For each $1 \leqslant i \leqslant n$ there exists a continuous $f_i : X_{\alpha_i} \to [0, 1]$ such that $f_i(b_{\alpha_i}) = 0$ and $f_i \restriction (X_{\alpha_i} \setminus U_{\alpha_i}) = 1$.

For each $i = 1, \ldots, n$, the map that sends $\vec{x} \in X$ to $f_i(x_{\alpha_i})$ is continuous since $f_i(x_{\alpha_i}) = f_i \circ \pi_{\alpha_i}(\vec{x})$. Hence if we define $f : X \to [0, 1]$ by $f(\vec{x}) = \max\{f_i(x_{\alpha_i}) : 1 \leqslant i \leqslant n\}$, then f is continuous. Note that $f(\vec{b}) = 0$. Take $\vec{a} \in A$. Then $\vec{a} \notin B$, implying there exists some i such that $a_{\alpha_i} \notin U_{\alpha_i}$. It follows that $f(a_{\alpha_i}) = 1$, so $f(\vec{a}) = 1$. Thus $f \restriction A = 1$. ■

Proof of Theorem 51

Let X be normal and let H and K be disjoint closed subsets thereof. Let $\langle r_n : n < \omega \rangle$ be a well-ordering of $\mathbb{Q} \cap [0,1]$ with $r_0 = 1$ and $r_1 = 0$. By normality, there exists an open set U of X such that $H \subset U$ and $\overline{U} \cap K = \varnothing$. Put $U_{r_0} = U$. There also exists $V \subset X$ open such that $H \subset V$ and $\overline{V} \subset U_{r_0}$. Put $U_{r_1} = V$.

Suppose for some $n > 1$ that $U_{r_0}, \ldots, U_{r_{n-1}}$ satisfy (a) and (b). There exist $0 \leqslant i, j \leqslant n-1$ such that

$$r_i = \max \{r_k : 0 \leqslant k \leqslant n-1 \wedge r_k < r_n\}$$

and

$$r_j = \min \{r_k : 0 \leqslant k \leqslant n-1 \wedge r_k > r_n\}.$$

Since $r_i < r_j$ we have $\overline{U}_{r_i} \subset U_{r_j}$, so there exists an open set W with $\overline{U}_{r_i} \subset W$ and $\overline{W} \subset U_{r_j}$. Put $U_{r_n} = W$; then $U_{r_0}, \ldots, U_{r_n}$ satisfy (a) and (b). By induction, define appropriate U_r for all rational $r \in (0,1)$.■

Proof of Theorem 52

(a → b) Let H and K be disjoint closed subsets of X and let U_r, $r \in \mathbb{Q} \cap (0,1)$, be defined as guaranteed by Theorem 51; put $U_0 = \varnothing$ and $U_1 = X$. For each $x \in X$ let $Q_x = \{r \in \mathbb{Q} \cap [0,1] : x \in U_r\}$. Certainly $0 \leqslant \inf Q_x \leqslant 1$.

Define $f \colon X \to [0,1]$ by $x \mapsto \inf Q_x$. If $x \in H$ then $x \in U_r$ for all $r \in \mathbb{Q} \cap (0,1)$, so $f(x) = 0$. If $x \in K$ then $x \in U_r \leftrightarrow r = 1$, so $f(x) = 1$.

Let $0 < a < 1$ and put $U = f^{-1}[[0,a)]$ and $V = f^{-1}[(a,0]]$. If U and V are open then the preimage of any subbasic open set is open, implying f is continuous. (Recall that preimages preserve unions and intersections, and an open set may be written as a union of finite intersections of subbasic open sets.)

Let $x \in U$; then $\inf Q_x < a$, so there exists $r_x \in \mathbb{Q} \cap (0,a)$ such that $x \in U_{r_x}$. But $y \in U_{r_x} \to \inf Q_y < a \to y \in U$, so $U_{r_x} \subset U$. Thus, $U = \bigcup_{x \in U} U_{r_x}$ is open.

Let $x \in V$; then $a < \inf Q_x$ so there exists $r_x \in \mathbb{Q} \cap (a,1)$ such that $r_x < \inf Q_x$. Certainly $x \in X \setminus \overline{U}_{r_x}$. Moreover,

$$y \in X \setminus \overline{U}_{r_x} \to r_x \leqslant \inf Q_y \to a < \inf Q_y \to y \in V.$$

Thus, $V = \bigcup_{x \in V} U_{r_x}$ is open.

(b → a) Note $f^{-1}[[0,1/3)]$ and $f^{-1}[(2/3,1]]$ are disjoint open sets containing H and K, respectively. ■

Proof of Corollary 53
Apply Theorem 52, noting singletons are closed in T_4 spaces. ■

Proof of Theorem 54

Proof of Theorem 55
$\rightarrow$ Let $A, B \subset Y \subset X$, with A and B disjoint and closed (in Y). Note that $\overline{A} \cap B = \overline{A} \cap (Y \cap B) = (\overline{A} \cap Y) \cap B = A \cap B = \varnothing$. Similarly $A \cap \overline{B} = \varnothing$, so A and B are separated in X. Then there exist disjoint open (in X) sets U and V containing A and B, respectively. It follows that $U \cap Y$ and $V \cap Y$ are disjoint open subsets of Y containing A and B, respectively.

$\leftarrow$ Let A and B be separated subsets of X and put $Y = X \setminus (\overline{A} \cap \overline{B})$; then $A, B \subset Y$ with $Y \subset X$ open. Note that

$$\overline{A}^Y \cap \overline{B}^Y = (\overline{A} \cap Y) \cap (\overline{B} \cap Y) = \overline{A} \cap \overline{B} \cap Y = \varnothing,$$

so there exist disjoint open (in Y) sets U and V containing $\overline{A}^Y$ and $\overline{B}^Y$, respectively. But Y is open, so U and V are open in X containing $A \subset \overline{A}^Y$ and $B \subset B^Y$, respectively. ■

Proof of Corollary 56
Just combine Theorems 35, 49 and 55. ■

Proof of Lemma 57
Fix $a \in A$. Then there exists $n_a \in \mathbb{N}$ such that $a \in U_{n_a}$. For each $1 \leqslant i \leqslant n_a$, choose W_i open in X containing a such that $W_i \cap V_i = \varnothing$. Put $W_a = W_1 \cap \cdots \cap W_{n_a}$. Construct such W_a for each $a \in A$ and let $U = \bigcup_{a \in A} (U_{n_a} \cap W_a)$. Certainly $A \subset U$.

Similarly, take $W_b = \bigcap_{i=1}^{n_b} W_i'$, where W_i' is open and $W_i' \cap U_i = \varnothing$. Put $V = \bigcup_{b \in B} (V_{n_b} \cap W_b)$. Certainly $B \subset V$.

Suppose on the contrary that $U \cap V \neq \varnothing$ and let $x \in U \cap V$. Then there exist $a \in A$ and $b \in B$ such that $x \in U_{n_a} \cap W_i$ for all $1 \leqslant i \leqslant n_a$ and $x \in V_{n_b} \cap W'_i$ for all $1 \leqslant i \leqslant n_b$. Without loss of generality we may assume $n_a \leqslant n_b$. Since $x \in W'_i$ for all $i \leqslant n_b$, $x \in W'_{n_a} \subset X \setminus \overline{U}_{n_a}$. Thus $x \notin U_{n_a}$, a contradiction. ■

Proof of Theorem 58

Let (X, d) be a metric space. We know X is normal. Let $A \subset X$ be closed. For each $n \in \mathbb{N}$, let $U_n = \cup\{B(a, 1/n) : a \in A\}$. Certainly U_n is open. Moreover, $A \subset \bigcap_{n\in\mathbb{N}} U_n$.

Let $x \in \bigcap_{n\in\mathbb{N}} U_n$. Then for each $n \in \mathbb{N}$ there exists $a_n \in A$ such that $x \in B(a_n, 1/n)$. This implies that $\lim_{n\to\infty} d(a_n, x) = 0$. So $(a_n)_{n=1}^{\infty} \to x$ and $x \in \overline{A} = A$. It follows that $\bigcap_{n\in\mathbb{N}} U_n \subset A$, whence A is G_δ. ■

Proof of Theorem 59

(c → d) Let H and K be disjoint closed subsets of X and let $f, g : X \to I$ be continuous functions such that $H = f^{-1}(0)$ and $K = g^{-1}(0)$. Since $H \cap K = \varnothing$, $f(x) + g(x) > 0$ for all $x \in X$. Consider the function $h : X \to I$ defined by

$$h(x) = \frac{f(x)}{f(x) + g(x)}.$$

Certainly h is continuous. Moreover, $h(x) = 0$ iff $f(x) = 0$ iff $x \in H$, so $H = h^{-1}(0)$. Additionally, $h(x) = 1$ iff $f(x) = f(x) + g(x)$ iff $g(x) = 0$ iff $x \in K$, so $K = h^{-1}(1)$.

(d → b) Let H be a closed subset of X. If $H = X$, take $U_n = X$ for each $n \in \mathbb{N}$. Otherwise, pick $x \in X \setminus H$ and let $f : X \to I$ be continuous such that $H = f^{-1}(0)$ and $\{x\} = f^{-1}(1)$.

For each $n \in \mathbb{N}$, put $U_n = f^{-1}[[0, 1/n)]$. Certainly each U_n is open. Moreover,

$$H = f^{-1}(0) = f^{-1}[\bigcap_{n\in\mathbb{N}} [0, 1/n)]$$

$$= \bigcap_{n\in\mathbb{N}} f^{-1}[[0, 1/n)] = \bigcap_{n\in\mathbb{N}} U_n$$

and $\overline{U}_{n+1} \subset f^{-1}[[0, 1/n+1]] \subset U_n$. It follows that

$$\bigcap_{n\in\mathbb{N}} \overline{U}_n \subset \bigcap_{n\in\mathbb{N}} U_n,$$

so

$$H = \bigcap_{n\in\mathbb{N}} U_n = \bigcap_{n\in\mathbb{N}} \overline{U}_n.$$

(b → a) Let A and B be disjoint closed sets. Let $W_1, W_2, \ldots$ be open sets such that $A = \bigcap_{n\in\mathbb{N}} W_n = \bigcap_{n\in\mathbb{N}} \overline{W}_n$. Put $V_n = X \setminus \overline{W}_n$. Then $B \subset \bigcup_{n\in\mathbb{N}} V_n$ and $\overline{V}_n \cap A = \varnothing$. Similarly, there are open sets U_n such that $A \subset \bigcup_{n\in\mathbb{N}} U_n$ and $\overline{U}_n \cap B = \varnothing$. By Lemma 57 A and B can be separated by disjoint open sets, so X is normal. Certainly our assumption implies closed sets are G_δ, so X is perfectly normal.

(a → c) Let $H \subset X$ be closed. Then there exist open sets U_n such that $H = \bigcap_{b\in\mathbb{N}} U_n$. By normality, for each $n \in \mathbb{N}$ there exists a continuous function $f_n : X \to I$ such that $f_n(x) = 0$ for all $x \in H$ and $f_n(x) = 1$ for all $x \in X \setminus U_n$. Put $f(x) = \sum_{n=1}^{\infty} f_n(x)/2^n$. Since the range of each f_n is $[0, 1]$, this is an absolutely convergent series of functions and therefore continuous. Moreover $f(x) = 0$ iff $f_n(x) = 0$ for all $n \in \mathbb{N}$, implying $x \in H$. ■

Proof of Theorem 60

Let X be perfectly normal and let $Y \subset X$. Let H be closed in Y; then $H = \overline{H}^Y = \overline{H}^X \cap Y$. By Theorem 59(b), there are open (in X) sets U_n, $n < \omega$, such that $\overline{H}^X = \bigcap_{n<\omega} U_n = \bigcap_{n<\omega} \overline{U}_n^X$. Then $H = \overline{H}^X \cap Y = (\bigcap_{n<\omega} U_n) \cap Y = (\bigcap_{n<\omega} \overline{U}_n^X) \cap Y$. Now $H = \bigcap_{n<\omega} (U_n \cap Y) \subset \bigcap_{n<\omega} \overline{U_n \cap Y}^Y = \bigcap_{n<\omega} (\overline{U_n \cap Y}^X \cap Y) \subset (\bigcap_{n<\omega} \overline{U}_n^X) \cap Y = \overline{H}^X \cap Y = H$, so all are equal. Thus Y is perfectly normal by Theorem 59(b). ■

Proof of Theorem 61

(a) Let X be a compact space and Y a space. Let $f\colon X \to Y$ be continuous and let $\mathcal{V} = \{V_\alpha : \alpha \in \Lambda\}$ be an open cover of $f[X]$. Then $\{f^{-1}[V_\alpha] : \alpha \in \Lambda\}$ is an open cover of X by continuity. Since X is compact there is a finite subset F of Λ such that $\{f^{-1}[V_\alpha] : \alpha \in F\}$

covers X. Thus

$$f[X] = f[\bigcup_{\alpha \in F} f^{-1}[V_\alpha]] = \bigcup_{\alpha \in F} f[f^{-1}[V_\alpha]] \subset \bigcup_{\alpha \in F} V_\alpha.$$

It follows that $\{V_\alpha : \alpha \in F\}$ is a finite subcover of $\mathcal{V}$, so $f[X]$ is compact.

(b) Let A be a closed subset of a compact space X; let $\mathcal{U}$ be an open cover of A by open subsets of X. Then $\mathcal{U} \cup \{X \setminus A\}$ is an open cover of X, so there exists a finite subcollection $\{U_1, \ldots, U_n\}$ of $\mathcal{U}$ such that $\{U_1, \ldots, U_n\} \cup \{X \setminus A\}$ covers X. It follows that A is covered by $\{U_1, \ldots, U_n\}$, so A is compact.

Similar proofs handle the Lindelöf cases. ■

Proof of Theorem 62

Let X be Hausdorff and let $Y \subset X$ be compact. Take $x \in X \setminus Y$. For each $y \in Y$ let U_y and V_y be disjoint open neighborhoods of x and y, respectively. Sicne Y is compact, there exist $y_1, \ldots, y_n \in Y$ such that $Y \subset \bigcup_{i=1}^n V_{y_i}$. Then $U = \bigcap_{i=1}^n U_{y_i}$ is an open neighborhood of x which does not meet Y. So $X \setminus Y$ is open, whence Y is closed. ■

Proof of Exercise 63

If $\beta < \gamma \in [0, \alpha]$, then $[0, \beta + 1)$ and $(\beta, \alpha]$ are disjoint open sets containing β and γ, respectively.

Certainly $[0, 0] = \{0\}$ is compact. Suppose $[0, \beta]$ is compact for all $\beta < \alpha$. If $\mathcal{U}$ is an open cover of $[0, \alpha]$, there exists $\gamma < \alpha$ such that $(\gamma, \alpha]$ is contained in some member of $\mathcal{U}$; we need only finitely more members of $\mathcal{U}$ to cover $[0, \gamma]$. ■

Proof of Theorem 64

Let X be compact, Y Hausdorff, and $f\colon X \to Y$ a continuous bijection. Let $H \subset X$ be closed. Since X is compact, H is compact. Thus $f[H]$ is compact, hence closed. So f is a closed map and therefore a homeomorphism. ■

Proof of Exercise 65

$X/A = \{\{x\} : x \in (0, 2\pi)\} \cup \{A\}$ with quotient map $p\colon X \to X/A$ defined by $x \mapsto \{x\}$ iff $x \in (0, 2\pi)$ and $x \mapsto A$ iff $x \in A$. The topology

of X/A is

$$\mathcal{T}_p = \{U \subset X/A : p^{-1}[U] \text{ is open in } X\}.$$

Define $f\colon X/A \to S^1$ by

$$f(\theta) = (\cos\theta, \sin\theta) \text{ for } \theta \in X/A \setminus \{A\}$$

and $f[A] = (1, 0)$. Certainly f is well-defined.

To show injectivity let $\theta, \psi \in X/A$. Then

$$f(\theta) = f(\psi) \to (\cos\theta, \sin\theta) = (\cos\psi, \sin\psi) \to \theta = \psi.$$

For surjectivity, let $\langle x, y\rangle \in S^1$; then $x = \cos\theta$ and $y = \sin\theta$ for some $\theta \in [0, 2\pi]$.

It remains to show f is continuous. By Theorem 43, we need only show $g = f \circ p\colon [0, 2\pi, \to]S^1$ is continuous. Observe that $g(x) = \langle \cos x, \sin x\rangle$ for all $x \in [0, 2\pi]$, hence is continuous. So f is continuous too.

Finally, note that X/A is compact since it is the quotient—hence, continuous—image of the compact space X. Since X/A is compact and S^1 is T_2, f is a homeomorphism by Theorem 64. ■

Proof of Theorem 66

Suppose there is no such x_0. Then for each $x \in X$ there is $\varepsilon_x > 0$ and $y_x \in X$ such that $f(y_x) > f(x) + \varepsilon_x$. The collection

$$\{f^{-1}[(-\infty, f(x) + \varepsilon_x)] : x \in X\}$$

is an open cover of X, so there exist $x_1, \ldots, x_n \in X$ such that

$$X \subset \bigcup_{i=1}^{n} f^{-1}[(-\infty, f(x_i) + \varepsilon_{x_i})].$$

Let j be such that $f(x_j) + \varepsilon_{x_j} = \max\{f(x_i) + \varepsilon_{x_i} : i = 1, \ldots, n\}$. Now each $x \in X$ is in $f^{-1}[(-\infty, f(x_i) + \varepsilon_{x_i})]$ for some i, whence $f(x) < f(x_i) + \varepsilon_{x_i} \leqslant f(x_j) + \varepsilon_{x_j}$. But $f(y_{x_j}) > f(x_j) + \varepsilon_{x_j}$, a contradiction. ■

Proof of Theorem 67

(a) Let X be compact Hausdorff. Let $x \in X$ and $H \subset X$ closed not containing x. For each $y \in H$ there exist disjoint open neighborhoods

U_y and V_y of x and y, respectively. Then there exist $y_1, \ldots, y_n \in H$ such that $H \subset \bigcup_{i=1}^n V_{y_i}$. Put $U = \bigcap_{i=1}^n U_{y_i}$ and $V = \bigcup_{i=1}^n V_{y_i}$. Then U and V are disjoint open neighborhoods of x and H, respectively, whence X is regular.

(b) Now let H and K be disjoint closed subsets of X. For each $y \in K$ there exist disjoint open neighborhoods U_y and V_y of H and y, respectively. Then there exist $y_1, \ldots, y_n \in K$ such that $K \subset \cup_{i=1}^n V_{y_i}$. Put $U = \bigcap_{i=1}^n U_{y_i}$ and $V = \bigcup_{i=1}^n V_{y_i}$. Then U and V are disjoint open sets containing H and K, respectively, whence X is normal. ■

Proof of Theorem 68

Let X be a regular Lindelöf space and let A and B be disjoint closed subsets of X. For each $a \in A$ there exists an open neighborhood U_a of a such that $\overline{U}_a \cap B = \varnothing$. Similarly, for each $b \in B$ there exists an open neighborhood V_b of b such that $\overline{V}_b \cap A = \varnothing$.

Certainly $\{U_a : a \in A\}$ and $\{V_b : b \in B\}$ are open covers of A and B, respectively. Since A and B are Lindelöf, there exist countable subcovers $\{u_n : n \in \mathbb{N}\}$ and $\{V_n : n \in \mathbb{N}\}$ thereof, respectively.

Now $A \subset \bigcup_{n\in\mathbb{N}} U_n$ and $B \subset \bigcup_{n\in\mathbb{N}} V_n$. Moreover, $\overline{U}_n \cap B = \varnothing = \overline{V}_n \cap A$ for all $n \in \mathbb{N}$. By Lemma 57 A and B may be separated, whence X is normal. ■

Proof of Example 69

Recall $\{(a, b] : a, b \in \mathbb{R} \wedge a < b\}$ is a base (of clopen sets) for $\mathbb{S}$.

(a) To see that $\mathbb{S}$ is regular, let $x \in \mathbb{S}$ and let $H \subset \mathbb{S}$ be closed not containing x. There is a basic open set U containing x and missing H; U and $\mathbb{S} \setminus U$ are disjoint open sets containing x and H, respectively.

To see that $\mathbb{S}$ is Lindelöf, let $\mathcal{U} = \{(a_\alpha, b_\alpha] : \alpha \in \Lambda\}$ be a cover of $\mathbb{S}$; put $A = \bigcup \{(a_\alpha, b_\alpha) : \alpha \in \Lambda\}$. Since $\mathbb{R}$ is hereditarily Lindelöf, there is a countable subcollection $\mathcal{U}'$ of $\mathcal{U}$ which covers A. Let $\Lambda' \subset \Lambda$ such that $\mathbb{S} \setminus A = \{b_\alpha : \alpha \in \Lambda'\}$. Observe

$$\forall \alpha, \beta \in \Lambda' (b_\alpha \neq b_\beta \to (a_\alpha, b_\alpha] \cap (a_\beta, b_\beta] = \varnothing).$$

Since $(a_\alpha, b_\alpha]$ meets $\mathbb{Q}$ for each $\alpha \in \Lambda'$, $\mathbb{S} \setminus A$ is countable.

We show that $\mathbb{S}$ is perfectly normal; first note that it is normal by Theorem 68. Let U be open in $\mathbb{S}$ and let V be the interior of U in $\mathbb{R}$. Since $\mathbb{R}$ is perfectly normal, V is F_σ in $\mathbb{R}$ and also $\mathbb{S}$ (observe that the Sorgenfrey topology is finer than the Euclidean topology). By the argument above $U \setminus V$ is countable, whence U is F_σ.

(b) Note that $Y = \{\langle x, -x\rangle : x \in \mathbb{S}\}$ is an uncountable closed discrete subspace of $\mathbb{S}^2$. By Theorem 61(b), $\mathbb{S}^2$ cannot be Lindelöf. By Lemma 247, $\mathbb{S}^2$ is not normal since $\mathbb{Q} \times \mathbb{Q}$ is dense in $\mathbb{S}^2$ and $2^{\mathfrak{c}} > \mathfrak{c}$. ■

Proof of Theorem 70

Let $\mathcal{U}$ be an open cover of X. For all $x \in X$, there exists $U_x \in \mathcal{U}$ and $\varepsilon_x > 0$ such that $B(x, \varepsilon_x) \subset U_x$. Take $\mathcal{B} = \{B(x, \varepsilon_x/2) : x \in X\}$ and let $\mathcal{B}'$ be a finite subcover thereof. Put $\lambda = \min\{\varepsilon_x/2 : B(x, \varepsilon_x/2) \in \mathcal{B}'\}$. $\mathcal{B}'$ is finite, so $\lambda > 0$.

Now let $y \in X$. We want to show that $B(y, \lambda)$ is contained in some member of $\mathcal{U}$. Since $\mathcal{B}'$ is a cover, there exists $x \in X$ such that $y \in B(x, \varepsilon_x/2)$. Suppose $z \in B(y, \lambda)$. Then $d(x, z) \leqslant d(x, y) + d(y, z) < \varepsilon_x/2 + \varepsilon_x/2 = \varepsilon_x$. Thus $B(y, \lambda) \subset B(x, \varepsilon_x) \subset U_x \in \mathcal{U}$. ■

Proof of Lemma 71

Let $\mathcal{U}$ be a collection of open sets in $X \times Y$ covering $\{x_0\} \times Y$. Note that $\{x_0\} \times Y$ is compact, being homeomorphic to Y. For each $y \in Y$, let $W_y \in \mathcal{U}$ such that $(x_0, y) \in W_y$.

For each $y \in Y$, choose a basis element of $X \times Y$, say $U_y \times V_y$, such that $(x_0, y) \in U_y \times V_y \subset W_y$. There there is a finite collection $\{U_{y_i} \times V_{y_i} : 1 \leqslant i \leqslant N\}$ convering $\{x_0\} \times Y$. Put $O = \bigcap_{i=1}^N U_{y_i}$ and $\mathcal{V} = \{W_{y_i} : i = 1, \ldots, N\}$. Then

$$\{x_0\} \times Y \subset O \times Y \subset \bigcup_{i=1}^N (U_{y_i} \times V_{y_i}) \subset \bigcup_{i=1}^N W_{y_i} = \cup\mathcal{V}.$$

■

Proof of Theorem 72

Let X and Y be compact and let $\mathcal{U}$ be an open cover of $X \times Y$. Let $x \in X$. Since $\{x\} \times Y \subset \cup\mathcal{U}$, there is a finite subcollection $\mathcal{V}_x = \{U_i^x : 1 \leqslant i \leqslant m(x)\}$ of $\mathcal{U}$ and an open neighborhood O_x of x such

that $O_x \times Y \subset \cup \mathcal{V}_x$. Since X is compact, $\{O_x : x \in X\}$ has a finite subcover $\{O_{x_i} : 1 \leqslant i \leqslant n\}$. Put $\mathcal{V} = \{U_i^{x_j} : 1 \leqslant i \leqslant m(x_j), 1 \leqslant j \leqslant n\}$. Certainly $\mathcal{V}$ is a finite subcollection of $\mathcal{U}$.

Take $(x, y) \in X \times Y$. Because $\{O_{x_j} : 1 \leqslant j \leqslant n\}$ covers X, $x \in O_{x_j}$ for some j. Moreover, $(x, y) \in O_{x_j} \times Y \subset \bigcup_{i=1}^{m(x_j)} U_i^{x_j}$, so there exists i such that $(x, y) \in U_i^{x_j} \in \mathcal{V}$. So $\mathcal{V}$ is a cover, whence $X \times Y$ is compact.■

Proof of Theorem 73

$\rightarrow$ Let X be a space and let $\mathcal{F}$ be a collection of closed sets with the finite intersection property such that $\bigcap \mathcal{F} = \varnothing$. Then $\{X \setminus F : F \in \mathcal{F}\}$ is an open cover of X with no finite subcover, whence X is not compact.

$\leftarrow$ Let X be a space in which every collection of closed sets with the finite intersection property has a nonempty intersection. Let $\mathcal{U}$ be an open cover of X and observe that $\{X \setminus U : U \in \mathcal{U}\}$ cannot have the finite intersection property. There exists a finite subcollection $\mathcal{U}'$ of $\mathcal{U}$ such that $\bigcap \{X \setminus U : U \in \mathcal{U}'\} = \varnothing$. It follows that $\mathcal{U}'$ is a finite subcover of $\mathcal{U}$. ■

Proof of Theorem 74

(a $\rightarrow$ b) Let $G \subset X$ such that $G \cap F \neq \varnothing$ for all $F \in \mathcal{F}$; put

$$\mathcal{F}' = \{F' \subset X : \exists F \in \mathcal{F}(G \cap F \subset F')\}.$$

It is straightforward to check that $\mathcal{F}'$ is a filter with $\mathcal{F} \subset \mathcal{F}'$ and $G \in \mathcal{F}'$. Since $\mathcal{F}$ is an ultrafilter, we must have $\mathcal{F} = \mathcal{F}'$.

(b $\rightarrow$ c) Let $G \subset X$ such that $G \cap F = \varnothing$ for some $F \in \mathcal{F}$. Then $F \subset X \setminus G$, so $X \setminus G \in \mathcal{F}$.

(c $\rightarrow$ a) Let $\mathcal{F}'$ be a filter on X such that $\mathcal{F} \subset \mathcal{F}'$ and fix $A \in \mathcal{F}'$. By assumption, either $A \in \mathcal{F}$ or $X \setminus A \in \mathcal{F}$. If $X \setminus A \in \mathcal{F}$ then $A \cap (X \setminus A) = \varnothing \in \mathcal{F}'$, a contradiction. Thus $\mathcal{F}' \subset \mathcal{F}$. ■

Proof of Theorem 75

Let $\mathfrak{P}$ be the set of all collections $\mathcal{G} \subset \mathcal{P}(X)$ having the finite intersection property such that $\mathcal{F} \subset \mathcal{G}$; note that $\mathfrak{P}$ is partially ordered

by inclusion. Let $\mathbb{H}$ be a chain in $\langle \mathfrak{P}, \subset \rangle$ and put $\mathcal{H} = \bigcup \mathbb{H}$. Certainly $\mathcal{F} \subset \mathcal{H}$. Moreover, $\mathcal{H}$ has the finite intersection property since any finite subcollection thereof must be contained in some member of $\mathbb{H}$. Thus $\mathcal{H} \in \mathfrak{P}$, so $\mathbb{H}$ is bounded above in $\langle \mathfrak{P}, \subset \rangle$. By Zorn's Lemma, there is a maximal $\mathcal{G} \in \mathfrak{P}$.

We now show that $\mathcal{G}$ is a filter.

(1) Let $F_1, F_2 \in \mathcal{G}$. Since $\mathcal{G}$ has the finite intersection property, we have $F \cap (F_1 \cap F_2) \neq \varnothing$ for all $F \in \mathcal{G}$. Thus $\mathcal{G} \cup \{F_1 \cap F_2\}$ has the finite intersection property and contains $\mathcal{G}$. By the maximality of $\mathcal{G}$ we must have $F_1 \cap F_2 \in \mathcal{G}$.

(2) Let $F \in \mathcal{G}$ and $G \subset X$ scuh that $F \subset G \subset X$. For any $F' \in \mathcal{G}$ we have $\varnothing \neq F' \cap F \subset F' \cap G$, so $\mathcal{G} \cup \{G\}$ has the finite intersection property and contains $\mathcal{G}$. By the maximality of $\mathcal{G}$ we must have $G \in \mathcal{G}$.

(3) That $\varnothing \notin \mathcal{G}$ follows from $\mathcal{G}$ having the finite intersection property.

To see that $\mathcal{G}$ is an ultrafilter, let $F \subset X$ such that $F \cap G \neq \varnothing$ for all $G \in \mathcal{G}$. Then $\mathcal{G} \cup \{F\}$ has the finite intersection property and contains $\mathcal{G}$, so $F \in \mathcal{G}$. By Theorem 74(b), $\mathcal{G}$ is an ultrafilter. ■

Proof of Theorem 76

(a → b) Let $\mathcal{F}$ be a filter on X and put $\mathcal{G} = \{\overline{F} : F \in \mathcal{F}\}$. Certainly $\mathcal{G}$ has the finite intersection property; by Theorem 73, there exists $x \in \bigcap \mathcal{G}$. Then $x \in \overline{F}$ for all $F \in \mathcal{F}$, so $\mathcal{F}$ clusters at x.

(b → c) Let $\mathcal{F}$ be an ultrafilter on X and let x be a cluster point of $\mathcal{F}$. If U is a neighborhood of x then $U \cap F = \varnothing$ for all $F \in \mathcal{F}$, so $U \in \mathcal{F}$ by Theorem 74(b). It follows that every neighborhood of x belongs to $\mathcal{F}$, so $\mathcal{F}$ converges to x.

(c → a) Suppose X is not compact and let $\{U_\alpha : \alpha \in \Lambda\}$ be an open cover of X with no finite subcover. Then $\{X \setminus U_\alpha : \alpha \in \Lambda\}$ has the finite intersection property and is contained in an ultrafilter $\mathcal{F}$ by Theorem 75. Fix $x \in X$; there exists $\alpha \in \Lambda$ such that $x \in U_\alpha$. But $X \setminus U_\alpha \in \mathcal{F}$ implies $U_\alpha \notin \mathcal{F}$, so $\mathcal{F}$ does not converge to x. ■

Proof of Theorem 77

Let X_α be compact for all $\alpha \in \Lambda$. Put $X = \prod_{\alpha\in\Lambda} X_\alpha$ and let $\mathcal{F}$ be an ultrafilter on X. We want to show that $\mathcal{F}_\alpha = \{\pi_\alpha[F] : F \in \mathcal{F}\}$ is an ultrafilter on X_α for all $\alpha \in \Lambda$.

For convenience fix $\alpha \in \Lambda$, let $\pi = \pi_\alpha$, and put $\mathcal{G} = \mathcal{F}_\alpha$.

(1) Let $G_1, G_2 \in \mathcal{G}$; there exist $F_1, F_2 \in \mathcal{F}$ such that $G_1 = \pi[F_1]$ and $G_2 = \pi[F_2]$. Thus $G_1 \cap G_2 = \pi[F_1] \cap \pi[F_2] \supset \pi[F_1 \cap F_2]$, so $F_1 \cap F_2 \subset \pi^{-1}[G_1 \cap G_2]$. Since $F_1 \cap F_2 \in \mathcal{F}$, we have $\pi^{-1}[G_1 \cap G_2] \in \mathcal{F}$. It follows that $G_1 \cap G_2 = \pi[\pi^{-1}[G_1 \cap G_2]] \in \mathcal{G}$.

(2) Let $G \in \mathcal{G}$ and $G' \subset X_\alpha$ such that $G \subset G'$; there exists $F \in \mathcal{F}$ such that $G = \pi[F]$. Let $F' = \pi^{-1}[G']$ and note $F \subset F'$ (so $F' \in \mathcal{F}$). It follows that $G' = \pi[F'] \in \mathcal{G}$.

(3) That $\varnothing \notin \mathcal{G}$ follows from $\varnothing \notin \mathcal{F}$.

To see that $\mathcal{G}$ is an ultrafilter, let $G \subset X_\alpha$; by Theorem 74(c) either $F = \pi^{-1}[G] \in \mathcal{F}$ or $X \setminus F \in \mathcal{F}$. It follows that $G = \pi[F] \in \mathcal{G}$ or $X_\alpha \setminus G = \pi[X \setminus F] \in \mathcal{G}$.

For each $\alpha \in \Lambda$, there exists $x_\alpha \in X_\alpha$ such that $\mathcal{F}_\alpha$ converges to x_α. We want to show that $\mathcal{F}$ converges to $\vec{x} = \langle x_\alpha : \alpha \in \Lambda\rangle$. Let $U = \prod_{\alpha\in\Lambda} U_\alpha$ be a basic open neighborhood of $\vec{x}$; let $A \subset \Lambda$ be the set of $\alpha \in \Lambda$ with $U_\alpha \neq X_\alpha$. For all $\alpha \in A$, $\mathcal{F}_\alpha$ converges to x_α so $U_\alpha \in \mathcal{F}_\alpha$. Hence, for all $\alpha \in A$ there is some $F_\alpha \in \mathcal{F}$ such that $\pi_\alpha[F_\alpha] = U_\alpha$. This implies $\pi_\alpha^{-1}[F_\alpha] \supset F_\alpha$, so $U \supset \bigcap_{\alpha\in A} F_\alpha$. But $\bigcap_{\alpha\in A} F_\alpha \in \mathcal{F}$, we have $U \in \mathcal{F}$. ■

Proof of Theorem 78

$\rightarrow$ Any open cover (countable or not) has a finite subcover, hence a countable subcover.

$\leftarrow$ Any open cover has a countable subcover, which in turn has a finite subcover. ■

Proof of Theorem 79

(a) Let A be a countably infinite subset of X countably compact. Suppose on the contrary that A has not limit in X. For each $a \in A$ pick U_a open such that $U_a \cap A = \{a\}$. Since A is closed $\{U_a : a \in A\} \cup$

$\{X \setminus A\}$ is a countable open cover of X with no finite subcover, a contradiction.

(b) $\rightarrow$ This is 79(a).

$\leftarrow$ Let X be a T_1-space which is not countably compact. Let $\{U_n : n < \omega\}$ be a countable open cover with no finite subcover. Without loss of generality we may assume that $U_n \subsetneq U_{n+1}$ for each $n < \omega$. For eacn $n < \omega$ pick $x_n \in U_n \setminus \bigcup_{i<n} U_i$ and put $A = \langle x_n : n < \omega \rangle$. Take $x \in X$ and $n < \omega$ such that $x \in U_n$. Observe that $U_n \cap A \subset \{x_0, \ldots, x_n\}$. Since singletons are closed, $(U_n \setminus \{x_0, \ldots, x_n\}) \cup \{x\}$ is an open neighborhood of x containing no point of A other than perhaps x itself. It follows that A is an infinite subset of X without a limit point. ■

Proof of Theorem 80

$\rightarrow$ Let $\langle B_n : n < \omega \rangle$ be a decreasing local base at x and for each $n < \omega$ pick $a_n \in A \cap B_n$; certainly $\langle a_n \rangle \rightarrow x$.

$\leftarrow$ If there is a sequence $\langle a_n \rangle$ in A converging to x then every neighborhood of x meets A, whence $x \in \overline{A}$. ■

Proof of Theorem 81

$\rightarrow$ This is Theorem 31.

$\leftarrow$ Let $A \subset X$ and $x \in \overline{A}$; by Theorem 80 there is a sequence $\langle a_n \rangle$ in A converging to x. By assumption $\langle f(x_n) \rangle \rightarrow f(x)$, so $f(x) \in \overline{f[A]}$. By Theorem 25(f), f is continuous. ■

Proof of Theorem 82

Let $\langle X, d \rangle$ be a metric space and fix $x \in X$. Set $\mathcal{B}_x = \{B(x, 1/n) : n \in \mathbb{N}\}$ and let $U \subset X$ be open containing x. By Corollary 34, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$. Pick $n \in \mathbb{N}$ such that $1/n < \varepsilon$. Then $B(x, 1/n) \subset B(x, \varepsilon) \subset U$, whence $\mathcal{B}_x$ is a local base at x. ■

Proof of Exercise 83

Note $X/A = \{A\} \cup \{(i, y) : i \in \mathbb{N} \wedge 0 < y \leqslant 1\}$. Points of the form $\langle i, y \rangle$

for $y > 0$ have their usual neighborhoods in the interval $\{i\} \times (0, 1]$, and a basic open neighborhood of A has the form

$$\{A\} \cup \bigcup_{i \in \mathbb{N}} (\{i\} \times (0, \varepsilon_i)),$$

where $0 < \varepsilon_i \leqslant 1$ for all i.

We wish to show that X/A is not first-countable. Indeed, suppose $\{U_i : i \in \mathbb{N}\}$ is a countable base at A. If q is the quotient map, then $q^{-1}[U_i]$ contains $(i, 0)$ and is open in X, so for all i there is some $y_i > 0$ such that $x_i = (i, y_i) \in U_i$. Then $X \setminus \{x_i : i \in \mathbb{N}\}$ is a saturated open set in X, hence its image is an open neighborhood of A which does not contain any x_i, hence not any U_i, so we get a contradiction.

Since X/A is not first-countable, it cannot be metrizable. ■

Proof of Theorem 84

Let X be second-countable and let $\mathcal{B} = \{B_n : n < \omega\}$ be a countable base for X.

(a) Note that if $Y \subset X$ then $\{B_n \cap Y : n < \omega\}$ is a countable base for the subspace topology on Y by Theorem 23.

(b) It suffices to show that X is separable. For each $n < \omega$ pick $x_n \in B_n$ and let $D = \{x_n : n < \omega\}$. Let $U \subset X$ be a nonempty open set. There exists $n < \omega$ such that $B_n \subset U$, whence $x_n \subset U$.

(c) It suffices to show X is Lindelöf. Let $\mathcal{U}$ be an open cover of X. For each $n < \omega$, if there is $U \in \mathcal{U}$ with $B_n \subset U$ pick one and call it V_n. If there is no such U, leave V_n undefined. Let $\mathcal{V}$ be the collection of those V_n which are defined. Let $x \in X$; then $x \in U$ for some $U \in \mathcal{U}$. Since $\mathcal{B}$ is a base, there exists $B_n \in \mathcal{B}$ with $x \in B_n \subset U$. It follows that V_n is defined and $x \in B_n \subset V_n$. ■

Proof of Theorem 85

Let $\langle X, d\rangle$ be a metric space.

(a → d) By Theorem 84.

(d → c) By definition.

(c → a) Let D be a countable dense subset of X. Put

$$\mathcal{B} = \{B(x, 1/n) : x \in D \wedge n \in \mathbb{N}\};$$

certainly $\mathcal{B}$ is countable. Let $U \subset X$ be open and $x \in U$. Then there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$. Pick $n \in \mathbb{N}$ such that $2/n < \varepsilon$ and let $y \in B(x, 1/n) \cap D$. Certainly $x \in B(y, 1/n)$. Take $z \in B(y, 1/n)$. Then

$$d(x, z) \leqslant d(x, y) + d(y, z) < 1/n + 1/n = 2/n < \varepsilon,$$

so $z \in B(x, \varepsilon)$. It follows that $x \in B(y, 1/n) \subset B(x, \varepsilon) \subset U$, so $\mathcal{B}$ is a base.

(a → e) By Theorem 84.

(e → b) By definition.

(b → a) Let $\mathcal{V}_n$ be a countable subcover of $\{B(x, 1/n) : x \in X\}$ for each $n \in \mathbb{N}$. Put $\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{V}_n$; certainly $\mathcal{B}$ is countable. Let $W \subset X$ be open and $x \in W$. There exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset W$. Choose $n \in \mathbb{N}$ such that $2/n < \varepsilon$. Then there exists $y \in X$ such that $x \in B(y, 1/n) \in \mathcal{V}_n$. Take $z \in B(y, 1/n)$. Then

$$d(x, z) \leqslant d(x, y) + d(y, z) < 1/n + 1/n = 2/n < \varepsilon,$$

so $z \in B(x, \varepsilon)$. It follows that $x \in B(y, 1/n) \subset B(x, \varepsilon) \subset W$, so $\mathcal{B}$ is a base. ■

Proof of Theorem 86

Let $X = \prod_{n \in \mathbb{N}} X_n$, where each X_n is first-countable. Take $\vec{x} \in X$; $\vec{x} = (x_n)_{n \in \mathbb{N}}$. For each $n \in \mathbb{N}$, let $\mathcal{B}_n$ be a countable local base at x_n. Put

$$\mathcal{B} = \{\prod_{i=1}^{m} B_i \times \prod_{i=m+1}^{\infty} X_i : B_i \in \mathcal{B}_i, m \in \mathbb{N}\}.$$

Certainly $\mathcal{B}$ is countable.

Let $U = \prod_{i=1}^{m} U_i \times \prod_{i=m+1}^{\infty} X_i$ be a basic open neighborhood of $\vec{x}$. For each $i \leqslant m$, take $B_i \in \mathcal{B}_i$ such that $B_i \subset U_i$. Then $\vec{\in} \prod_{i=1}^{m} B_i \times \prod_{i=m+1}^{\infty} X_i \subset U$, so $\mathcal{B}$ is a local base at $\vec{x}$.

The proof for each X_n second-countable is similar. ■

Proof of Example 87
Let $\{U_n : n < \omega\}$ be collection of open neighborhoods of $\vec{0}$; we show this cannot be a local base at $\vec{0}$. For all $n < \omega$, there exists a basic open set O_n such that $\vec{0} \in O_n \subset U_n$; say $O_n = \prod\{O_{n,\alpha} : \alpha \in \Lambda\}$. Then $O_{n,\alpha}$ is open in 2 and $O_{n,\alpha} = 2$ for all but finitely many α.

Note there exists $\beta \in \Lambda$ such that $O_{n,\beta} = 2$ for all $n < \omega$; put $V = \pi_\beta^{-1}(0)$. Let $\vec{x} \in 2^\Lambda$ such that $x_\beta = 1$ and $x_\alpha = 0$ for all $\alpha \neq \beta$. Then $\vec{x} \in O_n \subset U_n$ for each $n < \omega$ but $\vec{x} \notin V$. Since $U_n \not\subset V$ for all $n < \omega$, there is no countable local base at $\vec{0}$. ■

Proof of Theorem 88
Let $X = \prod_{n\in\mathbb{N}} X_n$, with each X_n separable. For each $n \in \mathbb{N}$ let D_n be a countable dense subset of X_n. Fix $\vec{x} = (x_n)_{n\in\mathbb{N}} \in X$ and put

$$D = \{(d_n)_{n\in F} \times (x_n)_{n\notin F} : F \in [\mathbb{N}]^{<\omega}, d_n \in D_n\}.$$

Certainly D is countable.

Let $U = \prod_{i\in A} U_i \times \prod_{i\notin A} X_i$ be a basic open set, so A is finite. Pick $d_i \in D_i \cap U_i$ for each $i \in A$. Put $\vec{y} = (y_i)_{i\in A} \times (x_i)_{i\notin A}$. Then $\vec{y} \in D \cap U$, so D is dense. ■

Proof of Theorem 89
(a) Let A be an infinite subset of X compact. Suppose on the contrary that A has no limit point in X. Then for each $x \in X$ there exists an open neighborhood U_x of x such that $U_x \cap (A \setminus \{x\}) = \varnothing$. Note that $\mathcal{U} = \{U_x : x \in X\}$ is an open cover of X and let $\mathcal{V}$ be a finite subcover thereof. But each $V \in \mathcal{V}$ contains at most one point of A implying A is finite, a contradiction.

(b) Let A be an uncountable subset of a Lindelöf space X. Suppose A has no complete accumulation point in X. Then for each $x \in X$ there exists an open neighborhood U_x of x such that $U_x \cap A$ is countable. Note that $\mathcal{U} = \{U_x : x \in X\}$ is an open cover of X and let $\mathcal{V}$ be a countable subcover thereof. But this implies $A = \bigcup_{V\in\mathcal{V}}(V \cap A)$ is countable, a contradiction. ■

Proof of Corollary 90
Recall that separable metric spaces are hereditarily Lindelöf and apply

Theorem 89(b). ■

Proof of Theorem 91

(a) Let X be a space which is not countably compact. Let $\{U_n : n < \omega\}$ be a countable open cover of X with no finite subcover; without loss of generality we may assume that $U_n \subsetneq U_{n+1}$ for each $n < \omega$. Take a sequence $\langle x_n \rangle_{n<\omega}$ in X with $x_n \in U_{n+1} \setminus U_n$. Let $x \in X$ and pick $k < \omega$ such that $x \in U_k$. Then U_k is an open neighborhood of x containing only finitely many terms in the sequence. It follows that $\langle x_n \rangle$ has no infinite convergent subsequence.

(b) $\rightarrow$ Let X be first-countable and countably compact. Let $\langle x_n \rangle_{n<\omega}$ be an infinite sequence in X and suppose on the contrary there is no $x \in X$ whose every neighborhood contains infinitely many of the x_n. Put $A_n = \{x_m : m \geqslant n\}$; fix $x \in X$. Then there is a nieghborhood U of x and $n < \omega$ such that $\{k : x_k \in U\} \subset n$, implying $x \notin \overline{A}_n$. Thus $\{X \setminus \overline{A}_n : n < \omega\}$ is a countable open cover of X with no finite subcover, a contradiction.

So, let $x \in X$ such that every neighborhood of x contains infinite many terms of the sequence. Let $\langle V_n : n < \omega \rangle$ be a countable decreasing local base at x. Pick a subsequence $\langle x_{n_i} \rangle$ such that $x_{n_i} \in V_i$ for all $i < \omega$ and $i < j \rightarrow n_i < n_j$. Certainly $\langle x_{n_i} \rangle \rightarrow x$.

$\leftarrow$ This is 91(a). ■

Proof of Theorem 92

Let $\langle X, d \rangle$ be a metric space, not necessarily separable.

(a $\rightarrow$ c) By Theorems 91(b), 78 and 82.

(c $\rightarrow$ b) By Theorem 91(a).

(b $\rightarrow$ a) We claim that $\{B(x, 1/n) : x \in X\}$ has a finite subcover for each $n \in \mathbb{N}$. Suppose not. Take $x_0 \in X$. Then $B(x_0, 1/n) \neq X$, so pick $x_1 \in X \setminus B(x_0, 1/n)$. If $x_0, \ldots, x_k$ have been chosen, pick $x_{k+1} \in X \setminus \bigcup_{i \leqslant k} B(x_i, 1/n)$. Put $A = \langle x_n : n < \omega \rangle$; for each $p < q < \omega$ we have $d(x_p, x_q) \geqslant 1/n$. Thus A has no limit point, a contradiction.

For each $n \in \mathbb{N}$ let $F_n \subset X$ be finite such that $\{B(x, 1/n) : x \in F_n\}$ covers X. Note that for each $x \in X$ there exists $y \in F_n$ such that $d(x, y) < 1/n$. It follows that $D = \bigcup_{n\in\mathbb{N}} F_n$ is a countable dense subset of X. Thus X is separable, hence Lindelöf by Theorem 85. By Theorem 78, X is compact. ■

Proof of Exercise 93

Let $\langle \alpha_n \rangle$ be a sequence in ω_1; by Corollary 6 there exists $\alpha < \omega_1$ such that $\alpha_n \in [0, \alpha]$ for all $n < \omega$. Each $\beta \in [0, \alpha]$ has countably many predecessors, so $[0, \alpha]$ must be first-countable. By Theorem 52(b), $[0, a]$ is sequentially compact. Thus $\langle \alpha_n \rangle$ has an infinite convergent subsequence in $[0, \alpha]$, hence in ω_1.

To see that ω_1 is not compact, consider $\{[0, \alpha] : \alpha < \omega_1\}$. ■

Proof of Theorem 94

Let $X^* = X \cup \{\infty\}$ and $\mathcal{B} = \mathcal{T} \cup \{X^* \setminus K : K \subset X \wedge K \text{ compact}\}$; let $\mathcal{T}^*$ be the topology generated by $\mathcal{B}$.

Let $\mathcal{U}$ be an open cover of X^*. Then there exists $U_\infty \in \mathcal{U}$ such that $\infty \in U_\infty$, which implies that $X^* \setminus U_\infty$ is compact. So there exists $\mathcal{V} = \{U_1, \ldots, U_n\} \subset \mathcal{U}$ which covers $X^* \setminus U_\infty$. Hence $\mathcal{V} \cup \{U_\infty\}$ is a finite subcover of $\mathcal{U}$.

Let $x \in X$. Since X is locally compact, there exists a compact neighborhood U_x of x. It follows that $X \setminus U_x$ is an open neighborhood of ∞. Then U_x° and $X \setminus U_x$ are disjoint open sets containing x and ∞, respectively. Also, if $x, y \in X$, these two points can be separated by disjoint open sets since X is Hausdorff. Thus X^* is Hausdorff. ■

Proof of Theorem 95

Claim A. *Every subspace of a completely regular space is completely regular.*

Proof of Claim A. Let X be a completely regular space and $Y \subset X$. Let C be closed in Y and $y \in Y \setminus C$. Then $y \notin C = \operatorname{cl}_Y C = \operatorname{cl}_X C \cap Y$, so $y \notin \operatorname{cl}_X C$. Since X is completely regular, there exists $f\colon X \to [0, 1]$ continuous such that $f(y) = 0$ and $f[\operatorname{cl}_X C] = \{1\}$. Define $g\colon Y \to [0, 1]$ by $y \mapsto (f \restriction Y)(y)$. Then g is continuous, $g(y) = 0$, and $g[C] = \{1\}$. Thus Y is completely regular. □

Since compact Hausdorff spaces are completely regular, this follows from the one-point compactification of a locally compact Hausdorff space and Proof A. ■

Proof of Lemma 96

Let $x \in U$ with U open in X. We need to show there is a compact neighborhood K of x contained in U.

Since X is locally compact, there is a compact neighborhood N of x. Then $V = N^\circ \cap U$ is open in X and contains x. Also, $N \setminus V$ is closed in N, hence in X. By Theorem 95, X is completely regular. Let $f\colon X \to [0,1]$ be continuous such that $f(x) = 0$ and $f[N \setminus V] = \{1\}$. Let $K = f^{-1}[[0,1/2]] \cap N$. Then K is closed in N, hence compact. Also, $x \in f^{-1}[[0,1/2]] \cap V$ which is open in X and, since $V \subset N$, is containd in K. So K is a compact neighborhood of x. It remains to prove that $K \subset U$. Let $y \in K$. Then $y \in N$. Since $f(y) \in [0,1/2]$, $y \notin N \setminus V$. But $y \in N$, so $y \in V \subset U$. ■

Proof of Lemma 97

Suppose that $\bigcap_{n\in\mathbb{N}} H_n$ is empty. Then $\{H_1 \setminus H_n : n \in \mathbb{N}\}$ is a (countable) open cover of H_1. Let $\{H_1 \setminus H_n : n \in F\}$, with $F \subset \mathbb{N}$ finite, be a subcover thereof. If $k = \max F$, we have $H_1 \setminus H_k = H_1$. But this implies $H_k = \varnothing$, a contradiction. ■

Proof of Theorem 98

Put $D = \bigcap_{n\in\mathbb{N}} G_n$ and let $U \subset X$ be nonempty and open. Since G_1 is dense, pick $x_1 \in G_1 \cap U$. Let H_1 be a compact neighborhood of x_1 contained in $G_1 \cap U$. Since G_2 is dense, pick $x_2 \in G_2 \cap H_1^\circ$. Let H_2 be a compact neighborhood of x_2 contained in $G_2 \cap H_1^\circ$. Continuing in this way, construct for each $n \in \mathbb{N}$ a nonempty compact H_n such that $H_1 \subset U$ and $H_n \subset G_n \cap H_{n-1}^\circ$. Put $K = \bigcap_{n\in\mathbb{N}} H_n$. Then $K \subset U \cap (\bigcap_{n\in\mathbb{N}} G_n) = U \cap D$. By Theorem 98, $K \neq \varnothing$ so $U \cap D \neq \varnothing$. ■

Proof of Lemma 99

Let X be a Baire space and let $O \subset X$ be open. Let $\{U_n : n \in \mathbb{N}\}$ be a collection of open sets in O with $\overline{U_n}^O = O$ for each $n \in \mathbb{N}$. We show that $\overline{\bigcap_{n\in\mathbb{N}} U_n}^O = O$. Since O is open in X, U_n is open in X for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, set $V_n = U_n \cup X \setminus O^\circ$. Thus V_n is open in X

for every $n \in \mathbb{N}$.

We claim each V_n is dense in X. Fix $n \in \mathbb{N}$ and let V be a nonempty open set in X. If $V \cap O \neq \varnothing$, then $V \cap U_n \neq \varnothing$ since U_n is dense in O. If $V \cap O = \varnothing$, then $V \subset X \setminus O$, so $V \subset X \setminus O^\circ$ implying $V \subset V_n$. Therefore $V \cap V_n \neq \varnothing$, so V_n is dense. Since X is Baire, we have $\overline{\bigcap_{n\in\mathbb{N}} V_n} = X$.

Let $V \subset O$ be a nonempty open set. We have $V \cap (\bigcap_{n\in\mathbb{N}} V_n) \neq \varnothing$. But $V \cap X \setminus O^\circ = \varnothing$ so V must meet $\bigcap_{n\in\mathbb{N}} U_n$. Thus $\bigcap_{n\in\mathbb{N}} U_n$ is dense in O, so O is Baire. ■

Proof of Theorem 100

Let X be a Baire space with $X = \bigcup_{n\in\mathbb{N}} X_n$. Suppose on the contrary that $\overline{X}_n^\circ = \varnothing$ for all $n \in \mathbb{N}$. Then $X \setminus \overline{X}_n$ is dense and open in X for each $n \in \mathbb{N}$, so $\bigcap_{n\in\mathbb{N}} X \setminus \overline{X}_n$ is dense (and nonempty) in X. But

$$\bigcap_{n\in\mathbb{N}} X \setminus X_n \subset \bigcap_{n\in\mathbb{N}} X \setminus X_n = X \setminus \bigcup_{n\in\mathbb{N}} X_n = \varnothing,$$

a contradiction.

Now let U be open in X, so $U = \bigcup_{n\in\mathbb{N}}(U \cap X_n)$. By Lemma 99, U is Baire. By the previous argument, there exists $n \in \mathbb{N}$ such that

$$\varnothing \neq \operatorname{int}_U (\overline{U \cap X_n}^U) = \operatorname{int}_U(\overline{X}_n \cap U) \subset \overline{X}_n^\circ \cap U.$$

Thus, every open set meets $\bigcup_{n\in\mathbb{N}} \overline{X}_n^\circ$. ■

Proof of Theorem 101

(a → b) Suppose $C = A \cup B$, where A and B are nonempty separated sets in X. Put $U = X \setminus \operatorname{cl}_X B$ and $V = X \setminus \operatorname{cl}_X A$. Then $C \cap U = A$ and $C \cap V = B$ are disjoint open subsets of C, so C is not connected.

(b → a) Now suppose C is not connected. Then $C = A \cup B$, where A and B are disjoint nonempty open subsets of C. Let U and V be open in X such that $A = U \cap C$ and $B = V \cap C$. Since $A \subset X \setminus V$, $\operatorname{cl}_X A \cap B = \varnothing$. Similarly, $\operatorname{cl}_X B \cap A = \varnothing$. Thus C is the union of two nonempty separated sets in X. ■

Proof of Theorem 102

Let A be connected and suppose on the contrary that B is not connected. Then $B = H \cup K$, where H and K are nonempty separated sets in X.

Furthermore, $H \cap A$ and $K \cap A$ are separated sets whose union is A. Since A is connected, we may assume without loss of generality that $K \cap A = \varnothing$. Then $A \subset H$ so $B \subset \overline{A} \subset \overline{H}$, contradicting the fact that $K \cap \overline{H} = \varnothing$. ■

Proof of Theorem 103

$\rightarrow$ Suppose X is not convex; there exist $a < b$ in X such that $(a, b) \not\subset X$. Fix $x \in (a, b)$ such that $x \notin X$. Then we may write $X = A \cup B$, where $A = X \cap (-\infty, x)$ and $B = (x, \infty)$ are disjoint nonempty open sets in X.

$\leftarrow$ Suppose X is not connected; then $X = A \cup B$ with A and B nonempty separated sets in $\mathbb{R}$. Without loss of generality, there exist $a \in A$ and $b \in B$ such that $a < b$. Let $x = \sup(A \cap [a, b])$. Note $x \in \overline{A}$, so $x \notin B$.

If $x \notin A$, then $x \notin X$. If $x \in A$, then $x \notin \overline{B}$ so there exists $y \in (x, b)$ with $y \notin B$. But (x, b) misses A, so $y \notin X$. ■

Proof of Theorem 104

Let $f\colon X \to Y$ be a continuous surjection. Suppose $Y = U \cup V$, where U and V are nonempty disjoint open sets. Then $X = f^{-1}(U) \cup f^{-1}(V)$, so X is not connected.

Now suppose X is path connected and take $x, y \in Y$. Choose $a \in f^{-1}(p)$ and $b \in f^{-1}(q)$. Let g be a path in X from a to b. Then $f \circ g$ is a path in Y from x to y, whence Y is path connected. ■

Proof of Theorem 105

Let $\{A_\alpha : \alpha \in \Lambda\}$ be a collection of connected subspaces of a space X such that $A_\alpha \cap A_\beta \neq \varnothing$ for any $\alpha, \beta \in \Lambda$. Put $A = \cup\{A_\alpha : \alpha \in \Lambda\}$ and suppose on the contrary that A is not connected. Then there exist nonempty separated subsets U and V of X such that $A = U \cup V$. For each $\alpha \in \Lambda$, put $U_\alpha = U \cap A_\alpha$ and $V_\alpha = V \cap A_\alpha$. Then $A_\alpha = U_\alpha \cup V_\alpha$ for each $\alpha \in \Lambda$ with U_α and V_α separated. But each A_α is connected, so either $U_\alpha = \varnothing$ or $V_\alpha = \varnothing$. Without loss of generality we may assume $U_\alpha = \varnothing$ for some $\alpha \in \Lambda$, so $A_\alpha = V_\alpha$. For each $\beta \in \Lambda$ we have

$$A_\alpha \cap A_\beta = V_\alpha \cap A_\beta \subset V \cap A_\beta = V_\beta,$$

so $V_\beta \neq \varnothing$. It follows that $U_\beta = \varnothing$ for each $\beta \in \Lambda$ so $U = \varnothing$, a contradiction.

Now let $\{A_\alpha : \alpha \in \Lambda\}$ be a collection of path connected subspaces of a space X such that $A_\alpha \cap A_\beta \neq \varnothing$ for any $\alpha, \beta \in \Lambda$. Put $A = \cup\{A_\alpha : \alpha \in \Lambda\}$ and take $x, y \in A$. Choose $\alpha, \beta \in \Lambda$ such that $x \in A_\alpha$ and $y \in A_\beta$, and pick $z \in A_\alpha \cap A_\beta$. Let $f_\alpha : \mathbb{I} \to A_\alpha$ be a path from x to z and $f_\beta : \mathbb{I} \to A_\beta$ a path from z to y. Then $f : \mathbb{I} \to A$ defined by

$$f(t) = \begin{cases} f_\alpha(2t) & \text{if } 0 \leqslant t \leqslant 1/2 \\ f_\beta(2t-1) & \text{if } 1/2 \leqslant t \leqslant 1 \end{cases}$$

is a path from x to y. ■

Proof of Theorem 106

(a) For any $x \in X$ and $y \in Y$ we know $\{x\} \times Y$ and $X \times \{y\}$ are connected. For each $x \in X$ and $y \in Y$, let $P(x, y) = (\{x\} \times Y) \cup (X \times \{y\})$. Since $(\{x\} \times Y) \cap (X \times \{y\})$ is nonempty, $P(x, y)$ is connected for each (x, y).

Note that $\cup\{P(x, y) : (x, y) \in X \times Y\} = X \times Y$. If $x, x' \in X$ and $y, y' \in Y$, then $P(x, y) \cap P(x', y') \supset \{(x', y)\}$. Thus $X \times Y$ is connected by Theorem 105.

(b) Let $X = \prod_{\alpha \in \Lambda} X_\alpha$, with each X_α connected. Suppose on the contrary that X is not connected. Then there exist disjoint nonempty open sets U and V such that $X = U \cup V$. Take $\vec{x} \in U$. Let $B = \prod_{\alpha \in F} U_\alpha \times \prod_{\alpha \notin F} X_\alpha$ be a basic open set with $\vec{x} \in B \subset U$.

Fix $\vec{y} \in X$. Then $C = \prod_{\alpha \in F} X_\alpha \times \prod_{\alpha \notin F} \{y_\alpha\}$ is connected and $C \cap B \neq \varnothing$. Moreover $U \cap C \neq \varnothing$, so $y \in C \subset U$. But this implies $V = \varnothing$, a contradiction. ■

Proof of Theorem 107

Let X be a path connected space. Suppose on the contrary that X is not connected. Then there are disjoint nonempty open sets U and V such that $X = U \cup V$. Take $a \in U$ and $b \in V$ and let $f : \mathbb{I} \to X$ be a path from a to b. Put $K = f(\mathbb{I})$. Then $U \cap K$ and $V \cap K$ are disjoint nonempty open subsets of K whose union is K, whence K is

not connected. But K is the image of a connected space, so we have a contradiction. ■

Proof of Theorem 108
$\rightarrow$ If f is continuous, then $\pi_\alpha \circ f$ is continuous as a composition of continuous functions.

$\leftarrow$ Now suppose that $\pi_\alpha \circ f$ is continuous for all $\alpha \in \Lambda$. Let $U = \prod_{\alpha \in F} U_\alpha \times \prod_{\alpha \notin F} X_\alpha$ be a basic open set, so F is finite. Then

$$f^{-1}[U] = f^{-1}[\bigcap_{\alpha \in F} \pi_\alpha^{-1}[U_\alpha]] = \bigcap_{\alpha \in F} (\pi_\alpha \circ f)^{-1}[U_\alpha]$$

is open as a finite intersection of open sets. ■

Proof of Theorem 109
Let $X = \prod_{\alpha \in \Lambda} X_\alpha$, with each X_α path connected, and take $\vec{a}, \vec{b} \in X$. For each $\alpha \in \Lambda$, let $f_\alpha : \mathbb{I} \to X_\alpha$ be a path from a_α to b_α. Define $f : \mathbb{I} \to X$ by $f(t) = (f_\alpha(t))\alpha \in \Lambda$. Then f is continuous because $\pi_\alpha \circ f = f_\alpha$ is continuous for each $\alpha \in \Lambda$. Moreover $f(0) = (f_\alpha(0))_{\alpha \in \Lambda} = \vec{a}$ and $f(1) = (f_\alpha(1))_{\alpha \in \Lambda} = \vec{b}$, so f is a path from $\vec{a}$ to $\vec{b}$. ■

Proof of Lemma 110
Let $f\colon X \to Y$ be continuous and let $G = \{\langle x, f(x)\rangle : x \in X\}$. Define $h : G \to X$ by $(x, f(x)) \mapsto x$. Certainly h is a bijection. Furthermore $h = \pi_1 \restriction G$, so h is continuous. Note that $h^{-1} : X \to G \subset X \times Y$ by $x \mapsto (x, f(x))$, so $(\pi_1 \circ h^{-1}) = \mathrm{id}_X$ and $(\pi_2 \circ h^{-1}) = f$ are continuous. It follows that h^{-1} is continuous. ■

Proof of Example 111
Let $G = \{(x, \sin 1/x) : 0 < x \leqslant 1\}$. Then $G \cong (0, 1]$, so G is connected. It follows that the topologist's sine curve $\overline{G} = (\{0\} \times [-1, 1]) \cup G$ is connected.

Now suppose on the contrary that S is path connected. Let $f : \mathbb{I} \to S$ be a path from $(0, 0)$ to $(1/\pi, 0)$. Then $f(t) = (x(t), y(t))$, where $x = \pi_1 \circ f$ and $y = \pi_2 \circ f$ are continuous. Note that $x[\mathbb{I}]$ must be a connected, hence convex, subset of $\mathbb{R}$. Since $x(0) = 0$ and $x(1) = 1/\pi$, there exists $0 < t_1 < 1$ such that $x(t_1) = 2/3\pi$. Similarly there exists

$0 < t_2 < t_1$ such that $x(t_2) = 2/5\pi$. Continuing this way, we define a sequence $(t_n)_{n\in\mathbb{N}}$ satisfying $0 < \cdots < t_n < \cdots < t_1 < 1$ such that $x(t_n) = 2/(2n+1)\pi$. Note $y(t_n) = (-1)^n$.

Since the sequence $(t_n)_{n\in\mathbb{N}}$ is decreasing and bounded below we have $t_n \to t_0$ for some t_0. Since y is continuous, $y(t_n) \to y(t_0)$. But $\lim_{n\to\infty}(-1)^n$ does not exist so $y(t_n)$ cannot converge, a contradiction. ■

Proof of Theorem 112

(a) Trivial.

(b) Certainly, for if two (path) components met their union would be a larger (path) connected set.

(c) The closure of a connected set is connected. ■

Proof of Lemma 113

Let X be a locally (path) connected space and let C be a (path) component of X. For each $x \in C$ let N_x be a (path) connected neighborhood thereof. Certainly each $N_x \subset C$, whence C is open. ■

Proof of Theorem 114

Let X be locally (path) connected and let $\mathcal{B}$ be the collection of all open and (path) connected subsets of X. Let $W \subset X$ be open and take $x \in W$. Let $C_W(x)$ be the (path) component of x in W. We want to see that $C_W(x)$ is open in W; by Lemma 113 it suffices to show that W is locally (path) connected.

Let $y \in W$ and $U \subset W$ open containing y. As W is open in X, U is also open in X. Since X is locally (path) connected, there exists $N_y \subset U$ (path) connected such that $y \in N_y^\circ$. Thus, W is locally (path) connected.

Now suppose X has a base consisting of (path) connected subsets. If $W \subset X$ is open and $x \in X$, there exists a (path) connected basic open set B such that $x \in B \subset W$. It follows immediately that X is locally (path) connected. ■

Proof of Theorem 115
Suppose X is locally path connected and fix $x \in X$. Let $C(x)$ be the component of x in X and $P(x)$ its path component. Then $P(x) \subset C(x)$ since $P(x)$ is connected and contains x. $P(x)$ is open by Lemma 113. Likewise, $X \setminus P(x)$ is open as a union of path components. It follows that $P(x)$ is a nonempty clopen subset of $C(x)$, whence $P(x) = C(x)$.■

Proof of Theorem 116
If H is clopen and $H \cap C(x) \neq \varnothing$, then $C(x) \subset H$ since $X = H \cup X \setminus H$ and $C(x)$ is connected. Thus $C(x) \subset Q(x)$. ■

Proof of Lemma 117
Let $U \subset X$ be open containing H. Since $H \subset U$, we have $X \setminus U \subset X \setminus H = \bigcup_{K \in \mathcal{K}} X \setminus K$. Being closed, $X \setminus U$ is compact so there exists a finite subcollection $\mathcal{K}'$ of $\mathcal{K}$ such that $X \setminus U \subset \bigcup_{K \in \mathcal{K}'} X \setminus K$. It follows that $H = \cap \mathcal{K} \subset \cap \mathcal{K}' \subset U$. ■

Proof of Theorem 118
Fix $x \in X$; by Theorems 112(a) and 116, it suffices to show $Q(x)$ is connected. Suppose on the contrary that $Q(x)$ is not connected. Let A and B be disjoint nonempty clopen subsets of $Q(x)$ with $Q(x) = A \cup B$. As $Q(x)$ is closed in X, both A and B are closed in X. Since X is normal, being compact Hausdorff, there exist disjoint open sets U and V in X containing A and B, respectively. By Lemma 117, there exist clopen (in X) sets $F_1, \ldots, F_n$ such that $Q(x) \subset F_1 \cap \cdots \cap F_n \subset U \cup V$; put $F = F_1 \cap \cdots \cap F_n$, $U' = U \cap F$, and $V' = V \cap F$. Since F is clopen in X and $F = U' \cup V'$, we have that U' and V' are clopen in X.

Without loss of generality, suppose $x \in A$. Then $x \in U'$, so $Q(x) \subset U'$. Thus $Q(x) \cap B = \varnothing$, a contradiction. ■

Proof of Theorem 119
Let $\langle K \rangle$ be a decreasing sequence of compact, connected subsets of a Hausdorff space X and put $K = \bigcap_{n<\omega} K_n$. Certainly K is a closed subset of K_0, so K is compact.

Suppose on the contrary that K is not connected; let A and B be disjoint nonempty closed subsets of K such that $K = A \cup B$. Note that A and B are closed in K_0. Since K_0 is normal, being compact Hausdorff, there exist disjoint open sets U and V (in K_0) containing

A and B, respectively. By Lemma 117 there exists $n < \omega$ such that $K_n \subset U \cup V$. We may write $K_n = U' \cup V'$, where $U' = U \cap K_n$ and $V' \cap K_n$ are disjoint nonempty open (in K_n) sets. Thus K_n is not connected, a contradiction. ■

Proof of Theorem 120

Suppose on the contrary there exists $h \in H$ such that $C(h) \cap \partial H = \varnothing$. Note that ∂H is closed, and by Theorem 118 we have $C(h) = Q(h)$. By Lemma 117 there is a finite collection, whose intersection we denote by F, of clopen subsets of H such that

$$C(h) = Q(h) \subset F \subset H \setminus \partial H = H^\circ.$$

Observe that F is a nonempty clopen subset of X, being a closed subset of H and an open subset of H°. Thus $X = F \subset H$, a contradiction. ■

Proof of Lemma 121

Fix $k < \omega$. If $K_k = \varnothing$ we may simply take $C_k = X$, so assume K_k is nonempty. Choose $j < \omega$ such that $j \neq i$ and K_j is nonempty. Since X is normal, being compact Hausdorff, there exist disjoint open sets U and V containing K_j and K_k, respectively. Let $x \in K_j$ and let C_k be the component of x in $\overline{U}$. Note that C_k is compact and connected, $C_k \cap K_k = \varnothing$, and $C_k \cap K_k \neq \varnothing$. By Theorem 120, we have $C_k \cap \partial\overline{U} \neq \varnothing$; let $y \in C_k \cap \partial\overline{U}$. Since $\mathcal{K}$ covers X, there exists $\ell < \omega$ with $y \in K_\ell$. Now $\ell \neq j$ since $K_j \subset U$ and $\ell \neq k$ since $\overline{U} \cap V = \varnothing$. ■

Proof of Theorem 122

Suppose on the contrary that $\mathcal{K}$ is a partition of X into closed subsets with $1 < |\mathcal{K}| < \aleph_1$. Immediately we must have $|\mathcal{K}| = \aleph_0$, for if $\mathcal{K}$ is finite we may write X as a union of disjoint clopen subsets. Let $\{K_n : n < \omega\}$ be an enumeration of $\mathcal{K}$ without repetition.

Applying Lemma 121 with $k = 0$, we get a compact connected subspace C_0 of X missing K_0 and meeting at least two members of $\mathcal{K}$. Put $\mathcal{K}_0 = \{C_0 \cap K_n : n < \omega\}$. By Lemma 121 there exists a compact connected subspace C_1 of C_0 which misses K_1 and meets at least two members of $\mathcal{K}_0$. Continuing in this way, for each $n < \omega$ we get a compact connected C_n missing K_n such that $C_n \supset C_{n+1}$. By Lemma 117, we have that $C = \bigcap_{n<\omega} C_n$ is nonempty. But $C \cap K_n = \varnothing$ for each $n < \omega$, a contradiction. ■

Proof of Lemma 123
Let $\mathcal{U} = \{U_\alpha : \alpha \in \Lambda\}$, and for each $x \in X$ let V_x be an open neighborhood of x such that $F_x = \{\alpha \in \Lambda : V \cap U_\alpha \neq \varnothing\}$ is finite.

(a) If $I \subset \Lambda$ and $x \in \overline{\{U_\alpha : \alpha \in I\}}$, then

$$x \in \overline{\bigcup \{U_\alpha : \alpha \in I \cap F_x\}} = \bigcup \{\overline{U}_\alpha : \alpha \in I \cap F_x\} \subset \bigcup \{\overline{U}_\alpha : \alpha \in I\}.$$

(b) There is a finite subset F of X such that $\{V_x : x \in F\}$ covers X. Then $\Lambda = \bigcup \{F_x : x \in F\}$, which is finite. ■

Solution of Exercise 124
(a) Take $\mathcal{U} = \{(n, n+2) : n \in \mathbb{Z}\}$.

(b) Take $\mathcal{U} = \{(0, 1/n) : n \in \mathbb{N}\} \cup \{\mathbb{R}\}$.

(c) Take $\mathcal{U} = \{(1/n + 1, 1/n : b \in \mathbb{N}\} \cup \{[0, 1]\}$.

(d) Take $\mathcal{U} = \{[0, 1/n] : n \in \mathbb{N}\}$. ■

Proof of Theorem 125
Let X be a paracompact T_2-space.

(a) Fix $x \in X$ and $H \subset X$ closed not containing x. For each $y \in H$ let O_y and V_y be open containing x and y, respectively. Note that $\mathcal{U} = \{X \setminus H\} \cup \{V_y : y \in H\}$ is an open over of X; let $\mathcal{V}$ be a locally finite open refinement thereof. Take $\mathcal{W} = \{V \in \mathcal{V} : V \cap H \neq \varnothing\}$. By Lemma 123(a) $\mathcal{W}$ is closure-preserving; put $W = \bigcup \mathcal{W}$ and note W is open containing H.

Fix $A \in \mathcal{W}$ and note $A \subset V_y$ for some $y \in H$. Since $x \in O_y$ and $O_y \cap V_y = \varnothing$, we have $x \notin \overline{A}$. It follows that $x \notin \overline{W} = \bigcup \{\overline{A} : A \in \mathcal{W}\}$.

(b) Let H and K be disjoint closed sets. By 125(a), for each $y \in K$ there exist disjoint open sets O_y and V_y containing H and y, respectively. Let $\mathcal{V}$ be a locally finite open refinement of $\mathcal{U} = \{X \setminus K\} \cup \{V_y : y \in K\}$. Put $W = \bigcup \{A \in \mathcal{V} : A \cap K \neq \varnothing\}$ and note that $X \setminus \overline{W}$ and W are disjoint open sets containing H and K, respectively. ■

Proof of Theorem 126
Let X be regular Lindelöf and let $\mathcal{A}$ be an open cover of X. For each $x \in X$ there exists $A_x \in \mathcal{A}$ and W_x open in X such that $x \in W_x \subset \overline{W}_x \subset A_x$. Let $\mathcal{W} = \{W_{x_n} : x_n \in X \wedge n < \omega\}$ be a countable subcover of $\{W_x : x \in X\}$. For each $n < \omega$ put $V_n = A_{x_n} \setminus (\bigcup_{i<n} \overline{W}_{x_i})$; take $\mathcal{V} = \{V_n : n < \omega\}$.

Fix $y \in X$ and put $m = \min\{n < \omega : y \in W_{x_n}\}$. Note $y \in A_{x_m}$ and $y \notin \overline{W}_{x_i}$ for $i < m$, so $y \in V_m$. To see that $\mathcal{V}$ is locally finite, observe that $V_n \cap W_{x_m} = \varnothing$ when $n > m$. ■

Proof of Theorem 127
$\rightarrow$ Certainly a finite subcover of an open cover is a locally finite open refinement of the same.

$\leftarrow$ It suffices to show that any locally finite collection of subsets of X is finite. (To do so we only need the fact that X is countably compact.) Let $\mathcal{U} = \{U_n \subset X : n < \omega\}$ be locally finite and put

$$C_n = \bigcup \{\overline{U}_k : k \geqslant n\} = \overline{\bigcup \{U_k : k \geqslant n\}}.$$

Observe that $C_1 \supset C_2 \supset \cdots$ and that each C_n is closed and nonempty. ■

Proof of Theorem 128

Proof of Theorem 129

Proof of Theorem 130
Let D be the set of all functions $f\colon I \to I$ of the form

$$f(t) = \begin{cases} q_1 & \text{if } t \in [0, p_1) \\ q_2 & \text{if } t \in [p_1, p_2) \\ \vdots & \\ q_n & \text{if } t \in [p_{n-1}, 1] \end{cases},$$

where $p_i, q_i \in \mathbb{Q} \cap I$, $p_1 < p_2 < \cdots < p_{n-1}$, and $n \in \mathbb{N}$. Certainly this is a countable dense subset of I^I. ■

Proof of Theorem 131
Let D be a dense subset of $I^{\mathfrak{c}^+}$. For each $\alpha \in \mathfrak{c}^+$, put $D_\alpha = \pi_\alpha^{-1}[[0, 1/2)] \cap D$. If $\alpha \neq \beta \in \mathfrak{c}^+$ then $\pi_\alpha^{-1}[(1/2, 1]] \cap D_\beta \neq \varnothing$, so $D_\alpha \neq D_\beta$. It follows that $|D| \geqslant \mathfrak{c}^+$. ■

Proof of Theorem 132
Suppose $\mathcal{S}$ is an uncountable collection of finite sets. Since

$$\mathcal{S} = \bigcup_{n<\omega} \{A \in \mathcal{S} : |A| = n\},$$

there exists $n \in \mathbb{N}$ and an uncountable $\mathcal{S}' \subset \mathcal{S}$ such that $|A| = n$ for all $A \in \mathcal{S}'$. We prove the following via induction on n:

($*$) If $\mathcal{S}'$ is an uncountable collection of sets and $n \in \mathbb{N}$ such that $|A| = n$ for all $A \in \mathcal{S}'$, then there exists an uncountable $\mathcal{S}''' \subset \mathcal{S}'$ and a set r such that $A \cap B = r$ for all $A \neq B \in \mathcal{S}'''$.

Well, if $n = 1$ then all members of $\mathcal{S}'$ are distinct, so let $\mathcal{S}''' = \mathcal{S}'$ and $r = \varnothing$.

Suppose $k \in \mathbb{N}$ and ($*$) holds whenever $n = k$; we show ($*$) holds when $n = k + 1$. Suppose $\mathcal{S}'$ is uncountable and $|A| = k + 1$ for all $A \in \mathcal{S}'$. For each $A \in \mathcal{S}'$, enumerate $A = \{A(1), \ldots, A(k + 1)\}$ and let $A_{\leqslant k} = \{A(1), \ldots, A(k)\}$.

(1) If $\{A_{\leqslant k} : A \in \mathcal{S}'\}$ is countable, there exists an uncountable $S''' \subset \mathcal{S}'$ such that $A_{\leqslant k} = B_{\leqslant k}$ and $A(k + 1) \neq B(k + 1)$ for all $A \neq B \in \mathcal{S}'''$. Let $r = A_{\leqslant k}$ for some (any) $A \in \mathcal{S}'''$.

(2) If $\{A_{\leqslant k} : A \in \mathcal{S}'\}$ is uncountable, then by the induction hypothesis there exists an uncountable $\mathcal{S}'' \subset \mathcal{S}'$ and a set r such that $A_{\leqslant k} \cap B_{\leqslant k} = r$ for all $A \neq B \in \mathcal{S}''$.

If $\{A(k + 1) : A \in \mathcal{S}''\}$ is countable, then there is an uncountable $\mathcal{S}''' \subset \mathcal{S}''$ and x such that $A(k+1) = x$ for all $A \in \mathcal{S}'''$; then $\mathcal{S}'''$ and $r \cup \{x\}$ work. If $\{A(k + 1) : A \in \mathcal{S}''\}$ is uncountable, then there is an uncountable $\mathcal{S}''' \subset cS''$ such that the $A(k + 1)$'s are

distinct as A ranges over $\mathcal{S}'''$. Note that $A(k+1) \notin r$ for any $A \in \mathcal{S}'''$. Hence $A \cap B = r$ for any $A \neq B \in \mathcal{S}'''$. ■

Proof of Theorem 133

Let $X = \prod_{\alpha<\kappa} X_\alpha$. Suppose for some finite $F \subset \kappa$ that $X_F = \prod_{\alpha\in F} X_\alpha$ does not have the *ccc*. Then there exists an uncountable collection $\{U_i : i \in \Lambda\}$ of pairwise-disjoint basic open subsets of X_F. If for each $i \in \Lambda$, $U_i = \prod_{\alpha\in F U^i_\alpha}$ with U^i_α open in X_α, then

$$\{\prod_{\alpha\in F} U^i_\alpha \times \prod_{\alpha\in\kappa\setminus F} X_\alpha : i \in \Lambda\}$$

is an uncountable collection of pairwise-disjoint open subsets of X. Thus, X does not have the *ccc*.

Now, suppose X does not have the *ccc*. Then there exists an uncountable collection $\{U_i : i \in \Lambda\}$ of pairwise-disjoint basic open subsets of X. For each $i \in \Lambda$, let $F_i \subset \kappa$ be the support of U_i. By Theorem 132, there exists an uncountable subcollection Λ' of Λ and $R \subset \kappa$ such that $F_i \cap F_j = R$ for all distinct $i, j \in \Lambda'$. Observe that $R \neq \varnothing$, else the U_i would not be pairwise-disjoint. If $i \neq j \in \Lambda'$, there exists $\alpha \in R$ such that $\pi_\alpha[U_i] \cap \pi_\alpha[U_j] = \varnothing$. It follows that $\{\prod_{\alpha\in R} \pi_\alpha[U_i] : i \in \Lambda'\}$ is an uncountable collection of pairwise-disjoint open subsets of $\prod_{\alpha\in R} X_\alpha$, whence not every finite subproduct has the *ccc*. ■

Proof of Corollary 134

Proof of Lemma 135

Suppose $\mathcal{H}$ is a discrete collection of closed sets. If two members of $\mathcal{H}$ intersect, then obviously $\mathcal{H}$ can't be discrete. Let $\mathcal{H}' \subset \mathcal{H}$, and suppose $p \notin \bigcup \mathcal{H}'$. Let U be an open neighborhood of P meeting at most one member, say H_0, of $\mathcal{H}$. If $p \in H_0$, then $H_0 \notin \mathcal{H}'$, so U misses $\bigcup \mathcal{H}'$. If $H_0 \in \mathcal{H}'$, then $p \notin H_0$ and then $V = U \setminus H_0$ is an open neighborhood of p missing $\bigcup \mathcal{H}'$. Thus p is not a limit point of $\bigcup \mathcal{H}'$, and it follows that $\bigcup \mathcal{H}'$ is closed. ■

Proof of Theorem 136

Suppose X is paracompact Hausdorff. Let $\mathcal{H}$ be a discrete collection of closed subsets of X. For each $x \in X$, let U_x be a neighborhood of x intersection at most one member of $\mathcal{H}$. Let U'_x be an open set containing x whose closure misses

(1) $\bigcup\{H' \in \mathcal{H} : H' \neq H\}$ if there exists $H \in \mathcal{H}$ such that $U_x \cap H \neq \varnothing$, or

(2) $\bigcup \mathcal{H}$ if there is no such H.

Let $\mathcal{V}$ be a locally finite refinement of $\{U_x \cap U'_x : x \in X\}$. Note that the closure of each member of $\mathcal{V}$ meets at most one member of $\mathcal{H}$.

Define, for $H \in \mathcal{H}$,

$$W_H = X \setminus \overline{\bigcup\{V \in \mathcal{V} : \overline{V} \cap H = \varnothing\}}$$
$$= X \setminus \bigcup\{\overline{V} : V \in \mathcal{V} \wedge \overline{V} \cap H = \varnothing\}.$$

W_H is an open set containing H. Suppose $H' \neq H$ and $x \in W_H \cap W_{H'}$. Let $V \in \mathcal{V}$ contain x; then $\overline{V}$ meets at most one member of $\mathcal{H}$. Therefore, either $\overline{V} \cap H = \varnothing$ or $\overline{V} \cap H' = \varnothing$. Therefore, either $x \notin W_H$ or $x \notin W_{H'}$, a contradiction. So $W_H \cap W_{H'} = \varnothing$. ■

Proof of Corollary 137

Proof of Example 138

We first prove that X is not collectionwise normal. Note that $\{\{e_\alpha\} : \alpha \in A\}$ is a discrete collection of singleton sets if and only if E is a closed set, and its subspace topology is discrete. If $x \notin E$, then $\{x\}$ is an open set missing E; thus E is closed. Now fix $e_\alpha \in E$. Let $U_\alpha = \{f \in X : f[\{\alpha\}] = 1\}$. U_α is an open set containing $\{e_\alpha\}$ and missing $\{e_\beta\}$ for each $\beta \neq \alpha$. Thus E is discrete.

Now suppose $e_\alpha \in V_\alpha$ for all $\alpha \in A$, where V_α is open in X. There is an open set O_α in the product topology of $2^{\mathcal{P}(A)}$ such that $e_\alpha \in O_\alpha \subset V_\alpha$. The product topology is *ccc*, so the collection $\{O_\alpha : \alpha \in A\}$ is not pairwise disjoint, hence $\{V_\alpha : \alpha A\}$ is not pairwise disjoint. So X is not collectionwise normal.

We now show that X is normal. Since $X \setminus E$ consists of isolated points, it is not difficult to see that if any two disjoint subsets of E can be separated by disjoint open sets, then any two disjoint closed subsets of X can be separated by disjoint open sets. So let H and K be disjoint subsets of E. Let $A_H = \{\alpha \in A : e_\alpha \in H\}$. Let $U = \{f \in X : f[A_H] = 1\}$ and $V = \{f \in X : f[A_H] = 0\}$. Then

$$e_\alpha \in H \to \alpha \in A_H \to e_\alpha[A_H] = 1 \to e_\alpha \in U,$$

and

$$e_\alpha \in K \to \alpha \notin A_H \to e_\alpha[A_H] = 0 \to e_\alpha \in V.$$

So U and V are disjoint open sets containing H and K. ■

Proof of Theorem 139

Let $\langle X, d\rangle$ be a metric space. For each $x \in X$ and $K \subset X$ closed not containing x, put $d(x, K) = \inf\{d(x, y) : y \in K\}$. Some neighborhood of x misses K, so $d(x, K) > 0$. For each pair H, K of disjoint closed subsets of X, let $U(H, K) = \bigcup_{x\in H} B(x, d(x, K)/2)$. Since $U(H, K) \cap U(K, H) = \varnothing$, $H \subset U(H, K) \subset \overline{U(H, K)} \subset X \setminus K$. If $H \subset H'$ and $K' \subset K$, with H', K' disjoint closed subsets of X, observe

$$\bigcup_{x\in H} B(x, d(x, K')/2) \subset U(H', K').$$

But $d(x, K') \geqslant d(x, K)$, so

$$U(H, K) \subset \bigcup_{x\in H} B(x, d(x, K')/2) \subset U(H', K').$$

■

Proof of Lemma 140

Assume X is monotonically normal. Let H, K be disjoint closed subsets of X. By assumption, there exists a monotone normality operator U for X. Define $U'(H, K) = U(H, K) \setminus \overline{U(K, H)}$. Clearly, U' inherits the properties of a monotone normality operator from U. Furthermore,

$$U'(H, K) \cap U'(K, H) =$$

$$[U(H, K) \setminus \overline{U(K, H)}] \cap [U(K, H) \setminus \overline{U(H, K)}] = \varnothing.$$

■

Proof of Theorem 141
Let X be a monotonically normal space and let U be the monotone normality operator guaranteed by Lemma 140. Let $\mathcal{H}$ be a discrete collection of closed subsets of X. Note by Lemma 135 that $\mathcal{H}$ is pairwise-disjoint and $\bigcup \mathcal{H}'$ is closed for any subcollection $\mathcal{H}'$ of $\mathcal{H}$.

We want to find a collection $\mathcal{U}$ of pairwise-disjoint open sets U_H such that $U_H \supset H$ for each $H \in \mathcal{H}$. Put $U_H = U(H, \bigcup \mathcal{H} \setminus \{H\})$ for each $H \in \mathcal{H}$.

Let $K \neq H \in \mathcal{H}$. Then

$$U_H \cap U_K = U(H, \bigcup \mathcal{H} \setminus \{H\}) \cap U(K, \bigcup \mathcal{H} \setminus \{K\})$$

$$\subset U(H, K) \cap U(K, H) = \varnothing,$$

so $\mathcal{U} = \{U_H : H \in \mathcal{H}\}$ is the desired collection. ■

Proof of Theorem 142
Suppose X is T_1.

(a → b) Note singletons in X are closed and let U be a monotone normality operator satisfying the condition of Lemma 140. For each $x \in X$ and open neighborhood W of x, let $W_x = U(\{x\}, X \setminus W)$. Then W_x is an open neighborhood of x. Suppose $x, y \in X$ and W, V are open neighborhoods of x and y, respectively, with $x \notin V$ and $y \notin W$. Then $X \setminus W \supset \{y\}$ and $\{x\} \subset X \setminus V$. So

$$W_x \cap V_y = U(\{x\}, X \setminus W) \cap U(\{y\}, X \setminus V) \subset U(\{x\}, \{y\}) \cap U(\{y\}, \{x\}) = \varnothing.$$

(b → c) Trivial.

(c → a) For disjoint closed H and K, define

$$U(H, K) = \bigcup \{U_x : x \in H \wedge U \cap K = \varnothing\}.$$

For each $x \in H$ there exists a (basic) open neighborhood thereof which misses K, so that $H \subset U(H, K) \subset \overline{U(H, K)}$. We show $\overline{U(H, K)} \subset X \setminus K$. Suppose $p \in K$. Let V be an open set containing p which misses H. We may assume $V_p \subset V$. If $x \in H$ and $U \cap K = \varnothing$, then we can have neither $x \in V_p$ nor $y \in U_x$, so $V_p \cap U_x = \varnothing$. So $V_p \cap U(H, K) = \varnothing$, so $p \notin \overline{U(H, K)}$. ■

Proof of Theorem 143

Let X be a monotonically normal T_1-space. For each $x \in X$ and each open neighborhood U' of x, let U'_x be defined as in Theorem 142(b). Let $Y \subset X$. For each open $U \subset Y$, there exists U' open in X such that $U = U' \cap Y$. For each $y \in Y$ and each open neighborhood U of y, put $U_y = U'_y \cap Y$.

Let $x, y \in Y$. If $U_x \cap V_y \neq \varnothing$, then $Y \cap (U'_x \cap V'_y) \neq \varnothing$, so $U'_x \cap V'_y \neq \varnothing$. It follows that $x \in V'$ or $y \in U'$. But $x \in V'$ implies $x \in V$ and $y \in U'$ implies $y \in U$, so $U_x \cap V_y \neq \varnothing$ implies $x \in V$ or $y \in U$. By Theorem 142, Y is monotonically normal. ■

Proof of Theorem 144

Let X be linearly ordered by $<$. We will show X is monotonically normal by way of Theorem 142. Let $\mathcal{B}$ be the base of all open intervals (a, b) with $a, b \in X \cup \{-\infty, \infty\}$. Let $\prec$ be any well-ordering of X.

For $a < x < b$, define a_x as a if $(a, x) = \varnothing$, otherwise let a_x be the $\prec$-least element of (a, x). Similarly, define b_x to be x if $(x, b) = \varnothing$, and the $\prec$-least element of (x, b) otherwise. Now, let $x \in (a, b) \in \mathcal{B}$ and $y \in (c, d) \in \mathcal{B}$. Put $(a, b)_x = (a_x, b_x)$ and $(c, d)_y = (c_y, d_y)$.

Assume $(a, b)_x \cap (c, d)_y \neq \varnothing$. Without loss of generality we may assume $d \geqslant c$. Suppose that $x \notin (c, d)$ and $y \not\in (a, b)$. Then $x \leqslant c$ and $y \geqslant b$. Also, $(a, b)_x \cap (c, d)_y = (a_x, b_x) \cap (c_y, d_y) \neq \varnothing$ implies $c_y < b_x$. So we have $x \leqslant c < c_y < b_x < b \leqslant y$. But b_x is the $\prec$-least element of (x, b) so $b_x \prec c_y$, and c_y is the $\prec$-least element of (c, y) so $c_y \prec b_x$, a contradiction. Thus $y \in (a, b)$ or $x \in (c, d)$, so X is monotonically normal. ■

Proof of Theorem 145

(a) Note $\pi_f \circ e_{\mathcal{F}} = f$ for all $f \in \mathcal{F}$.

(b) let $x, y \in X$ such that $x \neq y$. Since $\mathcal{F}$ separates points, there exists $f \in \mathcal{F}$ such that $f(x) \neq f(y)$. It follows that

$$e_{\mathcal{F}}(x) = \langle f(x) : f \in \mathcal{F} \rangle \neq \langle f(y) : f \in \mathcal{F} \rangle = e_{\mathcal{F}}(y).$$

(c) Note $\mathcal{F}$ separates points; it suffices to show $e_{\mathcal{F}}$ is closed. Let $H \subset X$ be closed and suppose on the contrary that $e_{\mathcal{F}}[H]$ is not closed. Fix $y \in \overline{e_{\mathcal{F}}[H]} \setminus e_{\mathcal{F}}[H]$ and $x \in X \setminus H$ such that $e_{\mathcal{F}}(x) = y$. Since $x \notin H$

and $\mathcal{F}$ separates points from closed sets, theere exists $f \in \mathcal{F}$ such that $f(x) \notin \overline{f[H]}$. But this means $y = e_{\mathcal{F}}(x) \notin \prod_{f \in \mathcal{F}} \overline{f[H]} \supset \overline{e_{\mathcal{F}}[H]}$, a contradiction. ■

Proof of Theorem 146

(a → b) X is normal by Theorem 49 and second-countable by Theorem 85.

(b → c) X is hereditarily Lindelöf by Theorem 84 and normal by Theorem 68; we first show it is perfectly normal.

Let H be a closed subset of X. Then for each $x \in X \setminus H$, there is an open neighborhood U_x of x such that $\overline{U}_x \cap H = \varnothing$. Let $\mathcal{U}$ be a countable subcover of $\{U_x : x \in X \setminus H\}$. Since $X \setminus H = \bigcup \{\overline{U} : U \in \mathcal{U}\}$, we have $H = \bigcap \{X \setminus \overline{U} : U \in \mathcal{U}\}$.

Now let $\mathcal{B} = \{B_n : n < \omega\}$ be a base for X. By Theorem 59(c), for each $n < \omega$ there is a map $f_n \colon X \to [0,1]$ such that $f_n^{-1}(0) = X \setminus B_n$; put $\mathcal{F} = \{f_n : n < \omega\}$. If $H \subset X$ is closed and $x \in X \setminus H$, there exists $n < \omega$ such that $x \in B_n$ and $H \subset X \setminus B_n$. It follows that $f_n(x) \notin \overline{f[H]}$, so $\mathcal{F}$ separates points from closed sets. By Theorem 145, X is homeomorphic to a subspace of $I^{\mathbb{N}}$.

(c → a) Note $I^{\mathbb{N}}$ is separable by Theorem 88 and metrizable by Corollary 41, so any subspace of $I^{\mathbb{N}}$ is separable and metrizable. ■

Proof of Theorem 147

→ For any H closed in X and any $x \in X \setminus H$, there is a map $f_{x,H} \colon X \to [0,1]$ which separates x from H; let $\mathcal{F}$ be the collection of these maps. By Theorem 145, X is homeomorphic to a subspace of $[0,1]^{|\mathcal{F}|}$.

← I^{κ} is completely regular by Theorem 50, so any subspace is also. ■

Proof of Theorem 148

Let X be a regular space.

(a → b) By definition.

(b → c) Let $\mathcal{U}$ be an open cover of X. For each $x \in X$, pick $U_x \in \mathcal{U}$ and $W_x \subset X$ open such that $x \in W_x \subset \overline{W}_x \subset U_x$. Let $\mathcal{V}$ be a locally finite refinement of $\{W_x : x \in X\}$; put $\mathcal{B} = \{\overline{V} : V \in \mathcal{V}\}$. That $\mathcal{B}$ is a closed refinement of $\mathcal{U}$ is immediate.

Fix $x \in X$ and let $O \subset X$ be an open neighborhood of x meeting finitely many members of $\mathcal{V}$. It follows that $\{V \in \mathcal{V} : O \cap \overline{V} \neq \varnothing\}$ is finite, so $\mathcal{B}$ is locally finite.

(a → c) Let $\mathcal{U}$ be an open cover of X, and let $\mathcal{A}$ be a locally finite closed refinement thereof. Let $\mathcal{V}$ be those open subsets of X which meet only finitely many members of $\mathcal{A}$; note that every point in X has a neighborhood meeting only finitely many members of $\mathcal{A}$, so $\mathcal{V}$ is an open cover X. Let $\mathcal{B}$ be a locally finite closed refinement of $\mathcal{V}$. For each $A \in \mathcal{A}$ pick $U_A \in \mathcal{U}$ such that $A \subset U_A$; define

$$W_A = X \setminus \bigcup \{B \in \mathcal{B} : B \cap A = \varnothing\}.$$

Put $O_A = W_A \cap U_A$ and let $\mathcal{O} = \{O_A : A \in \mathcal{A}\}$; certainly $\mathcal{O}$ is an open refinement of $\mathcal{U}$.

Fix $x \in X$ and let O be an open neighborhood of x meeting only finitely many members of $\mathcal{B}$, say $B_1, B_2, \ldots, B_n$. Let I be those members of $\mathcal{A}$ meeting at least one of $B_1, B_2, \ldots, B_n$; note I is finite since $\mathcal{B}$ refines $\mathcal{V}$. Note $O \subset B_1 \cup B_2 \cup \cdots \cup B_n$; if $A \in \mathcal{A} \setminus I$ then A misses $B_1, B_2, \ldots, B_n$, so O misses W_A. It follows that O misses O_A for each $A \in \mathcal{A} \setminus I$, whence O can meet only finitely many members of $\mathcal{O}$. ■

Proof of Theorem 149

Let $\mathcal{U}$ be an open cover of X and let $\mathcal{V} = \bigcup_{n \in \mathbb{N}} \mathcal{V}_n$ be a σ-locally finite open refinement thereof. For each $n \in \mathbb{N}$ let $W_n = \bigcup \mathcal{V}_n$, and for each $V \in \mathcal{V}_n$ let $S_n(V) = V \setminus \bigcup_{i<n} W_i$. Let $\mathcal{C}_n = \{S_n(V) : V \in \mathcal{V}_n\}$ for each $n \in \mathbb{N}$, and put $\mathcal{C} = \bigcup_{n \in \mathbb{N}} \mathcal{C}_n$.

Fix $x \in X$ and let $n_x = \min\{n \in \mathbb{N} : x \in W_n\}$. Choose $V_x \in \mathcal{V}_{n_x}$ such that $x \in V_x$. Then $x \in S_{n_x}(V_x) \in \mathcal{C}$, so $\mathcal{C}$ refines $\mathcal{U}$.

For each $i \leqslant n_x$, choose an open neighborhood O_i of x meeting only finitely many members of $\mathcal{V}_i$. Note that $O = \bigcup_{i=1}^{n_x} (V_x \cap O_i)$ is an open neighborhood of x meeting only finitely many members of

$\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_{n_x}$. But V_x misses $S_n(V)$ whenever $n > n_x$, so O misses every member of $\mathcal{C}_n$ whenever $n > n_x$. It follows that $\mathcal{C}$ is locally finite.

By Theorem 148, X is paracompact. ■

Proof of Lemma 150

For each $x \in X$, let $B(x) = B_d(x, 1/4n)$.

Fix $n \in \mathbb{N}$ and $x \in X$; we show $B(x)$ meets at most one element of $\mathcal{V}_n$. Suppose there are $\alpha_1 < \alpha_2 < \kappa$ such that $B(x) \cap V_{\alpha_i,n} \neq \varnothing$ for $i = 1, 2$. For $i = 1, 2$ there is $y_i \in H_{\alpha_i,n}$ such that $B(x) \cap B(y_i) \neq \varnothing$. Choose $z_i \in B(x) \cap B(y_i)$ for $i = 1, 2$. Then

$$\begin{aligned} d(y_1, y_2) &\leqslant d(y_1, x) + d(x, y_2) \\ &\leqslant d(y_1, z_1) + d(z_1, x) + d(x, z_2) + d(z_2, y_2) < 1/n. \end{aligned}$$

It follows that $y_2 \in B_d(y_1, 1/n) \subset U_{\alpha_1}$. This is a contradiction, since $y_2 \in U_{\alpha_2} \setminus \bigcup_{\alpha<\alpha_2} U_\alpha$.

We now show that $\mathcal{V}$ covers X. Let $\alpha_x = \min\{\alpha : x \in U_\alpha\}$. Then $x \in U_{\alpha_x} \setminus \bigcup_{\alpha<\alpha_x} U_\alpha$ and $\operatorname{dist}_d(x, X \setminus U_{\alpha_x}) > 0$, where

$$\operatorname{dist}_d(x, A) = \inf\{d(x, a) : a \in A\}.$$

Choose $m \in \mathbb{N}$ such that $\operatorname{dist}_d(x, X \setminus U_{\alpha_x}) > 1/m$. Then

$$x \in H_{\alpha_x,m} \subset V_{\alpha_x,m} \in \mathcal{V}_m \subset \mathcal{V}.$$

By construction $V_{\alpha,n} \subset U_\alpha$ for each $\alpha < \kappa$ and $n \in \mathbb{N}$. Thus $\mathcal{V}$ is a σ-locally finite open refinement of $\mathcal{U}$. ■

Proof of Corollary 151

Combine Theorem 48, Theorem 49, Theorem 149, and Lemma 150. ■

Proof of Lemma 152

Let $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}_n$ be a σ-locally finite base of a regular space X. Let $H \subset X$ be closed. For all $n \in \mathbb{N}$, put

$$U_n = \bigcup\{B : B \in \mathcal{B}_n \wedge \overline{B} \subset X \setminus H\}.$$

Note that $\overline{U}_n \subset X \setminus H$.

Put $O_n = X \setminus \overline{U}_n$; note $H \subset O_n$ for all $n \in \mathbb{N}$. Let $x \in \bigcap_{n\in\mathbb{N}} \overline{O}_n$ and suppose $x \notin H$; there exists $U \subset X$ open such that $x \in U \subset \overline{U} \subset X \setminus H$. Pick $n \in \mathbb{N}$ and $B \in \mathcal{B}_n$ such that $x \in B \subset U_x$. It follows that $\overline{B} \subset X \setminus H$, so $x \in U_n$. But $\overline{O}_n \subset X \setminus U_n$ implies $x \notin \overline{O}_n$, a contradiction. By Theorem 59(b), X is perfectly normal. ■

Proof of Lemma 153

We first show that d is a metric on X. For each $n \in \mathbb{N}$ and $B \in \mathcal{B}_n$, $|f_{n,B}(x) - f_{n,B}(y)| \leqslant 1/n \leqslant 1$. So $\sup_{f\in\mathcal{F}} \{|f(x) - f(y)|\} \leqslant 1$, whence d is well-defined.

Certainly $d(x, y) = d(y, x)$ and $x = y$ implies $d(x, y) = 0$. Suppose $x \neq y$. Then there exists $n \in \mathbb{N}$ and $B \in \mathcal{B}_n$ such that $x \in B$ and $y \notin B$, implying $f_{n,B}(y) = 0$ and $f_{n,B}(x) > 0$; we have

$$0 < f_{n,B}(x) \leqslant \sup_{f\in\mathcal{F}} \{|f(x) - f(y)|\} = d(x, y).$$

Let $x, y, z \in X$. For each $f \in \mathcal{F}$ we have

$$|f(x) - f(z)| \leqslant |f(x) - f(y)| + |f(y) - f(z)| \leqslant d(x, y) + d(y, z),$$

so $d(x, z) \leqslant d(x, y) + d(y, z)$.

Let $\mathcal{T}$ be the given topology on X, and let $\mathcal{T}_d$ be the topology on X generated by d; we first show $\mathcal{T}_d \subset \mathcal{T}$. Fix $x \in X$ and $\varepsilon > 0$; we find $O \in \mathcal{T}$ with $x \in O \subset B_d(x, \varepsilon)$. Choose $n_0 \in \mathbb{N}$ such that $1/n_0 < \varepsilon/2$.

Claim. *For all $1 \leqslant n \leqslant n_0$, there exists $O_n \in \mathcal{T}$ such that $x \in O_n$ implies $\sup_{B\in\mathcal{B}_n} |f_{n,B}(x) - f_{n,B}(y)| \leqslant \varepsilon/2$ for all $y \in O_n$.*

Proof of Claim. Let $1 \leqslant n \leqslant n_0$; then there is an open neighborhood U of x meeting only finitely many members of $\mathcal{B}_n$, say $B_1, \ldots, B_m$. Using f_{n,B_i}, choose an open neighborhood U_i of x such that

$$|f_{n,B_i}(x) - f_{n,b_i}(y)| < \varepsilon/2$$

for all $y \in U_i$. Put $O_n = \bigcap_{i=1}^m (U \cap U_i)$; for all $y \in O_n$ we have $|f_{n,B}(x) - f_{n,B}(y)| < \varepsilon/2$ for all $B \in \mathcal{B}_n$. It follows that

$$\sup_{B\in\mathcal{B}_n} \{|f_{n,B}(x) - f_{n,B}(y)|\} \leqslant \varepsilon/2$$

for all $y \in O_n$. □

Put $O = \bigcap_{n=1}^{n_0}$; certainly O is an open neighborhood of x. Let $y \in O$. For $n > n_0$, $|f_{n,B}(x) - f_{n,B}(y)| < 1/n < 1/n_0 < \varepsilon/2$ for all $B \in \mathcal{B}_n$. For $n \leqslant n_0$, $O \subset O_n$ gives $|f_{n,B}(x) - f_{n,B}(y)| \leqslant \varepsilon/2$ for all $B \in \mathcal{B}_n$. Thus

$$d(x,y) = \sup_{f \in \mathcal{F}} \{|f(x) - f(y)|\} \leqslant \varepsilon/2 < \varepsilon$$

for all $y \in O$, so $O \subset B_d(x,\varepsilon)$.

It remains to show $\mathcal{T} \subset \mathcal{T}_d$. Let $B \in \mathcal{B}$ and $x \in B$; there exists $n \in \mathbb{N}$ such that $f_{n,B}(x) > 0$. Let $\varepsilon = f_{n,B}(x)$. Note that $y \in B_d(x,\varepsilon)$ implies $|f_{n,B}(x) - f_{n,B}(y)| < \varepsilon$. It follows that $f_{n,B}(y) \neq 0$, so $y \in B$. Hence, $B_d(x,\varepsilon) \subset B$. ■

Proof of Theorem 154

$\rightarrow$ Let X be metrizable; recall that metrizable spaces are regular and paracompact. For each $n \in \mathbb{N}$ define $\mathcal{U}_n = \{B(x,1/n) : x \in X\}$ and let $\mathcal{V}_n$ be a locally finite open refinement thereof. Let $\mathcal{V} = \bigcup_{n\in\mathbb{N}} \mathcal{V}_n$; certainly $\mathcal{V}$ is σ-locally finite.

Fix $x \in X$ and $\varepsilon > 0$. Choose $m \in \mathbb{N}$ such that $2/m < \varepsilon$. Pick $V \in \mathcal{V}_m$ containing x and fix $y \in V$. Recall that $\mathcal{V}_m$ refines $\mathcal{U}_m$, so $V \subset B(z,1/m)$ for some $z \in X$. It follows that

$$d(x,y) \leqslant d(x,z) + d(z,y) \leqslant 1/m + 1/m = 2/m < \varepsilon,$$

so $y \in B(x,\varepsilon)$.

$\leftarrow$ This is Lemma 153. ■

Proof of Theorem 155

Let X be a locally metrizable paracompact T_2-space; by Theorem 125, X is regular. For each $x \in X$ let N_x be a metrizable neighborhood of x; put $\mathcal{U} = \{N_x^\circ : x \in X\}$. Let $\mathcal{V}$ be a locally finite open refinement of $\mathcal{U}$. For each $V \in \mathcal{V}$, let d_V be a metric generating the topology on V. Each $V \in \mathcal{V}$ is open in X, so any ball $B_{d_V}(x,\varepsilon) \subset V$ with $x \in V$ and $\varepsilon > 0$ will be open in X. For each $n \in \mathbb{N}$ let $\mathcal{B}_n = \{B_{d_V}(x,1/n) : V \in \mathcal{V} \wedge x \in V\}$ and let $\mathcal{B}'_n$ be a locally finite open refinement thereof. Put $\mathcal{B} = \bigcup_{n\in\mathbb{N}} \mathcal{B}'_n$; certainly $\mathcal{B}$ is σ-locally finite.

Now let $U \subset X$ be open with $x \in U$. We know finitely many members of $\mathcal{V}$ contain x, say $V_1, \ldots, V_k$. For each $1 \leqslant i \leqslant k$ $U \cap V_i$ is an open

neighborhood of x in V_i, so there exists $\varepsilon > 0$ such that $B_{d_{V_i}}(x, \varepsilon) \subset U \cap V_i$ for all $1 \leqslant i \leqslant k$. Choose $n \in \mathbb{N}$ such that $1/n < \varepsilon/2$. Recall $\mathcal{B}'_n$ is a cover, so there exists $B \in \mathcal{B}'_n \subset \mathcal{B}$ containing x. But $\mathcal{B}'_n$ is a refinement of $\mathcal{B}_n$, so there is $V \in \mathcal{V}$ and $y \in V$ such that $x \in B \subset B_{d_V}(y, 1/n)$. Since $x \in V$, $V = V_i$ for some $1 \leqslant i \leqslant k$. It follows that

$$x \in B \subset B_{d_V}(y, 1/n) = B_{d_{V_i}}(y, 1/n) \subset B_{d_{V_i}}(x, \varepsilon) \subset U,$$

hence $\mathcal{B}$ is a basis. By Theorem 154, X is metrizable. ■

Proof of Lemma 156

Since X is compact Hausdorff, X is regular. Let

$$\mathcal{B} = \{X \setminus \overline{\bigcup \mathcal{V}} : \mathcal{V} \subset \mathcal{U} \text{ is finite}\}.$$

Clearly $\mathcal{B}$ is countable.

Let $U \subset X$ be open and let $x \in U$. For all $y \in X \setminus U$, there exists $U_y \in \mathcal{U}$ with $y \in U_y$ and $x \notin \overline{U_y}$. Observe that the U_y cover $X \setminus U$, so there exist finitely many, say $U_1, \ldots, U_n$, which cover $X \setminus U$. Then $x \in \bigcap_{i=1}^n (X \setminus \overline{U}_i) \subset U$. But $\bigcap_{i=1}^n (X \setminus \overline{U}_i) = X \setminus \overline{\bigcap_{i=1}^n U_i} \in \mathcal{B}$, so $\mathcal{B}$ is a base. It follows that X is second-countable, hence separable metrizable. ■

Proof of Theorem 157

Suppose X is compact Hausdorff.

$\leftarrow$ Assume X has a G_δ diagonal. There exist open subsets U_n, $n < \omega$, of $X \times X$ such that $\Delta = \bigcap_{n<\omega} U_n$. We show X is metrizable by Lemma 156. Because X is compact Hausdorff, X is regular. So for each $n < \omega$ and $x \in X$ there exists an open $B_n(x) \subset X$ containing x, with $\overline{B_n(x)} \times \overline{B_n(x)} \subset U_n$. For each $n < \omega$ let $\mathcal{V}_n$ be a finite subcover of $\{B_n(x) : x \in X\}$. Let $\mathcal{B} = \bigcup_{n<\omega} \mathcal{V}_n$. Clearly $\mathcal{B}$ is countable. And if $x \neq y \in X$ then there exists $n < \omega$ such that $\langle x, y\rangle \notin U_n$. So there exists $B \in \mathcal{V}_n$ such that $\langle x, x\rangle \subset \overline{B} \times \overline{B} \subset U_n$ and so $y \notin \overline{B}$. ■

Proof of Theorem 158

Suppose X has a countable network $\{N_i : i < \omega\}$ and $f\colon X \to Y$ is a continuous surjection. Claim $\{f[N_i] : i < \omega\}$ is a countable network for

Y. Well suppose V is open in Y and $y \in V$. We find $i < \omega$ such that $y \in f[N_i] \subset V$. Since f is surjective, there exists $x \in f^{-1}(y)$. Since $\{N_i : i < \omega\}$ is a network for X and $f^{-1}[V]$ is an open subset of X containing x, there exists $i < \omega$ such that $x \in N_i \subset f^{-1}[V]$. Then $y = f(x) \in f[N_i] \subset f[f^{-1}[V]] \subset V$. ■

Proof of Theorem 159

(a → b) Let $\langle X, \mathcal{T} \rangle$ be a T_1-space with a countable network $\mathcal{N}$. Put $\mathcal{S} = \mathcal{N} \cup \{X \setminus N : N \in \mathcal{N}\}$ and $\mathcal{B} = \{\bigcap \mathcal{F} : \mathcal{F} \subset \mathcal{S} \text{ is finite}\}$. Let $\mathcal{T}_{\mathcal{N}}$ be the topology on X generated by $\mathcal{B}$.

Certainly $\langle X, \mathcal{T}_{\mathcal{N}} \rangle$ is second-countable. Let $x \in X$ and $H \subset X$ closed in $\langle X, \mathcal{T}_{\mathcal{N}} \rangle$ not containing x. Pick $U \in \mathcal{B}$ such that $x \in U \subset X \setminus H$. Recall that $U = \bigcup \mathcal{F}$ for some finite $\mathcal{F} \subset \mathcal{S}$. Since each $F \in \mathcal{F}$ is clopen, U is clopen. It follows that U and $X \setminus U$ are disjoint open sets containing x and H, respectively. Thus $\langle X, \mathcal{T}_{\mathcal{N}} \rangle$ is regular, hence separable metrizable.

Let $U \in \mathcal{T}$. Then $U = \bigcup \mathcal{N}'$ for some $\mathcal{N}' \subset \mathcal{N}$. Since $\mathcal{N}' \subset \mathcal{B}, U \in \mathcal{T}_{\mathcal{N}}$. Therefore, $f \colon \langle X, \mathcal{T}_{\mathcal{N}} \rangle \to \langle X, \mathcal{T} \rangle$ defined by $x \mapsto x$ is continuous.

(b → a) Recall that separable metric spaces are second-countable. Since any countable base is necessarily a countable network, X has a countable network by Theorem 158. ■

Proof of Theorem 160

Suppose X is regular and has a countable network $\mathcal{N}$. We show $X \times X$ has a G_δ diagonal Δ by showing $(X \times X) \setminus \Delta$ is the countable union of closed sets.

Clearly $\mathcal{L} = \{\overline{N} \times \overline{M} : N, M \in \mathcal{N} \wedge \overline{N} \times \overline{M} \subset (X \times X) \setminus \Delta\}$ is a countable collection of closed subsets of $X \times X$. We claim $(X \times X) \setminus \Delta = \bigcup \mathcal{L}$. Well, suppose $\langle x, y \rangle \in (X \times X) \setminus \Delta$. Since $X \times X$ is regular there exists a basic open set $B_1 \times B_2$ with $\langle x, y \rangle \in \overline{B}_1 \times \overline{B}_2 \subset (X \times X) \setminus \Delta$. There exists $N, M \in \mathcal{N}$ such that $x \in N \subset B_1$ and $y \in M \subset B_2$, so that $\langle x, y \rangle \in \overline{N} \times \overline{M} \subset \overline{B}_1 \times \overline{B}_2 \subset (X \times X) \setminus \Delta$. ■

Proof of Theorem 161

Suppose X is compact metrizable and Y is a Hausdorff continuous image of X. Every compact metrizable space has a countable basis.

In particular, X has a countable network. By Theorem 158, Y has a countable network. As the continuous image of a compact space, Y is compact. Since Y is also Hausdorff by assumption, we have Y is regular. By Theorem 160, Y has a G_δ diagonal. This, together with the fact that Y is compact Hausdorff, implies Y is metrizable by Theorem 157. ■

Proof of Theorem 162

Let $f\colon X \to Y$ be a continuous surjection.

(a → b) Suppose $y \in Y$. Let $U \subset X$ be an open set containing $f^{-1}(y)$. Then $V = Y \setminus f[X \setminus U]$ is open in Y containing y. Since $V \cap f[X \setminus U] = \varnothing$, we have $f^{-1}[V] \subset U$.

(b → c) Let $U \subset X$ be open and let $y \in f^*[U]$. There exists an open $V \subset Y$ such that $y \in V$ and $f^{-1}[V] \subset U$. So $f^{-1}(y') \subset U$ for all $y' \in V$. So $V \subset f^*[U]$, implying $f^*[U]$.

(c → a) Suppose $H \subset X$ is closed and y is a limit point of $f[H]$. By assumption, $f^*[X \setminus H]$ is open; it clearly misses $f[H]$, so $y \notin f^*[X \setminus H]$. Hence exists $x \in H$ with $f(x) = y$. So $y \in f[H]$. This proves $f[H]$ is closed. ■

Proof of Theorem 163

Let $f\colon X \to Y$ be a perfect surjection, where X is separable metrizable. Then X has a countable base $\mathcal{B}$. Without loss of generality, we may assume $\mathcal{B}$ is closed under finite unions. Now let $\mathcal{C} = \{f^*[B] : B \in \mathcal{B}\}$, where f^* is as defined in Theorem 162(c). So $\mathcal{C}$ is a countable collection of open sets.

We prove that $\mathcal{C}$ is a base for Y. Let $y \in U$, where U is open in Y. Then $f^{-1}(y)$ is a compact subset of the open set $f^{-1}[U]$. Since $\mathcal{B}$ is closed under finite unions, there is some $B \in \mathcal{B}$ with $f^{-1}(y) \subset B \subset f^{-1}[U]$. Then $y \in f^*[B] \subset U$. Hence $\mathcal{C}$ is a countable base for Y.

Since a regular space with a countable base is separable and metrizable, it remains to prove that Y is regular. Each point of X is closed, and f is a closed map, so it follows that each point of Y is closed; so Y is T_1. Now suppose $y \in Y$ and H is a closed subset of Y not containing y. Since X is normal, there are disjoint open sets U and V containing

$f^{-1}(y)$ and $f^{-1}[H]$, respectively. Then $f^*[U]$ and $f^*[V]$ are disjoint open sets containing y and H, respectively. Thus Y is regular. ■

Proof of Theorem 164

(a → b) Let $A_1, A_2, \ldots$ be a decreasing sequence of nonempty closed sets with $\operatorname{diam}\langle A_n\rangle \to 0$; fix $x_n \in A_n$ for all $n \in \mathbb{N}$. Note $\langle x_n\rangle$ is a Cauchy sequence. By completeness, $\langle x_n\rangle \to x$ for some $x \in X$. Since $\langle x_n\rangle_{n\geqslant k} \to x$ for each $k \in \mathbb{N}$, x is a limit point of $\{x_n : n \geqslant k\} \subset A_k$. Since A_k is closed, $x \in A_k$ for all k.

(b → a) Let $\langle x_n\rangle$ be a Cauchy sequence, and for each n set $A_n = \{x_i : i \geqslant n\}$; certainly $A_1, A_2, \ldots$ is a decreasing sequence of nonempty closed sets. Fix $\varepsilon > 0$; there is some $k \in \mathbb{N}$ such that $d(x_i, x_j) \leqslant \varepsilon$ whenever $i, j \geqslant k$. It follows that $\operatorname{diam} A_k = \operatorname{diam}\{x_i : i \geqslant k\} \leqslant \varepsilon$, so $\operatorname{diam}\langle A_n\rangle \to 0$.

Pick $x \in \bigcap_{n\in\mathbb{N}} A_n$ and let U be a neighborhood of x. There is $\varepsilon > 0$ such that $B(x,\varepsilon) \subset U$. Pick A_n where $\operatorname{diam} A_n < \varepsilon$. Then $x \in A_n \subset B(x,\varepsilon)$. But A_n contains all but finitely many members of $\langle x_n\rangle$, so $B(x,\varepsilon)$ does also. Thus, $\langle x_n\rangle \to x$. ■

Proof of Corollary 165

Let d be a metric on the compact space X. By Theorem 73, any decreasing sequence of nonempty closed sets has nonempty intersection. By Theorem 164, d is complete. ■

Proof of Theorem 166

Let d be a complete metric on the space X. Let $\mathcal{O} = \{O_n : n \in \mathbb{N}\}$ be a countable collection of dense open subsets of X; we want to show that $\bigcap \mathcal{O}$ is dense.

Let V be a nonempty open set. Since O_1 is dense, $V \cap O_1 \neq \varnothing$. Pick $x_1 \in V \cap O_1$ and $0 < \varepsilon_1 < 1$ such that $\overline{B(x_1,\varepsilon_1)} \subset V \cap O_1$. Since O_2 is dense, $B(x_1,\varepsilon_1) \cap O_2 \neq \varnothing$. Again, pick $x_2 \in B(x_1,\varepsilon_1) \cap O_2$ and $0 < \varepsilon_2 < 1/2$ such that $\overline{B(x_2,\varepsilon_2)} \subset B(x_1,\varepsilon_1) \cap O_2$. Inductively, for each $n \in \mathbb{N}$ we may pick $x_n \in B(x_{n-1},\varepsilon_{n-1}) \cap O_n$ and $0 < \varepsilon_n < 1/n$ such that $\overline{B(x_n,\varepsilon_n)} \subset B(x_{n-1},\varepsilon_{n-1}) \cap O_n$.

By Theorem 164, fix $x \in \bigcap_{n\in\mathbb{N}} \overline{B(x_n,\varepsilon_n)}$. Since $x \in O_n$ for all $n \in \mathbb{N}$ we have $x \in \bigcap \mathcal{O}$. Moreover, $\overline{B(x_1,\varepsilon_1)} \subset V \cap O_1$, so $x \in V$. ■

Proof of Theorem 167

$\rightarrow$ Assume $d \restriction A \times A$ is complete and let $A \subset X$. Fix $x \in \overline{A}$. There exists $\langle x_n \rangle \to x$ where $x_n \in A$ for all $n \in \mathbb{N}$; certainly $\langle x_n \rangle$ is Cauchy in A. It follows that $\langle x_n \rangle \to y$ for some $y \in A$. Since X is Hausdorff we must have $y = x$, whence A is closed.

$\leftarrow$ Let d be a complete metric on X and let $A \subset X$ be closed. Let $\langle x_n \rangle$ be a Cauchy sequence of A; certainly $\langle x_n \rangle$ is Cauchy in X. Since d is complete, there exists $x \in X$ such that $\langle x_n \rangle \to x$ in X. Certainly x is a limit point of A, so $x \in \overline{A} = A$. ■

Proof of Theorem 168

First a useful fact. Let $\langle X, d \rangle$ be a complete metric space and define $d' \colon X \times X \to \mathbb{R}$ by

$$d'(x, y) = \begin{cases} M & \text{if } d(x, y) \geqslant M, \\ d(x, y) & \text{otherwise.} \end{cases}$$

By Lemma 39, d' is equivalent to d; we want to show d' is complete.

Let $\langle x_n \rangle$ be a d'-Cauchy sequence. Then for each $\varepsilon > 0$ there exists $k \in \mathbb{N}$ such that $d'(x_i, x_j) \leqslant \varepsilon$ whenever $i, j \geqslant k$. If $\varepsilon < M$, then $d(x_i, x_j) \leqslant \varepsilon$ for all $i, j \geqslant k$. If $\varepsilon \geqslant M$, set $\varepsilon' = M/2$; there exists $k' \in \mathbb{N}$ such that $d'(x_i, x_j) \leqslant \varepsilon'$ for $i, j \geqslant k'$. It follows that $d(x_i, x_j) \leqslant \varepsilon' < \varepsilon$ for all $i, j \geqslant k'$. So, $\langle x_n \rangle$ is d-Cauchy and hence converges. It follows that d' is complete.

Let $\langle X_n \rangle$ be a collection of completely metrizable spaces. Then for each $n \in \mathbb{N}$ there exists a complete metric d_n which generates the topology on X_n. Define for each $n \in \mathbb{N}$ an equivalent metric d'_n such that $d'_n(x, y) = 1/2^n$ if $d_n(x, y) \geqslant 1/2^n$ and $d'_n(x, y) = d_n(x, y)$ otherwise. By Theorem 40, $d(\vec{x}, \vec{y}) = \max_{n \in \mathbb{N}} \{d'_n(x_n, y_n)\}$ is a metric on the product of the X_n.

Let $\langle \vec{x}_n \rangle$ be a d-Cauchy sequence, where $\vec{x}_n = \langle \vec{x}_n(1), \vec{x}_n(2), \ldots \rangle$. Fix $n \in \mathbb{N}$. We show that $\langle \vec{x}_n(i) \rangle_{i=1}^{\infty}$ is a d'_n-Cauchy sequence. Fix $\varepsilon > 0$; since $\langle \vec{x}_n \rangle$ is d-Cauchy, there exists $k \in \mathbb{N}$ such that $d(\vec{x}_i, \vec{x}_j) \leqslant \varepsilon$ for all $i, j \geqslant k$. It follows that $\max \{d'_m(\vec{x}_i(m), \vec{x}_j(m)) : m \in \mathbb{N}\} = d(\vec{x}_i, \vec{x}_j) \leqslant \varepsilon$ for $i, j \geqslant k$; in particular, $d'_n(\vec{x}_i(n), \vec{x}_j(n)) \leqslant \varepsilon$ for $i, j \geqslant k$.

Thus for each $n \in \mathbb{N}$. $\langle \vec{x}_i(n) \rangle_{i=1}^{\infty} \to x_n$ for some $x_n \in X_n$. Hence $\langle \vec{x}_i \rangle \to \langle x_n \rangle = \vec{x}$. ■

Proof of Lemma 169

(a) Let $U \subset X$ be open in X metrizable. Define $f\colon U \to f[U] \subset X \times \mathbb{R}$ so that $x \mapsto (x, 1/d(x, X \setminus U))$. Since $X \setminus U$ is closed, we have $d(x, X \setminus U) = 0$ iff $x \in X \setminus U$. So $d(x, X \setminus U) > 0$ for all $x \in U$. It follows that f is well-defined.

(1) Clearly f is one-to-one and onto its range.

(2) f is continuous as each coordinate function is (noting d is continuous). Furthermore $f^{-1}(x, y) = x$ is also continuous, so f is a homeomorphism.

(3) Let $\langle f(x_n) \rangle \subset f[U]$ be a converging sequence to some $y \in X \times \mathbb{R}$. We show $y \in f[U]$. We have $f(x_n) = (x_n, 1/d(x_n, X \setminus U)) \to y = (y_1, y_2)$, so $x_n \to y_1$ and $1/d(x_n, X \setminus U) \to y_2$. By continuity of the distance function: $x_n \to y_1$ implies $1/d(x_n, X \setminus U) \to 1/d(y_1, X \setminus U)$, thus we should have $y_1 \in U$ and $1/d(y_1, X \setminus U) = y_2$ since $\mathbb{R}$ is Hausdorff. Therefore

$$y = (y_1, y_2) = (y_1, 1/d(y_1, X \setminus U)) \in f[U],$$

so $f[U]$ is closed.

(b) Now suppose G is G_δ, so $G = \bigcap_{n \geqslant 1} G_i$ with each G_i open in X metrizable. Define $h\colon G \to h[G] \subset X \times \mathbb{R}^{\mathbb{N}}$ so that

$$x \mapsto \langle x, 1/d(x, X \setminus G_1), 1/d(x, X \setminus G_2), \ldots \rangle.$$

A similar argument to 169(a) shows that h is a homeomorphism. Now let $\langle h(x_n) \rangle \subset h[G]$ be a sequence converging to some $y \in X \times \mathbb{R}^{\mathbb{N}}$. So

$$h(x_n) = \langle x_n, 1/d(x_n, X \setminus G_1), 1/d(x_n, X \setminus G_2), \ldots \rangle \\ \to y = \langle y_0, y_1, y_2, \ldots \rangle.$$

Then $x_n \to y_0$ and $1/d(x_n, X \setminus G_i) \to y_i$ for all $i \geqslant 1$. Now $x_n \to y_0$ implies

$$1/d(x_n, X \setminus G_i) \to 1/d(y_0, X \setminus G_i),$$

so $y_0 \in G_i$ and $1/d(y_0, X \setminus G_i) = y_i$ for all $i \geqslant 1$. Hence $y_0 \in G$ and

$$y = \langle y_0, 1/d(y_0, X \setminus G_1), 1/d(y_0, X \setminus G_2), \ldots \rangle \in h[G],$$

therefore $h[G]$ is closed. ■

Proof of Corollary 170
Let X be a G_δ-subset of the completely metrizable space Y. By Lemma 169(b), X is homeomorphic to a closed subset of $Y \times \mathbb{R}^{\mathbb{N}}$. But $Y \times \mathbb{R}^{\mathbb{N}}$ is completely metrizable by Theorem 168, so X is completely metrizable by Theorem 167. ■

Proof of Lemma 171
Let $y \in Y$. We need to show there is an open-in-X set containing y and contained in Y. By local compactness, there is an open-in-Y subset U of Y with $y \in U$ and $\overline{U}^Y = K$ compact. There is an open-in-X subset U' of X such that $U' \cap Y = U$. Since Y is dense in X, U is dense in U'. Now $U = U' \cap Y \subset K$ and K, begin compact, is closed in the Hausdorff space X. Thus $U' \subset \overline{U}^X \subset K \subset Y$, so $U' \subset Y$, and y is in the open-in-X set U'. Hence Y is open in X. ■

Proof of Corollary 172
Let X be locally compact and metrizable, and let $\tilde{X}$ be its completion. By Lemma 171 X is open, hence G_δ, in $\tilde{X}$. By Corollary 170, X is completely metrizable. ■

Proof of Theorem 173
That $d(f,g) = d(g,f)$, $d(f,g) \geqslant 0$, and $d(f,g) = 0$ iff $f = g$ are straightforward. Let $f, g, h \in C^*(X)$. For each $x \in X$, $|f(x) - h(x)| \leqslant |f(x) - g(x)| + |g(x) - h(x)|$. Hence

$$d(f,h) = \sup_{x \in X} |f(x) - h(x)| \leqslant \sup_{x \in X} (|f(x) - g(x)| + |g(x) - h(x)|)$$

$$\leqslant \sup_{x \in X} |f(x) - g(x)| + \sup_{x \in X} |g(x) - h(x)| = d(f,g) + d(g,h).$$

It follows that d is a metric on $C^*(X)$.

Now we prove d is complete. Suppose $\langle f_n \rangle$ is d-Cauchy. Let $x \in X$ and consider $\langle f_n(x) \rangle$. If $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $d(f_i, f_j) \leqslant \varepsilon$

for all $i, j \geqslant N$. But then $|f_i(x) - f_j(x)| \leqslant \varepsilon$ too, so $\langle f_n(x) \rangle$ is Cauchy, whence it converges to some $g(x) \in \mathbb{R}$; let this define the function g. Note that

$$(\forall x \in X)(\forall i \geqslant N)[|f_i(x) - g(x)| = \lim_{j \in \mathbb{N}} |f_i(x) - f_j(x)| \leqslant \varepsilon].$$

It follows that g is bounded since f_i is, and that $\langle f_n \rangle$ converges uniformly to g, so g is continuous and hence $g \in C^*(X)$. Finally it also follows that $d(f_n, g) \to 0$, so $\langle f_n \rangle \to g$ in the metric topology. ■

Proof of Theorem 174

Let d' be the metric on $C^*(X)$. Then

$$d'(j(x), j(y)) = d'(f_x, f_y) = \sup_{z \in X} |f_x(z) - f_y(z)|$$

$$= \sup_{z \in X} |(d(z, x) - d(z, a)) - (d(z, y) - d(z, a))|$$

$$= \sup_{z \in X} |d(z, x) - d(z, y)|.$$

Since $d(z, x) \leqslant d(z, y) + d(x, y)$, we have $d(z, x) - d(z, y) \leqslant d(x, y)$. Similarly, $-(d(z, x) - d(z, y)) \leqslant d(x, y)$. Hence $\sup_{z \in X} |d(z, x) - d(z, y)| \leqslant d(x, y)$. Now, if $z = y$ then $d(z, x) - d(z, y) = d(x, y)$. Hence

$$d'(j(x), j(y)) = d(x, y),$$

so j is an isometry. ■

Proof of Corollary 175

Proof of Theorem 176

Let $\langle X, d \rangle$ be a metric space and let $\langle \tilde{X}_1, \tilde{d}_1 \rangle$ and $\langle \tilde{X}_2, \tilde{d}_2 \rangle$ be complete metric spaces such that

(1) X is dense in $\tilde{X}_1$ and $\tilde{X}_2$, and

(2) $\tilde{d}_1 \restriction X \times X = \tilde{d}_2 \restriction X \times X = d$.

We want to show that $\langle \tilde{X}_1, \tilde{d}_1 \rangle$ and $\langle \tilde{X}_2, \tilde{d}_2 \rangle$ are isometric.

Define a function $h \colon \tilde{X}_1 \to \tilde{X}_2$ as follows:

(1) If $x \in X \subset \tilde{X}_1$, take $h(x) = x$.

(2) If $x \in \tilde{X}_1 \setminus X$, let $\langle x_n \rangle$ be a sequence in X converging to x (when considered as a sequence in $\tilde{X}_1$.; note that such a sequence exists because $\tilde{X}_1$ is complete metric and X is dense therein).

    Note that $\langle x_n \rangle$ is $\tilde{d}_2$-Cauchy, so it coverges to some $y \in \tilde{X}_2$. Put $h(x) = y$.

Certainly h is well-defined for all $x \in X$. To see this for $x \in \tilde{X}_1 \setminus X$, suppose $\langle x_n \rangle$ and $\langle y_n \rangle$ are two sequences in X converging to $x \in \tilde{X}_1$. Since both coverge to x, it must be the case that $d(x_n, y_n) \to 0$. This is true in $\tilde{X}_2$, so $\langle x_n \rangle$ and $\langle y_n \rangle$ must coverge to the same point in X_2 too.

Now let $x, y \in \tilde{X}_1$ and let $\langle x_n \rangle$ and $\langle y_n \rangle$ be sequences in X converging to x and y, respectively. Then

$$\tilde{d}_1(x, y) = \lim_{n\to\infty} \tilde{d}_1(x_n, y_n) = \lim_{n\to\infty} d(x_n, y_n)$$

$$= \lim_{n\to\infty} \tilde{d}_2(x_n, y_n) = \tilde{d}_2(h(x), h(y)),$$

so h is an isometry. ■

Proof of Theorem 177

For each $n \geqslant 1$, we have C_n is a union of 2^n disjoint compact intervals. Hence C_n is compact and nonempty for each $n \geqslant 1$. Futhermore, we have $C_{n+1} \subset C_n$ for each $n \geqslant 1$. Hence, by Lemma 97, we conclude that $\mathbb{C} = \bigcup_{n\in\mathbb{N}} C_n$ is a compact nonempty set. It is clear that $\mathbb{C}$ is metrizable as a subspace of a metrizable space.

Let us consider the collection $\mathcal{B} = \{\mathbb{C} \cap I_\sigma : \exists n \in \mathbb{N}(\sigma \in 2^n)\}$. Let $n \geqslant 1$ and $\sigma \in 2^n$ be given. We have that $\mathbb{C} \cap I_\sigma$ is closed as an intersection of closed sets. Furthermore, we have $C_n \setminus I_\sigma$ is closed, hence $\mathbb{C} \setminus (\mathbb{C} \cap I_\sigma) = \mathbb{C} \cap (C_n \setminus I_\sigma)$ is closed. Then $\mathbb{C} \cap I_\sigma$ is also open in $\mathbb{C}$, so $\mathcal{B}$ is a countable collection of clopen sets in $\mathbb{C}$.

Let U be any open set in $\mathbb{C}$, and let $x \in U$. There is $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \cap C \subset U$. Choose $n \geqslant 1$ such that $1/3^n < \varepsilon$. Since $x \in C_n$, there is $\sigma \in 2^n$ such that $x \in I_\sigma$. Observe that $\operatorname{diam} I_\sigma = 1/3^n$. Hence $I_\sigma \subset (x-\varepsilon, x+\varepsilon)$, which implies that $x \in \mathbb{C} \cap I_\sigma \subset \mathbb{C} \cap (x-\varepsilon, x+\varepsilon) \subset U$.

It follows that $\mathcal{B}$ is a base for $\mathbb{C}$, whence $\mathbb{C}$ has a countable base of clopen sets.

Let $n \geqslant 1$ and $\sigma \in 2^n$ be given. Let $a, b \in \mathbb{R}$ such that $I_\sigma = [a, b]$. By construction we have $a \in I_{\sigma_0} \subset C_m$ and $b \in I_{\sigma_1} \subset C_m$ for all $m \geqslant 1$, where

$$\sigma_0 = \sigma \frown \underbrace{00\ldots0}_{m \text{ times}} \text{ and } \sigma_1 = \sigma \frown \underbrace{11\ldots1}_{m \text{ times}}.$$

Hence, $\{a, b\} \subset \mathbb{C}$. This implies that $\{a, b\} \subset \mathbb{C} \cap I_\sigma$. Thus $|\mathbb{C} \cap I_\sigma| \geqslant 2$. This shows that any element of $\mathcal{B}$ has at least two points. Since $\mathcal{B}$ is a base for $\mathbb{C}$, we conclude that any nonempty open set in $\mathbb{C}$ has more than one point. Thus, $\mathbb{C}$ has no isolated points.

Let $x \in \mathbb{C}$ and $\varepsilon > 0$ be fixed. From above, we know that there is $\sigma \in 2^n$, for some $n \geqslant 1$, such that $x \in I_\sigma \subset (x - \varepsilon, x + \varepsilon)$. let K denote the middle third open interval such that

$$I_\sigma = I_{\sigma \frown 0} \cup K \cup I_{\sigma \frown 1}.$$

We have that $K \subset I \setminus C_{n+1}$. Hence $K \subset (x - \varepsilon, x + \varepsilon) \cap (I \setminus \mathbb{C})$. Thus $(x - \varepsilon, x + \varepsilon) \not\subset \mathbb{C}$, so $\mathbb{C}$ has empty interior in $\mathbb{R}$.

For each $n \geqslant 1$, we have

$$\mathbb{C} = C_n \cap \mathbb{C} = \bigcup_{\sigma \in 2^n} \mathbb{C} \cap I_\sigma.$$

Since for each $n \geqslant 1$ and $\sigma \neq \sigma' \in 2^n$ we have $I_\sigma \cap I_{\sigma'} = \varnothing$, for each $x \in \mathbb{C}$ there is a unique $\sigma_n^x \in 2^n$ such that $x \in I_{\sigma_n^x}$. For each $x \in C$ and $n \geqslant 1$, let us set $\sigma_n^x = x_1 \ldots x_n$. Observe that if $m < n$, then σ_n^x extends σ_m^x. Thus x_n is uniquely determined by any σ_j^x for $j \geqslant n$. Now define

$$f \colon \mathbb{C} \to 2^{\mathbb{N}} \text{ so that } x \mapsto \langle x_n \rangle.$$

Certainly f is well-defined. Let $\langle x_n \rangle \in 2^{\mathbb{N}}$ be given and consider the sequence of compact sets $\langle I_{x_1 \ldots x_n} \rangle$. By definition, we have

$$I_{x_1 \ldots x_{n+1}} \subset I_{x_1 \ldots x_n} \text{ and } \operatorname{diam} I_{x_1 \ldots x_n} = 1/3^n$$

for all $n \geqslant 1$. Thus by Lemma 97 we have $\bigcap_{n \in \mathbb{N}} I_{x_1 \ldots x_n} \neq \varnothing$. Clearly we have that $\bigcap_{n \in \mathbb{N}} I_{x_1 \ldots x_n} \subset \mathbb{C}$. Since $\operatorname{diam} I_{x_1 \ldots x_n} \to 0$, $\bigcap_{n \in \mathbb{N}} I_{x_1 \ldots x_n}$

is a single point. That is, there is a unique $x \in \mathbb{C}$ such that $\{x\} = \bigcap_{n\in\mathbb{N}} I_{x_1\ldots x_n}$. Hence $f(x) = \langle x_n \rangle$. Since x is uniquely determined for each given sequence $\langle x_n \rangle \in 2^{\mathbb{N}}$, we conclude that f is bijective. Thus $\mathbb{C}$ is uncountable. ■

Proof of Theorem 178
Let $\mathcal{B} = \{B_n : n \geqslant 1\}$ be a base of nonempty clopen sets in X.

Claim 1 Given any nonempty clopen set $A \in X$ and any $n \geqslant 1$, there is A_n nonempty clopen in X such that $A \setminus A_n \neq \varnothing$ and $A = A_n \cup (A \setminus A_n)$. Furthermore, either each of $A_n, A \setminus A_n$ is contained in B_n or completely misses B_n.

$\varnothing \subsetneq B_n \cap A \subsetneq A$: Then $A = A \cap B_n \cup A \setminus B_n$ with $A \cap B_n$ nonempty clopen in X and $A \setminus B_n \neq \varnothing$.

$A \cap B_n = \varnothing$ or $A \subset B_n$: Since X has no isolated points and A is nonempty open in X, there is $x, y \in A$ such that $x \neq y$. Choose $B_m \in \mathcal{B}$ such that $x \in B_m$, $y \in A \setminus B_m$. Thus $A = B_m \cup (A \setminus B_m)$.

Construction of Cantor Tree of clopen sets Divide $X = X_\varnothing$ into two clopen nonempty pieces X_0 and X_1 such that, unless B_1 is the whole space, then B_1 is one of the pieces. Then divide each of X_0, X_1 into two pieces—say $X_{00}, X_{01}, X_{10}, X_{11}$—such that each piece is either contained in B_2 or completely misses B_2. And so on.

Claim 3 Given an infinite sequence $\sigma = \langle \sigma_1, \sigma_2, \ldots \rangle \in 2^{\mathbb{N}}$, the intersection of the branch $X_\varnothing, X_{\sigma_1}, X_{\sigma_1\sigma_2}, \ldots$ is a single point.

Let $\sigma \in 2^{\mathbb{N}}$ be given. Since $X_\varnothing \supset X_{\sigma_1} \supset X_{\sigma_1\sigma_2} \supset \cdots$ are compact nonempty sets in X, then their intersection A is nonempty. Let $p \in A$ and $y \in X$ such that $y \neq p$. Then there exist $n, m \in \mathbb{N}$ such that $p \in B_n$, $y \in B_m$, and $B_n \cap B_m = \varnothing$. Thus $p \in X_{\sigma_1\ldots\sigma_n} \cap B_n$ and by part 1, $X_{\sigma_1\ldots\sigma_n} \subset B_n$. There $y \notin X_{\sigma_1\ldots\sigma_n}$ and so $y \notin A$, thus $A\{p\}$.

Claim 4 $\{X_{\sigma\restriction n} : n \in \mathbb{N}\}$ is a base at p.

Let U be open in X such that $p \in U$. Then there is $n \in \mathbb{N}$ such that $p \in B_n \subset U$. Thus $p \in X_{\sigma\restriction n} \cap B_n$ and by the construction, $X_{\sigma\restriction n} \subset B_n$.

Consider the map $f\colon 2^{\mathbb{N}} \to X$ defined by

$$f(\sigma) = X_{\sigma\restriction 0} \cap \bigcap_{n\in\mathbb{N}} X_{\sigma\restriction n}.$$

From Claim 3, f is well-defined.

Claim 5 f is one-to-one.

Let $\sigma, \tau \in 2^{\mathbb{N}}$ such that $\sigma \neq \tau$. Then there is $i \in \mathbb{N}$ such that $\sigma_i \neq \tau_i$. Thus $X_{\sigma\restriction(i+1)} \cap X_{\sigma\restriction(i+1)} = \varnothing$. It follows that $f(\sigma) \neq f(\tau)$.

Claim 6 f is onto.

Let $p \in X$. We construct $\sigma \in 2^{\mathbb{N}}$ as follows: $\sigma_1 = 0$ if $p \in X_0$ and $\sigma_1 = 1$ if $p \in X_1$. If $\sigma_1, \ldots, \sigma_n$ have been defined, take $\sigma_{n+1} = 0$ if $p \in X_{\sigma_1\ldots\sigma_n\frown 0}$ and $\sigma_{n+1} = 1$ if $p \in X_{\sigma_1\ldots\sigma_n\frown 1}$. Then $f(\sigma) = p$.

Claim 7 f is continuous.

Let $p \in X$ and $\sigma \in 2^{\mathbb{N}}$ such that $f(\sigma) = p$. It suffices to show that $f^{-1}[X_{\sigma\restriction n}]$ is open in $2^{\mathbb{N}}$ for each $n \in \mathbb{N}$. Since

$$f^{-1}[X_{\sigma\restriction n}] = \{\sigma_1\} \times \cdots \times \{\sigma_n\} \times \{2\} \times \{2\} \times \cdots$$

is basic open, f is continuous.

Claim 8 f is a homeomorphism.

Since $2^{\mathbb{N}}$ is compact and X is Hausdorff, f is a closed map; closed continuous bijections are homeomorphisms per Theorem 29.

Claim 9 X is homeomorphic to the Cantor ternary set $\mathbb{C}$.

By Theorem 177, $\mathbb{C}$ is compact Hausdorff without isolated points and has a countable base of nonempty clopen sets. Thus $\mathbb{C}$ is homeomorphic to $2^{\mathbb{N}}$. But X also is homeomorphic to $2^{\mathbb{N}}$, so $X \cong \mathbb{C}$. ■

Proof of Corollary 179

Proof of Theorem 180

Certainly $f[2^{\mathbb{N}}] \subset [0,1]$ and f is well-defined.

To see that f is continuous, let $\langle \vec{x}_n \rangle \to \vec{x}$ in $2^{\mathbb{N}}$. Let $\varepsilon > 0$ and consider $B(f(\vec{x}), \varepsilon)$. Pick $N \in \mathbb{N}$ such that $1/2^N < \varepsilon/2$, and pick $m \in \mathbb{N}$ such that $\pi_i(\vec{x}_n) = \pi_i(\vec{x})$ for all $i \leqslant N$ and for all $n > m$. Then $d(f(\vec{x}), f(\vec{x}_n)) \leqslant 1/2^{N+1} + 1/2^{N+1} + \cdots \leqslant 1/2^N < \varepsilon/2$ and so $f(\vec{x}) \in B(f(\vec{x}), \varepsilon)$.

To show f is onto, let $D = \{a/2^n : a \in \mathbb{Z} \wedge n \in \mathbb{N}\} \cap [0,1]$ be the set of all dyadic rationals in the unit interval. We will show D is dense in $[0,1]$, f is onto D, and f is onto $[0,1]$.

(1) Let $\varepsilon > 0$ and let $B = B(x, \varepsilon) \subset [0,1]$. Note we may pick $n \in \mathbb{N}$ such that $1/2^n < \varepsilon$. Furthermore pick $a \in \mathbb{N}$ such that $a/2^n \leqslant x < a + 1/2^n$, so that $d(x, a/2^n) < 1/2^n < \varepsilon$. It follows that $a/2^n \in B$, so D is dense.

(2) Let $a/2^n \in D$. Let $a = a_1 a_2 \ldots a_n$ be the binary representation of a, thus $a = a_1/2^{n-1} + a_2/2^{n-2} + \cdots + a_n/2^0$. Then $a/2^n = \sum_{i=1}^{n} a_i/2^i$ and $a/2^n = f((a_1, a_2, \ldots, a_n, 0, 0, \ldots))$. Thus $D \subset f[2^{\mathbb{N}}]$.

(3) By Corollary 179 $2^{\mathbb{N}} \cong \mathbb{C}$, which is compact by Theorem 177. Every continuous image of a compact space is again compact—Theorem 61—so $f[2^{\mathbb{N}}]$ is a compact subspace of the Hausdorff space $[0,1]$, hence closed by Theorem 62. Thus $[0,1] = \overline{D} \subset f[2^{\mathbb{N}}]$. ■

Proof of Theorem 181

By Theorem 180 there exists a continuous surjection $h\colon \mathbb{C} \to [0,1]$; so $\bar{h}\colon \mathbb{C}^2 \to [0,1]^2$ defined by $\langle s, t \rangle \mapsto \langle h(s), h(t) \rangle$ is also a continuous surjection. Using Corollary 179 we may get a continuous surjection $g\colon \mathbb{C} \to [0,1]^2$.

Define $f\colon [0,1] \to [0,1]^2$ as follows:

(1) If $x \in \mathbb{C}$, take $f(x) = g(x)$.

(2) If $x \notin \mathbb{C}$, there exists a maximal $(a, b) \subset [0,1] \setminus \mathbb{C}$ such that $x \in (a, b)$. Put $f(x) = (1-t)g(a) + tg(b)$, where $0 < t < 1$ and $x = (1-t)a + tb$.

Certainly f is surjective.

Let $x \in [0,1]$; if $x \notin \mathbb{C}$, then f is continuous at x by linearity. If $x \in \mathbb{C}$, let $\langle x_n \rangle$ be a sequence in $[0,1]$ converging to x. We have the following cases:

(1) If $\langle x_n \rangle_{n=k}^{\infty} \subset \mathbb{C}$ for some $k \in \mathbb{N}$ then

$$\langle f(x_n) \rangle_{n=k}^{\infty} = \langle g(x_n) \rangle_{n=k}^{\infty} \to g(x) = f(x),$$

whence $\langle f(x_n) \rangle \to f(x)$.

(2) If $\langle x_n \rangle$ does not have a tail in $\mathbb{C}$, then for each $n \in \mathbb{N}$ with $x_n \notin \mathbb{C}$ let $(a_n, b_n) \subset [0,1] \setminus \mathbb{C}$ be maximal containing x_n.

(a) If diam $(a_n, b_n) \to 0$, define a sequence $\langle x'_n \rangle$ as follows:

(1) If $x_n \in \mathbb{C}$, take $x'_n = x_n$.

(2) If $x_n \notin \mathbb{C}$, let x'_n be the endpoint of (a_n, b_n) to which x_n is closest. In a tie, say $x'_n = b_n$.

Certainly $\langle x'_n \rangle \to x$ and

$$\langle f(x'_n) \rangle = \langle g(x'_n) \rangle = g(x) = f(x).$$

Since diam $(a_n, b_n) \to 0$ we must have $d(g(a_n), g(b_n)) \to 0$ as well. But this implies $d(f(x_n), f(x'_n)) \to 0$, whence $\langle f(x_n) \rangle \to f(x)$.

(b) If diam $(a_n, b_n) \not\to 0$, then there is a maximal $(a, b) \subset [0,1] \setminus \mathbb{C}$ containing infinitely many terms of $\langle x_n \rangle$. If (a, b) contains a tail of $\langle x_n \rangle$, then $\langle f(x_n) \rangle \to f(x)$ by linearity.

If (a, b) does not contain a tail, we may assume without a loss of generality that $x = b$. Then infinitely many terms converge from the left and right. Let $\langle y_n \rangle$ be the subsequence of $\langle x_n \rangle$ consisting of those terms strictly less than x. Note that $\langle y_n \rangle \to x$, so $\langle f(y_n) \rangle \to f(x)$ by linearity.

Let $\langle z_n \rangle$ be the subsequence of $\langle x_n \rangle$ containing those terms not in $\langle y_n \rangle$. Note that $\langle z_n \rangle \to x$ and satisfies either (1) or Case (2a). Thus $\langle f(z_n) \rangle \to f(x)$, otherwise x would be an isolated point of $\mathbb{C}$.

It follows that $\langle f(x_n) \rangle \to f(x)$. ■

Proof of Theorem 182

Let $\langle X, < \rangle = \{x_n : n < \omega\}$ and $\langle Y, \prec \rangle = \{y_n : n < \omega\}$ be densely and linearly ordered sets with no first or last point. Put $X_0 = \{x_0\}$ and $Y_0 = \{y_0\}$. Let $f_0 \colon X_0 \to Y_0$. Suppose for some $n < \omega$ that $f_n \colon X_n \to Y_n$ is an order-preserving bijection with $\{x_0, \ldots, x_n\} \subset X_n$, $\{y_0, \ldots, y_n\} \subset Y_n$, and both X_n and Y_n finite.

If $x_{n+1} \in X_n$ put $X'_{n+1} = X_n$, $Y'_{n+1} = Y_n$, and $f'_{n+1} = f_n$. Otherwise, let $A = \{x \in X_n : x < x_{n+1}\}$ and $B = \{x \in X_n : x_{n+1} < x\}$. If $A = X_n$, pick $y \in Y \setminus Y_n$ such that $y' \prec y$ for all $y' \in Y_n$. If $B = X_n$, pick $y \in Y \setminus Y_n$ such that $y \; prec y'$ for all $y' \in Y_n$. Otherwise, put $a = \max A$ and $b = \min B$ and pick $y \in (f_n(a), f_n(b))$. Set $X'_{n+1} = X_n \cup \{x_{n+1}\}$ and $Y'_{n+1} = Y_n \cup \{y\}$, and define $f'_{n+1} \colon X'_{n+1} \to Y'_{n+1}$ so that $f'_{n+1} \restriction X_n = f_n$ and $f'_{n+1}(x_{n+1}) = y$.

If $y_{n+1} \in Y'_{n+1}$ put $X_{n+1} = X'_{n+1}$, $Y_{n+1} = Y'_{n+1}$, and $f_{n+1} = f'_{n+1}$. Otherwise, let $A = \{y \in Y'_{n+1} : y \prec y_{n+1}\}$ and $B = \{y \in Y'_{n+1} : y_{n+1} \prec y\}$. If $A = Y'_{n+1}$, pick $x \in X \setminus X'_{n+1}$ such that $x' < x$ for all $x' \in X'_{n+1}$. If $B = Y'_{n+1}$, pick $x \in X \setminus X'_{n+1}$ such that $x < x'$ for all $x' \in X'_{n+1}$. Otherwise, let $a = \max A$ and $b = \min B$ and pick $x \in ((f'_{n+1})^{-1}(a), (f'_{n+1})^{-1}(b))$. Set $X_{n+1} = X'_{n+1} \cup \{x\}$ and $Y_{n+1} = Y'_{n+1} \cup \{y_{n+1}\}$, and define $f_{n+1} \colon X_{n+1} \to Y_{n+1}$ so that $f_{n+1} \restriction X'_{n+1} = f'_{n+1}$ and $f_{n+1}(x) = y_{n+1}$.

Note that $f_{n+1} \colon X_{n+1} \to Y_{n+1}$ is an order-preserving bijection with $\{x_0, \ldots, x_{n+1}\} \subset X_{n+1}$, $\{y_0, \ldots, y_{n+1}\} \subset Y_{n+1}$, and both X_{n+1} and Y_{n+1} finite. Then $\bigcup_{n<\omega} f_n$ is an order-preserving bijection from X to Y. ■

Proof of Theorem 183

Let X be a connected linearly ordered space. Let $x, y \in X$ with $x \neq y$. Without loss of generality we may assume that $x < y$. If $(x, y) = \varnothing$, then $X = (-\infty, y) \cup (x, \infty)$, implying X is not connected. It follows that $(x, y) \neq \varnothing$, whence X is densely ordered.

Now, let A be a bounded set and suppose on the contrary that A has no least upper bound. Then for all $x \in A$ there exists some $y \in A$ with $x < y$. Also, for each upper bound b of A there exists another upper bound b' such that $b' < b$. Take $U = \bigcup \{(-\infty, y) : y \in A\}$ and $V = \bigcup \{(b, \infty) : b \text{ is an upper bound of } A\}$. Then $X = U \cup V$ with U, V open and disjoint, a contradiction. It follows that every bounded

subset of X has a least upper bound.

For the other direction, let X be a linearly ordered non-connected space in which every bounded subset has a least upper bound. Let $U, V \subset X$ be disjoint, nonempty, and open such that $X = U \cup V$. Without loss of generality we may assume there exists $x \in U$ and $y \in V$ such that $x < y$. Put $U' = (-\infty, y) \cap U$ and let b be a least upper bound of U'. Note that either $b \in U'$ or $b \in V$.

If $b \in U'$, then there exists $(a, c) \subset X$ with $b \in (a, c) \subset U'$. Since $(b, c) \cap V = \varnothing$ and $(b, c) \cap U = \varnothing$, it must be the case that $(b, c) = \varnothing$. Similarly, if $b \in V$ there exists $(a, c) \subset X$ with $b \in (a, c) \subset V$. Now $(a, b) \cap U' = \varnothing$ and $(a, b) \cap V = \varnothing$, since b is a least upper bound, so $(a, b) = \varnothing$. It follows that X is not densely ordered. ■

Proof of Theorem 184

Let X be a separable, connected, and linearly ordered space with no first or last point. Let D be a countable dense subset of X. Observe that D is densely ordered and has no first or last point. By Theorem 182, there exists an order-preserving bijection $g\colon D \to \mathbb{Q}$.

For each $x \in X$ put $L_x = \{d \in D : d < x\}$. Define $f\colon X \to \mathbb{R}$ so that $x \mapsto \sup g[L_x]$. Since $g[L_x]$ is bounded above, $\sup g[L_x]$ exists, so f is well-defined. Let $x_1, x_2 \in X$ with $x_1 \neq x_2$. Without loss of generality we may assume that $x_1 < x_2$. Since X is densely ordered and D is dense, there exist $d_1, d_2 \in D$ such that $x_1 < d_1 < d_2 < x_2$. Then

$$\sup g[L_{x_1}] \leqslant g(d_1) < g(d_2) \leqslant \sup g[L_{x_2}],$$

so $f(x_1) < f(x_2)$. Thus f is one-to-one and order-preserving.

Let $y \in \mathbb{R}$ and put $L = \{q \in \mathbb{Q} : q < y\}$. Then $g^{-1}[L] \subset D$ is bounded above, so $x = \sup g^{-1}[L] \in X$ exists. But $L_x = g^{-1}[L]$, so $f(x) = y$. It follows that f is onto. ■

Proof of Corollary 185

Let X be a separable compact connected linearly ordered space. First we show that X has a first and last point. Well, if X has neither then $X \cong \mathbb{R}$ by Theorem 184, so that X is not compact, a contradiction. Without loss of generality, suppose x_0 is a first point of X. If X has no last point, then the collection $\{[x_0, x) : x \in X\}$ is an open cover of X,

so it has a finite subcover. So there exists $x \in X$ such that $[x_0, x) = X$, a contradiction. Thus X has a last point x_1.

Now $X \setminus \{x_0, x_1\}$ has no first or last point. (For instance, if it has a first point x_2 then $(x_0, x_2) = \varnothing$. But this would contradict Theorem 183.) By Theorem 184, $(x_0, x_1) \cong \mathbb{R} \cong (0, 1)$; hence $X \cong [0, 1]$. ■

Proof of Lemma 186

Since X is $T_{31/2}$, it may be embedded into some product $\prod_{i \in I} \mathbb{R}_i$ of real lines. For instance, the map from X to $\mathbb{R}^{C(X)}$ given by $x \mapsto \langle f(x) \rangle_{f \in C(X)}$ is an embedding. We therefore consider X as a subspace of $\prod_{i \in I} \mathbb{R}_i$. Let

$$\mathcal{B} = \{\bigcap_{i \in F} \pi_i^{-1}[(c_i, d_i)] \cap X : F \in [I]^{<\omega} \wedge \forall i \in F (c_i < d_i \in \mathbb{R}_i \setminus \pi_i[X])\}.$$

If $F \subset I$ and $c_i < d_i \in \mathbb{R}_i \setminus \pi_i[X]$ for each $i \in F$ then

$$\bigcap_{i \in F} \pi_i^{-1}[(c_i, d_i)] \cap X = \bigcap_{i \in F} \pi_i^{-1}[[c_i, d_i]] \cap X,$$

so that the sets of $\mathcal{B}$ are clopen in X.

We now show $\mathcal{B}$ is a base for X. Suppose U is an open subset in X and $\langle x_i \rangle \in U$; note that $\langle x_i \rangle$ here is some point $\vec{x}$ in the product space, and *not* a sequence of points in U. We find a clopen subset of X containing $\langle x_i \rangle$, contained in U. There exists an open $U' \subset \prod_{i \in I} \mathbb{R}_i$ such that $U' \cap X = U$. There exists a basic product open set $\bigcap_{i \in F} \pi_i^{-1}[U_i']$ containing $\langle x_i \rangle$ and contained in U (here F is some finite subset of I and each U_i' is open in $\mathbb{R}_i$). For each $i \in F$ there exists $a_i, b_i \in \mathbb{R}_i$ such that $x_i \in (a_i, b_i) \subset U_i'$.

Since $|X| < |\mathbb{R}|$, $\pi_i[X]$ does not contain an interval for any $i \in I$. So for any $i \in I$ and $a < b \in \mathbb{R}_i$, there exists $r \in (a, b) \cap \mathbb{R}_i \setminus \pi_i[X]$. In particular, for each $i \in F$ there exist $c_i < d_i \in (a_i, b_i) \cap \mathbb{R}_i \setminus \pi_i[X]$ with $x_i \in (c_i, d_i)$. We have

$$\langle x_i \rangle \in \bigcap_{i \in F} \pi_i^{-1}[(c_i, d_i)] \cap X \subset \bigcap_{i \in F} \pi_i^{-1}[(a_i, b_i)] \cap X$$

$$\subset \bigcap_{i \in F} \pi_i^{-1}[U_i'] \cap X \subset U' \cap X = U.$$

■

Proof of Lemma 187
Let X be a space and let $\mathcal{A}$ and $\mathcal{B}$ be bases thereof with $|\mathcal{A}| = \kappa$. Fix $A \in \mathcal{A}$. Let $\mathcal{B}(A) = \{B \in \mathcal{B} : B \subset A\}$ and let

$$\mathcal{A}(A) = \{A' \in \mathcal{A} : \exists B \in \mathcal{B}(A)[A' \subset B]\}.$$

Since A is open and $\mathcal{B}$ is a base, we must have $\bigcup \mathcal{B}(A) = A$. Similarly, $\bigcup \mathcal{A}(A) = A$. Finally, for each $A' \in \mathcal{A}(A)$, let $B(A') \in \mathcal{B}(A)$ such that $A' \subset B(A')$. Put $\mathcal{C}(A) = \{B(A') : A' \in \mathcal{A}(A)\}$. Since

$$\bigcup \mathcal{A}(A) \subset \bigcup \mathcal{C}(A) \subset \bigcup \mathcal{B}(A),$$

we have $\bigcup \mathcal{C}(A) = A$. Moreover, $|\mathcal{C}(A)| \leqslant \kappa$.

Let $\mathcal{C} = \bigcup_{A \in \mathcal{A}} \mathcal{C}(A)$. Certainly $\mathcal{C} \subset \mathcal{B}$ is a base. By Lemma 202(a), $|\mathcal{C}| \leqslant \kappa$.

If κ is the least cardinal of a base for X, naturally $|\mathcal{C}| = \kappa$. ■

Proof of Theorem 188
Let X be a countable, regular and first-countable space with no isolated points. Observe that X is second-countable, hence metrizable and completely regular. By Lemma 186 and Lemma 187, X has a countable base $\mathcal{B} = \{B_n : n < \omega\}$ of clopen sets.

Claim A. *X embeds into 2^ω.*

Claim B. *X is homeomorphic to a dense subset of $\mathbb{C}$.*

Claim C. *Let $f \in 2^\omega$. Then $h\colon 2^\omega \to 2^\omega$ defined so that $g \mapsto f + g$ is a homeomorphism.*

Claim D. *Let $E \subset \mathbb{C}$ be the collection of endpoints of all I_σ, where $\sigma \in 2^{<\omega}$. Then X is homeomorphic to a subset of $\mathbb{C}$ disjoint from E.*

Claim E. *Let $D \subset \mathbb{C}$ be dense and disjoint from E. Then D is densely ordered and has no first or last point.*

By the above claims, let X be densely embedded in $\mathbb{C}$ such that $X \cap E = \varnothing$. By Theorem 182 there exists an order-preserving bijection from X to $\mathbb{Q}$, so $X \cong \mathbb{Q}$.

Proof of Claim A. For each $n < \omega$, define $f_n \colon X \to 2$ so that $f_n[B_n] = \{0\}$ and $f_n[X \setminus B_n] = \{1\}$. Certainly f_n is continuous for each $n < \omega$. Put $\mathcal{F} = \{f_n : n < \omega\}$. and define $e_{\mathcal{F}} \colon X \to 2^\omega$ so that $x \mapsto \langle f_n(x) \rangle$. Note that $\pi_n \circ e_{\mathcal{F}} = f_n$ for each $n < \omega$, so $e_{\mathcal{F}}$ is continuous.

Let $x \in X$ and $H \subset X$ closed not containing x. Then there exists $n < \omega$ such that $x \in B_n \subset X \setminus H$. It follows that $f_n(x) = 0$ and $\overline{f_n[H]} = \{1\}$, so $\mathcal{F}$ separates points from closed sets. By Theorem 145, $e_{\mathcal{F}}$ is a homeomorphic embedding into 2^ω.

Note that we could have defined each f_n to map X onto the endpoints of $[0, 1]$ to more closely follow Theorem 145. □

Proof of Claim B. Since $2^\omega \cong \mathbb{C}$, there exists an embedding $f \colon X \to \mathbb{C}$ by Proof C. Observe that $\overline{f[X]} \subset \mathbb{C}$ is compact Hausdorff and has a countable base of clopen sets. Since $f[X]$ has no isolated points, neither does $\overline{f[X]}$. By Theorem 178, there is a homeomorphism $h \colon \overline{f[X]} \to \mathbb{C}$. Since $f[X]$ is dense in $\overline{f[X]}$, we have $h[f[X]]$ dense in $\mathbb{C}$. □

Proof of Claim C. Certainly h is a bijection. Let $U \subset 2^\omega$ be a basic open set. Then

$$U = \prod_{i \in F_0} \{0\} \times \prod_{i \in F_1} \{1\} \times \prod_{i \notin F_0 \cup F_1} 2,$$

where $F_0, F_1 \subset \omega$ are finite and disjoint. Thus

$$h[U] = \prod_{i \in F_0'} \{0\} \times \prod_{i \in F_0''} \{1\} \times \prod_{i \in F_1'} \{1\} \times \prod_{i \in F_1''} \{0\} \times \prod_{i \notin F_0 \cup F_1} 2,$$

where $F_j' = \{i \in F_j : f(i) = 0\}$ and $F_j'' = \{i \in F_j : f(i) = 1\}$ for $j = 0, 1$. It follows that h is open; by the same argument h^{-1} is open. □

Proof of Claim D. Let X be (densely) embedded in $\mathbb{C}$ and let h witness $\mathbb{C} \cong 2^\omega$. Pick $a \in 2^\omega \setminus \{f + g : f \in h[X] \wedge g \in h[E]\}$. Put $X_a = \{f + a : f \in h[X]\}$. Then $h[X] \cong X_a$ and $h[E] \cap X_a = \varnothing$. It follows that $X \cong h^{-1}[X_a]$ and $E \cap h^{-1}[X_a] = \varnothing$. □

Proof of Claim E. Suppose on the contrary that D has a first point, say d. Since $D \cap E = \varnothing$, we must have $d > 0$; thus $(-\infty, d)$ is a nonempty

open set of $\mathbb{C}$ missing D, a contradiction. By a similar argument, D has no last point.

Now let $d_1, d_2 \in D$ with $d_1 < d_2$. Then there exists $\sigma \in 2^{<\omega}$ such that $d_1 \in I_\sigma$ but $d_2 \notin I_\sigma$. It follows that (d_1, d_2) contains the right endpoint of I_σ. Since (d_1, d_2) is nonempty, $(d_1, d_2) \cap D \neq \varnothing$, whence D is densely ordered. ■

Proof of Corollary 189

Proof of Theorem 190

First, let X be a compact linearly ordered space. Suppose on the contrary that there exists some $Y \subset X$ without a least upper bound. Put $\mathcal{U} = \{(-\infty, y) : y \in Y\}$ and $\mathcal{V} = \{(x, \infty) : x \in X \wedge \forall y \in Y (y < x)\}$. Certainly $\mathcal{U} \cup \mathcal{V}$ is an open cover of X, so there exist finite subcollections $\mathcal{U}'$ and $\mathcal{V}'$ of $\mathcal{U}$ and $\mathcal{V}$, respectively, such that $\mathcal{U}' \cup \mathcal{V}'$ covers X.

If $Y \neq \varnothing$ then $\mathcal{U}' \neq \varnothing$, so we may assume

$$\mathcal{U}' = \{(-\infty, y_1), \ldots, (-\infty, y_n)\}$$

with $y_1 < \cdots < y_n$. It follows that $Y \subset (-\infty, y_n)$ which implies $y_n \notin Y$, a contradiction. On the other hand, if $Y = \varnothing$ we may assume $\mathcal{V}' = \{(x_1, \infty), \ldots, (x_m, \infty)\}$ with $x_1 < \cdots < x_m$. Then $X \subset (x_1, \infty)$ so $x_1 \notin X$, a contradiction. It follows that every $Y \subset X$ must have a least upper bound and, by a similar argument, a greatest lower bound.

Now let X be a linearly ordered space in which every subset has a least upper bound and a greatest lower bound. Observe that X has a least element, say 0, and a greatest element, say 1.

Let $\mathcal{U}$ be an open cover of X and let A be the collection of all $x \in X$ such that $[0, x]$ can be covered by finitely many members of $\mathcal{U}$. Put $a = \sup A$. We claim that $a \in A$ and $a = 1$.

That $a \in A$ is immediate, for there must be some $x \in A$ and $U \in \mathcal{U}$ such that $(x, a] \subset U$. Suppose on the contrary that $a < 1$. If a has no immediate successor, there exists $U \in \mathcal{U}$ such that $(a, y] \subset U$ for some $y > a$. But this means $[0, y]$ can be covered by finitely many members

of $\mathcal{U}$, so a must have an immediate successor, say b. But some member of $\mathcal{U}$ must contain b, so $[0, b]$ can be covered by finitely many members of $\mathcal{U}$, contradicting the definition of a. It follows that $a = 1$, whence X is compact. ■

Proof of Theorem 191

Let $\hat{X} = \langle \{A \subset X : A \text{ is left-closed}\}, \subset \rangle$. By Theorem 190, it suffices to show that every subset of $\hat{X}$ has a least upper bound and a greatest lower bound. Let $\mathcal{A} \subset \hat{X}$.

Claim. *$\overline{\bigcup \mathcal{A}}$ is the least upper bound of $\mathcal{A}$.*

Proof of Claim. Clearly $\overline{\bigcup \mathcal{A}}$ is an upper bound of $\mathcal{A}$. Note $b < a \in \overline{\bigcup \mathcal{A}} \to b < a' \in A$ for some $A \in \mathcal{A}$, which implies $b \in A \subset \overline{\bigcup \mathcal{A}}$. Thus $\overline{\bigcup \mathcal{A}}$ is left-closed and a member of $\hat{X}$.

Let $B \subsetneq \overline{\bigcup \mathcal{A}}$ be left-closed. We show that B is not an upper bound of $\mathcal{A}$. Suppose it were. Then $B \supset A$ for every $A \in \mathcal{A}$. But B is closed, so then $B \supset \overline{\bigcup \mathcal{A}}$, a contradiction. □

Claim. *$\bigcap \mathcal{A}$ is the greatest lower bound of $\mathcal{A}$.*

Proof of Claim. If $b < a \in \bigcap \mathcal{A}$, then $b < a \in A$ for all $A \in \mathcal{A}$, so $b \in A$ for all $A \in \mathcal{A}$ and $b \in \bigcap \mathcal{A}$; thus $\mathcal{A} \in \hat{X}$. Certainly $\bigcap \mathcal{A}$ is a lower bound for $\mathcal{A}$. If $\bigcap \mathcal{A}$ is a proper subset of a left-closed set B, then there exists $b \in B$ with $\bigcap \mathcal{A} < b$. If B were a lower bound for $\mathcal{A}$, then $b \in B \subset A$ for every A, whence $b \in \bigcap \mathcal{A}$, contradiction. □

Claim. *X is dense in $\hat{X}$.*

Proof of Claim. Let $(A, B) \subset \hat{X}$ be nonempty. There is $C \in \hat{X}$ such that $A \subset C \subset B$. There exists $x \in C$ with $A < x$ such that $A \subset \{a \in X : a \leqslant x\} \subset C \subset B$. ■

Proof of Theorem 192

TODO: Find the corollaries that prove this. ■

Proof of Theorem 193

(a) Since C and D are unbounded, there exists an increasing sequence $\langle x_i \rangle$ with $x_{2i} \in C$ and $x_{2i+1} \in D$ for each $i < \omega$. Then $x = \sup \langle x_i \rangle$

is a limit point of both C and D. Since C and D are closed, we have $x \in C \cap D$.

(b) Using the unboundedness of the sets C_n, we construct a strictly increasing sequence $\langle x_i \rangle$ of points in ω_1 as follows. Let $x_0 \in C_0$ and $x_1 \in C_1$. Let $x_2 \in C_0$, $x_3 \in C_1$, and $x_4 \in C_2$. Let $x_5 \in C_0$, $x_6 \in C_1$, $x_7 \in C_2$, and $x_8 \in C_3$. Continue in this manner. Let $x = \sup \langle x_i \rangle$. Then for each $n < \omega$, $\langle x_i \rangle$ has a subsequence in C_n which limits to x. By closedness of the sets C_n, we have $x \in \bigcap_{n<\omega} C_n$. ■

Proof of Theorem 194

Let $H \subset \omega_1$ be the of limit ordinals in ω_1. If $\alpha \in \omega_1 \setminus H$ then $\alpha = 0$ or $\alpha = \beta + 1$ for some $\beta < \omega_1$. So α is isolated. Claim any open superset of H contains all but ω-many points of ω_1. Well, let $O \subset \omega_1$ be open containing H. Then $H \cap (\omega_1 \setminus O) = \varnothing$ and $\omega_1 \setminus O$ is closed, so it can't be unbounded and miss H. Thus $\omega_1 \setminus O$ is countable. Let $\{O_n : n < \omega\}$ be a collection of open sets, each containing H. For each $n < \omega$ there exists $\alpha_n < \omega_1$ such that $O_n \supset [\alpha_n, \omega_1)$. Then $\bigcap_{n<\omega} O_n \supset [\alpha, \omega_1)$ where $\alpha = \sup_{n<\omega} \alpha_n$. So $\bigcap_{n<\omega} O_n \neq H$. So H is not G_δ and hence ω_1 is not perfectly normal. ■

Proof of Theorem 195

Let C be a closed unbounded set in ω_1. Because C is well-ordered, there is a least $c_0 \in C$. Send c_0 to 0. Similarly, there is a least $c_1 \in C \setminus \{c_0\}$; send c_1 to 1. Now, if c_β has been defined for all $\beta < \alpha$, where $\alpha < \omega_1$, put $c_\alpha = \min(C \setminus \{c_\beta : \beta < \alpha\})$ and send c_α to α.

Let f be the map just constructed. Certainly f is bijective. Let $(\beta, \alpha] \subset \omega_1$ be basic open. We claim that $f^{-1}[(\beta, \alpha]] = (c_\beta, c_\alpha] \cap C$ is basic open in C. Note that if α is a limit ordinal, then $c_\alpha = \min(C \setminus \{c_\beta : \beta < \alpha\}) = \sup\{c_\beta : \beta < \alpha\}$ is a limit of $C \cap (0, c_\alpha)$. If α is not a limit, $(c_{\alpha-1}, c_\alpha] \cap C = \{c_\alpha\} = f^{-1}[\{\alpha\}]$. The claim follows. ■

Proof of Theorem 196

Suppose α is a limit of C not in C. Suppose $\beta < \alpha$. Then $(\beta, \alpha] \cap C \neq \varnothing$. There is $\gamma \in (\beta, \alpha) \cap C$. Then $f(\beta) < \gamma < \alpha$. Therefore $\alpha \in C$, a contradiction, so C is closed.

Let $\gamma < \omega_1$. We need to show there is an $\alpha \in C$ such that $\alpha > \gamma$. Pick

$\alpha_0 > \gamma$. Pick $\alpha_1 > \sup\{f(\beta) : \beta < \alpha_0\}$. Pick $\alpha_2 > \sup\{f(\beta) : \beta < \alpha_1\}$. Having already picked α_n, pick $\alpha_{n+1} = \{f(\beta) : \beta < \alpha_n\}$. Let $\alpha = \sup\{\alpha_n : n < \omega\}$. Suppose $\beta < \alpha$. Then $\beta < \alpha_n$ for some n, hence $f(\beta) < \alpha_{n+1} < \alpha$. So $\alpha \in C$. ■

Proof of Theorem 197

Let f be pressing down. Suppose $|f^{-1}(\alpha)| \leqslant \omega$ for all α. Define $g\colon \omega_1 \to \omega_1$ such that for all α, $g(\alpha) > \sup \bigcup_{\beta \leqslant \alpha} f^{-1}(\beta)$. Let C be as in Theorem 196: $C = \{\alpha : \forall \beta < \alpha (g(\beta) < \alpha)\}$. There is an $\alpha \in C \cap S$. Let $\beta = f(\alpha)$. Then $\beta < \alpha$, but $g(\beta) > \alpha$, a contradiction to $\alpha \in C$. ■

Proof of Proposition A

Suppose not. For every $S \subset \omega_1$, either S or $\omega_1 \setminus S$ is nonstationary, implying either S or $\omega_1 \setminus S$ contains a club. We will show why this is not possible.

Identify ω_1 with a subset of $[0, 1]$. Either $[0, 1/2]$ or $[1/2, 1]$ contains a club. Without loss of generality we may assume $[1/2, 1]$ contains a club. Consider $[1/2, 3/4]$ and $[3/4, 1]$. One of them must contain a club: say $[3/4, 1]$ is nonstationary. Then $[0, 1/2] \cup [3/4, 1]$ is nonstationary, and $[1/2, 3/4]$ contains a club. Similarly, $[1/2, 5/8]$ or $[5/8, 3/4]$ contains a club. Inductively, get $I_1 \supset I_2 \supset I_3 \supset \cdots$ such that for all n, I_n contains a club with $\operatorname{diam} I_n = 1/2^{n-1}$. Therefore $\bigcap I_n$ contains a club, but $|\bigcap I_n| = 1$. ■

Proof of Theorem 198

Let $\mathcal{U} = \{[0, \alpha] \cap S : \alpha < \omega_1\}$ be an open cover of S and let $\mathcal{V}$ be an open refinement thereof. Let C be the set of all points α in ω_1 that are limit points of S. Then C is closed unbounded. Let $S' = C \cap S$; it is easily check that S' is stationary.

For each $\alpha \in S'$, there exists $V \in \mathcal{V}$ and $f(\alpha) \in S$ such that $f(\alpha) < \alpha$ and $[f(\alpha), \alpha] \cap S \subset V$. By the Pressing Down Lemma, there exists $\beta \in S$ such that $f(\alpha) = \beta$ for uncountably many $\alpha \in S'$.

It suffices to show that β is a member of more than finitely many members of $\mathcal{V}$. To that end, suppose on the contrary that β is a member of only finitely many members of $\mathcal{V}$, say $V_{\alpha_1}, \ldots, V_{\alpha_n}$. Let $\gamma < \omega_1$ such that $\gamma > \sup \bigcup_{i=1}^{n} V_{\alpha_i}$. Then there exists $\delta \in S'$ such that $\delta > \gamma$ and $f(\delta) = \beta$. It follows that $[\beta, \delta] \cap S \not\subset V_{\alpha_i}$ for $i = 1, \ldots, n$. Thus $\beta \in V_\delta$

with $V_\delta \neq V_{\alpha_i}$ for all $1 \leqslant i \leqslant n$, a contradiction.

Since β is contained in more than finitely many members of $\mathcal{V}$, $\mathcal{V}$ is not point-finite. It follows that $\mathcal{U}$ has no locally-finite open refinement, whence S is not paracompact.

The same proof shows that there is no point-finite or even point-countable open refinement of $\mathcal{U}$, and hence a stationary set S is not even metacompact or metalindelöf. ■

Proof of Theorem 199

Proof of Theorem 200

Let $S \subset \omega_1$. Metrizability of S implies its paracompactness, and by Theorem 198 paracompactness of S implies S is nonstationary. It remains to prove that if S is nonstationary, then it is metrizable. If S is nonstationary, there exists a club C such that $C \cap S = \varnothing$. By Theorem 199, $\omega_1 \setminus C = \bigcup_{\alpha \in I} M_\alpha$ with the M_α convex open disjoint. As a countable subspace of a regular first-countable space, each M_α is regular and second-countable. So each M_α is metrizable by a metric d_α with max 1. It is easily check that the function d given by $d(x, y) = d_\alpha(x, y)$ if $x, y \in M_\alpha$, $d(x, y) = 2$ if $x\ in M_\alpha$ and $y \in M_\beta$ with $\alpha \neq \beta$, is a metric on $\omega_1 \setminus C$ which induces the subspace topology on $\omega_1 \setminus C$. ■

Proof of Example 201

Proof of Lemma 202

(a) This is essentially equivalent to $\kappa = |\kappa \times \kappa|$. Suppose it is true for all cardinals less than κ. Define a well-order $\prec$ on $\kappa \times \kappa$ as follows. Let $\langle \alpha, \beta \rangle$ and $\langle \gamma, \delta \rangle$ be distinct points in $\kappa \times \kappa$. Define $\langle \alpha, \beta \rangle \prec \langle \gamma, \delta \rangle$ if and only if $\max\{\alpha, \beta\} < \max\{\gamma, \delta\}$, or $\max \alpha, \beta = \max\{\gamma, \delta\}$ and $\langle \alpha, \beta \rangle$ is less than $\langle \gamma, \delta \rangle$ is the lexicographic order. It is not difficult to check that this is a well-ordering. Let $\langle \gamma, \delta \rangle \in \kappa \times \kappa$ and suppose $\max\{\gamma, \delta\} = \mu$. The set of predecessors of $\langle \gamma, \delta \rangle$ has cardinality at most $|\mu \times \mu| = |\mu|$ by the induction hypothesis, which is less than κ. But κ is the only ordinal such that cardinality of every predecessor is less than κ, while

the cardinality of the whole set is not. Hence $|\kappa \times \kappa| = \kappa$.

For each $\alpha < \kappa$, let $|A_\alpha| \leqslant \kappa$. Let $f_\alpha \colon \kappa \to A_\alpha$ be onto. Then $F \colon \kappa \times \kappa \to \bigcup_{\alpha<\kappa} A_\alpha$ by $F(\alpha, \beta) = f_\alpha(\beta)$ is onto. Hence $|\bigcup_{\alpha<\kappa} A_\alpha| \leqslant |\kappa \times \kappa| \leqslant \kappa$.

(b) Suppose $\kappa = \lambda^+$, but that $\mu = \operatorname{cf} \kappa < \kappa$. Note that $\mu \leqslant \lambda$. Let $f \colon \mu \to \kappa$ be unbounded in κ. Then $\{f(\alpha) : \alpha < \mu\}$ is a collection of $\leqslant \lambda$-many sets each of cardinality $\leqslant \lambda$, so its union has cardinality $\leqslant \lambda$. But its union is κ, contradiction.

(c) Suppose $\mu = \operatorname{cf} \lambda$ is not regular. Then $\nu = \operatorname{cf} \mu < \mu$. Let $f \colon \nu \to \mu$ be cofinal and nondecreasing, and $g \colon \mu \to \lambda$ cofinal and nondecreasing. It is easy to check that $g \circ f \colon \nu \to \lambda$ is cofinal, so $\operatorname{cf} \lambda \leqslant \nu$, contradiction. ■

Proof of Theorem 203

Let κ be an infinite cardinal.

(a → b) Suppose $A \subset \kappa$ with $|A| < \kappa$. Let $\lambda = |A|$ and enumerate $A = \{a_\gamma : \gamma < \lambda\}$. Define $f \colon \lambda \to \kappa$ by $\gamma \mapsto \alpha_\gamma$. Then $\sup A = \sup_{\gamma<\lambda} f(\gamma) < \kappa$.

(b → c) Suppose $|A_\gamma| < \kappa$ for $\gamma < \lambda < \kappa$. Without loss of generality, we may assume $A_\gamma \subset \kappa$ for each $\gamma < \lambda$ and that the A_γ are pairwise-disjoint. By assumption, $\sup A_\gamma < \kappa$ for each $\gamma < \lambda$. Then $\sup_{\gamma<\lambda}(\sup A_\gamma) < \kappa$, so

$$|\bigcup_{\gamma<\lambda} A_\gamma| \leqslant |\sup_{\gamma<\lambda}(\sup A_\gamma)| \leqslant \sup_{\gamma<\lambda}(\sup A_\gamma) < \kappa.$$

(c → a) Suppose $\lambda < \kappa$ and let $f \colon \lambda \to \kappa$. For $\gamma < \lambda$, define $A_\gamma = f(\gamma)$. (Recall that an ordinal *is* the set of its predecessors.) Then $|\bigcup_{\gamma<\lambda} A_\gamma| < \kappa$, implying f is bounded in κ. ■

Proof of Theorem 204

Proof of Theorem 205
Suppose κ is regular and $S \subset \kappa$ is stationary. Let $\mathcal{U} = \{[0, \alpha] \cap S : \alpha \in S\}$, and suppose $\mathcal{V}$ is a locally finite open refinement of $\mathcal{U}$. For each $\alpha \in S$ there exists $\beta_\alpha < \alpha$ such that $[\beta_\alpha, \alpha] \cap S \subset V_\alpha \in \mathcal{V}$. By Theorem 204(c) there exists β with $\beta_\alpha = \beta$ for κ-many $\alpha \in S$. On the other hand, only finitely many members of $\mathcal{V}$ contain β. There exists $\alpha \in S$ such that $\beta_\alpha = \beta$ and $\alpha > \sup \bigcup \{V \in \mathcal{V} : \beta \in V\}$. But there exists $V' \in \mathcal{V}$ such that $[\beta, \alpha] \subset V'$, contradicting $\alpha > \sup \bigcup \{V \in \mathcal{V} : \beta \in V\}$. ■

Proof of Theorem 206
Suppose X is connected and $\mathcal{U}$ is an open cover of X. Let $\mathcal{U}(U)$ denote those $V \in \mathcal{U}$ for which there is a finite linked chain between U and V.

Claim. *$O = \bigcup \mathcal{U}(U)$ is connected.*

Proof of Claim. Clearly O is open. We first show it is also closed. Suppose x is a limit point of O. Since $\mathcal{U}$ is an open cover, there exists $U_x \in \mathcal{U}$ such that $x \in U_x$. Then $U_x \cap O \neq \varnothing$, so there exists $V \in \mathcal{U}(U)$ such taht $U_x \cap V \neq \varnothing$. Thus we may extend a finite linked chain between U and V to a finite linked chain between U and U_x, so that $x \in U_x \subset O$. □

Since X is connected and O is clopen, we must have $O = X$. Let $U_1, U_2 \in \mathcal{U}$. There exist $V_1, V_2 \in \mathcal{U}(U)$—note that $\mathcal{U}(U)$ covers X—such that $V_1 \cap U_1$ and $V_2 \cap U_2$ are both nonempty. Extend chains between V_1 and U, and V_2 and U, to a chain between U_1 and U_2.

If U and V are nonempty disjoint clopen sets $U \cup V = X$, then $\mathcal{U} = \{U, V\}$ is an open cover of X which is not connected. ■

Proof of Theorem 207

Proof of Lemma 208
(a → b) Trivial.

(b → c) Trivial.

(c → a) Trivial.

(c → d) Let $\{x_\alpha : \alpha < \omega_1\} \subset X$ such that $x_\alpha \notin \overline{\{x_\beta : \beta > \alpha\}}$ for all $\alpha \in \omega_1$. Then for each $\alpha \in \omega_1$ there exists an open neighborhood U_α of x_α such that $U_\alpha \cap \{x_\beta : \beta > \alpha\} = \varnothing$. Take $\mathcal{U} = \{U_\alpha : \alpha < \omega_1\}$. If $\mathcal{V} \subset \mathcal{U}$ is countable, $\bigcup \mathcal{V}$ may contain only countably many members of $\{x_\alpha : \alpha < \omega\}$, so $\bigcup \mathcal{V} \neq \bigcup \mathcal{U}$.

(d → c) Let $\mathcal{U}$ be a collection of open subsets of X such that $\bigcup \mathcal{V} \neq \bigcup \mathcal{U}$ for all countable $\mathcal{V} \subset \mathcal{U}$. Pick $U_0 \in \mathcal{U}$ and $x_0 \in U_0$. Let $0 < \alpha < \omega_1$ and suppose $U_\beta \in \mathcal{U}$ and $x_\beta \in U_\beta$ have been defined for all $\beta < \alpha$. Since $\bigcup \{U_\beta : \beta < \alpha\}$ is a proper subset of $\bigcup \mathcal{U}$, there exists $U_\alpha \in \mathcal{U}$ such that $U_\alpha \setminus \bigcup \{U_\beta : \beta < \alpha\} \neq \varnothing$. Pick $x_\alpha \in U_\alpha \setminus \bigcup \{U_\beta : \beta < \alpha\}$ and observe that for each $\beta < \alpha$, U_β is a neighborhood of x_β not containing x_α. It follows that $\{x_\alpha : \alpha < \omega_1\}$ is a subset of X such that $x_\alpha \notin \overline{\{x_\beta : \beta > \alpha\}}$ for all $\alpha \in \omega_1$. ■

Proof of Theorem 209

Proof of Theorem 210

(a) Let X be a *ccc* nonseparable linearly ordered space. Define $x \sim y$ if and only if the interval from x to y is separable. Note that equivalence classes are convex. We claim that the set $S = \{[x] : x \in X\}$ of equivalence classes with the natural order, denoted by $\prec$, satisfies the desired conditions.

It follows easily that S is *ccc* because X is. To see that S is densely ordered, suppose $([x], [y])$ is empty. Then (x, y) is separable and $[x] = [y]$, a contradiction.

To complete the proof, we need to show that S has no separable open intervals. First we show that every equivalence class $[x]$ is separable. It follows from the *ccc* of X that $\operatorname{cf}[x]$ is countable, and similarly so is the coinitiality of $[x]$. Let us assume $[x]$ has a greatest point but no least point; other cases can be handled similarly. Let $\langle y_n \rangle$ be a decreasing sequence of points of $[x]$ which is coinitial in $[x]$. Then $[x] = \bigcup_{n<\omega} (y_n, z]$, where z is the greatest point of $[x]$. Since each (y_n, z) is separable, so is $[x]$. Now suppose $([x], [y])$ is separable, say with dense set $D = \{[z_i] : i < \omega\}$. For each i, let E_i be a dense (in X) subset of

$[z_i]$. It follows that $(x, y) \cap [\bigcup_{i<\omega} E_i]$ is dense in (x, y), hence $[x] = [y]$, a contradiction.

(b) Let X be a *ccc* nonseparable linearly ordered space. By 210(a), we may also assume X is densely ordered and that no nonempty open interval in X is separable. Let $\hat{X}$ be the compactification of X as in Theorem 191.

We claim that $\hat{X}$ is densely ordered. Suppose not. Let $a, b \in \hat{X}$ with $(a, b) = \varnothing$. If $a \in \hat{X} \setminus X$ and $b \in X$, then $(-\infty, a) \cap X$ is a left-closed subset of X. Note that in X, every neighborhood of b contains an interval $(c, b]$, where $c < b$. But then $c \in (-\infty, a)$ and since X is densely ordered, $(c, b) \cap (= \infty, a) \neq \varnothing$. So b is a limit point of $(-\infty, a)$, a contradiction to it being left-closed.

If $a, b \in \hat{X} \setminus X$, then $(-\infty, a) \cap X = (-\infty, b) \cap X$, so these subsets are identified with the same element of $\hat{X}$, forcing $a = b$. If $a \in X$ and $b \in \hat{X} \setminus X$, then $(-\infty, a] \cap X = (-\infty, b) \cap X$, giving $a = b$ as above. So $\hat{X}$ is densely ordered.

By Theorem 190, every subset of $\hat{X}$ has a least upper bound. Because $\hat{X}$ is densely ordered and every bounded subset of $\hat{X}$ has a least upper bound, $\hat{X}$ is connected by Theorem 183. Throw out the first and last points of $\hat{X}$ to get a space X' that is connected, *ccc*, and has a subspace that is not separable. It follows that X' is not homeomorphic to $\mathbb{R}$ and is thus a Suslin line. ■

Proof of Theorem 211

Since $\mathbb{C} = \bigcap_{n \in \mathbb{N}} C_n$ and each C_n is a finite union of closed sets, then C_n is closed and thus $\mathbb{C}$ is closed. $\mathbb{C}$ is compact since it is a closed subset of the compact space I. $\mathbb{C}$ is metrizable since it is a subspace of the metrizable space $\mathbb{R}$.

Claim 1 : $I_\sigma \cap \mathbb{C}$ is clopen. Let $\mathcal{B} = \{I_\sigma \cap \mathbb{C} : \sigma \in 2^{<\omega}\}$. For a given σ, $I_\sigma \cap \mathbb{C}$ is closed since it is the intersection of two closed subsets. Also, $C_n \setminus I_\sigma$ is closed since C_n is a finite union of disjoint closed intervals one of which is I_σ, so $\mathbb{C} \setminus (\mathbb{C} \cap I_\sigma) = \mathbb{C} \cap (C_n \setminus I_\sigma)$ is closed, which gives us $\mathbb{C} \cap I_\sigma$ is open.

Claim 2: $\mathcal{B}$ is a base for $\mathbb{C}$, and $\mathbb{C}$ has empty interior in $\mathbb{R}$. Let U be any open set in $\mathbb{C}$ and let $x \in U$. There some $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \cap \mathbb{C} \subset U$. Pick some $n \geqslant 1$ with $1/3^n < \varepsilon$. Since $x \in C_n$, $x \in I_\sigma$ for some σ of length n. Notice $\operatorname{diam} I_\sigma = 1/3^n$. So $x \in \mathbb{C} \cap I_\sigma \subset \mathbb{C} \cap (x-\varepsilon, x+\varepsilon) \subset U$.

Now if U were open in $\mathbb{R}$ and contained in $\mathbb{C}$, then as above we would have $I_\sigma \subset U \subset \mathbb{C}$ for some σ. But the middle third of I_σ is disjoint from $\mathbb{C}$, contradiction. Thus $\mathbb{C}$ has empty interior in $\mathbb{R}$.

Claim 3: $\mathbb{C}$ is uncountable. For each $f\colon \omega \to 2$, there is exactly one point x_f in the intersection of the branch coded by f; that is, $\{x_f\} = \bigcap_{n<\omega} I_{f\restriction n} \cap \mathbb{C}$. Suppose $f, g \in 2^\omega$ and $f \neq g$. Let n be minimal such that $f(n) \neq g(n)$. Then $x_f \in I_{f\restriction n+1}$, $x_g \in I_{g\restriction n+1}$, and $I_{f\restriction n+1} \cap I_{g\restriction n+1} = \varnothing$. Hence $x_f \neq x_g$. Thus the map $f \mapsto x_f$ is one-to-one, and so $\mathbb{C}$ is uncountable (in fact, it has the same cardinality as the real line).

Claim 4: $\mathbb{C}$ has no isolated points. Let $\sigma \in 2^n$. If f and g are distinct members of 2^ω such that $f \restriction n = g \restriction n = \sigma$, then x_f and x_g are distinct points in $I_\sigma \cap \mathbb{C}$. So every base element has at least two points, hence $\mathbb{C}$ has no isolated points. ■

Proof of Theorem 212

Let X be a space with a base $\mathcal{B} = \{B_\sigma : \sigma \in 2^{<\omega}\}$ of clopen sets isomorphic to the Cantor tree such that the intersection of any branch is a single point. For each $c \in \mathbb{C}$, there is a unique $\sigma_c \in 2^\omega$ such that $\{c\} = \bigcap_{n<\omega} I_{\sigma_c\restriction n}$. By assumption $\bigcap_{n<\omega} B_{\sigma_c\restriction n}$ contains a single element, say x_c. Define $h\colon \mathbb{C} \to X$ by $c \mapsto x_c$.

Claim 1: h is surjective. Suppose not. Then there exists $x \in X$ and $\sigma \in 2^{<\omega}$ such that $x \in B_\sigma$ but $x \notin B_{\sigma\frown 0} \cup B_{\sigma\frown 1}$. Then $B_\sigma \setminus (B_{\sigma\frown 0} \cup B_{\sigma\frown 1})$ is an open neighborhood of x not containing any base elements, a contradiction.

Claim 2: h is injective. Suppose note. Then there exists $\sigma, \tau \in 2^\omega$ and $x \in X$ such that $\sigma \neq \tau$ but

$$\bigcap_{n<\omega} B_{\sigma\restriction n} = \{x\} = \bigcap_{n<\omega} B_{\tau\restriction n}.$$

Let $k < \omega$ such that $\sigma \restriction k = \tau \restriction k$ but $\sigma(k) \neq \tau(k)$; put $\upsilon = \sigma \restriction k$. Then $x \in B_{\upsilon\frown 0} \cap B_{\upsilon\frown 1}$, so there exists $u \in 2^{<\omega}$ such that $x \in B_u \subset B_{\upsilon\frown 0} \cap B_{\upsilon\cap 1}$. Thus $\upsilon\frown 0$ and $\upsilon\frown 1$ are initial segments of u, a contradiction. ■

Proof of Theorem 213

Proof of Theorem 214

(a) Define a Cantor tree of closed subintervals of $[0,1]$ as follows: $J_\varnothing = [0,1]$, and if $J_\sigma = [\ell, r]$ has been defined for $\sigma \in 2^n$, let $J_{\sigma\frown 0}$ and $J_{\sigma\frown 1}$ be $[\ell, m]$ and $[m, r]$, respectively, where m is the midpoint of $[\ell, r]$. Note that each branch of this tree is a single point, and every point of $[0,1]$ is the intersection of some branch. The map that sends the intersection of a branch of the Cantor tree to the intersection of the corresponding branch of this tree for $[0,1]$ is easily seen to be continuous and onto.

It is not a bijection—it can't be, else it'd be a homeomorphism—but it is two-to-one on a countable set and one-to-one otherwise.

(b) Let f be a continuous function from $\mathbb{C}$ onto $[0,1]$. For $\vec{x} \in \mathbb{C}^\omega$, let $f(\vec{x}) = \langle f(x_n)\rangle \in [0,1]^\omega$. Then f is clearly continuous and onto. Since $\mathbb{C}^\omega \cong \mathbb{C}$, we are done.

(c) If K is a closed subset of $\mathbb{C}$, it is easily seen that $K \times \mathbb{C}$ is compact, has no isolated points, and has a countable base of clopen sets. Hence $K \times \mathbb{C}$ is homeomorphic to $\mathbb{C}$. The projection onto the first coordinate is a continuous map onto K.

(d) Let M be any compact metric space. Then M is homeomorphic to a (compact, hence closed) subset N of the Hilbert cube. Let $f\colon \mathbb{C} \to [0,1]^\omega$ be continuous and onto. Let $K = f^{-1}[N]$. Then K is a closed subset of $\mathbb{C}$, so by 214(c), there is a continuous surjection $g\colon \mathbb{C} \to K$. Then $f \circ g\colon \mathbb{C} \to N$ is continuous and onto. Since $N \cong M$ we are done. ■

Proof of Exercise 215

Let $q_1, q_2, \ldots$ be an enumeration of the rationals with $q_1 = 0$. We define clopen subsets $\{B_\sigma : \sigma \in \omega^{<\omega}\}$ of $\mathbb{P}$ such that

(1) $B_\varnothing = \mathbb{P}$;

(2) for each σ, the collection $\{B_{\sigma\frown n} : n < \omega\}$ is a clopen partition of B_σ; and

(3) if σ has length n, then $\operatorname{diam} B_\sigma \leqslant 1/n$.

If we do the above, the resulting B_σ's will form a base which under $\supset$ is a tree isomorphic to $\langle \omega^{<\omega}, \leqslant \rangle$. There is one extra thing we need to be concerned about: how do we know the intersectino of each branch is a single point? (It could be empty.)

To start, of course we set $B_\varnothing = \mathbb{P}$. The collection $\{(n, n+1) \cap \mathbb{P} : n \in \mathbb{Z}\}$ is a countable clopen partition of $\mathbb{P}$, so we may let $\{B_n : n < \omega\}$ index it. As we continue, each B_σ will have the form $(a, b) \cap \mathbb{P}$ for some rationals a, b. We want to make sure that each rational is included as an endpoint at some stage. Note that since $q_1 = 0$, q_1 is taken care of. We will make sure q_n gets taken care of at or preceding the construction of the nth level of the tree.

So, suppose B_σ has been defined for all σ of length $\leqslant n$. Let σ have length n. We show how to construct $B_{\sigma\frown n}$ for $n < \omega$. We have assumed $B_\sigma = (a, b) \cap \mathbb{P}$ for some rationals a, b. Let $\langle a_k \rangle$ be a decreasing sequence of rationals in (a, b) converging to a, and $\langle b_k \rangle$ an increasing sequence of rationals in (a, b) converging to b. We may assume we make the choice such that $a_0 < b_0$, and that the intervals (a_0, b_0), (a_{n+1}, a_n), and (b_n, b_{n+1}) for $n < \omega$ have diameter $\leqslant 1/n + 1$. We may also assume that if the nth q_n lies between a and b, then q_n is one of the a_n's or b_n's. Note that if q_n was not already an endpoint of some chosen interval, then it will be taken care of at this stage. Finally, let $\{B_{\sigma\frown n} : n < \omega\}$ be a listing of

$$\{(a_0, b_0) \cap \mathbb{P}\} \cup \{(a_{n+1}, a_n) \cap \mathbb{P} : n < \omega\} \cup \{(b_n, b_{n+1}) \cap \mathbb{P} : n < \omega\}.$$

Clearly the B_σ's satisfy our requirements above. We need to see that the intersection of each branch is a singleton. Note that each branch corresponds to a decreasing sequence $I_0 \cap \mathbb{P}, I_1 \cap \mathbb{P}, \ldots$ where

the endpoints of the I_n's are rational and the diameters go to 0. In the whole real line, $\bigcap_{n<\omega} I_n$ is a single point, say x. If x is irrational, we are done. Suppose x is rational. At some stage, say n, x becomes an endpoint of one of the intervals. So it is either an endpoint of I_n, or an endpoint of some disjoint interval. In either case, $x \notin \bigcap_{n<\omega} I_n$, and so x must be irrational. ■

Proof of Theorem 216

(a → b) By Exercise 215.

(b → c) Suppose X has a base $\mathcal{B}$ such that $\langle \mathcal{B}, \supset \rangle \cong \langle \omega^{<\omega}, \leqslant \rangle$ and the intersection of each branch of $\langle \mathcal{B}, \supset \rangle$ is a single point. We show X

(1) has a countable base of clopen sets, implying it is separable;

(2) is nowhere locally compact; and

(3) is completely metrizable.

(1) $\omega^{<\omega}$ is countable.

(2) Let $\mathcal{B} = \{B_\sigma : \sigma \in \omega^{<\omega}\}$ be an enumeration of $\mathcal{B}$ such that $B_\sigma \supset B_\tau$ if and only if τ extends σ. Note that for each $\sigma \in \omega^{<\omega}$, $\{B_{\sigma\frown j} : j < \omega\}$ partitions B_σ: each $B_{\sigma\frown j}$ is contained in B_σ since $\sigma\frown j$ extends σ. We now show $\{B_{\sigma\frown j} : j < \omega\}$ covers B_σ. Suppose $x \in B_\sigma$. If $i \neq j < \omega$, then the subsets $B_{\sigma\frown i}$ and $B_{\sigma\frown j}$ of B_σ are incomparable under $\supset$. Thus B_σ contains more than one point. Let $y \in B_\sigma$ with $y \neq x$. Since $\mathcal{B}$ is a base, there exists $B \in \mathcal{B}$ with $x \in B$ and $y \notin B$ and there exists $B_\tau \subset B \cap B_\sigma$ with $x \in B_\tau$. B_τ is a proper subset of B_σ containing x, so τ properly extends σ and $x \in B_{\tau\restriction n+1} \in \{B_{\sigma\frown j} : j < \omega\}$. Thus $\{B_{\sigma\frown j} : j < \omega\}$ covers B_σ. Finally, if $i \neq j < \omega$ we have $B_{\sigma\frown i} \cap B_{\sigma\frown j} = \varnothing$. Otherwise there exists $B_\tau \subset B_{\sigma\frown i} \cap B_{\sigma\frown j}$. Then τ extends both $\sigma\frown i$ and $\sigma\frown j$, which is impossible.

Now we are ready to show X is nowhere locally compact. Suppose U is a neighborhood. Then U has nonempty interior. There exists $B_\sigma \subset U^\circ \subset U$. If U is compact then since B_σ is closed, we have B_σ is compact. But $\{B_{\sigma\frown j} : j < \omega\}$ is an open cover of B_σ with no finite subcover.

(3) For $x, y \in X$ let $L_{x,y} = \{i \in \omega : \exists \sigma \in \omega^i (x, y \in B_\sigma)\}$. Define $d\colon X \times$

$X \to [0, \infty)$ by

$$d(x,y) = \begin{cases} 1/\max L_{x,y} & \text{if } L_{x,y} \text{ is bounded and } \max L_{x,y} > 0, \\ 1 & \text{if } L_{x,y} \text{ is bounded and } \max L_{x,y} = 0, \\ 0 & \text{if } L_{x,y} \text{ is unbounded.} \end{cases}$$

d **is a metric.** If $x = y$, then x and y may not be separated by sets in $\mathcal{B}$, so $L_{x,y}$ is unbounded, so $d(x,y) = 0$. Conversely, if $x \neq y$ then there exists $\sigma, \tau \in \omega^{<\omega}$ with $x \in B_\sigma$, $y \in B_\tau$, and $B_\sigma \cap B_\tau = \varnothing$. Then x and y are separated at level $\max\{\operatorname{dom}\sigma, \operatorname{dom}\tau\}$, so $L_{x,y}$ is bounded. Hence $d(x,y) \neq 0$.

Trivially, $d(x,y) = d(y,x)$.

The triangle inequality is trivial to see if $d(x,y) = 0$, so suppose $d(x,y) = 1/n$. Then x and z or y and z are contained in disjoint basic sets at level $n+1$ (say x and z). So there is a max level $m \leqslant n$ at which x and z are together. Thus $d(x,y) = 1/n \leqslant 1/m = d(x,z) \leqslant d(x,z) + d(y,z)$.

Now we show that the topology induced by d coincides with the topology induced by $\mathcal{B}$ (the original topology on X.) It is easy to see that $B(x, 1/n) = B_\tau$, where $\tau \in \omega^{n+1}$ is such that $x \in B_\tau$. So a metric ball contains a set in $\mathcal{B}$. Conversely, if $\varnothing \neq \sigma \in \omega^{<\omega}$ and $x \in B_\sigma$, then $B(x, 1/\operatorname{dom}\sigma) = B_{\sigma\frown i} \subset B_\sigma$, with i such that $x \in B_{\sigma\frown i}$. If $\sigma = \varnothing$ then $B_\sigma = X$ so any metric ball is contained in it.

d **is complete.** Suppose $\langle x_n \rangle \in X^\omega$ is a Cauchy sequence with respect to d. There exists an increasing sequence $\langle m_n \rangle \in \omega^\omega$ such that for each $n, k \geqslant m_n$ impies $d(x_{m_n}, x_k) < 1/2^{n+1}$. The balls $B(x_{m_n}, 1/2^n)$, $n < \omega$, are nested (decreasing). Indeed, if $n < \omega$ and $x \in B(x_{m_{n+1}}, 1/2^{n+1})$ then we have

$$d(x, x_{m_n}) \leqslant d(x, x_{m_{n+1}}) + d(x_{m_n}, x_{m_{n+1}}) < 1/2^{n+1} + 1/2^{n+1} = 1/2^n,$$

whence $x \in B(x_{m_n}, 1/2^n)$. Thus the balls $B(x_{m_n}, 1/2^n)$ correspond to a brance in the tree $\langle \mathcal{B}, \supset \rangle$, which intersects to a point x. Let $\varepsilon > 0$ and let $n \in \mathbb{N}$ such that $2/2^n < \varepsilon$. For $k \geqslant m_n$ we have

$$d(x, x_k) \leqslant d(x_{m_n}, x) + d(x_{m_n}, x_k) < 1/2^n + 1/2^{n+1} < 2/2^n < \varepsilon,$$

so x is the limit of $\langle x_n \rangle$.

(c → a) Suppose X is nowhere locally compact, completely metrizable, and has a countable base $\mathcal{B}$ of clopen sets.

Claim. *For every nonempty open $U \subset X$ and $n \in \mathbb{N}$ there exists a partition of U into ω-many nonempty clopen sets, each with diameter less than $1/n$.*

Let $U \subset X$ nonempty open and $n \in \mathbb{N}$. Using the fact that X is nowhere locally compact, $\mathcal{U}_n = \{B \in \mathcal{B} : \operatorname{diam} B < 1/n \wedge B \subset U\}$ is an open cover of U with no finite subcover. Enumerate $\mathcal{U}_n = \{B_0, B_1, \ldots\}$. After throwing away the empty elements in $\mathcal{U}_n^* = \{B_n \setminus \bigcup_{i<n} B_i : n < \omega\}$, we have the desired partition.

Using the claim, recursively construct a tree of clopen sets isomorphic to $\langle \omega^{<\omega}, \leqslant \rangle$. Each branch of this tree intersects to at most one point since its nodes have diameters shrinking to 0. The intersection of each branch is nonempty: create a Cauchy sequence by selecting a sequence up the branch. It converges by completeness, to a point which must be in the intersection of the branch. Construct a homeomorphism between X and the space of irrationals by mapping branches to branches, as in Proof 213. ■

Proof of Corollary 217

Proof of Theorem 218

Let T be a tree of height ω such that L_α is finite for each $\alpha < \omega$. For each $t \in T$, put $S_t = \{s \in T : t < s\}$. Since T has height ω, there must be infinitely many nodes in T.

Pick $b_0 \in L_0$ so that S_{b_0} is infinite. This is possible since L_0 is finite. Suppose for some $n < \omega$ that $b_0 < \cdots < b_n$ have been defined so that $b_i \in L_i$ and S_{b_n} is infinite. Pick $b_{n+1} \in S_{b_n} \cap L_{n+1}$ such that $S_{b_{n+1}}$ is infinite. This is possible since $S_{b_n} \cap L_{n+1}$ is finite.

Certainly $b = \{b_n : n < \omega\}$ is an infinite branch of T. ■

Proof of Theorem 219

We construct the tree T by induction, along with a function $\theta\colon T \to \mathbb{Q}$ which is increasing in the sense that $x < y \in T$ implies $\theta(x) < \theta(y)$. This function will guarantee that there are no branches all the way through the tree, and will also tell us some things about the structure of T. (See the comment after the proof.)

For each $\alpha < \omega_1$, we let T_α denote the set of nodes of T of height $< \alpha$. To start, let $T_\omega = \omega^{<\omega}$. Let $\theta(\varnothing) = 0$. The first level L_1 of T_ω—that is, the set of immediate successors of $\varnothing$—is countable infinite, so we can let $\theta \restriction L_1$ be a bijection from L_1 to the positive rationals. Given $t \in L_1$ with $\theta(t) = q_t$, let θ send the immediate successors of t one-to-one and onto the set of rationals $> q_t$. And so on.

From here on, we denote $\theta(t)$ by q_t.

Next we define $T_{\omega+\omega}$ as follows. For each $s \in T_\omega$ and $q \in \mathbb{Q}$ with $q > q_s$, choose a branch $b(s,q)$ of T_ω with $s \in b(s,q)$ and $q = \sup\{q_t : t \in b(s,q)\}$. This is possible by the construction of θ on T_ω. Now let $\{T(s,q) : s \in T_\omega, q > q_s\}$ be a collection of copies of $\omega^{<\omega}$ disjoint from each other and from T_ω; we'll put $T(s,q)$ above the nodes of $b(s,q)$. let $T_{\omega+\omega} = T_\omega \cup \bigcup_{s \in T_\omega, q > q_s} T(s,q)$. Give the copies of $\omega^{<\omega}$ their usual order, and if $x \in T_\omega$ and $y \in T(s,q)$, define $x < y$ if and only if $x \in b(s,q)$. To help see the picture, note that if $\varnothing_{s,q}$ is the copy in $T(s,q)$ of the empty node of $\omega^{<\omega}$, then the set of predecessors of $\varnothing_{s,q}$ in $T_{\omega+\omega}$ is precisely $b(s,q)$. So $\varnothing_{s,q}$ stands at the top of $b_{s,q}$ at level ω, and $T(s,q)$ extends above it. Define $\theta(\varnothing_{s,q}) = q$, and extend θ to $T(s,q)$ level by level simlar to the way it was defined on T_ω.

Now suppose α is a limit ordinal greater than $\omega + \omega$, and T_β and $\theta \restriction T_\beta$ has been defined for all limit ordinals $\beta < \alpha$ such that the following holds:

($*$) If $\gamma < \delta < \beta$, s is in the γth level of T_β and $q \in \mathbb{Q}$ is such that $q > q_s$, then there is a successor t of s in the δth level of T_β with $q_t = q$.

It is easily checked that ($*$) holds for T_ω and $T_{\omega+\omega}$.

If α is a limit of limit ordinals, we simply let $T_\alpha = \bigcup_{\beta<\alpha} T_\beta$ and $\theta \restriction T_\alpha = \bigcup_{\beta<\alpha} \theta \restriction T_\beta$. It is easy to check that ($*$) holds with $\beta = \alpha$.

It remains to construct T_α for the limit ordinal α when there is a maximal limit ordinal $\beta < \alpha$, so $\alpha = \beta + \omega$. The construction in this

case is similar to the construction of $T_{\omega+\omega}$ from T_ω. First we show that for each $s \in T_\beta$ and $q > q_s$, there is a branch $b(s,q)$ of T_β which meets every level of T_β below β and such that $q = \sup\{q_t : t \in b(s,q)\}$. let $\langle \gamma_n \rangle$ be an increasing sequence of ordinals with supremum β, such that γ_0 is the level of $s_0 = s$. Let $\langle q_n \rangle$ be a sequence of rationals with $q_0 = q_s$ and $q = \sup\{q_n : n < \omega\}$. By $(*)$, there is a successor s_1 of s_0 at level γ_1 with $q_{s_1} = q_1$, then a successor s_2 of s_1 at level γ_2 with $q_{s_2} = q_2$, and so on. Let $b(s,q)$ be the branch determined by the sequence $\langle s_n \rangle$. Let $\{T(s,q) : s \in T_\beta \wedge q > q_s\}$ be a collection of copies of $\omega^{<\omega}$ disjoint from each other and from T_β and let $T_\alpha = T_\beta \cup \bigcup_{s \in T_\beta, q > q_s} T(s,q)$. Define the order on T_α and θ to T_α in the same way this was done for $T_{\omega+\omega}$ vis-à-vis T_ω.

This defines T_α for all $\alpha < \omega_1$. Let $T = \bigcup_{\alpha<\omega_1} T_\alpha$ and for each $t \in T$ let $\theta(t) = q_t \in \mathbb{Q}$ be as defined in the construction. Then it is immediate from the construction that T has height ω_1, every level of T is countable, and as noted earlier, since $s < t \to q_s < q_t$ there are no uncountable branches. ■

Comment. Let $\theta(t) = q_t$ be as in the proof, and for each $q \in \mathbb{Q}$ let $A_q = \{t \in T : q_t = q\}$. Note that each A_q is an antichain of T, so $T = \bigcup_{q \in \mathbb{Q}} A_q$ is the union of countably many antichains, at least one of which must be uncountable. So this tree is definitely not Suslin. Indeed, any Aronszajn tree which is the union of countably many antichains is called a *special* Aronszajn tree.

It is consistent with the axioms of set theory not only that there are no Suslin trees, but that every Aronszajn tree is special; for example, this holds under the axiom $\mathsf{MA}(\omega_1)$.

Proof of Theorem 220

Let X be a Suslin line. Define separable closed subsets C_α, $\alpha < \omega_1$, of X as follows. Let $C_0 = \varnothing$. If α is a limit ordinal and C_β has been defined for all $\beta < \alpha$, let $C_\alpha = \overline{\bigcup_{\beta<\alpha} C_\beta}$. If $\alpha = \gamma + 1$ and C_γ has been defined, let L_γ be the collection of convex components of $X \setminus C_\gamma$. For each $I \in L_\gamma$, choose a countable sequence of points converging to each endpoint of I. Let C'_γ be the collection of these chosen points for all $I \in L_\gamma$. Then let $C_\alpha = \overline{C_\gamma \cup C'_\gamma}$. Since the closure of the union of

countably many separable sets is separable, and C'_γ is countable, we see that each C_α is separable.

Now let L_α be the set of all convex components of $X \setminus C_\alpha$, and let $T = \bigcup_{\alpha<\omega_1} L_\alpha$. We claim that T ordered by $\supset$ is a Suslin tree.

Let $I \in T$. Then there is a unique α such that I is a convex component of $X \setminus C_\alpha$. Since the C_α's get bigger with α, if $I \subset J \in T$, then J is a convex component of $X \setminus C_\beta$ for some $\beta < \alpha$. Furthermore, for each $\beta < \alpha$ there is a unique such J, call it J_β. Then $\{J_\beta : \beta < \alpha\}$ is the set of predecessors of I and has order type α, so I is a member of level α of T. Since each C_α is separable, $X \setminus C_\alpha \neq \varnothing$, so T has a node at every level $\alpha < \omega_1$, and hence T has height ω_1.

Let L_α denote the αth level of T. Suppose $I \in L_\alpha$, $J \in L_\beta$, $\alpha \leqslant \beta$, and I and J are incomparable. There is a unique $J' \in L_\alpha$ with $J \subset J'$. Since I and J are incomparable, $I \neq J'$, but this means that $I \cap J' = \varnothing$, and $I \cap J = \varnothing$. Thus an antichain in T corresponds to a pairwise-disjoint collection of open sets, so every antichain of T is countable.

Note that by construction, each I in T has at least two immediate successors. We need to show that T has no uncountable chain. We show something more general: if T is a tree with no uncountable antichains, and each node of T has at least two immediate successors, then T has no uncountable chain. Suppose it did. Then there is a chain $\{t_\alpha : \alpha < \omega_1\}$ with $t_\alpha \in L_\alpha$. Let s_α be an immediate successor of $t_{\alpha+1}$ which is not equal to $t_{\alpha+1}$. It is easy to check that $\{s_\alpha : \alpha < \omega_1\}$ is an uncountable antichain, a contradiction. ■

Proof of Theorem 221

Let T be a Suslin tree and let $\prec$ be some linear order on T. Let X be the set of all branches of T. For each $b \in X$ let $b(\alpha)$ be the member of b at level α of the tree. Given $b_1 \neq b_2 \in X$, let α be least such that $b_1(\alpha) \neq b_2(\alpha)$ and define $b_1 < b_2$ if and only if $b_1(\alpha) \prec b_2(\alpha)$. Then clearly $<$ linearly orders X.

Claim. *X is ccc.*

Proof of Claim. Let $\mathcal{U} = \{(a_i, b_i) : i \in I\}$ be a collection of nonempty pairwise disjoint open intervals in X. For each $i \in I$ let $c_i \in (a_i, b_i)$, the let α be minimal such that $c_i(\alpha) \neq a_i(\alpha), b_i(\alpha)$ and let $t_i = c_i(\alpha + 1)$.

Then $\{t_i : i \in I\}$ is an antichain in T: if $i \neq j \in I$ and c is a branch containing t_i and t_j then $c \in (a_i, b_i) \cap (a_j, b_j)$, a contradiction. So I is countable by the *ccc* in T. $\square$

Claim. *X is nonseparable.*

Proof of Claim. Let $C \subset X$ be countable. Let $\kappa = \sup\{\operatorname{ht} b : b \in C\} + 1$. Then there exists $t \in T$ at level κ and three branches $a < b < c \in X$, each containing t. Then $a(\alpha) = b(\alpha)$ for each $\alpha < \kappa$ so any $d \in C$ lies on the same side of a and b, hence $d \notin (a, b)$. So (a, b) is a nonempty open set missing C, so C is not dense in X. ■

Proof of Theorem 222

Let S be a Suslin line and let $T = \bigcup_{\alpha<\omega_1} L_\alpha$ be as in the hint for Theorem 220. For each α, pick $I_\alpha \in L_\alpha$ and let $I^0_\alpha, I^1_\alpha \in L_{\alpha+1}$ be disjoint subsets of I_α.

Let $\alpha, \beta < \omega_1$ with $\alpha < \beta$. Certainly I_β has to miss either I^0_α or I^1_α. Thus either I^0_β misses I^0_α or I^1_β misses I^1_α, so $(I^0_\alpha \times I^1_\alpha) \cap (I^0_\beta \times I^1_\beta) = \varnothing$. It follows that $\{I^0_\alpha \times I^1_\alpha : \alpha < \omega_1\}$ is a pairwise-disjoint collection. ■

Proof of Theorem 223

If there is a Suslin line, then by Theorem 210(a), there is a densely ordered *ccc* linearly ordered space X such that no nonempty open interval is separable. In the proof of Theorem 210(b), a Suslin line Y is constructed from such a linearly ordered space such that X is dense in Y. It follows that Y has no nonempty separable open intervals either. Now let $\mathcal{C} = \{C_\alpha : \alpha < \omega_1\}$ be the strictly increasing sequence of separable closed subsets of Y as constructed in the proof of Theorem 220. For each $\alpha < \omega_1$, C_α is closed so if it failed to be nowhere dense, it would contain an open interval of Y, and this interval would be separable because C_α is. This is a contradiction, so C_α is nowhere dense.

We claim that $Y = \bigcup_{\alpha<\omega_1} C_\alpha$. Suppose otherwise, and let $p \in Y \setminus \bigcup_{\alpha<\omega_1} C_\alpha$. Then for each α, $p \notin C_\alpha$, so there is a convex component I_α of $Y \setminus C_\alpha$ containing p. In the proof of Theorem 220, it was shown that any pair I, J of convex components of a pair of C_α's is either disjoint or comparable. So $\{I_\alpha : \alpha < \omega_1\}$ is an uncountable chain in the Suslin tree constructed there, which is a contradiction. Thus Y is a Suslin line which is the union of ω_1-many nowhere dense sets. ■

Proof of Theorem 224
Let $D_0, D_1, \ldots$ be countably many dense subsets of a partial order P. Choose $d_0 \in D_0$. Since D_1 is dense, there is $d_1 \in D_1$ with $d_1 \leqslant d_0$. Similarly, there is $d_2 \in D_2$ with $d_2 \leqslant d_1$, and so on. Let $F = \{p \in P : \exists n < \omega(d_n \leqslant p)\}$. It is easy to check that F is a filter in P. By construction, $F \cap D_n \neq \varnothing$ for all n. ■

Proof of Theorem 225
Let $\mathcal{U}$ be a collection of $\leqslant \kappa$-many dense open sets in the compact Hausdorff *ccc*-space X. Let $\mathcal{O}(X)$ be the collection of nonempty open subsets of X, ordered by $\subset$. Note that incompatible elements of $\mathcal{O}(X)$ are disjoint, so since X has the *ccc*, so does $\mathcal{O}(X)$.

For each $U \in \mathcal{U}$, let $D_U = \{V \in \mathcal{O}(X) : \overline{V} \subset U\}$. Since X is regular, it is easy to check that each D_U is dense in the partial order $\mathcal{O}(X)$. By $\mathsf{MA}(\kappa)$, there is a filter G is $\mathcal{O}(X)$ which meets each D_U. Let $g_U \in G \cap D_U$. Since G is a filter, whenever $U_1, \ldots, U_n$ are in $\mathcal{U}$, there is $h \in \mathcal{O}(X)$ with $h \subset g_{U_i}$, for each $1 \leqslant i \leqslant n$. It follows that $\{g_U : U \in \mathcal{U}\}$ has the finite intersection property, and so does $\{\overline{g}_U : U \in \mathcal{U}\}$. Since X is compact, $\bigcap_{U \in \mathcal{U}} \overline{g}_U \neq \varnothing$. Since $g_U \in D_U$, $\overline{g}_U \subset U$, and so $\bigcap \mathcal{U} \neq \varnothing$.■

Proof of Corollary 226
Let $X = [0,1]$, and for each $x \in [0,1]$, let $U_x \in X \setminus \{x\}$. Then $\mathcal{U} = \{U_x : x \in [0,1]\}$ is a collection of 2^ω-many dense open subsets of the interval, and of course $\bigcap \mathcal{U} = \varnothing$. ■

Proof of Corollary 227
By Theorem 223, if there is a Suslin line, then there is a Suslin line X which is a union of ω_1-many nowhere dense sets, say $\{A_\alpha : \alpha < \omega_1\}$. Then $\{X \setminus \overline{A}_\alpha : \alpha < \omega_1\}$ is a collection of ω_1-many dense open sets with empty intersection. ■

Proof of Lemma 228
Suppose $\mathsf{MA}(\omega_1)$, X is a *ccc* space, and $\{U_\alpha : \alpha < \omega_1\}$ is an uncountable collection of (distinct) nonempty open subsets of X.

We claim there exists $\alpha_0 < \omega_1$ such that $\bigcup_{\beta > \alpha_0} U_\beta \subset \overline{\bigcup_{\beta > \alpha} U_\beta}$ for each $\alpha > \alpha_0$. If not, there exists $\alpha_0 > 0$ such that $\bigcup_{\beta > 0} U_\beta \setminus \overline{\bigcup_{\beta > \alpha_0} U_\beta} \neq \varnothing$. If $\delta < \omega_1$ and α_γ has been defined for all $\gamma < \delta$, pick $\alpha_\delta = \sup_{\gamma < \delta} \alpha_\gamma$ if δ is a limit, and pick $\alpha_\delta > \alpha_{\delta - 1}$ such that $\bigcup_{\beta > \alpha_{\delta-1}} U_\beta \setminus \overline{\bigcup_{\beta > \alpha_\delta} U_\beta} \neq \varnothing$

if δ is a successor. Then

$$\{\bigcup_{\beta>\alpha_\delta} U_\beta \setminus \overline{\bigcup_{\beta>\alpha_{\delta+1}} U_\beta} : \delta < \omega_1\}$$

violates the *ccc* in X.

Let $\mathbb{P} = \{O \in \mathcal{O}(X) : O \subset \bigcup_{\beta>\alpha_0} U_\beta\}$. For each $\alpha > \alpha_0$ let

$$D_\alpha = \{O \in \mathbb{P} : \exists \beta > \alpha (O \subset U_\beta)\}.$$

Then D_α is dense in $\mathbb{P}$; let $O \in \mathbb{P}$. Then $O \subset \bigcup_{\beta>\alpha_0} U_\beta \subset \overline{\bigcup_{\beta>\alpha} U_\beta}$ so there exists $\beta > \alpha$ such that $O \cap U_\beta \neq \varnothing$. Then $O \cap U_\beta \in D_\alpha$ and $O \cap U_\beta \subset O$ as desired.

Note that X is *ccc* implies $\mathcal{O}(X)$ is *ccc* implies $\mathbb{P}$ is *ccc*. By $\mathsf{MA}(\omega_1)$ there exists a filter G on $\mathbb{P}$ such that $G \cap D_\alpha \neq \varnothing$ for all $\alpha > \alpha_0$. Since G is closed upward, for each $\alpha > \alpha_0$ there exists $\beta > \alpha$ such that $U_\beta \in G$. Let $\gamma_0 > \alpha_0 + 1$ such that $U_{\gamma_0} \in G$. If $\delta < \omega_1$ and an increasing sequence $\langle \gamma_\beta \rangle_{\beta<\delta}$ has been defined so that $U_{\gamma_\beta} \in G$ for each $\beta < \delta$, let $\gamma_\alpha > \sup_{\beta<\delta} \gamma_\beta$ such that $U_{\gamma_\delta} \in G$. Then $\{U_{\gamma_\delta} : \delta < \omega_1\}$ is an uncountable subcollection of $\{U_\alpha : \alpha < \omega_1\}$ with the finite intersection property. ■

Proof of Corollary 229

Immediate from Lemma 228. ■

Proof of Theorem 230

Suppose X, Y have the *ccc* and $\{U_\alpha : \alpha \in \Lambda\}$ is an uncountable collection of pairwise disjoint nonempty open subsets of $X \times Y$. Assume the U_α's are basic; $U_\alpha = V_\alpha \times W_\alpha$. By Corollary 229, X has property K. There exists an uncountable $A \subset \Lambda$ such that $\{V_\alpha : \alpha \in A\}$ is linked. Then $\{W_\alpha : \alpha \in A\}$ is uncountable, so there exists an uncountable $B \subset A$ such that $\{W_B : \beta \in B\}$ is linked. Let $\beta \neq \gamma \in B$. Then $U_\beta \cap U_\gamma = (V_\beta \times W_\beta) \cap (V_\gamma \times W_\gamma) \neq \varnothing$ since $V_\beta \cap V_\gamma \neq \varnothing$ and $W_\beta \cap W_\gamma \neq \varnothing$, a contradiction. The theorem follows from Theorem 132. ■

Proof of Lemma 231

(a) Let $\langle X, \mathcal{T} \rangle$ be a regular Lindelöf space and suppose X is hereditarily Lindelöf. Certainly X is normal. Let $H \subset X$ be closed and for each

$x \in X \setminus H$ let $U_x \in \mathcal{T}$ such that $x \in U_x \subset \overline{U}_x \subset X \setminus H$. Since $X \setminus H$ is Lindelöf, there exist $x_0, x_1, \ldots \in X$ such that $\{U_{x_i} : i < \omega\}$ covers $X \setminus H$. It follows that $H = \bigcap_{i<\omega} X \setminus \overline{U}_{x_i}$.

Now suppose X is perfectly normal. Let $\mathcal{U} \subset \mathcal{T}$ be uncountable and put $H = X \setminus \bigcup \mathcal{U}$. Let $\{V_i : i < \omega\} \subset \mathcal{T}$ such that $H = \bigcap_{i<\omega} V_i$. Observe that each $X \setminus V_i$ is closed, hence Lindelöf. For each $i < \omega$, let $\mathcal{V}_i \subset \mathcal{U}$ be a countable cover of $X \setminus V_i$. Put $\mathcal{V} = \bigcup_{i<\omega} \mathcal{V}_i$. Then $\mathcal{V} \subset \mathcal{U}$ is countable with $\bigcup \mathcal{V} = \bigcup \mathcal{U}$, whence X is hereditarily Lindelöf.

(b) Let $\langle X, \mathcal{T} \rangle$ be a compact Hausdorff space and suppose X is first-countable. Let $x \in X$ and let $\{U_i : i < \omega\}$ be a local base at x. If $y \in X \setminus \{x\}$ there must exist some $i < \omega$ such that $y \notin U_i$, so $\{x\} = \bigcap_{i<\omega} U_i$.

Now suppose every point of X is G_δ. Let $x \in X$ and $\{V_i : i < \omega\} \subset \mathcal{T}$ with $\{x\} = \bigcap_{i<\omega} V_i$. Put $U_0 = V_0$. If $n < \omega$ and U_n has been defined, let $U_{n+1} \in \mathcal{T}$ such that $x \in U_{n+1} \subset \overline{U}_{n+1} \subset U_n \cap V_{n+1}$. Then $U_0 \supset \overline{U}_1 \supset U_1 \supset \cdots$ and $\{x\} = \bigcap_{i<\omega} U_i = \bigcap_{i<\omega} \overline{U}_i$. Let U be an open neighborhood of x. Since $X \setminus U$ is compact and covered by $\{X \setminus \overline{U}_i : i < \omega\}$, there exists $k < \omega$ such that $X \setminus U \subset X \setminus \overline{U}_k$. It follows that $U_k \subset U$, whence $\{U_i : i < \omega\}$ is a local base at x. ■

Proof of Theorem 232

Assume $\mathsf{MA}(\omega_1)$. Suppose X is compact Hausdorff hereditarily Lindelöf but not hereditarily separable. Then there exists a subspace S of X which is not separable. Then $\overline{S}$ is not separable: suppose $\{x_n : n < \omega\}$ is dense in $\overline{S}$. By Lemma 231, X is first-countable. For each $n < \omega$ let $\{U_i^n : i < \omega\}$ be a local base at x_n. For each $n, i < \omega$ there exists $s_i^n \in S \cap U_i^n$. Then $\{s_i^n : n, i < \omega\}$ is countable and dense in S.

Claim 1: There are points $x_\alpha \in \overline{S}$, for all $\alpha < \omega_1$, such that x_α is not a member of $\mathrm{cl}_{\overline{S}}\{x_\beta : \beta < \alpha\} = \overline{\{x_\beta : \beta < \alpha\}}$. Let $x_0 \in \overline{S}$. Assuming $\alpha < \omega_1$ and x_β, $\beta < \alpha$, have been chosen appropriately, $\{x_\beta : \beta < \alpha\}$ is countable and thus there exists $x_\alpha \in \overline{S} \setminus \mathrm{cl}_{\overline{S}}\{x_\beta : \beta < \alpha\}$.

Let $Y = \overline{\{x_\alpha : \alpha < \omega_1\}}$.

Claim 2: For each $y \in Y$ there exists $\alpha < \omega_1$ such that $y \in \overline{\{x_\beta : \beta < \alpha\}}$. If $y \in Y$ then, letting $\{U_n : n < \omega\}$ be a local base

at y and $x_{\alpha_n} \in U_n$ for each n, we have $y \in \overline{\{x_\beta : \beta < \sup_{n<\omega} \alpha_n\}}$.

Y is compact Hausdorff and therefore Y is regular, so by Claim 1 for each $\alpha < \omega_1$ there exists an open $U_\alpha \subset Y$ such that $x_\alpha \in U_\alpha \subset \overline{U}_\alpha \subset Y \setminus \overline{\{x_\beta : \beta < \alpha\}}$. Note that Y is *ccc* since it is hereditarily Lindelöf. By Lemma 228 there exists a cofinal subset A of ω_1 such that $\{U_\alpha : \alpha \in A\}$ has the finite intersection property. Then $\{\overline{U}_\alpha : \alpha \in A\}$ has the finite intersection property. Since Y is compact we have $\bigcap_{\alpha \in A} \overline{U}_\alpha \neq \varnothing$. This contradicts Claim 2, as a point in this intersection cannot be in $\overline{\{x_\beta : \beta < \alpha\}}$ for any $\alpha < \omega_1$. ■

Proof of Lemma 233

Let $\mathcal{B}$ be a countable base for $\mathbb{R}$. Let P be all pairs of the form $\langle \vec{C}, F \rangle$, where $\vec{C}$ is a finite sequence of members of $\mathcal{B}$, and F is a finite subset of κ. For $\langle \vec{C}', F' \rangle, \langle \vec{C}, F \rangle \in P$, define $\langle \vec{C}', F' \rangle \leqslant \langle \vec{C}, F \rangle$ if and only if $\vec{C}'$ extends $\vec{C}$, $F' \supset F$, and for each $i \in \operatorname{dom} \vec{C}' \setminus \operatorname{dom} \vec{C}$, we have $C_i \subset \bigcap_{\alpha \in F} U_\alpha$.

Claim 1: $\langle P, \leqslant \rangle$ is a partially ordered set. We need to prove transitivity. Suppose $\langle \vec{C}'', F'' \rangle \leqslant \langle \vec{C}', F' \rangle \leqslant \langle \vec{C}, F \rangle$. Then $\vec{C}''$ extends $\vec{C}'$ extends $\vec{C}$ and $F'' \supset F' \supset F$, so $\vec{C}''$ extends $\vec{C}$ and $F'' \supset F$. If $i \in \operatorname{dom} \vec{C}'' \setminus \operatorname{dom} \vec{C}$, then either $i \in \operatorname{dom} \vec{C}'' \setminus \operatorname{dom} \vec{C}'$ in which case $C_i \subset \bigcap_{\alpha \in F'} U_\alpha \subset \bigcap_{\alpha \in F} U_\alpha$, or $i \in \operatorname{dom} \vec{C}' \setminus \operatorname{dom} \vec{C}$ in which case $C_i \subset \bigcap_{\alpha \in F} U_\alpha$. Thus $\langle \vec{C}'', F'' \rangle \leqslant \langle \vec{C}, F \rangle$.

Claim 2: $\langle P, \leqslant \rangle$ has the *ccc*. Suppose $\langle \vec{C}_\alpha, F_\alpha \rangle$ is in P for each $\alpha < \omega_1$. We need to show that there is $\alpha \neq \beta$ such that $\langle \vec{C}_\alpha, F_\alpha \rangle$ and $\langle \vec{C}_\beta, F_\beta \rangle$ are compatible. Since $\mathcal{B}$ is countable, so is the collection of finite sequences from $\mathcal{B}$, so there are $\alpha \neq \beta$ such that $\vec{C}_\alpha = \vec{C}_\beta = \vec{C}$. Then it is easy to see that $\langle \vec{C}, F_\alpha \cup F_\beta \rangle$ is less than or equal to both $\langle \vec{C}_\alpha, F_\alpha \rangle$ and $\langle \vec{C}_\beta, F_\beta \rangle$, and hence they are compatible.

Now we define some dense sets. For each $B \in \mathcal{B}$ and $k < \omega$, let

$$D_{B,k} = \{\langle \vec{C}, F \rangle \in P : \exists i > k (C_i \subset B)\}.$$

The $D_{B,k}$'s serve two purposes. One, they will make sure that the generic filter G contains elements whose first coordinate is a sequence

of arbitrarily long length, and thus G will determine an infinite sequence $\langle C_0^G, C_1^G, \ldots\rangle$ of members of $\mathcal{B}$. Two, they make sure that $\bigcup_{i>n} C_i^G$ is dense in $\mathbb{R}$ for each n.

Also, for each $\alpha < \kappa$, let

$$E_\alpha = \{\langle \vec{C}, F\rangle \in P : \alpha \in F\}.$$

The E_α's will make sure that for each α, there is some n so that $\bigcup_{i>n} C_i \subset U_\alpha$.

Let us show that the $D_{B,k}$'s and the E_α's are indeed dense in P. Let $\langle \vec{C}, F\rangle \in P$. Then $\langle \vec{C}, F \cup \{\alpha\}\rangle \leqslant \langle \vec{C}, F\rangle$, so E_α is dense. Now fix $B \in \mathcal{B}$ and $k < \omega$. Suppose $\vec{C} = \langle C_0, C_1, \ldots, C_n\rangle$. Let $m > \max\{n, k\}$ and choose C_i for $i = n+1, \ldots, m$ such that $C_i \subset \bigcap_{\alpha\in F} U_\alpha$, and also $C_m \subset B \cap \bigcap_{\alpha\in F} U_\alpha$. This is possible since $\bigcap_{\alpha\in F} U_\alpha$ is dense open. Let $\vec{C}' = \langle C_0, C_1, \ldots, C_m\rangle$. Then $\langle \vec{C}', F\rangle \in D_{B,k}$ and $\langle \vec{C}', F\rangle \leqslant \langle \vec{C}, F\rangle$. So $D_{B,k}$ is dense.

Let G be a filter in P meeting all E_α's, $\alpha < \kappa$, and all $D_{B,k}$'s, $B \in \mathcal{B}$ and $k < \omega$.

Claim 3: If $\langle \vec{C}, F\rangle$ and $\langle \vec{C}', F'\rangle$ are in G, then the sequence $\vec{C}$ extends $\vec{C}'$ or vice-versa. Well, if $\langle \vec{C}, F\rangle$ and $\langle \vec{C}', F'\rangle$ are in G, then there is some $\langle \vec{C}'', F''\rangle$ in G such that $\langle \vec{C}'', F''\rangle$ is less than or equal to both $\langle \vec{C}, F\rangle$ and $\langle \vec{C}', F'\rangle$, and hence the sequence $\vec{C}''$ extends both $\vec{C}$ and $\vec{C}'$. The claim follows.

Now, since G meets all $D_{B,k}$'s, there are arbitrarily long sequences appearing as first coordinates of members of G, so we can define an infinite sequence $\vec{C}_G = \langle C_0^G, C_1^G, \ldots\rangle$ of members of $\mathcal{B}$ by defining C_i^G to be the ith term of $\vec{C}$ for some $\langle \vec{C}, F\rangle \in G$ with $i \in \operatorname{dom} \vec{C}$. It follows from Claim 3 that it doesn't matter which member of G we use to define C_i^G. Note that $\vec{C}_G = \bigcup\{\vec{C} : \exists F(\langle \vec{C}, F\rangle \in G)\}$, where $\vec{C}$ is viewed as a set of ordered pairs.

Claim 4: For each $n < \omega$, $\bigcup_{i>n} C_i^G$ is dense in $\mathbb{R}$. If $\langle \vec{C}, F\rangle \in G \cap D_{B,n}$, then we have $C_i \subset B$ for some $i > n$. It follows that every basic open set $B \in \mathcal{B}$ contains C_i^G for some $i > n$, from which it easily follows that $\bigcup_{i>n} C_i^G$ is dense in $\mathbb{R}$.

Claim 5: For each $\alpha < \kappa$, there is some $n < \omega$ such that $\bigcup_{i>n} C_i^G \subset U_\alpha$. Fix $\alpha < \kappa$, and let $\langle \vec{C}, F\rangle \in G \cap E_\alpha$. Then $\alpha \in F$. Let $\vec{C} = \langle C_0, C_1, \ldots, C_n\rangle$. Suppose $i > n$, and let $\langle \vec{C}', F'\rangle \in G$ such that $i \in \operatorname{dom} \vec{C}'$. There is some $\langle \vec{C}'', F''\rangle \in G$ with $\langle \vec{C}'', F''\rangle$ less than or equal to both $\langle \vec{C}', F'\rangle$ and $\langle \vec{C}, F\rangle$. Then $C_i^G = C_i''$ and $C_i'' \subset U_\alpha$ because $\langle \vec{C}'', F''\rangle \leqslant \langle \vec{C}, F\rangle$ and $\alpha \in F$.

Finally, from Claims 4 and 5, it follows that $\bigcap_{n<\omega} (\bigcup_{i>n} C_i^G)$ is a dense G_δ subset of $\mathbb{R}$ which is contained in $\bigcap_{\alpha<\kappa} U_\alpha$. ■

Proof of Theorem 234

Assume MA, and let $\mathcal{F}$ be a collection of $< 2^\omega$-many first category subsets of $\mathbb{R}$. Let $\kappa = |\mathcal{F}|$. For each $F \in \mathcal{F}$, let $\mathcal{N}(\mathcal{F})$ be a countable collection of nowhere-dense sets covering F. Since the closure of a nowhere-dense set is nowhere-dense, we may assume each member of $\mathcal{N}(\mathcal{F})$ is closed. Let

$$\mathcal{U} = \{\mathbb{R} \setminus N : \exists F \in \mathcal{F}[N \in \mathcal{N}(\mathcal{F})]\}.$$

Then $\mathcal{U}$ is a collection of κ-many dense open sets. By Lemma 233, there is a dense G_δ-set $G \subset \bigcap \mathcal{U}$. Then $\mathbb{R} \setminus G$ is the union of countably many nowhere dense sets and contains $\bigcup \mathcal{F}$. Hence $\bigcup \mathcal{F}$ is first category. ■

Proof of Theorem 235

Since the set of all functions in ω^ω is both $<^*$-dominating and $<^*$-unbounded, certainly both $\mathfrak{b}$ and $\mathfrak{d}$ are $\leqslant \mathfrak{c}$. Also, every $<^*$-dominating family is obviously $<^*$-unbounded, so $\mathfrak{b} \leqslant \mathfrak{d}$. So it remains to prove $\omega_1 \leqslant \mathfrak{b}$. Let $\{f_n : n < \omega\}$ be a countable subset of ω^ω. Define $g(n) = 1 + \max\{f_i(n) : i \leqslant n\}$. Then $g(n) > f_k(n)$ for all $n \geqslant k$. This shows that every countable subset of ω^ω is $<^*$-bounded, and it follows that $\omega_1 \leqslant \mathfrak{b}$. ■

Proof of Theorem 236

Let $\kappa < \mathfrak{c}$, and consider a collection of functions $\{f_\alpha \in \omega^\omega : \alpha < \kappa\}$. Let $\mathbb{P} = \langle \omega^{<\omega}, [\kappa]^{<\omega}\rangle$ be ordered by $\preccurlyeq$, where $\langle g, G\rangle \preccurlyeq \langle h, H\rangle$ if and only if

(1) $g \supset h$,

(2) $G \supset H$, and

(3) for $n \in |h| \setminus |h|$ and $\alpha \in H$, $g(n) > f_\alpha(n)$.

To check that $\preccurlyeq$ is a partial order on $\mathbb{P}$, let $\langle g, G\rangle \preccurlyeq \langle h, H\rangle \preccurlyeq \langle i, I\rangle$. One may easily see that $g \supset h \supset i$ and $G \supset H \supset I$. For $n \in |h| \setminus |i|$, if $n \in |g| \setminus |h|$, then $g(n) > f_\alpha(n)$ for all $\alpha \in H \supset I$; otherwise $n \in |h| \setminus |i|$, and thus $g(n) > f_\alpha(n)$ for all $\alpha \in I$.

To check that $\langle \mathbb{P}, \preccurlyeq\rangle$ has the *ccc*, let $A \in [bP]^{>\omega}$. Since $\omega^{<\omega}$ is countable, fix $\langle g, G_0\rangle, \langle g, G_1\rangle \in A$ such taht $G_0 \neq G_1$. It's then easy to verify that $\langle g, G_0 \cup G_1\rangle$ extends both, so A is not an antichain.

Let $D_{\alpha,n} = \{\langle g, G\rangle : |g| \geqslant n \wedge \alpha \in G\}$. For each $\langle h, H\rangle \in \mathbb{P}$, $\langle h, H\rangle \succcurlyeq \langle h', H \cup \{\alpha\}\rangle \in D_{\alpha,n}$ such that $h' \supset h$, $|h'| \geqslant n$ and for each $i \in |h'| \setminus |h|$ and $\beta \in H$, $h'(i) > f_\beta(i)$. Thus each $D_{\alpha,n}$ is dense.

Apply $\mathsf{MA} \to \mathsf{MA}(\kappa)$ to obtain a filter L intersecting each $D_{\alpha,n}$. It then follows that $f = \bigcup\{\ell(0) : \ell \in L\}$ is a function in ω^ω due to the compatibility of each $\ell(0)$ and the fact that L intersects $D_{0,n}$ for each $n < \omega$.

Let $\alpha < \kappa$, and choose $\ell_n \in L \cap D_{\alpha,n}$ for each $n < \omega$. It follows that $f = \bigcup\{\ell_n(0) : n < \omega\} \in \omega^\omega$. Also, for $i < |\ell_0(0)|$, choose $\ell \in L$ such that $\ell \succcurlyeq \ell_0, \ell_i$. Then since $i \in |\ell_i| \setminus |\ell_0| \subset |\ell| \setminus |\ell_0|$, it follows that $f(i) = \ell(0)(i) > f_\alpha(i)$.

Thus $\{f_\alpha : \alpha < \kappa\}$ is not $<^*$ *−unbounded*, and $\mathfrak{b} > \kappa$. Therefore $\mathfrak{b} \geqslant \mathfrak{c}$, and the result follows. ■

Proof of Theorem 237

Let $\{g_\alpha \in \omega^\omega : \alpha < \mathfrak{b}\}$ be $<^*$-unbounded, and define $f_0 = g_0$. For each limit ordinal $\gamma < \mathfrak{b}$ note that $\{f_\alpha : \alpha < \gamma\}$ is not $<^*$-unbounded; define f_γ such that $f_\alpha <^* f_\gamma$ for each $\alpha < \gamma$ and $g_\gamma <^* f_\gamma$. For a successor ordinal $\gamma + 1$, define $f_{\gamma+1}$ such that $f_{\gamma+1}(n) = f_\gamma(n) + g_{\gamma+1}(n) + 1$ for each $n < \omega$, noting $f_\gamma, g_{\gamma+1} <^* f_{\gamma+1}$.

By construction, $\{f_\alpha : \alpha < \mathfrak{b}\}$ is $<^*$-increasing. Let $f \in \omega^\omega$. Since $\{g_\alpha : \alpha < \mathfrak{b}\}$ is $<^*$-unbounded, pick α such that $g_\alpha \not<^* f$. Since $g_\alpha \leqslant^* f_\alpha$, it follows that $f_\alpha \not<^* f$, and thus $\{f_\alpha : \alpha < \mathfrak{b}\}$ is also $<^*$-unbounded. ■

Proof of Theorem 238

Assume $\mathfrak{b} = \mathfrak{d}$. Let $\{g_\alpha : \alpha < \mathfrak{d}\}$ be $<^*$-dominating; put $f_0 = g_0$. For each limit ordinal $\gamma < \mathfrak{d} = \mathfrak{b}$, the collection $\{f_\alpha : \alpha < \gamma\}$ is not $<^*$-unbounded, so define f_γ such that $f_\alpha <^* f_\gamma$ for each $\alpha < \gamma$, and $g_\gamma <^* f_\gamma$. For each successor ordinal $\gamma + 1$, define $f_{\gamma+1}$ such that

$f_{\gamma+1}(n) = f_\gamma(n) + g_{\gamma+1}(n) + 1$ for each $n < \omega$, and note that $f_\gamma <^* f_{\gamma+1}$ and $g_{\gamma+1} <^* f_{\gamma+1}$.

Certainly $\{f_\alpha : \alpha < \mathfrak{d}\}$ is $<^*$-increasing. To show this collection is $<^*$-dominating, let $g \in \omega^\omega$. Because $\{g_\alpha : \alpha < \mathfrak{d}\}$ is $<^*$-dominating, there exists $\alpha < \mathfrak{d}$ such that $g <^* g_\alpha$. Thus $g <^* g_\alpha <^* f_\alpha$, so $\{f_\alpha : \alpha < \mathfrak{d}\}$ is $<^*$-dominating. ■

Proof of Theorem 239

Let $\lambda = \min A$ and $\mu = \min B$.

$\boldsymbol{\mathfrak{d} \leqslant \lambda}$ Let $\{K_\alpha : \alpha < \lambda\}$ be a collection of λ-many compact sets covering ω^ω. Then $\pi_n[K_\alpha]$ is compact, and thus finite, for each $\alpha < \lambda$ and $n < \omega$. For each $\alpha < \lambda$ define $f_\alpha \in \omega^\omega$ by $f_\alpha(n) = 1 + \max \pi_n[K_\alpha]$. Certainly $|\{f_\alpha : \alpha < \lambda\}| \leqslant \lambda$. To see that $\{f_\alpha : \alpha < \lambda\}$ is $<^*$-dominating, note that $g \in K_\alpha$ implies $g < f_\alpha$.

$\boldsymbol{\lambda \leqslant \mu}$ Every dominating family $\mathcal{A}$ of compact subsets of ω^ω covers ω^ω: if $g \in \omega^\omega$ then certainly $\{g\}$ is compact, so there exists $A \in \mathcal{A}$ such that $\{g\} \subset A$ (that is, $g \in A$).

$\boldsymbol{\mu \leqslant \mathfrak{d}}$ Let $\mathcal{F} \subset \omega^\omega$ such that $\mathcal{F}$ is $<^*$ dominating and $|\mathcal{F}| = \mathfrak{d}$. For each $f \in \mathcal{F}$ and $k, \ell \in \omega$ define

$$K_{f_{k,\ell}} = \{g \in \omega^\omega : n \geqslant k \to g(n) \leqslant f(n) \wedge n < k \to g(n) \leqslant \ell\}.$$

Note $K_{f_{k,\ell}}$ is compact. Let $\mathcal{K} = \{K_{f_{k,\ell}} : f \in \mathcal{F} \wedge k, \ell \in \omega\}$. Then $|\mathcal{K}| \leqslant |\mathcal{F}| = \mathfrak{d}$. Let $K \subset \omega^\omega$ be compact. Define $g \in \omega^\omega$ by $g(n) = \max \pi_n[K]$. There exists $f \in \mathcal{F}$ and $k \in \omega$ such that $g(n) < f(n)$ when $n \geqslant k$. Let $\ell = \max\{g(n) : n < k\}$. Then $g \in K_{f_{k,\ell}}$, $K \subset K_{f_{k,l}}$. ■

Proof of Proposition C

Note that an equivalent definition follows: $A \subset \mathbb{P}$ is concentrated about the rationals if and only if for any closed subset C in the real line which misses the rationals, $A \cap C$ is countable.

There exists a closed uncountable subset of the irrationals. If $\mathbb{Q} = \langle q_n : n < \omega \rangle$, the collection $\mathcal{U} = \{(q_n - 1/2^n, q_n + 1/2^n) : n < \omega\}$ does not cover $\mathbb{R}$, so $\mathbb{R} \setminus \bigcup \mathcal{U}$ is an example. Alternately, consider 2^ω as a subset of ω^ω—it must be compact and thus closed in every T_2 superspace. ■

Proof of Proposition D

Let $f_\alpha \in \omega^\omega$ for $\alpha < \mathfrak{b}$ such that $\{f_\alpha : \alpha < \mathfrak{b}\}$ is $<^*$-unbounded and $<^*$-increasing, and let $\theta\colon \omega^\omega \to I \cap \mathbb{P}$ be a homeomorphism. Let $A = \theta[\{f_\alpha : \alpha < \mathfrak{b}\}] \in [\mathbb{P}]^{\mathfrak{b}}$.

Let $C \subset \mathbb{P}$ be closed in the reals. Since $C \cap I$ is compact, it follows that $\theta^{-1}[C] = \theta^{-1}[C \cap I]$ is compact in ω^ω, so there exists $g \in \omega^\omega$ satisfying $g >^* f$ for each $f \in \theta^{-1}[C]$. Then there exists $\beta < \mathfrak{b}$ such that $g \not>^* f_\alpha$ for all $\beta \leqslant \alpha < \mathfrak{b}$. Therefore $|\theta^{-1}[C]| \leqslant |\{f_\alpha : \alpha < \beta\}| < \mathfrak{b}$, and thus $|A \cap C| < \mathfrak{b}$. Thus A is $(< \mathfrak{b})$-concentrated about $\mathbb{Q}$.

Now suppose that $\kappa > \omega$, and that $A \in [\mathbb{P}]^\kappa$ is $(< \kappa)$-concentrated about $\mathbb{Q}$. Let $\theta\colon \omega^\omega \to \mathbb{P}$ be a homeomorphism. Let $g \in \omega^\omega$, and note that

$$K_{g,i,j} = \{f \in \omega^\omega : n < i \to f(n) < j \wedge n \geqslant i \to f(n) < g(n)\}$$

is compact, so $\theta[K_{g,i,j}]$ is compact and thus closed in the reals. As a result, $|A \cap \theta[K_{g,i,j}]| < \kappa$. Since $\bigcup \{K_{g,i,j} : i, j < \omega\} = \{f \in \omega^\omega : f <^* g\}$, it follows that $|A \cap \theta[\{f \in \omega^\omega : f <^* g\}]| < \kappa \times \omega = \kappa$ so there exists $x \in A$ such that $\theta^{-1}(x) \not<^* g$. Therefore $\theta^{-1}[A]$ is $<^*$-unbounded, so $\kappa \geqslant \mathfrak{b}$. ■

Proof of Theorem 240

Immediate from Proposition D. When $\mathfrak{b} = \omega_1$, then there exists $A \in [\mathbb{P}]^{\omega_1}$ which is $(< \omega_1)$-concentrated (therefore concentrated) about $\mathbb{Q}$. And if there exists some uncountable $A \subset \mathbb{P}$ concentrated about $\mathbb{Q}$, then any $A' \in [A]^{\omega_1}$ is concentrated (therefore $(< \omega_1)$-concentrated about $\mathbb{Q}$, showing $\mathfrak{b} = \omega_1$. ■

Proof of Example 241

Let everything be defined as in the statement.

To see that X is Lindelöf, let $\mathcal{U}$ be an open cover of X; let $\mathcal{V} \subset \mathcal{U}$ be countable covering $\mathbb{Q}$. Since $X \setminus \bigcup \mathcal{V}$ is countable, there exists a countable $\mathcal{W} \subset \mathcal{U}$ covering $X \setminus \bigcup \mathcal{V}$. Then $\mathcal{V} \cup \mathcal{W}$ is a countable subcover of $\mathcal{U}$.

To see that X is regular let $x \in X$ and let $H \subset X$ be closed not containing X. If $x \in A$, then $\{x\}$ and $X \setminus \{x\}$ separate x and H. If $x \in \mathbb{Q}$, there is a neighborhood $(a, b) \cap X$ of x missing H with $a, b \in \mathbb{P}$.

Then $X \setminus [(a,b) \cap X]$ is an open set containing H, so x and H may be separated.

Finally, consider $K = \{(a,a) : a \in A\} \subset X \times \mathbb{P}$. For $(x,y) \in (X \times \mathbb{P}) \setminus K$, there exist disjoint Euclidean neighborhoods U and V of x and y, respectively. So $(U \cap X) \times (V \cap \mathbb{P})$ is an open neighborhood of (x,y) missing K. Thus K is closed.

Furthermore, each $a \in A$ is isolated in X, so $\{\{a\} \times \mathbb{P} : a \in A\}$ is an open cover of K which is uncountable and pairwise disjoint, so K is discrete and uncountable and not Lindelöf. Therefore $X \times \mathbb{P}$ is not Lindelöf. ■

Proof of Theorem 242

(a) Let $H = \mathbb{Q} \times \{0\}$ and $K = \mathbb{P} \times \{0\}$. Then H and K are closed and disjoint. Suppose U and V are open and disjoint with $H \subset U$ and $K \subset V$. For each $p \in \mathbb{P}$ there exists $n_p \in \mathbb{N}$ such that $D(p, n_p) \subset V$. For each $n \in \mathbb{N}$ let $A_n = \{p \in \mathbb{P} : n_p = n\}$. Then $\bigcup_{n \in \mathbb{N}} A_n = \mathbb{P}$. Since $\mathbb{P}$ is a Baire space there exists $n \in \mathbb{N}$ such that A_n is dense in some interval (a,b) of $\mathbb{R}$. Let $q \in (a,b) \cap \mathbb{Q}$ and $\langle p_i \rangle$ an increasing sequence of irrationals in $(a,b) \cap A_n$ convering to q. Then any basic neighborhood of $\langle q, 0 \rangle$ must meet a basic open neighborhood $D(p_i, n)$ for some i. So $U \cap V \neq \varnothing$.

(b) $\mathcal{H} = \{\{\langle a, 0 \rangle\} : a \in A\}$ is a discrete collection of closed sets. Suppose $\mathcal{U} = \{U_H : H \in \mathcal{H}\}$ is a disjoint collection of open sets separating $\mathcal{H}$. Then we may identify each U_H with a unique $\langle q_H, r_H \rangle \in U_H$ where $q_H, r_H \in \mathbb{Q}$.

(c) Let $A \subset \mathbb{R}$ such that every subset of A is G_δ in A. Let $H \subset A$ and put $K = A \setminus H$. Let $\{H_n : n < \omega\}$ witness H is F_σ. For each $n < \omega$, set $U_n = \bigcup \{D(x,1) : x \in H_n\}$.

Certainly $H \times \{0\} \subset \bigcup_{n<\omega} U_n$. Fix $n < \omega$ and let $y \in K$. Let $(a,b) \subset \mathbb{R}$ be an interval containing y and missing H_n and pick $k \in \mathbb{N}$ such that $D(y,k) \cap [D(a,1) \cap D(b,1)] = \varnothing$. If $x \in H_n$, then $x \leqslant a$ or $x \geqslant b$, so $D(y,k) \cap D(x,1) = \varnothing$ as well. It follows that $\langle y, 0 \rangle \not\overline{U}_n$, so $\overline{U}_n \cap (K \times \{0\}) = \varnothing$ for each $n < \omega$.

Similarly there is a collection $\{V_n : n < \omega\}$ of open sets such that $K \times \{0\} \subset \bigcup_{n<\omega} V_n$ and $\overline{V}_n \cap (H \times \{0\}) = \varnothing$ for all $n < \omega$. By

Lemma 57, $H \times \{0\}$ and $K \times \{0\}$ may be separated, so $X(A)$ is normal. ■

Proof of Theorem 243

Suppose A is an uncountable subspace of reals and $2^\omega < 2^{\omega_1}$. Then A ha a countable base $\{B_n : n < \omega\}$. Define $f \colon \tau_A \to 2^\omega$ by

$$f[U](n) = \begin{cases} 1 & \text{if } B_n \subset U, \\ 0 & \text{else.} \end{cases}$$

Suppose $U \neq V \in \tau_A$. Then $\{n < \omega : B_n \subset U\} \neq \{n < \omega : B_n \subset V\}$. Without loss of generality we may assume $n \in \{n < \omega : B_n \subset U\} \setminus \{n < \omega : B_n \subset V\}$. Then $f[U](n) = 1 \neq 0 = f[V](n)$, so $\{f\}U \neq f[V]$. So f is an injection.

Let $\mathcal{G}_\delta$ be the family of G_δ-subsets of A. For each $G \in \mathcal{G}_\delta$, choose a sequence of open-in-τ_A sets $\langle U_n \rangle$ such that $G = \bigcap_{n<\omega} U_n$. Clearly $G \mapsto \langle f[U_n] \rangle$ is an injection from $\mathcal{G}_\delta$ into $(2^\omega)^\omega$.

We have $|\mathcal{G}_\delta| \leqslant (2^\omega)^\omega = 2^\omega \leqslant 2^{\omega_1} \leqslant \mathcal{P}(A)$, so not every subset of A is G_δ. ■

Proof of Theorem 244

Let $A \subset \mathbb{R}$ have cardinality $\kappa < \mathfrak{c}$ and let $B \subset A$. Let $\mathcal{C}$ be a countable base for A. The plan is to use $\mathsf{MA}(\kappa)$ to show that there is a sequence $\langle C_0, C_1, \ldots \rangle$ of members of $\mathcal{C}$ such that every $x \in B$ is in infinitely many C_i's, while every $y \in A \setminus B$ is in only finitely many C_i's. If we do this, then note that $B = \bigcap_{n<\omega} (\bigcup_{i>n} C_i)$, and hence B is a G_δ-set in the space A.

To this end, let P be all pairs of the form $\langle \vec{C}, F \rangle$ where $\vec{C}$ is a finite sequence of members of $\mathcal{C}$ and F is a finite subset of $A \setminus B$. For $\langle \vec{D}, G \rangle, \langle \vec{C}, F \rangle \in P$, define $\langle \vec{D}, G \rangle \leqslant \langle \vec{C}, F \rangle$ if and only if $\vec{D}$ extends $\vec{C}$, $G \supset F$, and for each $i \in \operatorname{dom} \vec{D} \setminus \operatorname{dom} \vec{C}$ we have $C_i \cap F = \varnothing$. That $\langle P, \leqslant \rangle$ is a partially ordered set, and has the *ccc*, follows from essentially the same argument as in Proof 234.

Now we define some dense sets. For each $b \in B$ and $k < \omega$, let

$$D_{b,k} = \{\langle \vec{C}, F \rangle \in P : \exists i > k (b \in C_i)\}.$$

Also, for each $a \in A \setminus B$, let

$$E_a = \{\langle \vec{C}, F\rangle \in P : a \in F\}.$$

That these sets are dense in P follows as easily as in Proof 234.

Let $\mathcal{G}$ be a filter in P meeting all E_a's, $a \in A \setminus B$, and all $D_{b,k}$'s, $b \in B$ and $k < \omega$. Since $\mathcal{G}$ meets all $D_{b,k}$'s, there are arbitrarily long sequences appearing as first coordinates of member of $\mathcal{G}$, so we can define an infinite sequence $\vec{C}_{\mathcal{G}} = \langle C_0^{\mathcal{G}}, C_1^{\mathcal{G}}, \ldots\rangle$ of members of $\mathcal{C}$ by defining $C_i^{\mathcal{G}}$ to be the ith of $\vec{C}$ for some $\langle \vec{C}, F\rangle \in \mathcal{G}$ such that $i \in \operatorname{dom} \vec{C}$. Note that

$$\vec{C}_{\mathcal{G}} = \bigcup \{\vec{C} : \exists F(\langle \vec{C}, F\rangle \in \mathcal{G})\},$$

where $\vec{C}$ is viewed as a set of ordered pairs.

Fix $b \in B$. Since $\mathcal{G}$ meets $D_{b,k}$ for eack $k < \omega$, it follows that $b \in C_i^{\mathcal{G}}$ for infinitely many i. Now fix $a \in A \setminus B$. There is some $\langle \vec{C}, F\rangle \in \mathcal{G} \cap E_a$. Then $a \in F$. Let $\vec{C} = \langle C_0, C_1, \ldots, C_n\rangle$. Suppose $i > n$ and let $\langle \vec{D}, G\rangle \in \mathcal{G}$ such that $i \in \operatorname{dom} \vec{D}$. There is some $\langle \vec{E}, H\rangle \in \mathcal{G}$ with $\langle \vec{J}, H\rangle$ less than or equal to both $\langle \vec{D}, G\rangle$ and $\langle \vec{C}, F\rangle$. Then $C_i^{\mathcal{G}} = J_i$ and $a \notin J_i$ because $\langle \vec{J}, H\rangle \leqslant \langle \vec{C}, F\rangle$ and $a \in F$. It follows that $a \notin \bigcup_{i>n} C_i^{\mathcal{G}}$, and hence a is in only finitely many terms of $\vec{C}_{\mathcal{G}}$. Now the result follows as indicated in the first paragraph. ■

Proof of Corollary 245

Proof of Corollary 246

Assume MA, and let $\kappa < 2^\omega$ be an infinite cardinal. There is a subset A of the real line of cardinality κ. By MA and Theorem 244, every subset of A is G_δ in A. There is a countable base for A and every open subset of A is a union of members of this base, so there are no more than 2^ω-many open sets in A. Each G_δ-set is by definition a countable intersection of open sets, so there are no more than $(2^\omega)^\omega = 2^{\omega\times\omega} = 2^\omega$ many G_δ-sets in A. So the cardinality of the power set of A, which is 2^κ, is no more than 2^ω. And of course it is at least that, so $2^\kappa = 2^\omega$.■

Proof of Lemma 247

Let X be a normal space, let $D \subset X$ be closed discrete, and let $G \subset X$ be dense. For $H \subset D$, let U_H and V_H be disjoint open sets containing H and $D \setminus H$, respectively. Define $f\colon \mathcal{P}(D) \to \mathcal{P}(G)$ by $f[H] = U_H \cap G$.

Let $H, K \subset D$ with $H \neq K$. Without loss of generality, we may assume $H \setminus K$ is nonempty. Then $G \cap (U_H \cap V_K)$ is a nonempty subset of $f[H]$ missing $f[K]$, so $f[H] \neq f[K]$. Since f is one-to-one, $2^{|D|} \leqslant 2^{|G|}$.

This proves a more general version of the lemma. Apply this with G a countable dense set to obtain the lemma as stated. ■

Proof of Corollary 248

Suppose $2^\omega < 2^{\omega_1}$ and let X be a normal, separable space. Let $\mathcal{H}$ be a discrete collection of closed sets in X. For each $H \in \mathcal{H}$, choose $x_H \in H$ and put $D = \{x_H : H \in \mathcal{H}\}$. Then D is closed discrete, so $2^{|D|} \leqslant 2^\omega$, implying $|D| \leqslant \omega$ and $|\mathcal{H}| \leqslant \omega$.

Enumerate $\mathcal{H} = \{H_i : i < \omega\}$. For each $i < \omega$, observe that H_i and $(\bigcup \mathcal{H}) \setminus H_i$ are disjoint closed sets. Since X is normal, for each $i < \omega$ there exist disjoint open sets U_i and V_i containing H_i and $(\bigcup \mathcal{H}) \setminus H_i$, respectively. For each $i < \omega$, put $W_i = U_i \cap (\bigcap_{j<i} V_j)$. Certainly $H_i \subset W_i$ for each $i < \omega$. Moreover, if $i < j < \omega$ then $W_i \cap W_j = \varnothing$ since $W_i \subset U_i$ and $W_j \subset V_i$. ■

Proof of Theorem 249

Certainly it suffices to show that there is an almost-disjoint family of subsets of some countable set, such as $\mathbb{Q}$, of cardinality 2^ω. For each $x \in \mathbb{R}$, let S_x be the terms of a sequence of rationals converging to x. Then $x \neq y \to S_x \cap S_y$ is finite, so $\{S_x : x \in \mathbb{R}\}$ is an almost-disjoint family of subsets of $\mathbb{Q}$ of cardinality 2^ω. ■

Proof of Proposition E

Proof of Lemma 250

Suppose $\{A_n : n < \omega\}$ is an almost-disjoint family of subsets of ω. Note that each finite union $\bigcup_{i \leqslant n} A_i$ has infinite complement, for otherwise A_{n+1} would have to intersect some A_i, $i \leqslant n$, in an infinite set. Now

for each n choose

$$x_n \in \omega \setminus (\{x_i : i < n\} \cup \bigcup_{i \leqslant n} A_i).$$

Let $X = \{x_n : n < \omega\}$. Then X is infinite but $X \cap A_n$ is finite for each n. Hence $\{A_n : n < \omega\}$ is not maximal. ■

Proof of Theorem 251

Let $\mathcal{A} = \{A_\alpha : \alpha < \kappa\}$ be an almost-disjoint family (of infinite subsets of ω) of cardinality κ, where $\omega \leqslant \kappa < 2^\omega$. Let P be all pairs of the form $\langle \vec{x}, F\rangle$, where $\vec{x}$ is a finite sequence of members of ω and F is a finite subset of κ. For $\langle \vec{y}, G\rangle$ and $\langle \vec{x}, F\rangle$ in P, define $\langle \vec{y}, G\rangle \leqslant \langle \vec{x}, F\rangle$ if and only if $\vec{y}$ extends $\vec{x}$, $G \supset F$, and for each $i \in \operatorname{dom} \vec{y} \setminus \operatorname{dom} \vec{x}$ we have $x_i \notin \bigcup_{\alpha \in F} A_\alpha$.

To see that $\langle P, leq\rangle$ is a poset we need to show transitivity. Suppose $\langle \vec{z}, H\rangle \leqslant \langle \vec{y}, G\rangle \leqslant \langle \vec{x}, F\rangle$. Then $\vec{z}$ extends $\vec{y}$ extends $\vec{x}$ and $H \supset G \supset F$, so $\vec{z}$ extends $\vec{x}$ and $H \supset F$. If $i \in \operatorname{dom} \vec{z} \setminus \operatorname{dom} \vec{x}$, then either

(1) $i \in \operatorname{dom} \vec{z} \setminus \vec{y}$ in which case $x_i \notin \bigcup_{\alpha \in F} A_\alpha$ since $x_i \notin \bigcup_{\alpha \in G} A_\alpha$ and $F \subset G$, or

(2) $i \in \operatorname{dom} \vec{y} \setminus \vec{x}$ in which case $x_i \notin \bigcup_{\alpha \in F} A_\alpha$.

Thus $\langle \vec{z}, H\rangle \leqslant \langle \vec{x}, F\rangle$.

To see that $\langle P, \leqslant\rangle$ has the *ccc*, suppose $\langle \vec{x}_\alpha, F_\alpha\rangle \in P$ for each $\alpha < \omega_1$. There are only countably many finite sequences in ω, so there exist $\alpha \neq \beta < \omega_1$ with $\vec{x}_\alpha = \vec{x}_\beta = \vec{x}$. Certainly $\langle \vec{x}, F_\alpha \cup F_\beta\rangle \leqslant \langle \vec{x}_\alpha, F_\alpha\rangle$ and $\langle \vec{x}, F_\alpha \cup F_\beta\rangle \leqslant \langle \vec{x}_\beta, F_\beta\rangle$, so $\langle \vec{x}_\alpha, F_\alpha\rangle$ and $\langle \vec{x}_\beta, F_\beta\rangle$ are compatible.

For each $k < \omega$, put $D_k = \{\langle \vec{x}, F\rangle \in P : |\{x_i : i \in \operatorname{dom} \vec{x}\}| > k\}$. For each $\alpha < \kappa$, put $E_\alpha = \{\langle \vec{x}, F\rangle \in P : \alpha \in F\}$. Let $\langle \vec{x}, F\rangle \in P$. Then $\langle \vec{x}, F \cup \{\alpha\}\rangle \leqslant \langle \vec{x}, F\rangle$, so E_α is dense in P for each $\alpha < \kappa$. Fix $k < \omega$ and suppose $\vec{x} = \langle x_0, x_1, \ldots, x_n\rangle$. Take

$$\{x_{n+1}, \ldots, x_{n+k+1}\} \subset \omega \setminus (\bigcup_{\alpha \in F} A_\alpha \cup \{x_i : i \in \operatorname{dom} \vec{x}\})$$

so that $x_i \neq x_j$ whenever $n+1 \leqslant i < j \leqslant n+k+1$. This is possible since any finite subcollection of $\mathcal{A}$ must leave infinitely many members of ω

uncovered. Put $\vec{y} = \langle x_0, \ldots, x_n, x_{n+1}, \ldots, x_{n+k+1} \rangle$. Then $\langle \vec{y}, F \rangle \in D_k$ and $\langle \vec{y}, F \rangle \leqslant \langle \vec{x}, F \rangle$, so D_k is dense in P.

Let $\mathcal{G}$ be a filter in P meeting E_α and D_k for each $\alpha < \kappa$ and each $k < \omega$. If $\langle \vec{x}, F \rangle, \langle \vec{y}, G \rangle \in \mathcal{G}$ then there exists $\langle \vec{z}, H \rangle \in \mathcal{G}$ less than or equal to both of them, whence $\vec{z}$ extends both $\vec{y}$ and $\vec{x}$. It follows that $\vec{y}$ extends $\vec{x}$, or vise-versa.

Let $\vec{x}_{\mathcal{G}} = \langle x_0^{\mathcal{G}}, x_1^{\mathcal{G}}, \ldots \rangle$, where $x_i^{\mathcal{G}}$ is the ith term of $\vec{x}$ for some $\langle \vec{x}, F \rangle \in G$ with $i \in \operatorname{dom} \vec{x}$. Put $A = \{x_i^{\mathcal{G}} : i < \omega\}$. Since G meets every D_k, we know $|A| = \omega$.

Finally, fix $\alpha < \kappa$ and let $\langle \vec{x}, F \rangle \in \mathcal{G} \cap E_\alpha$, so $\alpha \in F$. Let $\vec{x} = \langle x_0, x_1, \ldots, x_n \rangle$. Suppose $i > n$ and let $\langle \vec{y}, G \rangle \in \mathcal{G}$ such that $i \in \operatorname{dom} \vec{y}$. Take $\langle \vec{z}, H \rangle \in \mathcal{G}$ less than or equal to both $\langle \vec{y}, G \rangle$ and $\langle \vec{x}, F \rangle$. Then $x_i^{\mathcal{G}} = z_i \not\in A_\alpha$, so $|A \cap A_\alpha| < \omega$. It follows that $\mathcal{A} \cup \{A\}$ is an almost-disjoint family, hence $\mathcal{A}$ is not maximal. ■

Proof of Theorem 252

Consider the basic open neighborhood $b(x_A, 0) = \{x_A\} \cup A$ of x_A. Since every open set U containing x_A contains $b(x_A, n)$ for some n, U therefore contains all but at most finitely many points of $b(x_A, 0)$. Thus $b(x_A, 0)$ is compact. Since all points other than the x_A's are isolated, we have that $\psi(\mathcal{A})$ is locally compact.

To see Hausdorff, suppose $A \neq B \in \mathcal{A}$. Then $A \cap B$ is finite, so there is some $n < \omega$ such that $b(x_A, n) \cap b(x_B, n) = \varnothing$. Since all "non-$x_A$'s" are isolated, it easily follows that $\psi(\mathcal{A})$ is Hausdorff (this is the only place in the proof where "almost-disjoint" is needed).

Let $D = \{x_A : A \in \mathcal{A}\}$. Every point outside of D is isolated, so D is closed. Also, $b(x_A, 0) \cap D = \{x_A\}$, so D is discrete. Thus D is closed discrete. ■

Proof of Theorem 253

Let X be countably compact, and suppose $f \colon X \to \mathbb{R}$ is continuous. Then $f[X]$ is a countably compact subset of $\mathbb{R}$. But countable compactness and compactness are equivalent in metric spaces, so $f[X]$ is compact, hence bounded. Thus X is pseudocompact. ■

Proof of Theorem 254

Let $\mathcal{A}$ be an infinite MAD family of infinite subsets of ω. Let $f \colon \psi(\mathcal{A}) \to \mathbb{R}$ be continuous and suppose on the contrary that f has unbounded

range. Without loss of generality, we may assume there exists an increasing sequence $\langle x_n \rangle$ in ω such that $\langle f(x_n) \rangle$ is increasing and unbounded. Let $X = \{x_i : i < \omega\}$. Since $\mathcal{A}$ is MAD, there exists $A \in \mathcal{A}$ with $|X \cap A| = \omega$. Let $\langle a_n \rangle$ be the order-preserving indexing of A. Then $\langle a_n \rangle \to x_A$ but $\langle f(a_n) \rangle \not\to f(x_A)$, a contradiction. ■

Proof of Lemma 255

By CH, $\mathcal{A}$ has cardinality ω_1, so we can let $\{A'_\alpha : \alpha < \omega_1\}$ be a listing of $\mathcal{A}$ in type ω_1. Let $A_\omega = A'_\alpha$ where α is least such that $A'_\alpha \subset \{x_n : n < \omega\}$; since $\{x_n : n < \omega\}$ is the rationals, there will always be such an A'_α (indeed, there are ω_1-many α which satisfy this).

Now suppose A_β has been defined for each $\beta < \alpha$, where $\alpha < \omega_1$. Let $A_\alpha = A'_\delta$, where δ is at least such that $A'_\delta \subset \{x_\beta : \beta < \alpha\}$ and $A'_\delta \neq A_\beta$ for any $\beta < \alpha$. This defined A_α for all $\alpha < \omega_1$.

Certainly the condition holds by construction; it remains to prove that $\{A_\alpha : \omega \leqslant \alpha < \omega_1\} = \mathcal{A}$, which will be the case if for every $\gamma < \omega_1$ there is some $\alpha < \omega_1$ such that $A_\alpha = A'_\gamma$. To this end, fix $\gamma < \omega_1$. There will be some $\rho < \omega_1$ such that $A'_\gamma \subset \{x_\beta : \beta < \rho\}$. Then at any step $\alpha \geqslant \rho$, $A_\alpha = A'_\gamma$ if γ is least such that A'_γ has not already been chosen. Since the induction goes on for ω_1-many steps and there are only countably many ordinals $\mu < \gamma$, A'_γ must be chosen at some stage. Hence $\{A_\alpha : \omega \leqslant \alpha < \omega_1\} = \mathcal{A}$. ■

Proof of Lemma 256

For limit α, let τ_α be the topology generated by $\bigcup_{\beta<\alpha} \tau_\beta$. It is straightforward to check that 256(a) to 256(c) hold with $\beta = \alpha$.

If $\alpha = \beta + 1$, then $X_\alpha = X_\beta \cup \{x_\beta\}$, and the task is to define a neighborhood base $\mathcal{B}_\beta$ at x_β so that the topology generated by $\tau_\beta \cup \mathcal{B}_\beta$ satisfy 256(a) to 256(c). Enumerate $\{\gamma < \alpha : x_\beta \in \overline{A}_\gamma\}$ as $\{\gamma_i : i < \omega\}$. Let $x_{0,0} \in B(x_\beta, 1) \cap A_{\gamma_0}$ and let $C_0 = \{x_{0,0}\}$; here, $B(x, \varepsilon)$ is the ball about x of radius ε with respect to the Euclidean metric. If C_i and $x_{j,i}$ have been defined for $i \leqslant n$ and $j \leqslant i$, for each $j \leqslant n+1$ let $x_{j,n+1} \in B(x_\beta, 1/2^{n+1}) \cap A_{\gamma_j}$. Then let $C_{n+1} = C_n \cup \{x_{j,n+1} : j \leqslant n+1\}$.

Let $C = \{x_{j,n} : n < \omega \wedge j \leqslant n\}$. Every ball about x_β contains $x_{j,n}$ for all sufficiently large n, and hence contains all but finitely many members of C. So $C = \{c_i : i < \omega\}$ is a sequence which converges to x_β in the Euclidean topology. Let K_i be a compact open set in X_β containing

c_i with $\operatorname{diam} K_i \leqslant 1/i+1$; this is possible since the topology of X_β is a locally compact 0-dimensional topology finer than the Euclidean topology. Then let sets $U_{\beta,m} = \{x_\beta\} \cup \bigcup_{i \geqslant m} K_i$ for $m < \omega$ be a local base $\mathcal{B}_\beta$ at x_β in X_α.

Let τ_α be the topology on X_α generated by $\tau_\beta \cup \mathcal{B}_\beta$. ■

Proof of Theorem 257

That $\langle \mathbb{R}, \tau_{\omega_1} \rangle$ is locally compact Hausdorff follows from Lemmas 256(a) and 256(b). Certainly $\{X_\alpha : \alpha < \omega_1\}$ is an open cover with no countable subcover, so $\langle \mathbb{R}, \tau_{\omega_1} \rangle$ is not Lindelöf.

Let $Y \subset \mathbb{R}$ be uncountable and let D be a countable dense (in the Euclidean topology) subset thereof. Since $|\overline{D}| > \omega$—with $\overline{D}$ the Euclidean closure—there exists $\gamma < \omega_1$ such that $D = A_\gamma$. Let $\delta \in (\gamma, \omega_1)$. By Lemma 256(c),

$$x_\delta \in \overline{A}_\gamma = Y \to x_\delta \in \operatorname{cl}_{\tau_{\delta_1}} A_\delta \to x_\delta \in \operatorname{cl}_{\tau_{\omega_1}} A_\gamma.$$

It follows that $|Y \setminus \operatorname{cl}_{\tau_{\omega_1}} A_\gamma| \leqslant \omega$, so $D \cup (Y \setminus \operatorname{cl}_{\tau_{\omega_1}} A_\gamma)$ is a countable dense (in τ_{ω_1}) subset of Y. Thus, $\langle \mathbb{R}, \tau_{\omega_1} \rangle$ is hereditarily separable. ■

Proof of Lemma 258

Let X be a locally compact, dense subset of a T_2-space Z, let $x \in X$, and let N_x be a compact neighborhood of x in X. Let U_x be an open subset of Z satisfying $N_x^\circ = X \cap U_x$. We have N_x° is dense in U_x so that

$$U_x \subset \overline{U}_x = \overline{N_x^\circ} \subset \overline{N}_x = N_x,$$

where closures are in Z. Because each point of X is contained within a set open in Z contained within X, X is open in Z. ■

Proof of Corollary 259

Assume αX is a compactification of a locally compact space X. By definition, αX is a T_2-space in which homeomorphism $\alpha\colon X \to \alpha X$ embeds X as a dense subset $\alpha[X]$ of αX. This implies $\alpha[X]$ is also locally compact, so that $\alpha[X]$ is open in αX by Lemma 258. ■

Proof of Theorem 260

Define $f\colon \alpha X \to \omega X$ by $p \mapsto (\omega \circ \alpha^{-1})(p)$ if $p \in \alpha[X]$ and $p \mapsto \infty$ if $p \in \alpha X \setminus \alpha[X]$. For each $x \in X$ we have

$$(f \circ \alpha)(x) = f(\alpha(x)) = (\omega \circ \alpha^{-1})(\alpha(x)) = \omega(x),$$

so that $f \circ \alpha = \omega$. Now we prove f is continuous. To that end, let U be open in ωX. We show $f^{-1}[U]$ is open in αX.

If $\infty \notin U$ then U is an open subset of $\omega[X]$, so

$$f^{-1}[U] = (\omega \circ \alpha^{-1})^{-1}[U]$$

is open in $\alpha[X]$. By Corollary 259, $f^{-1}[U]$ is also open in αX.

If $\infty \in U$, then

$$\begin{aligned} f^{-1}[U] &= \alpha X \setminus f^{-1}[\omega X \setminus U] \\ &= \alpha X \setminus f^{-1}[\omega[X] \setminus U] = \alpha X \setminus (\omega \circ \alpha^{-1})^{-1}[\omega[X] \setminus U]. \end{aligned}$$

By definition of the one-point compactification ωX, we know that $\omega[X] \setminus U$ is compact. Note that the preimage $(\omega \circ \alpha^{-1})^{-1}$ is the image of the continuous inverse of $\omega \circ \alpha^{-1}$. So $(\omega \circ \alpha^{-1})^{-1}[\omega[X] \setminus U]$ is compact, thus it is closed in αX. Again we see that $f^{-1}[U]$ is open. ■

Proof of Theorem 261

Let $\gamma X = X \cup (\alpha X \setminus \alpha[X])$. Define $\theta : \gamma X \to \alpha X$ such that

$$\theta(x) = \begin{cases} \alpha(x) & : \quad x \in X \\ x & : \quad x \in \alpha X \setminus \alpha[X] \end{cases}$$

and declare U to be open in γX if and only if $\theta[U]$ is open in αX. Since θ is a bijection, it follows that θ is a homeomorphism by definition, and X is dense in γX. Finally, if $\gamma : X \to \gamma X$ is the inclusion map, $\theta \circ \alpha = \gamma$ trivially, so $\alpha X \approx \gamma X$. ■

Proof of Lemma 262

Suppose f and g agree on the dense subset D of X, but $f \neq g$. Pick $p \in X$ such that $f(p) \neq g(p)$. Since Y is Hausdorff, we can take disjoint open sets O_1 and O_2 containing $f(p)$ and $g(p)$, respectively. By continuity, there are open sets U_f and U_g contiaining p such that $f(U_f) \subset O_1$ and $g(U_g) \subset O_2$. Since D is dense, there is a point $d \in D \cap U_f \cap U_g$. But then $f(d) \in O_1$ and $g(d) \in O_2$, so $f(d) \neq g(d)$, contradiction. ■

Proof of Theorem 263

Let αX and γX be compactifications of a space X such that $\alpha : X \to \alpha X$ and $\gamma : X \to \gamma X$ are identity maps. The forwards implication is immediate.

Suppose $\alpha X \geqslant \gamma X$ and $\gamma X \geqslant \alpha X$. Let $f : \alpha X \to \gamma X$ and $g : \gamma X \to \alpha X$ be continuous such that $f \circ \alpha = \gamma$ and $g \circ \gamma = \alpha$. If $x \in X$ then

$$f(g(x)) = f(g(\gamma(x))) = f(\alpha(x)) = \gamma(x) = x.$$

Thus $f \circ g$ and $i_{\gamma X}$ agree on a dense subset of γX, so $f \circ g = i_{\gamma X}$. By a similar argument $g \circ f = i_{\alpha X}$, so f and g are inverses. It follows that f is a homeomorphism, whence $\alpha X \approx \gamma X$ ■

Proof of Theorem 264

(a) For each subbasic open set $\pi_f^{-1}[U]$ of $I^{\mathcal{F}}$ where U is open in I, note $f^{-1}[U]$ is open in X by the continuity of f. For each $x \in f^{-1}(U)$, $e_{\mathcal{F}}(x)(f) = f(x) \in U$, and thus $e_{\mathcal{F}}(x) \in \pi_f^{-1}[U]$. Thus $e_{\mathcal{F}}$ is continuous.

(b) Let $x \neq y$. Since $\mathcal{F}$ separates points, choose $f \in \mathcal{F}$ such that $f(x) \neq f(y)$. It then follows from

$$e_{\mathcal{F}}(x)(f) = f(x) \neq f(y) = e_{\mathcal{F}}(y)(f)$$

that $e_{\mathcal{F}}(x) \neq e_{\mathcal{F}}(y)$.

(c) By (a), $e_{\mathcal{F}}$ is continuous. As X is T_1, points are closed, and thus $\mathcal{F}$ separates points, showing $e_{\mathcal{F}}$ is one-to-one by (b). And of course, $e_{\mathcal{F}}$ is onto its range $e_{\mathcal{F}}[X]$.

We conclude the proof by showing that $e_{\mathcal{F}}$ is a closed map. Let H be closed, and $x' \notin e_{\mathcal{F}}[H]$. As $e_{\mathcal{F}}[H]$ is a bijection, choose $x \notin H$ with $e_{\mathcal{F}}(x) = x'$.

Since $\mathcal{F}$ separates points from closed sets, choose $f \in \mathcal{F}$ such that $f(x) \notin \overline{f[H]}$. So let U be an open set separating $f(x)$ from $f[H]$.

$x'(f) = e_{\mathcal{F}}(x)(f) = f(x) \in U$, so $x' \in \pi_f^{-1}[U]$. Furthermore for each $h' \in e_{\mathcal{F}}[H]$ where $h' = e_{\mathcal{F}}[h]$, $h'(f) = f(h) \notin U$, so $h' \notin \pi_f^{-1}[U]$. Thus x' is not a limit point of $e_{\mathcal{F}}[H]$, showing $e_{\mathcal{F}}[H]$ is closed. ■

Proof of Corollary 265

Proof of Corollary 266

Proof of Theorem 267
Let αX be a compactification of X. Without loss of generality, assume α is the identity on X. Define $\mathcal{F} = \{f \restriction X | f \in C(\alpha X, I)\}$, and $\varphi : \alpha X \to e_{\mathcal{F}} X$ by $\varphi(a) = e_{C(\alpha X, I)}(a) = \langle f(a) \rangle_{f \in C(\alpha X, I)}$. ■

Proof of Theorem 268
Define $H : I^{\mathcal{C}(X,I)} \to I^{\mathcal{C}(Y,I)}$ by $\pi_g H = \pi_{g \circ h}$. Then H is continuous and for any $x \in X$ and $g \in C(Y, I)$ we have

$$\begin{aligned}\pi_g(H \circ \beta_X)(x) = \pi_g H(\langle f(x) \rangle) &= (g \circ h)(x) \\ &= g(h(x)) = \pi_g \beta_Y(h(x)) = \pi_g(\beta_Y \circ h)(x),\end{aligned}$$

so that $H \circ \beta_X = \beta_Y \circ h$. By continuity of H we have

$$\begin{aligned}H[\beta X] = H[\mathrm{cl}_{I^{C(X,I)}} \beta_X(X)] &\subseteq \mathrm{cl}_{I^{C(Y,I)}} H[\beta_X(X)] \\ &= \mathrm{cl}_{I^{C(Y,I)}} \beta_Y[h(X)] \subseteq \beta Y.\end{aligned}$$ ■

Proof of Theorem 269
By the previous theorem, there exists a continuous $F' : I^{\mathcal{C}(X,I)} \to I^{\mathcal{C}(Y,I)}$ such that $F' \circ \beta_X = \beta_Y \circ f$ and $F' \restriction \beta_X X : \beta_X X \to \beta_Y Y$. Since Y is already compact, β_Y^{-1} is a homeomorphism from $\beta_Y Y$ to Y.

Let $F = \beta_Y^{-1} \circ (F' \restriction \beta_X X)$. Then for each $x \in X$,

$$F(\beta_X(x)) = \beta_Y^{-1}(F'(\beta_X(x))) = \beta_Y^{-1}(\beta_Y(f(x))) = f(x)$$ ■

Proof of Corollary 270

Proof of Theorem 271
Let $f : \omega_1 \to I^\kappa$ be continuous. For each $\alpha < \kappa$ there exist $\beta_\alpha < \omega_1$ and $x_\alpha \in I$ such that $(\pi_\alpha \circ f)(\beta) = x_\alpha$ for all $\beta > \beta_\alpha$. Define $F : \omega_1 + 1 \to I^\kappa$ so that

$$(\pi_\alpha \circ F)(\beta) = \begin{cases} (\pi_\alpha \circ f)(\beta) & \text{if } \beta \neq \omega_1 \\ x_a & \text{if } \beta = \omega_1 \end{cases}$$

for all $\alpha < \kappa$ and $\beta < \omega_1 + 1$.

Certainly $F \restriction \omega_1 = f$. Let $U \subset I$ be open. If $x_\alpha \in U$, then $(\pi_\alpha \circ F)^{-1}(U) = (\pi_\alpha \circ f)^{-1}(U)$. Otherwise, $(\pi_\alpha \circ F)^{-1}(U) = (\pi_\alpha \circ f)^{-1}(U) \cup (\beta_\alpha, \omega_1]$. Thus F is continuous, so $\omega_1 + 1 \approx \beta\omega_1$.

Alternatively... Let $p \neq q$ be in the remainder $\alpha X \setminus X$. Then there exist open sets U, V containing p, q whose closures are disjoint. But then

$$[\overline{U} \cap \omega_1] \cap [\overline{V} \cap \omega_1] = \varnothing$$

are closed unbounded sets, and should intersect. Contradiction. ■

Proof of Theorem 272

Let X be a separable compact Hausdorff space with countable dense subset D. There exists a bijection $f : \omega \to D$; f is continuous since ω is a discrete space. Using theorem 19, let $F : \beta\omega \to X$ be a continuous extension of f. Clearly, $F(\beta\omega)$ is dense in X (for $F(\beta\omega) \supseteq F(\omega) = f(\omega) = D$). Furthermore, $F(\beta\omega)$ is closed in X since $\beta\omega$ is compact, F is continuous, and X is Hausdorff. So we have $F(\beta\omega) = \overline{F(\beta\omega)} = X$, illustrating $\beta\omega$ maps continuously onto separable compact Hausdorff space X. ■

Proof of Corollary 273

Proof of Lemma 274

Define $f : X \to 2^{\mathcal{P}(\omega)}$ by

$$f(x)(A) = \begin{cases} 1 & \text{if } x \in \overline{\{d_i : i \in A\}} \\ 0 & \text{else} \end{cases}.$$

We show f is injective, proving that $|X| \leqslant 2^{\mathfrak{c}}$. To that end, suppose $x \neq y \in X$. Since X is Hausdorff, there exists an open set U containing x whose closure misses y. Let $A = \{i \in \omega : d_i \in U\}$. Then $x \in \overline{\{d_i : i \in A\}}$ but $y \notin \overline{\{d_i : i \in A\}}$. So $f(x)(A) = 1$ and $f(y)(A) = 0$. So $f(x) \neq f(y)$. ■

Proof of Corollary 275

Proof of Theorem 276
Let $\iota : X \to Z$ be the inclusion map, so $Z = \iota X$. Recall $\beta = e_{\mathcal{C}(X,I)} : X \to I^{\mathcal{C}(X,I)}$. Define $g : \iota X \to \beta X$ by $g(z)(f) = f^*(z)$. Note g is continuous as f^* is continuous, and $g \circ \iota = \beta$ as for each $x \in X$ and $f \in \mathcal{C}(X, I)$,

$$(g \circ \iota)(x)(f) = g(\iota(x))(f) = f^*(\iota(x)) = f^*(x)$$

$$= f(x) = e_{\mathcal{C}(X,I)}(x)(f) = \beta(x)(f).$$

Thus $\iota X = Z \geqslant \beta X$, so $Z \approx \beta X$. ■

Proof of Theorem 277
Let H and K be disjoint closed subsets of a normal space X. By theorem 7500.25, there exists a continuous function $f : X \to I$ with $f(H) = \{0\}$ and $f(K) = \{1\}$. By theorem 19, there is a continuous extension $F : \beta X \to I$ of f. Then $F^{-1}(\{0\})$ and $F^{-1}(\{1\})$ are disjoint closed subsets of βX containing H and K, respectively, so $cl_{\beta X} H \cap cl_{\beta X} K = \varnothing$. ■

Proof of Theorem 278
Let $f : H \to I$ be a continuous function. Because H is a closed subset of normal X, f can be extended continuously to $\hat{f} : X \to I$ using the Tietze Extension Theorem. By theorem 19 (and the remark following it), $\hat{f}$ can be extended continuously to $F : \beta X \to I$. Define $f^* : \mathrm{cl}_{\beta X} H \to I$ to be the restriction of F to $\mathrm{cl}_{\beta X} H$. Because $\mathrm{cl}_{\beta X} H$ is compact and Hausdorff, H is dense in $\mathrm{cl}_{\beta X} H$, and f extends continuously to f^*, $\mathrm{cl}_{\beta X} H \approx \beta H$ by theorem 26. ■

Proof of Corollary 279

Proof of Theorem 280
It suffices to show

$$\beta[0,1) \setminus [0,1) = \bigcap_{n \in \mathbb{N}} \mathrm{cl}_{\beta[0,1)}[1 - 1/n, 1),$$

since the intersection of a nested family of compact connected spaces is connected.

($\subseteq$): For a contradiction suppose $p \in \beta[0,1) \setminus [0,1)$ but $n \in \mathbb{N}$ such that $p \notin \mathrm{cl}_{\beta[0,1)}[1-1/n,1)$. There exists an open U in $\beta[0,1)$ with $p \in U$ and $U \cap \mathrm{cl}_{\beta[0,1)}[1-1/n,1) = \varnothing$. Then $U \cap [0,1) \subseteq [0,1-1/n]$. Since $p \notin [0,1)$ and $[0,1-1/n]$ is closed in $\beta[0,1)$, there exists an open set V in $\beta[0,1)$ with $p \in V$ and $V \cap [0,1-1/n] = \varnothing$. But $U \cap V$ is a nonempty open set in $\beta[0,1)$ which misses $[0,1)$, a contradiction.

($\supseteq$): Suppose $p \in [0,1)$. There exists $n \in \mathbb{N}$ such that $p < 1-1/n$. Then $\{p\}$ and $[1-1/n,1)$ are disjoint closed sets in $[0,1)$, so they have disjoint closures in $\beta[0,1)$, i.e., $p \notin \mathrm{cl}_{\beta[0,1)}[1-1/n,1)$.

Letting $(-\infty,-n] \cup [n,\infty)$ play the role of $[1-1/n,1)$,

$$\beta\mathbb{R}\backslash\mathbb{R} = \bigcap_{n\in\mathbb{N}} \mathrm{cl}_{\beta\mathbb{R}}((-\infty,-n]\cup[n,\infty)) = \bigcap_{n\in\mathbb{N}} \mathrm{cl}_{\beta\mathbb{R}}(-\infty,-n]\cup\mathrm{cl}_{\beta\mathbb{R}}[n,\infty) =$$

$$\bigcap_{n\in\mathbb{N}} \mathrm{cl}_{\beta\mathbb{R}}(-\infty,-n] \cup \bigcap_{n\in\mathbb{N}} \mathrm{cl}_{\beta\mathbb{R}}[n,\infty),$$

so that $\beta\mathbb{R} \setminus \mathbb{R}$ is the union of two connected components. ■

Proof of Theorem 281

$\neg$(c)$\rightarrow$ $\neg$(a): Let $G \subseteq X$ satisfy $G \notin \mathcal{F}$ and $X \setminus G \notin \mathcal{F}$. If there does not exist $F \in \mathcal{F}$ such that $F \cap G = \varnothing$, then let $\mathcal{F}' = \{F \cap G : F \in \mathcal{F}\}$ and observe $\mathcal{F}'$ is a filter and $\mathcal{F}' \supseteq \mathcal{F}$. Otherwise there exists $H \in \mathcal{F}$ where $H \cap G = \varnothing$, but then $H \setminus G = H$ and for each $F \in \mathcal{F}$, $\varnothing \neq F \cap H = F \cap (H \setminus G) \subseteq F \setminus G$. Thus $\mathcal{F}'' = \{F \setminus G : F \in \mathcal{F}\}$ is a filter and $\mathcal{F}'' \supseteq \mathcal{F}$.

$\neg$(a)$\rightarrow$ $\neg$(c): Let $\mathcal{F}' \supsetneq \mathcal{F}$ be a filter, and $G \in \mathcal{F}' \setminus \mathcal{F}$. Then $X \setminus G \notin \mathcal{F}'$, and thus $X \setminus G \notin \mathcal{F}$.

$\neg$(b)$\rightarrow$ $\neg$(a): $\mathcal{F} \cup \{G\}$ may be extended to a filter properly containing $\mathcal{F}$.

$\neg$(a)$\rightarrow$ $\neg$(b): Let $\mathcal{F}' \supsetneq \mathcal{F}$. Then there is a $G \in \mathcal{F}'$ which intersects every member of $\mathcal{F}$; therefore, G and $X \setminus G$ are not in $\mathcal{F}$. ■

Proof of Theorem 282

Proof of Theorem 283
Let $A \subset \omega$. To see (i), first note that if $x \in \omega$ then $x \in \overline{A}^{\beta\omega}$ iff $x \in A$. Moreover, if $x \in \beta\omega \setminus \omega$ then $x \in \overline{A}^{\beta\omega}$ iff every neighborhood of x meets A iff $A \in \mathcal{F}_x$. Thus, $\overline{A}^{\beta\omega} = A \cup \{p \in \beta\omega \setminus \omega : A \in \mathcal{F}_p\}$.

To see (ii), observe that $\beta\omega = \overline{A}^{\beta\omega} \cup \overline{\omega \setminus A}^{\beta\omega}$ with $\overline{A}^{\beta\omega} \cap \overline{\omega \setminus A}^{\beta\omega} = \varnothing$.

For (iii), let $U \subset \beta\omega$ be open and let $x \in U$. By regularity, there exists an open $V \subset \beta\omega$ with $x \in V \subset \overline{V}^{\beta\omega} \subset U$. Put $A = V \cap \omega$. If $x \in \omega$, then $x \in A$. If $x \in \beta\omega \setminus \omega$, then $A \in \mathcal{F}_x$. In either case, $x \in \overline{A}^{\beta\omega} \subset \overline{V}^{\beta\omega} \subset U$. ■

Proof of Theorem 284
Let X be the space defined above. Define $\varphi : X \to \beta\omega$ by $\varphi(x) = x$ for $x \in \omega$ and $\varphi(\mathcal{F}) = p$ for free ultrafilter $\mathcal{F}$ on ω and $p \in \beta\omega \setminus \omega$ satisfying $\mathcal{F}_p = \mathcal{F}$. By theorem 32, φ is a well-defined bijection. Let $A \subset \omega$. By theorem 33, $\{\overline{A}^{\beta\omega} : A \subset \omega\}$ forms a base for $\beta\omega$. We have

$$\varphi^{-1}(\overline{A}^{\beta\omega}) = \varphi^{-1}(A \cup \{p \in \beta\omega \setminus \omega : A \in \mathcal{F}_p\}) = A \cup \{\mathcal{F} : A \in \mathcal{F}\} = \overline{A}$$

thus demonstrating that the pre-image of a base element in $\beta\omega$ is a base element in X. By a similar argument, $\varphi(\overline{A}) = \overline{A}^{\beta\omega}$. We conclude φ is a homeomorphism, and $X \approx \beta\omega$. ■

Proof of Theorem 285
Let (a_n) be a nontrivial (not eventually constant) sequence of points in ω. Then (a_n) converges to no point in ω. Now suppose for a contradiction that (a_n) converges to $p \in \beta\omega \setminus \omega$. Then (a_n) is unbounded. Let (a_{n_k}) be a strictly increasing subsequence of (a_n). Then $p \in \mathrm{cl}_{\beta\omega}\{a_{n_{2k}} : k \in \omega\} \cap \mathrm{cl}_{\beta\omega}\{a_{n_{2k+1}} : k \in \omega\}$, a contradiction. ■

Proof of Lemma 286

Proof of Theorem 287
For each $A \subseteq \omega$, let $\mathcal{C}_A = \{A \cap n \subseteq \omega : n < \omega\}$.

Then for $A \neq B$, assume $a \in A \setminus B$. Then as $a \in A \cap i$ for all $i > a$, but $a \notin B \cap i$ for all $i < \omega$, we conclude that $\mathcal{C}_A, \mathcal{C}_B$ are almost disjoint. ■

Proof of Corollary 288
Let $\mathcal{A}$ be as in the previous theorem. Then $\mathcal{A}^* = \{A^* : A \in \mathcal{A}\}$ is a pairwise disjoint collection (Lem 36) of clopen sets (Thm 33) of size 2^ω. ■

Proof of Lemma 289
Let $D = \{d_n : n < \omega\}$ be a relatively discrete subset of a regular space X. For each $n < \omega$ let $W_n \subset X$ be open so that $W_n \cap D = \{d_n\}$. By regularity, for each $n < \omega$ there exists open $V_n \subset X$ with $d_n \in V_n \subset \overline{V_n} \subset W_n$. For each $n < \omega$, put $U_n = V_n \setminus (\cup_{i<n} \overline{V_i})$. Then $\{U_n : n < \omega\}$ is a pairwise disjoint family of open sets with $d_n \in U_n$ for each $n < \omega$.■

Proof of Theorem 290
Let $D \subset \beta\omega$ be countably infinite and relatively discrete. Let $f : D \to I$ be continuous. By Lemma 39, there is a family of disjoint open sets $\mathcal{U} = \{U_d : d \in D\}$ with $d \in U_d$ for each $d \in D$. Define $f^* : \omega \to I$ by $f^*(x) = f(d)$ if $x \in U_d$ for some $d \in D$, and $f^*(x) = 0$ otherwise. Because $\mathcal{U}$ is a family of disjoint sets, f^* is well-defined. Extend f^* continuously to $F : \beta\omega \to I$. Since $F(x) = f^*(x) = f(d)$ for every $x \in U_d \cap \omega$, and $U_d \cap \omega$ is dense in U_d, it follows that $F(d) = f(d)$ for every $d \in D$. Thus $F \restriction \overline{D}$ is a continuous extension of f to $\overline{D}$, hence $\overline{D} \approx \beta D$. ■

Proof of Corollary 291
The closure of any nontrivial convergent sequence $D = \{d_i : i \in \omega\}$ is the singleton $\{d\}$ consisting of the point d to which the sequence converges, and $\{d\} \not\approx \beta D$. ■

Proof of Lemma 292
Let X be an infinite Hausdorff space. If X is discrete, we are done. Otherwise, X has a limit point x^*. Let $x_0 \in X \setminus \{x^*\}$. Let U_0 be open in X with $x^* \in U_0$ and $x_0 \notin \overline{U}_0$. There exists $x_1 \in U_0 \setminus \{x^*\}$. Let $U_1 \subseteq U_0$ be open in X with $x^* \in U_1$ and $x_1 \notin \overline{U}_1$. Continue in this manner. Then for any $n \in \omega$, $(X \setminus \overline{U}_n) \cap U_{n-1}$ is an open set containing x_n but no x_i with $i \neq n$. So $\{x_n : n \in \omega\}$ is an infinite discret subspace of X. ■

Proof of Theorem 293
Let X be an infinite closed subset of $\beta\omega$. Let A be a countably infinite

discrete subspace of X. Clearly A is also discrete as a subspace of $\beta\omega$. By Theorem 40, we have $\beta\omega \simeq \mathrm{cl}_{\beta\omega} A \subseteq X$ ■

Proof of Lemma 294

Observe that $\subseteq^*$ is a partial order: Let $A \subseteq^* B$ and $B \subseteq^* C$. For $a \in A \setminus C$, then either $a \in B \setminus C$, or $a \notin B \setminus C$. In the latter case, as $a \notin C$, we have that $a \notin B$, and thus $a \in A \setminus B$. Therefore $|A \setminus C| \leqslant |A \setminus B| + |B \setminus C| < \omega$.

It follows then that $|\bigcap_{i<n} A_i| = \omega$; as this is true for $n = 0$, assume it is true for n. Then as $A_n \subseteq^* A_i$ for $i < n$, we have that

$$|A_n \setminus \bigcap_{i<n} A_i| = |\bigcup_{i<n} (A_n \setminus A_i)| < \omega$$

Since A_n is infinite, the induction hypothesis follows: $|A_n \cap \bigcap_{i<n} A_i| = \omega$.

Finally, pick $a_m \in \bigcap_{i<m} A_i$ such that $a_m > a_i$ for all $i < m$, and define $A = \{a_m : m < \omega\}$. Then for $n < \omega$, $a_m \in A_n$ for all $m > n$, and thus $|A \setminus A_n| < \omega$. ■

Proof of Theorem 295

Let G be a non-empty G_δ-set in ω^*. Then there exists $p \in G$ and $U_0, U_1, \ldots$ open in ω^* such that $G = \cap_{n<\omega} U_n$. Let $A_0 \subset \omega$ so that $p \in A_0^* \subset U_0$. If $A_n \subset \omega$ has been defined for some $n < \omega$, let $A_{n+1} \subset \omega$ so that $p \in A_{n+1}^* \subset A_n^* \cap U_{n+1}$. Then $A_0^* \supset A_1^* \supset \ldots$ and $G \supset \cap_{n<\omega} A_n^*$. It follows that $A_0 \supset^* A_1 \supset^* \ldots$, so there exists an infinite $A \subset \omega$ such that $A \subset^* A_n$ for each $n < \omega$. Thus $A^* \subset A_n^*$ for each $n < \omega$, so $A^* \subset G^\circ$. ■

Proof of Theorem 296

For each $n \in \omega$ let $A_n = \omega \cap [n, \infty)$. Enumerate $\mathcal{P}(\omega) = \{X_\alpha : \alpha < \omega_1\}$ so that $X_n = A_n$ for each $n < \omega$. Suppose $\omega \leqslant \alpha < \omega_1$ and $\{A_\beta : \beta < \alpha\}$ has been defined so that

(i) $A_\gamma \subseteq^* A_\beta$ for $\beta < \gamma < \alpha$, and

(ii) $A_\beta \subseteq^* X_\beta$ for each $\beta < \alpha$.

The sets $A_\beta \cap [n, \infty)$, $\beta < \alpha$, $n < \omega$, have the finite intersection property, so at least one of X_α and $\omega \setminus X_\alpha$ intersects every one of them. Choose one of the two that does, say X_α. Then $|X_\alpha \cap A_\beta| = \omega$ for each

$\beta < \alpha$ and $X_\alpha \cap A_\gamma \subseteq^* X_\alpha \cap A_\beta$ for $\beta < \gamma < \alpha$. By Lemma 44 there exists an infinite $A_\alpha \subseteq \omega$ such that $A_\alpha \subseteq^* X_\alpha \cap A_\beta$ for each $\beta < \alpha$.

Claim the set u in the hint is an ultrafilter. Well clearly it has the finite intersection property, so it generates a filter even if it isn't one already. By construction every subset of ω or its complement is in u, so that u is an ultrafilter. It is in ω^* since it includes the sets A_n.

Given any G_δ-set $G = \bigcap_{n\in\omega} U_n$ containing u, for each $n \in \omega$ let $\beta_n < \omega_1$ such that $A_{\beta_n} \subseteq U_n$ and let $\alpha = \sup\{\beta_n : n < \omega\} + 1$. Then $u \in A^*_\alpha \subseteq G$. ■

Proof of Lemma 297

Let X be an infinite compact T_2-space. By lemma 42, X contains an infinite relatively discrete subset $Y = \{y_i : i < \omega\}$. Because X is T_2 (and hence T_1), $\{y_i\}$ is closed in X for each $i < \omega$. This implies $X \setminus Y = \cap_{i<\omega} X \setminus \{y_i\}$ is G_δ.

Now, the closure $\overline{Y}$ of Y in compact X is also compact, so that $Y \neq \overline{Y}$ (otherwise, $\{\{y_i\} : i < \omega\}$ would be an open cover of Y for which there is no finite subcover). Let $x \in \overline{Y} \setminus Y \subseteq X \setminus Y$. Any open set containing x meets Y, so x is not an element of the interior of $X \setminus Y$. Therefore, $X \setminus Y$ is not a neighborhood of x, and x is not a P-point. ■

Proof of Corollary 298

Proof of Theorem 299

Suppose X is not countably compact. Then let $\mathcal{U} = \{U_n : n < \omega\}$ be an open cover such that $x_n \in U_n \setminus \bigcup_{i<n} U_i$. Then for each $x \in U_n$, note that U_n misses x_m for $m > n$, so x is not an infinite accumulation point of $\{x_n : n < \omega\}$.

If X is countably compact, choose $x_n \in X$. For $n < \omega$, let $C_n = \overline{\{x_i : i > n\}}$. Note that since $X \setminus C_n \subseteq X \setminus C_{n+1}$, and $X \setminus C_n \subsetneq X$, there is no finite subcollection of $\{X \setminus C_n : n < \omega\}$ covering X. Thus as X is countably compact, $\{X \setminus C_n : n < \omega\}$ is not an open cover.

So choose $x \in \bigcap_{n<\omega} C_n$. For any open neighborhood U of x and $n < \omega$, U intersects $\{x_i : i > n\}$. Thus U intersects infinitely many x_n, making x an infinite accumulation point. ■

Proof of Corollary 300

Proof of Lemma 301
Let $X, Y \subset \beta\omega$ with $X \cap Y = \omega$ and put $D = \{(x, x) : x \in \omega\}$. Certainly $D \subset X \times Y$ is discrete. If $(p, q) \notin D$ then there exist disjoint neighborhoods (in $\beta\omega$) U and V of p and q, respectively. Then $(U \cap X) \times (V \cap Y)$ is a neighborhood of (p, q) missing D, so D is closed. ■

Proof of Theorem 302
(a) There are at most $2^{\mathfrak{c}}$ (infinite) closed subsets of ω^* since it has a basis of size $\mathfrak{c}$ (note that there are actually $2^{\mathfrak{c}}$ by Corollary 38). Let $\{C_\alpha : \alpha < \kappa\}$ be an enumeration of the infinite closed subsets of ω^*, where $\kappa \leqslant 2^{\mathfrak{c}}$ is a cardinal. By Theorem 43, each C_α has cardinality $2^{\mathfrak{c}}$. Recursively define $X = \{x_\alpha : \alpha < \kappa\}$ and $\{y_\alpha : \alpha < \kappa\}$ so that $x_\alpha \in C_\alpha \setminus \{y_\beta : \beta < \alpha\}$ and $y_\alpha \in C_\alpha \setminus \{x_\beta : \beta \leqslant \alpha\}$ for each $\alpha < \kappa$.

(b) Let X be a subset of ω^* such that X and $\omega^* \setminus X$ meet every infinite closed subset of ω^*. Let Y be an infinite subset of $\omega \cup X$. By lemma 42, Y contains a countably infinite relatively discrete subset Z. The closure $\overline{Z}$ of Z in $\beta\omega$ contains a copy of $\beta\omega$ by theorem 43, and is uncountable. Let $x \in (\omega \cup X) \cap (\overline{Z} \setminus Z)$. As a limit point of Z, x is also a limit point of Y. So $\omega \cup X$ is countably compact by corollary 50. ■

Proof of Corollary 303

Proof of Theorem 304
Let $x \in \left(\prod_{n<\omega} X_n\right)^\omega$. Since X_0 is sequentially compact, it follows that there exists a subsequence x_0 of x such that $\pi_0(x_0)$ converges in X_0.

If x_n is a subsequence of x, then as X_{n+1} is sequentially compact, there exists a subsequence x_{n+1} of x_n such that $\pi_{n+1}(x_{n+1})$ converges in X_{n+1}. (In fact, $\pi_i(x_i)$ converges in X_i for all $i \leqslant n+1$.)

Define $f_n : \omega \to \omega$ such that $x_n(i) = x(f_n(i))$ and $\operatorname{ran}(f_n) \supseteq \operatorname{ran}(f_{n+1})$. It follows by REFERENCE that there exists an increasing $f : \omega \to \omega$ such that $\operatorname{ran}(f) \subseteq^* \operatorname{ran}(f_n)$ for all $n < \omega$.

Let $y \in \left(\prod_{n<\omega} X_n\right)^\omega$ such that $y(n) = x(f(n))$. It follows that $\pi_n(y)$ converges in X_n for all $n < \omega$, so y converges in $\prod_{n<\omega} X_n$. ■

Proof of Theorem 305
Recall $\beta\omega = \overline{e_{\mathcal{F}}(\omega)} \subset I^{\mathcal{F}}$, with $\mathcal{F} = \mathcal{C}(\omega, I)$. Since

$$|\mathcal{F}| = |\mathcal{C}(\omega, I)| = (2^\omega)^\omega = 2^{\omega\cdot\omega} = 2^\omega = |I|,$$

$\beta\omega$ is homeomorphic to a closed subset of I^I. As $\beta\omega$ is not sequentially compact, neither is I^I. ■

Proof of Theorem 306
We show that if $\kappa < \mathfrak{t}$ then every subset of ω^ω of size κ is $<^*$-bounded, so that $\mathfrak{b} \not< \mathfrak{t}$.

Let $\kappa < \mathfrak{t}$ and let $\{f_\alpha : \alpha < \kappa\} \subseteq \omega^\omega$. Let $g_0 \in \omega^\omega$ be increasing with $f_0 < g_0$. Suppose $\alpha \leqslant \kappa$ and an increasing g_β has been defined for each $\beta < \alpha$ so that $f_\gamma <^* g_\beta$ and $\operatorname{ran} g_\beta \subseteq^* \operatorname{ran} g_\gamma$ for $\gamma \leqslant \beta$. There exists an infinite $X \subseteq \omega$ such that $X \subseteq^* \operatorname{ran} g_\beta$ for each $\beta < \alpha$. Let g_α be increasing with range $\subseteq X$ skipping infinitely many members of X and $f_\alpha < g_\alpha$. Using the hint we have $f_\gamma < g_\gamma <^* g_\alpha$ for $\gamma \leqslant \alpha$. Continuing in the manner g_κ will be a $<^*$-upper bound for $\{f_\alpha : \alpha < \kappa\}$. ■

Proof of Theorem 307
Let $\kappa \in \mathfrak{c}$. Let $\{A_\alpha : \alpha < \kappa\}$ be a tower of length κ. We will show this tower has an infinite pseudo-intersection. Let P be the collection of all pairs of the form $(\vec{C}, F)$, where $\vec{C}$ is a finite sequence of distinct elements in ω, and F is a finite subset of κ. For $(\vec{C}, F), (\vec{C}', F') \in P$, define $(\vec{C}', F') < (\vec{C}, F)$ iff

- $\vec{C}'$ extends $\vec{C}$,
- $F' \supseteq F$, and
- for each $i \in \operatorname{dom}(\vec{C}') \setminus \operatorname{dom}(\vec{C})$, we have $C_i' \in \bigcap_{\alpha\in F} A_\alpha$.

($<$, partial order on P?) Reflexivity and antisymmetry of $<$ are trivially satisfied. For transitivity, let $(\vec{C}, F), (\vec{C}', F'), (\vec{C}", F") \in P$ such that $(\vec{C}", F") < (\vec{C}', F') < (\vec{C}, F)$. Clearly, $\vec{C}"$ extends $\vec{C}$ and $F" \supseteq F$. Let $i \in \operatorname{dom}(\vec{C}")\setminus\operatorname{dom}(\vec{C})$. If $i \in \operatorname{dom}(\vec{C}')$, then $C_i'' \in \bigcap_{\alpha\in F} A_\alpha$ because

$(\vec{C}', F') < (\vec{C}, F)$. If $i \notin \mathrm{dom}(\vec{C}')$, then $C_i'' \in \bigcap_{\alpha \in F'} A_\alpha \subseteq \bigcap_{\alpha \in F} A_\alpha]$ because $(\vec{C}", F") < (\vec{C}', F')$. So $<$ is a partial order on P.

($(P, <)$ has the ccc?) Let $X = \{(\vec{C}_\alpha, F_\alpha) : \alpha \in \Lambda\}$ be an uncountable collection of elements of P. There are only a countable number of finite sequences of distinct elements in ω, so there exist distinct $\alpha_1, \alpha_2 \in \Lambda$ such that $\vec{C}_{\alpha_1} = \vec{C}_{\alpha_2}$ and $F_{\alpha_1} \neq F_{\alpha_2}$. We have $(\vec{C}_{\alpha_1}, F_{\alpha_1} \cup F_{\alpha_2}) \in P$ such that $(\vec{C}_{\alpha_1}, F_{\alpha_1} \cup F_{\alpha_2}) < (\vec{C}_{\alpha_1}, F_{\alpha_1})$ and $(\vec{C}_{\alpha_1}, F_{\alpha_1} \cup F_{\alpha_2}) < (\vec{C}_{\alpha_2}, F_{\alpha_2})$, so that X is not an antichain. Therefore, every antichain of P is countable, and $(P, <)$ has the ccc.

Now, for each $i \in \omega$, define $D_i = \{(\vec{C}, F) \in P : i \in \mathrm{dom}(\vec{C})\}$. For each $\alpha \in \kappa$, define $E_\alpha = \{(\vec{C}, F) \in P : \alpha \in F\}$.

(E_α dense for each $\alpha \in \kappa$?) Let $(\vec{C}, F) \in P$ and $\alpha \in \kappa$. $(\vec{C}, F \cup \{\alpha\}) \in E_\alpha$ and $(\vec{C}, F \cup \{a\}) < (\vec{C}, F)$, so that E_α is dense in P.

(D_i dense for each $i \in \omega$?) Let $(\vec{C}, F) \in P$ and $i \in \omega$. Wlog, assume $i \notin \mathrm{dom}(\vec{C})$. Because $\{A_\alpha : \alpha < \kappa\}$ is a tower, the set $\bigcap_{\alpha \in F} A_\alpha$ is infinite. So we may (and do) extend $\vec{C}$ to $\vec{C}'$ for which $i \in \mathrm{dom}(\vec{C}')$ by selecting distinct elements of $\bigcap_{\alpha \in F} A_\alpha$ not already appearing in $\vec{C}$. $(\vec{C}', F) < (\vec{C}, F)$ and $(\vec{C}', F) \in D_i$, so that D_i is dense in P.

The collection $\mathcal{D} = \{E_\alpha : \alpha \in \kappa\} \cup \{D_i : i \in \omega\}$ is a collection of $\leqslant \kappa$-many dense sets, so by Martin's Axiom, there exists a filter G in P such that $G \cap D \neq \varnothing$ for every $D \in \mathcal{D}$. If $(\vec{C}, F), (\vec{C}', F') \in G$, then there is some $(\vec{C}", F") \in G$ such that $(\vec{C}", F") < (\vec{C}, F)$ and $(\vec{C}", F") < (\vec{C}', F')$, and hence the sequence $\vec{C}"$ extends both $\vec{C}$ and $\vec{C}'$. So either $\vec{C}$ extends $\vec{C}'$ or vice-versa. Define $\vec{C}$ and F so that $(\vec{C}, F) \in D_i \cap G$ for each $i \in \omega$.

($\{C_i : i \in \omega\}$, pseudo-intersection?) Let $\alpha \in \kappa$. Let $(\vec{C}, F) \in E_\alpha \cap G$. Suppose $i \notin \mathrm{dom}(\vec{C})$. There is $(\vec{C}', F') \in G \cap D_i$, and $(\vec{C}", F") \in G$ with $(\vec{C}", F") < (\vec{C}', F')$ and $(\vec{C}", F") < (\vec{C}, F)$. So $i \in \mathrm{dom}(\vec{C}"$ and $i \notin \mathrm{dom}(\vec{C})$. Since $(\vec{C}", F") < (\vec{C}, F)$, $C_i" \in \bigcap_{\beta \in F} A_\beta \subseteq A_\alpha$. ■

Proof of Theorem 308

Let $\kappa < \mathfrak{t}$ and let $X = \prod_{\alpha < \kappa} X_\alpha$ with each X_α sequentially compact. Let $(x_n)_{n<\omega}$ be a sequence in X. Since X_0 is sequentially compact, take $A_0 \subset \omega$ and $y_0 \in X_0$ so that $(x_n(0))_{n \in A_0} \to y_0$. Suppose for some $\beta < \kappa$ that for each $\alpha < \beta$ we have defined $A_\alpha \subset \omega$ and $y_\alpha \in X_\alpha$ so that $(x_n(\alpha))_{n \in A_\alpha} \to y_\alpha$. If $\beta = \gamma + 1$, let $A_\beta \subset A_\gamma$ and $y_\beta \in X_\beta$ so that

$(x_n(\beta))_{n\in A_\beta} \to y_\beta$. If β is a limit ordinal, let A'_β be an infinite pseudo-intersection of the tower $\{A_\alpha : \alpha < \beta\}$. Take $A_\beta \subset A'_\beta$ and $y_\beta \in X_\beta$ so that $(x_n(\beta))_{n\in A_\beta} \to y_\beta$. Let A be an infinite pseudo-intersection of $\{A_\alpha : \alpha < \kappa\}$. Then $(x_n)_{n\in A} \to (y_\alpha)_{\alpha<\kappa}$, so X is sequentially compact. ■

Proof of Lemma 309

For (a), let $x \in X$ and U be an open neighborhood of x. Since X is completely regular, there exists a continuous function $f_U : X \to I$ such that $f_U(x) = 0$ and $f_U(y) = 1$ for all $y \notin U$. So let $F_U = \{z : f_U(z) \leqslant 0.5\} \subseteq U$; note that this is a zero-set neighborhood of x. Likewise, let $G_U = \{z : f_U(z) < 0.5\} \subseteq U$; note that this is a cozero-set open neighborhood of x.

For (b), note that if F is a zero-set, then $F = f^{-1}[\{0\}]$ for some continuous $f : X \to \mathbb{R}$, so F is closed. Furthermore, letting $U_n = \{x : f(x) < 2^{-n}\}$ witnesses that F is a regular G_δ set.

For (c), assume the U_n are decreasing. If X is normal and $F = \bigcap_{n\in\omega} U_n = \bigcap_{n\in\omega} \overline{U}_n$, then let $f_n : X \to [0, 2^{-n}]$ be continuous and satisfy $f_n(x) = 0$ for all $x \in \overline{U_{n+1}}$ and $f_n(y) = 2^{-n}$ for all $x \notin U_n$. It follows that $f = \sum_{n<\omega} f_n$ is a continuous function from X to $[0, 2]$ such that $f^{-1}[\{0\}] = F$. ■

Proof of Lemma 310

Proof of Theorem 311

(a) Let $F_1, F_2 \in \mathcal{F}_p$. By Lemma 310(b), $F_1 \cap F_2$ is a zero-set. Suppose $p \notin \overline{F_1 \cap F_2}^{\beta\omega}$. Using Lemma 309(a), let G be a zero-set neighborhood of p satisfying $G \cap (\overline{F_1 \cap F_2}^{\beta X}) \neq \varnothing$. By Lemma 310(c), $G \cap X$ is a zero-set in X and, again by Lemma 310(b), $G \cap F_1$ and $G \cap F_2$ are zero-sets in X. We have $p \in (\overline{G \cap F_1}^{\beta X}) \cap (\overline{G \cap F_2}^{\beta X})$ but $(G \cap X) \cap F_1 \cap F_2 = (G \cap F_1) \cap (G \cap F_2)$. So $p \in \overline{F_1 \cap F_2}^{\beta X}$, and $F_1 \cap F_2 \in \mathcal{F}_p$. It is easy to check the other requirements that $\mathcal{F}_p$ be a z-filter.

Let $\mathcal{F}$ be a filter with $\mathcal{F}_p \subset \mathcal{F}$, and let $G \in \mathcal{F}$. Let $U \subset \beta X$ be open and contain p. Using Lemma 309(a), let F be a zero-set neighborhood of p contained in U. $F \cap X$ is a zero-set in X, and $\overline{F \cap X}^{\beta X} \supset \{p\}$,

so $F \in \mathcal{F}_p$. $U \cap G \supset F \cap G \neq \varnothing$, thus $p \in \overline{G}^{\beta X}$, $g \in \mathcal{F}_p$, and $\mathcal{F}_p$ is maximal.

(b) Let F be a zero-set in X. $\overline{F}^{\beta\omega} = F \cup \{p \in \beta X \setminus X : p \in \overline{F}^{\beta\omega}\} = F \cup \{p \in \beta X \setminus X : F \in \mathcal{F}_p\}$.

(c) Let $x \in \beta X$ and U be an open neighborhood of x. There exists a cozero-set V such that $x \in V \subseteq U$. Then $X \setminus V$ is a zero-set in X, and $\overline{X \setminus V} = \beta X \setminus V$. Thus $x \in \beta X \setminus \overline{X \setminus V} \subseteq U$.

(d) Let U be cozero in X. Note $\overline{X \setminus U} = X \setminus U \cup \{p \in \beta X \setminus X : X \setminus U \in \mathcal{F}_p\}$. As $\mathcal{F}_p$ is a z-ultrafilter, we have

$$\beta X \setminus \overline{X \setminus U} = U \cup \{p \in \beta X \setminus X : X \setminus U \notin \mathcal{F}_p\}$$

$$= U \cup \{p \in \beta X \setminus X : \exists F \subseteq U (F \in \mathcal{F}_p)\}$$

If $x \in \beta X \setminus \overline{F}$ for some zero-set F in X, then $U = X \setminus F$ is a cozero-set in F where $x \in \beta X \setminus \overline{F} = \beta X \setminus \overline{X \setminus U} = \operatorname{Ex} U$, so the $\operatorname{Ex} U$'s form a base.

(e) Let U be cozero in X. Suppose $p \notin \operatorname{Ex} U = \beta X \setminus (\overline{X \setminus U})$. Then $p \in \overline{X \setminus U}$, so any open neighborhood of p hits the complement of U, so p cannot be in an open set whose intersection lays inside U. ■

Proof of Theorem 312

Let X be a continuum. The reverse implication is immediate, for if H and K are proper subcontinua of X with $X = H \cup K$ we have $X \setminus H \subset K^\circ$.

For the forwards direction, suppose H is a proper subcontinuum of X with $H^\circ \neq \varnothing$ and put $Y = X \setminus H^\circ$. If Y is connected, then Y is a proper subcontinuum and $X = H \cup Y$. Otherwise, let $\mathcal{C}$ be the collection of components of Y. Pick $K \subsetneq Y$ clopen (in Y) and observe that $K = \cup\{C \in \mathcal{C} : K \cap C \neq \varnothing\}$. Since each $C \in \mathcal{C}$ meets ∂Y, and thus H, $H \cup K$ is a proper subcontinuum of X. Similarly, $H \cup (Y \setminus K)$ is a proper subcontinuum of X. It follows that X is not indecomposable.■

Proof of Lemma 313
Wlog, assume O is unbounded. Put $a_0 = 0$, and let $x_0 \in O$ such that $x_0 > a_0$. Choose $h_0 \in \mathbb{H} \setminus \overline{O}$ such that $h_0 > \max\{x_0, 1\}$. Because $h_0 \in \mathbb{H} \setminus \overline{O}$, there is an open interval (b_0, a_1) containing h_0 missing O. For each $n \in \omega$, select $x_n > a_n$, $h_n \in \mathbb{H} \setminus \overline{O}$ with $h_n > \max\{x_n, n+1\}$, and $b_n, a_{n+1} \in \mathbb{H} \setminus \overline{O}$ with $c_n \in (b_n, a_{n+1}) \subset \mathbb{H} \setminus \overline{O}$. ■

Proof of Lemma 314
(a) Let $U \subseteq \beta\mathbb{H}$ be open with $K \subseteq U$ and $\mathbb{H}^* \cap \mathrm{cl}_{\beta\mathbb{H}} U \subsetneq \mathbb{H}^*$. Let $O = U \cap \mathbb{H}$. Assume $0 \notin O$. Since $\mathbb{H} \setminus \overline{O}$ is unbounded, by Lemma 63 there are numbers $a_n < b_n < a_{n+1}$ so that $O \subseteq \bigcup_{n\in\omega}(a_n, b_n)$. Then $K \subseteq \mathrm{Ex} \bigcup_{n\in\omega}(a_n, b_n)$.

(b) Since $\mathrm{cl}_{\beta\mathbb{H}}(L_1 \cap L_2) = \mathrm{cl}_{\beta\mathbb{H}} L_1 \cap \mathrm{cl}_{\beta\mathbb{H}} L_2$ whenever L_1 and L_2 are closed subsets of $\mathbb{H}$, it is clear that u is a filter. By part (a), $K \subseteq \mathrm{cl}_{\beta\mathbb{H}} \bigcup_{n\in\omega}[a_n, b_n]$. Let $A \subseteq \omega$. If neither $A \in u$ nor $\omega \setminus A \in u$, then $\mathrm{cl}_{\beta\mathbb{H}} \bigcup_{n\in A}[a_n, b_n]$ and $\mathrm{cl}_{\beta\mathbb{H}} \bigcup_{n\in\omega\setminus A}[a_n, b_n]$ disconnect K. ■

Proof of Proposition F
If $p \in \mathrm{Ex}\, O \setminus X$, then there exists $F \in \mathcal{F}_p$ where $F \subseteq O$. Since $F \in \mathcal{F}_p$, $p \in \overline{F}^{\beta X} \subseteq \overline{O}^{\beta X}$. ■

Proof of Theorem 315
Let K, L be a proper subcontinuum, so let $U_A = \bigcup_{n\in A}(a_n, b_n)$ for $A \subseteq \omega$ and u_K be the ultrafilter of subsets of ω such that

$$K \subseteq \overline{U_A}$$

for each $A \in u_K$ as in the previous lemma.

We proceed by showing that $\mathbb{H}^* \setminus K$ is dense. Consider an arbitrary open basic set $\mathrm{Ex}\, V$. Then consider

$$A = \{n < \omega : V \cap (a_n, b_n) \neq \varnothing\}.$$

Then either A is finite, or there exists an infinite subset of A which is not in u_K. Either way, let $A' \subseteq A$ such that $A' \notin u_K$. Then

$$V \cap U_{A'}$$

has a distinct closure from $\overline{U_{\omega\setminus A'}} \supseteq K$.

Thus $\mathrm{Ex}(V \cap U_{A'})$ hits a point distinct from K, showing that $\mathbb{H}^* \setminus K$ is dense. ■

Proof of Lemma 316

Let $p \in \alpha X \setminus X$ and $x_0 \in X$. Let U, V be open disjoint neighborhoods of p, x_0 respectively in αX. Note $p \in \overline{U \cap X}$. Then by the continuity of f, $f(p) \in f(\overline{U \cap X}) \subseteq \overline{f(U \cap X)} = \overline{U \cap X}$, so $f(p) \neq x_0$. ■

Proof of Theorem 317

Note all the backwards implications are immediate.

For (a)→(b), let X be a G_δ subset of αX, and then let $f : \beta X \to \alpha X$ such that $f \restriction X$ is the identity map. Then there exist open in αX neighborhoods U_n for $n < \omega$ such that $X = \bigcap_{n<\omega} U_n$. Then as $X = f^{-1}[\bigcap_{n<\omega} U_n] = \bigcap_{n<\omega} f^{-1}[U_n]$, we see X is G_δ in βX.

For (b)→(c), let αX be any compactification of X. Then as $X = \bigcap_{n<\omega} U_n$ for U_n open in βX, we have $X^* = \bigcup_{n<\omega} K_n$ for K_n complements of U_n in βX. Since $f(X^*) = f(\bigcup_{n<\omega} K_n) = \bigcup_{n<\omega} f(K_n) = \alpha X \setminus X$, it follows that $\alpha X \setminus X$ is F_σ, so X is G_δ in αX.

For (c)→(d), if X is G_δ in αX, and X is a dense subset of Z, then X is also a dense subset of βZ, which is a compactification of X, so X is G_δ in βZ, and therefore also in Z. ■

Proof of Theorem 318

Let X be Čech complete. Using theorem 67, let αX be a compactification in which $X = \cap_{i\in\omega} U_i$ for open $U_i \subset \alpha X$, $i \in \omega$. For each $i \in \omega$, let $\mathcal{U}_i$ be an open cover of X satisfying $\overline{U}^{\alpha X} \subset U_i$ for every $U \in \mathcal{U}_i$.

Let $\mathcal{F}$ be a filter of closed sets such that for each n there is some $F_n \in \mathcal{F}$ with $F_n \subset U$ for some $U \in \mathcal{U}_n$. Because αX is compact, $\bigcap_{F\in\mathcal{F}} \overline{F}^{\alpha X} \neq \varnothing$. Let $x \in \bigcap_{F\in\mathcal{F}} \overline{F}^{\alpha X}$. For some $V_n \in \mathcal{U}_n$, there is an $F_n \in \mathcal{F}$ with $x \in \overline{F}_n^{\alpha X} \subset \overline{V}_n^{\alpha X} \subset U_n$ so that $x \in \bigcap_{n\in\omega} \overline{F}_n^{\alpha X} \subset \bigcap_{n\in\omega} U_n = X$.

Let $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \ldots$ be a sequence of open covers of X with the desired property. For each $n < \omega$ let $\mathcal{V}_n = \{V \subset \beta X : V \text{ open}, V \cap X \in \mathcal{U}_n\}$ and put $V_n = \bigcup \mathcal{V}_n$. Certainly $X \subset \bigcap_{n<\omega} V_n$.

Let $x \in \bigcap_{n<\omega} V_n$ and put $\mathcal{F}' = \{X \cap \overline{O} : x \in O, O \text{ open in } \beta X\}$. Note that if O_1 and O_2 are open neighborhoods of x we have $X \cap (\overline{O}_1 \cap \overline{O}_2) \supset$

$X \cap \overline{O_1 \cap O_2} \neq \varnothing$. Let $\mathcal{F}$ be the filter of closed sets generated by $\mathcal{F}'$. Fix $n < \omega$ and pick $V \in \mathcal{V}_n$ such that $x \in V$. Let $W \subset \beta X$ be open with $x \in W \subset \overline{W} \subset V$. Then $X \cap \overline{W} \in \mathcal{F}$, $X \cap V \in \mathcal{U}_n$ and $X \cap \overline{W} \subset X \cap V$, so $\bigcap \mathcal{F} \neq \varnothing$. It follows that

$$X \cap \bigcap \{\overline{O} : x \in O, O \text{ open in } \beta X\} = \bigcap \mathcal{F}' \neq \varnothing.$$

Since $\bigcap \{\overline{O} : x \in O, O \text{ open in } \beta X\} = \{x\}$ we have $X = \bigcap_{n<\omega} V_n$, so X is G_δ in βX. ■

Proof of Exercise 319

False.

Let $S = [0, 1)$ have the Sorgenfrey topology (where $S \setminus \{0\}$ is a dense copy of the Sorgenfrey line), and $R = (0, 1]$ have the reverse Sorgenfrey topology. These spaces embed in $S \times \{0\} \cup R \times \{1\}$, where for each $\langle x, 0\rangle$,

$$[x, x + \varepsilon) \times \{0\} \cup (x, x + \varepsilon) \times \{1\}$$

is a basic open set, and for each $\langle x, 1\rangle$,

$$(x - \varepsilon, x] \times \{1\} \cup (x - \varepsilon, x) \times \{0\}$$

is a basic open set.

A modification of the usual Heine-Borel theorem for the real line shows that this space is compact. Furthermore, any open set containing $S \times \{0\}$ misses at most countably many points of $R \times \{1\}$, so $S \times \{0\}$ is not G_δ.

True. Let X be the tangent disk space. If x is on the x-axis, let $D(x, \varepsilon)$ be the tangent disk neighborhood of x of radius ε, and if x is in the upper half plane, let $B(x, \varepsilon)$ be the standard Euclidean open ball about x of radius ε. For each $n \in \omega$, let

$$\mathcal{U}_n = \{D(x, 1/2^n) : x \in \mathbb{R} \times \{0\}\} \cup \{B((x_1, x_2), 1/2^{n+1}) : x_2 \geqslant 1/2^n\}.$$

Let $\mathcal{F}$ be a filter of closed sets such that for each n there is some $F_n \in \mathcal{F}$ with $F_n \subset U_n$ for some $U_n \in \mathcal{U}_n$.

Case 1. Some $F \in \mathcal{F}$ is compact. Then clearly $\bigcap \mathcal{F} \neq \varnothing$.

Case 2. No $F \in \mathcal{F}$ is compact. Then for each n, $U_n = D(x_n, 1/2^n)$ for some point x_n on the x-axis. ■

Proof of Theorem 320
(a)⇒(b): Suppose X is metrizable and Čech-complete. Then by Theorem 67 X is G_δ in its metric completion. By Completeness Fact (b), X is completely metrizable.

(b)⇒(a): Let d be a complete metric on X. For each $n \in \omega$ let $\mathcal{U}_n = \{B_d(x, 1/n) : x \in X\}$. Now suppose $\mathcal{F}$ is a filter of closed subsets of X such that for each $n \in \omega$ there is $F_n \in \mathcal{F}$ and $U \in \mathcal{U}_n$ such that $F_n \subseteq U$. Let $F'_n = \bigcap_{k \leqslant n} F_k$. Then the sets F'_n are nonempty closed decreasing with diameters going to 0. Since d is complete, we have $\bigcap F'_n \neq \varnothing$. Clearly there can be only one point x in this intersection. Let $F \in \mathcal{F}$. Again using completeness of d and the fact that $\mathcal{F}$ is a filter, $\bigcap(F'_n \cap F) \neq \varnothing$, so it must be that $x \in \bigcap(F'_n \cap F)$. Thus x witnesses $\bigcap \mathcal{F} \neq \varnothing$. By Theorem 68(b), X is Čech-complete. ■

Proof of Corollary 321

Proof of Theorem 322
As Y is Čech-complete, let $Y = \bigcap_{n<\omega} V_n$ witness that Y is G_δ in some compactification Z.

If $X = \bigcap_{n<\omega} U_n$ for U_n open in Y, then let U'_n be open in Z with $U_n = U'_n \cap Y$. Then $X = \bigcap_{n<\omega} U'_n \cap V_n \cap \overline{X}^Z$, showing that X is G_δ in its compactification $\overline{X}^Z$.

If X is closed in Y, then $X = K \cap Y = K \cap \bigcap_{n<\omega} V_n \subseteq K \cap \bigcap_{n<\omega} \overline{V_n}^Z$ for some closed (compact) subset K of Z. Thus X is G_δ in its compactification $K \cap \bigcap_{n<\omega} \overline{V_n}^Z = \overline{X}^Z$.

Proof of (b) missing. ■

Proof of Example 323
Let $U_0, U_1, \ldots$ be open in $(\omega+1)^{\omega_1}$ such that $\omega^{\omega_1} \subset \bigcap_{n<\omega} U_n$. For each $n < \omega$, let $V_n \subset U_n$ be a basic open set containing $\vec{0}$ and let $F_n \subset \omega_1$ be the support of V_n. Put $\beta = \sup(\cup_{n<\omega} F_n)$. Then $\prod_{\alpha=0}^{\beta}\{0\} \times \prod_{\alpha=\beta+1}^{\omega_1} \omega + 1 \subset \bigcap_{n<\omega} V_n \subset \bigcap_{n<\omega} U_n$, so $\omega^{\omega_1} \neq \bigcap_{n<\omega} U_n$. ■

Proof of Theorem 324
Let X be a Čech complete space. Let $\{D_n : n \in \omega\}$ be a countable

collection of dense open sets in X. Let $\{U_n : n \in \omega\}$ be a collection of open sets in βX with $X = \bigcap_{n\in\omega} U_n$. Since βX is a Baire space, $\bigcap_{n\in\omega}(\operatorname{Ex} D_n \cap U_n) \neq \varnothing$. So $\bigcap_{n\in\omega} D_n \neq \varnothing$. ■

Proof of Example 325

Consider any basic open set $\overline{A}^{\beta\omega} \cap \omega^*$. It follows that A is infinite, so therefore $\overline{A}^{\beta\omega} \cap \omega^*$ is closed and infinite. Thus X intersects $\overline{A}^{\beta\omega} \cap \omega^*$, showing that X is dense in ω^*.

Let U_n be open in ω^* with $X \subseteq U_n$ for $n < \omega$. Then $\omega^* \setminus U_n \subseteq \omega^* \setminus X$. Note then that $\omega^* \setminus U_n$ cannot be infinite, so U_n is co-finite. Then $\bigcap_{n<\omega} U_n$ is co-countable, but since X,$\omega^* \setminus X$ are uncountable, it follows that $X \neq \bigcap_{n<\omega} U_n$. ■

Proof of Theorem 326

Let X be a regular countably compact space. Let $\{U_n\}_{n\in\omega}$ be a collection of dense open sets. Let $V \subseteq X$ be open. Using regularity, let $V_0 \neq \varnothing$ be open with $V_0 \subseteq \overline{V_0} \subseteq (U_0 \cap V)$. For $n > 0$, let $V_n \neq \varnothing$ be open with $V_n \subseteq \overline{V_n} \subseteq (U_0 \cap V_{n-1})$. $\overline{V_0} \supseteq \overline{V_1} \supseteq \ldots$ is a decreasing chain of nonempty closed sets in a countably compact space, so $\bigcap_{n\in\omega} \overline{V_n} \neq \varnothing$. For each $n \in \omega$, $\overline{V_n} \subseteq U_n$ and $\overline{V_n} \subseteq V$, so $\bigcap_{n\in\omega} \overline{V_n} \subseteq \bigcap_{n\in\omega}(U_n \cap V)$, and $V \cap (\bigcap_{n\in\omega} U_n) = \bigcap_{n\in\omega}(U_n \cap V)$ is nonempty. We conclude $\bigcap_{n\in\omega} U_n$ is dense, and X is a Baire space. ■

Proof of Theorem 327

Suppose on the contrary that B is completely metrizable. Then B is a G_δ-set in $\mathbb{R}$, so there exist open sets $U_n \subset \mathbb{R}$, $n < \omega$, with $B = \bigcap_{n<\omega} U_n$. Observe that $X \setminus U_n$ is a closed set missing B, so $X \setminus U_n$ must be countable for each $n < \omega$. It follows that $X \setminus B = \bigcup_{n<\omega} X \setminus U_n$ is countable, a contradiction as $X \setminus B$ must be Bernstein also.

To see that B is Baire, let G'_n be a dense open subset of B for each $n < \omega$ and let $U' \subset B$ be open. For each $n < \omega$ let G_n be open in $\mathbb{R}$ such that $G'_n = G_n \cap B$, and let U be open in $\mathbb{R}$ such that $U' = U \cap B$. Observe that each G_n is dense in $\mathbb{R}$.

Since $U \cap G_0$ is open and nonempty, there is a closed interval $I_\varnothing$ contained therein with $\operatorname{diam} I_\varnothing < 1$. Suppose for some $n > 0$ that $\{I_\sigma : \sigma \in 2^{n-1}\}$ is a pairwise-disjoint collection of closed intervals with $\operatorname{diam} I_\sigma < 1/2^{n-1}$ and $I_\sigma \subset U \cap \bigcap_{i=0}^{n-1} G_i$. For each $\sigma \in 2^{n-1}$, $I_\sigma^\circ \cap G_n$

is nonempty and open, so pick disjoint closed intervals $I_{\sigma\frown 0}$ and $I_{\sigma\frown 1}$ contained therein with diameters less than $1/2^n$.

Put $C = \bigcap_{n<\omega} \cup\{I_\sigma : \sigma \in 2^n\}$. If $\sigma \in 2^\omega$ then $\operatorname{diam} I_{\sigma\restriction n} \to 0$, so $\cap\{I_{\sigma\restriction n} : n < \omega\}$ is a single point. It follows that $C \subset U \cap \bigcap_{n<\omega} G_n$ is a Cantor set. As B must meet C we have $U' \cap \bigcap_{n<\omega} G'_n \neq \varnothing$, whence B is Baire. ■

Proof of Theorem 328

(a): Suppose Y is Baire and X is a subspace of Y. For each $n \in \omega$ let D_n be dense open in X. In each case below we shall show that $\bigcap_{n\in\omega} D_n$ is dense in X.

(a)(i): Suppose X is open in Y. Let U be nonempty open in X. Since U is also (nonempty) open in Y and each $D_n \cup Y \setminus \overline{X}$ is dense open in Y, we have $U \cap \bigcap_{n\in\omega}(D_n \cup Y \setminus \overline{X}) \neq \varnothing$. Since $U \subseteq X$ we must have $U \cap \bigcap_{n\in\omega} D_n \neq \varnothing$.

(a)(ii): Suppose X is a dense G_δ in Y. There exist open sets $X_n \subseteq Y$ such that $X = \bigcap_{n\in\omega} X_n$. For each $n \in \omega$ there exists an open $D'_n \subseteq Y$ such that $D_n = D'_n \cap X$. Note for each $n \in \omega$ we have $D'_n \cap X_n$ dense open in Y. Let U be nonempty open in X. There exists an open $U' \subseteq Y$ such that $U = U' \cap X$. Since Y is Baire, $U' \cap \bigcap_{n\in\omega} D'_n \cap X_n \neq \varnothing$. So $U \cap \bigcap D_n \neq \varnothing$.

(b): Suppose Y is a dense Baire subspace of X. Let $\{D_n : n \in \omega\}$ be a collection of dense open subsets of X. We show $\bigcap_{n\in\omega} D_n$ is dense in X. Claim: $D_n \cap Y$ is dense (open) in Y, for each $n \in \omega$. Let V be nonempty open in Y. Then $V = V' \cap Y$ for some V' (nonempty) open in X. Then $D_n \cap V'$ is nonempty open in X, so $D_n \cap V = (D_n \cap V') \cap Y \neq \varnothing$ by density of Y in X. Done with claim. Let U be nonempty open in X. Then $U \cap Y$ is nonempty open in Y. Since Y is Baire and each $D_n \cap Y$ is dense open in Y, we have $(U \cap Y) \bigcap_{n\in\omega}(D_n \cap Y) \neq \varnothing$. So $U \cap \bigcap_{n\in\omega} D_n \neq \varnothing$. ■

Proof of Example 329

Let $X = \mathbb{R} \setminus (C \setminus E)$, where C is the Cantor set, and E is the set of endpoints of the Cantor set. Note that $\mathbb{R} \setminus C$ is a dense Baire subspace, so X is Baire. However, E is a countable subspace of metrizable X with no isolated points, and $E = X \cap C$, so E is a closed copy of the rationals. ■

Proof of Theorem 330

Let X be Baire, and let Y be Baire and second-countable. Let $\{B_n\}_{n\in\omega}$ be a base for Y, and let $\{O_n\}_{n\in\omega}$ be a collection of dense open subsets of $X \times Y$.

Define $U_{ij} = \pi_X(O_i \cap (X \times B_j))$. Clearly, U_{ij} is open in $X \times Y$. Let $W \subseteq X$ be open and nonempty. Since $W \times B_j$ is open in $X \times Y$ and O_i is dense, $O_i \cap (W \times B_j)$ is nonempty. Thus, for each $i, j \in \omega$, U_{ij} is dense and open in X.

Now, let $U \times V$ be a basic open subset of $X \times Y$. Because X is Baire, there is $x \in U \cap \bigcap_{i,j\in\omega} U_{ij}$. Define $V_i = \{y \in Y : (x, y) \in O_i\}$ for each $i \in \omega$. V_i is open in Y since for each $y \in V_i$, there exists a basic open set $U_y \times V_y \subseteq X \times Y$ satisfying $(x, y) \in U_y \times V_y \subseteq O_i$, so that $y \in V_y \subseteq V_i$. To show V_i is dense in Y, let B_j be a basic open subset of Y. Because $x \in U_{ij} = \pi_X(O_i \cap (X \times B_j))$, there exists $(x, y) \in O_i \cap (X \times B_j)$ so that $y \in V_i \cap B_j$.

Because Y is Baire, there is $y \in V \cap \bigcap_{i\in\omega} V_i$. So $(x, y) \in (U \times V) \cap \bigcap_{i\in\omega} O_i$, and $X \times Y$ is Baire. ■

Proof of Theorem 331

Let O_n be open in $X \times Y$ for $n < \omega$. Consider any basic open $U \times V$ in $X \times Y$. We need to prove $(U \times V) \cap \bigcap_{n\in\omega} O_n \neq \varnothing$.

Let $\{U_\alpha\}_\alpha$ be a maximal disjoint collection of open subsets of U such that for each α there exist V_α open in Y such that $U_\alpha \times \overline{V_\alpha} \subseteq O_0 \cap (U \times V)$.

Claim. $W_1 = \bigcup_\alpha U_\alpha$ *is dense in* U. Suppose N is any nonempty open subset of U. Since O_0 is dense open in $X \times Y$, $O_0 \cap (N \times V) \neq \varnothing$. So there exists a basic open set $A \times B \subset O_0 \cap (N \times V)$. By regularity of Y we may assume $A \times \overline{B} \subset O_0 \cap (N \times V)$. A can't be disjoint from W_1, as that would contradict maximality of $\{U_\alpha\}_\alpha$. Since $A \subset N$, we have $N \cap W_1 \neq \varnothing$, proving W_1 dense in U.

Now for each α, let $\{U_{\alpha\beta}\}_\beta$ be a maximal disjoint collection of open subsets of U_α such that for each β there exist $V_{\alpha\beta}$ open in Y such that $U_{\alpha\beta} \times \overline{V_{\alpha\beta}} \subseteq O_1 \cap (U_\alpha \times V_\alpha)$. By a proof similar to that of the Claim, $\bigcup_\beta U_{\alpha\beta}$ is dense in U_α. It follows that $W_2 = \bigcup_{\alpha,\beta} U_{\alpha\beta}$ is dense in U.

Continue in this way, defining $W_3, W_4, \ldots$ dense open in U. Since X is Baire, so is U, hence there exists $x \in \bigcap_{n\in\mathbb{N}} W_n$. Since $\{U_\alpha\}_\alpha$ is pairwise disjoint, there exists a unique α_0 with $x \in U_{\alpha_0}$. Similarly, there exists a

unique α_1 with $x \in U_{\alpha_0\alpha_1}$. And so on. So we obtain a sequence $\alpha_0, \alpha_1, \ldots$ with $x \in \bigcap_{n\in\omega} U_{\alpha_0\alpha_1\ldots\alpha_n}$. By construction, $V_{\alpha_0} \supset V_{\alpha_0\alpha_1} \supset \ldots$.

If Y is countably compact, there exists $y \in \bigcap_{n\in\omega} \overline{V}_{\alpha_0\alpha_1\ldots\alpha_n}$. Then $(x, y) \in (U \times V) \cap \bigcap_{n\in\omega} O_n$, so $\bigcap_{n\in\omega} O_n$ is dense. Hence $X \times Y$ is Baire.

If Y is Čech-complete, let $\mathcal{C}_n$, $n \in \omega$, be a sequence of open covers satisfying the conditions of Theorem 318(b). In the above construction, we may choose $\overline{V}_{\alpha_0\alpha_1\ldots\alpha_n} \subset C$ for some $C \in \mathcal{C}_n$, guaranteeing the existence of $y \in \bigcap_{n\in\omega} \overline{V}_{\alpha_0\alpha_1\ldots\alpha_n}$, and then the rest of the proof proceeds as before. ■

Proof of Theorem 332

Suppose X is Baire and let σ be a strategy for E. Let $U = \sigma(\varnothing)$ be E's first move. Let $\{U_\alpha\}_\alpha$ be a maximal disjoint collection of open subsets of U such that, for each α, there is $V_\alpha \subset U$ such that $U_\alpha = \sigma(V_\alpha)$. Then $W_1 = \bigcup_\alpha U_\alpha$ is dense open in U. For each α, let $\{U_{\alpha\beta}\}_\beta$ be a maximal disjoint collection of open subsets of U_α such that, for each β, there is $V_{\alpha\beta} \subset U_\alpha$ such that $U_{\alpha\beta} = \sigma(V_\alpha, V_{\alpha\beta})$. Then $W_2 = \bigcup_{\alpha,\beta} U_{\alpha\beta}$ is dense open in U. Define $W_3, W_4, \ldots$ dense open in U by continuing in this way. U is Baire, so there exists $x \in \bigcap_{n\in\mathbb{N}} W_n$. Since $\{U_\alpha\}_\alpha$ is pairwise-disjoint, there exists a unique α_0 with $x \in U_{\alpha_0}$. Similarly, there exists a unique α_1 with $x \in U_{\alpha_0\alpha_1}$. We obtain a sequence $\alpha_0, \alpha_1, \ldots$ with $x \in \bigcap_{n<\omega} U_{\alpha_0,\alpha_1,\ldots,\alpha_n}$. By construction, $V_{\alpha_0} \supset V_{\alpha_0\alpha_1} \supset \ldots$. Thus NE wins by playing $V_{\alpha_0\alpha_1\ldots\alpha_n}$ in round n, so σ is not a winning strategy for E.

Now suppose X is not Baire. Without loss of generality we may assume X is meagre, so $X = \bigcup_{n\in\mathbb{N}} X_n$ with each X_n nowhere-dense. Let $U_0 = X$ be E's first move. If NE plays V_n in round n, let E play $U_{n+1} = V_n \setminus \overline{X}_{n+1}$ in round $n+1$. Certainly this is a winning strategy for E. ■

Proof of Theorem 333

If X is regular and countably compact, then NE just needs to pick the sequence of V_n's so that $\overline{V}_n \subseteq U_n$ for each $n \in \omega$.

Now suppose X is Čech complete. Then $X = \bigcap_{n\in\omega} W_n$ for some decreasing sequence of open subsets W_n of βX. Suppose $n \in \omega$ and the game has been played properly through stage $n-1$, and E has selected a legal U_n. Player NE can let V_n be nonempty open in X with

$\mathrm{cl}_{\beta X} V_n \subseteq W_n \cap \mathrm{Ex}\, U_n$. If the game continues in this manner, then

$$\varnothing \neq \bigcap_{n\in\omega} \mathrm{cl}_{\beta X} V_n \subseteq X \cap \bigcap_{n\in\omega} \mathrm{cl}_{\beta X} V_n \subseteq \bigcap_{n\in\omega} U_n,$$

so that NE wins. Alternatively, let $\mathcal{U}_0, \mathcal{U}_1, \ldots$ be a sequence of open covers of X given by Theorem 68. If U_n has been chosen, then there exists $W \in \mathcal{U}_n$ such that $W \cap U_n \neq \varnothing$. Let V_n be nonempty open in X with closure contained in $W \cap U_n$. Continuing in this manner, the closures of V_n's form a filter with the desired property. An element in the intersection of this filter corresponds to an element in $\bigcap_{n\in\omega} U_n$. ■

Proof of Theorem 334

Note that $E \not\uparrow \mathrm{Ch}(B)$ and $E \not\uparrow \mathrm{Ch}(\mathbb{R} \setminus B)$ as Bernstein sets are Baire.

If σ is a strategy for player NE in $\mathrm{Ch}(B)$, then define a strategy τ for E in $\mathrm{Ch}(\mathbb{R} \setminus B)$ such that $\tau(\varnothing) = \mathbb{R}$ and

$$\tau(s^\frown\langle U\rangle) \subseteq \tau(s) \cap \sigma(s^\frown\langle U\rangle)$$

such that the diameter of $\tau(s^\frown\langle U\rangle)$ is less than $2^{-|s|}$ for all sequences s of open sets. Since E lacks a winning strategy in $\mathrm{Ch}(\mathbb{R} \setminus B)$, there exists a counterattack $\langle V_0, V_1, \ldots\rangle$ such that $(\mathbb{R} \setminus B) \cap \bigcap_{n<\omega} V_n \neq \varnothing$. Since the diameters of the sets played according to τ converge to 0, it follows that $(\mathbb{R} \setminus B) \cap \bigcap_{n<\omega} V_n = \{x\}$. Therefore since

$$V_n \subseteq \tau(\langle V_0, \ldots, V_{n-1}\rangle) \subseteq \sigma(\langle V_0, \ldots, V_{n-1}\rangle)$$

it follows that $\langle V_0, V_1, \ldots\rangle$ is a valid counterattack to σ such that $B \cap \bigcap_{n<\omega} V_n = \varnothing$. Thus σ is not a winning strategy.

Therefore $E \not\uparrow \mathrm{Ch}(B)$ and $NE \not\uparrow \mathrm{Ch}(B)$, so $\mathrm{Ch}(B)$ is undetermined. ■

Proof of Theorem 335

Proof of Theorem 336

Let $X = \prod_{\alpha<\kappa} X_\alpha$, with each X_α Choquet. We show NE has a winning strategy in $\mathrm{Ch}(X, \mathcal{B})$, where $\mathcal{B}$ is the collection of basic open sets. For each $\alpha < \kappa$ let σ_α be a winning strategy for NE in X_α. Let $U_n =$

$\prod_{\alpha<\kappa} U_{\alpha,n}$ denote E's play in round n, and let F_n be the support of U_n. For each $\alpha < \kappa$ and $n < \omega$, put

$$V_{\alpha,n} = \begin{cases} \sigma_\alpha(U_{\alpha,i_\alpha}, \dots, U_{\alpha,n}) & \text{if } \alpha \in F_n \\ X_\alpha & \text{if } \alpha \notin F_n \end{cases},$$

where $i_\alpha = \min\{i : \alpha \in F_i\}$. For each $n < \omega$ let $V_n = \prod_{\alpha<\kappa} V_{\alpha,n}$. Then

$$\bigcap_{n<\omega} V_n = \bigcap_{n<\omega} \prod_{\alpha<\kappa} V_{\alpha,n} = \prod_{\alpha<\kappa} \bigcap_{n<\omega} V_{\alpha,n} \neq \varnothing,$$

so NE has a winning strategy in $\mathrm{Ch}(X, \mathcal{B})$. ■

Proof of Corollary 337

Proof of Lemma 338

(a): Let $A \subseteq \omega_1$ uncountable. Let $\sigma \in \omega_1^{<\omega}$. A is cofinal in ω_1, so there exists $\alpha \in A$ with $\alpha > \operatorname{ran}\sigma$. Let $k \in \omega \setminus \operatorname{dom}\sigma$. Define $f \in \omega_1^\omega$ by $f \restriction \operatorname{dom}\sigma = \sigma$, $f(k) = \alpha$, and $f(n) = 0$ for all $n \in \omega \setminus (\operatorname{dom}\sigma \cup \{k\})$. Then $f \in [\sigma] \cap A^*$.

(b): Claim U_i is open in $\omega_1^\omega \times \omega_1^\omega$. Let $(f, g) \in U_i$. There exist $k_1, k_2 \in \omega$ such that $\min\{f(k_1), g(k_2)\} > \max\{f(i), g(i)\}$. Let $\sigma = f \restriction \{i, k_1\}$ and $\tau = g \restriction \{i, k_2\}$. Then $(f, g) \in [\sigma] \times [\tau] \subseteq U_i$. Claim U_i is dense in $\omega_1^\omega \times \omega_1^\omega$. Let $\sigma, \tau \in \omega_1^{<\omega}$. Assume $i \in \operatorname{dom}\sigma \cap \operatorname{dom}\tau$. Let $k \in \omega \setminus (\operatorname{dom}\sigma \cup \operatorname{dom}\tau)$. Let $f, g \in \omega_1^\omega$ so that f extends σ, g extends τ, and $\min\{f(k), g(k)\} > \max\{\sigma(i), \tau(i)\}$. Then $(f, g) \in U_i \cap ([\sigma] \times [\tau])$. Since A and B are disjoint, it is clear that $\bigcap_{i\in\omega} U_i \cap (A^* \times B^*) = \varnothing$. Note that $U_i \cap (A^* \times B^*)$ is open dense in $A^* \times B^*$ (we could modify the density argument above to get $(f, g) \in A^* \times B^*$ by choosing $f(k) \in A$ and $g(k) \in B$ with $\min\{g(k), f(k)\} > \operatorname{ran}\sigma \cup \operatorname{ran}\tau$ and setting $f(n) = 0$ for all $n \in \omega \setminus (\operatorname{dom}\sigma \cup \{k\})$), so this proves $A^* \times B^*$ is not Baire. ■

Proof of Lemma 339

First, we show C_U is closed. Let β be a limit point of C_U, let $\sigma \in \beta^{<\omega}$, and let $\alpha = \max\{\sigma(n) : n \in \operatorname{dom}(\sigma)\}$. Because β is a limit point of C_U and $(\alpha, \beta]$ is open, there is $\gamma \in C_U \cap (\alpha, \beta]$, so that $\sigma \in \gamma^{<\omega}$, $\gamma \in C_U$, and $\sigma_U \in \gamma^{<\omega} \subset \beta^{<\omega}$. We conclude $\beta \in C_U$, and C_U is closed.

Next, we show C_U is unbounded by showing it has an unbounded subset. Define $f : \omega_1 \to \omega_1$ by $f(\alpha) = \sup\{\max\{\sigma(n) : n \in \operatorname{dom}\sigma_U\} : \sigma \in \alpha^{<\omega}\}$. By MH7550.44, $C = \{\alpha : \forall\beta < \alpha, f(\beta) < \alpha\}$ is closed and unbounded so that $C^{'} = \{\alpha \in C : \alpha \text{ is a limit ordinal}\}$ is unbounded. To show $C^{'} \subset C_U$, let $\alpha \in C^{'}$ and $\sigma \in \alpha^{<\omega}$. Because α is a limit ordinal, $\max\{\sigma(n) : n \in \operatorname{dom}\sigma\} = \beta < \beta + 1 < \alpha$, so that $f(\beta + 1) < \alpha$ and $\sigma \in (\beta + 1)^{<\omega}$. This means $\max\{\sigma_U(n) : n \in \operatorname{dom}\sigma_U\} \leqslant f(\beta + 1) < \alpha$, implying $\sigma_U \in \alpha^{<\omega}$ and $\alpha \in C_U$. So $C^{'} \subset C_U$, and C_U is unbounded.■

Proof of Theorem 340

Let $G'_0, G'_1, \ldots$ be dense open subsets of A^*. Then there exist dense open subsets $G_0, G_1, \ldots$ of ω_1^ω such that $G'_n = G_n \cap A^*$. For each $n < \omega$ and $\sigma \in \omega_1^\omega$, let σ^{G_n} be an extension of σ such that $[\sigma^{G_n}] \subset G_n$ and put $C_n = \{\delta < \omega_1 : \forall\sigma \in \delta^{<\omega}(\sigma^{G_n} \in \delta^{<\omega}\}$.

Fix $\sigma \in \omega_1^\omega$. Put $C = \bigcap_{n<\omega} C_n$ and note C is club. Pick $\delta \in C \cap A$ such that $\sigma \in \delta^{<\omega}$ and let $(\delta_n)_{n<\omega}$ be cofinal in δ. Put $\tau_0 = \sigma$. If $\tau_i \in \delta^{<\omega}$ has been defined for some $i < \omega$, set $\tau_{i+1} = \tau_i^{G_i} \frown \delta_i$.

Let $f = \bigcup_{n<\omega} \tau_n$. Certainly $f^* = \delta$, so $f \in A^*$. Since $\sigma \subset f$, $f \in [\sigma]$. Also f extends each $\tau_n^{G_n}$, so $f \in G_n$ for all $n < \omega$. Thus

$$f^* \in A^* \cap [\sigma] \cap \bigcap_{n<\omega} G_n = (A^* \cap [\sigma]) \cap \bigcap_{n<\omega} G'_n.$$

■

Proof of Corollary 341

Proof of Theorem 342

Suppose X is first-countable. Let $x \in X$, and let $B_0, B_1, B_2, \ldots$ be a decreasing local base at x. Then a winning strategy for O is to play B_n in the nth round.

Suppose X is a W-space, $x \in X$, and $x \in \overline{A}$. Consider a strategy for P that chooses a point in A in every round. If O uses her winning strategy, the result will be a sequence of points in A which converge to x. ■

Proof of Example 343

Let $X = (\omega \times \omega) \cup \{\infty\}$ be the sequential fan. Note that each column $\{n\} \times \omega$ is a sequence converging to ∞. So in the nth round, P can

always choose (n, k_n) for some $k_n \in \omega$. But then if $f(n) = k_n + 1$, then $B(f)$ misses all of P's chosen points. So this is a winning strategy for P.

Now suppose $\infty \in \overline{A}$. If $A \cap (\{n\} \times \omega)$ is finite for each n, then as above we may construct a function f such that $B(f) \cap A = \varnothing$, a contradiction. So $A \cap (\{n\} \times \omega)$ is infinite for some n, and this infinite set converges to ∞. Thus X is Fréchet. ■

Proof of Theorem 344

Let $x \in X$, and let σ be a strategy for O in $G_{O,P}(X, x)$.

If X is countable but not first-countable at x, then let $\mathcal{B} = \{\sigma(s) : s \in X^{<\omega}\}$. $\mathcal{B}$ is obviously countable. Since it cannot be a base, there exists an open neighborhood U of x such that no $\sigma(s)$ is a subset of U. Then choose

$$p_n \in \sigma(\langle p_0, \ldots, p_{n-1}\rangle) \setminus U$$

and observe that the p_n are a legal counterattack defeating σ. Thus X is not a W-space.

Likewise, if D is a countable dense subspace of regular X, but X is not first-countable at x, then let $\mathcal{B} = \{\sigma(s) : s \in D^{<\omega}\}$. $\mathcal{B}$ is obviously countable. Since it cannot be a base, there exists an open neighborhood U of x such that no $\sigma(s)$ is a subset of $\overline{U}$. Then choose

$$p_n \in D \cap \left(\sigma(\langle p_0, \ldots, p_{n-1}\rangle) \setminus \overline{U}\right)$$

and observe that the p_n are a legal counterattack defeating σ. Thus X is not a W-space. ■

Proof of Theorem 345

Suppose X is a W-space and $A \subseteq X$. Let $a \in A$. Let σ be a winning strategy for player O in the game $G_{O,P}(X, a)$. Let $\rho : A^{<\omega} \to \tau(A)$ be defined by $\rho(\langle a_0, ..., a_{n-1}\rangle) = \sigma(\langle a_0, ..., a_{n-1}\rangle) \cap A$. Clearly ρ is a winning strategy for player O in the game $G_{O,P}(A, a)$.

Suppose $X = \prod_{n\in\omega} X_n$ where each X_n is a W-space. Let $(x_n) \in X$. For each $n \in \omega$ let σ_n be a winning strategy for player O in the game $G_{O,P}(X_n, x_n)$. Let $\sigma(\varnothing) = X$. Suppose $i \in \omega$ and player P has chosen $(p_n^i) \in \sigma(\langle (p_n^0), ..., (p_n^{i-1})\rangle)$. Let $\sigma(\langle (p_n^0), ..., (p_n^i)\rangle) = \prod_n U_n$ where $U_n = \sigma_n(\langle p_n^0, ..., p_n^{i-1}\rangle)$ if $n \leqslant i$ and $U_n = X_n$ otherwise. Continuing the game

in this manner we get $p_n^i \to x_n$ for each $n \in \omega$, so that $(p_n^i) \to (x_n)$. Thus σ is a winning strategy for player O in the game $G_{O,P}(X, (x_n))$.■

Proof of Theorem 346

Let $X = \sum_{\alpha<\kappa} X_\alpha$ with each X_α a W-space and base point $\vec{x} = (x_\alpha)_{\alpha<\kappa}$. Let σ_α be a winning strategy for O in $G_{O,P}(X_\alpha, x_\alpha)$. We construct a winning strategy for O in $G_{O,P}(X, \vec{x})$.

Put $\sigma(\varnothing) = X$. If P responds with $\vec{x}(0)$, where $supp(\vec{x}(0)) = S_0 = \{\alpha_{0,i} : i < \omega\}$, put $\sigma(\langle \vec{x}(0) \rangle) = \pi^{-1}_{\alpha_{0,0}}(\sigma_{\alpha_{0,0}}(x(0)_{\alpha_{0,0}})$. If P plays $\vec{x}(n)$ at the end of round n, where $supp(\vec{x}(n)) = S_n = \{\alpha_{n,i} : i < \omega\}$, put

$$\sigma(\langle \vec{x}(0), \ldots, \vec{x}(n) \rangle) = \bigcap \{\pi_\alpha^{-1}(\sigma_\alpha(F)) : F \in \mathcal{F}_n(\alpha), \alpha = \alpha_{j,i}, 0 \leqslant i, j \leqslant n\},$$

where $\mathcal{F}_n(\alpha)$ is the collection of all subsequences of $\langle x(0)_\alpha, \ldots, x(n)_\alpha \rangle$.

We claim $x(n)_\alpha \to x_\alpha$ for each $\alpha < \kappa$. If $\alpha \notin S_n$ for all $n < \omega$, then $x(n)_\alpha = x_\alpha$. Otherwise $\alpha = \alpha_{k,\ell}$. If $n_0 = \max\{k, \ell\}$ and $n \geqslant n_0$, then $x(n+1)_\alpha \in \sigma_\alpha(\langle x(n_0)_\alpha, \ldots, x(n)_\alpha \rangle)$. Thus $x(n)_\alpha \to x_\alpha$, so σ is a winning strategy. ■

Proof of Theorem 347

Assume (a). Every countable subspace is first-countable by 94(a). Let $x \in \overline{A}$, for some $A \subset X$. By 94(a), $\overline{A}$ is a W-space. Let σ be a winning strategy in $G_{O,P}(\overline{A}, X)$ for O. Put $U_0 = \sigma(\varnothing)$ and $x_0 \in U_0 \cap A$. Having defined U_i and x_i for $i < n$, put $U_n = \sigma(x_0, \ldots, x_{n-1})$ and $x_n \in U_n \cap A$. Finally, put $B = \{x_n : n \in \omega\}$. B is countable and $x \in \overline{B}$, so X is countably tight.

Now assume (b). Let $x \in \overline{A_n}$ for all $n \in \omega$. Because X is countably tight, for all $n \in \omega$ there exists countable $B_n \subset A_n$ with $x \in \overline{B_n}$. Because $\left(\bigcup_{n\in\omega}\right) \cup \{x\}$ is countable, it is also first countable by assumption. Let $\{C_n : n \in \omega\}$ be a countable base at x with $C_n \supset C_{n+1}$ for each $n \in \omega$. Let $x_n \in C_n \cap B_n$. For any open neighborhood N of x, we have $N \supset C_m \supset \{x_m, x_{m+1}, \ldots\}$ for some $m \in \omega$, so $x_n \to x$.

Note first that X is Fréchet by setting $A_n = A$ in (c).

Let T_x be the collection of open neighborhoods of $x \in X$. Assume that for $s \in \omega^{<\omega}$ and $t \leqslant s$, $U_t \in T_x$. Then let

$$X_s = \{\sigma(\langle U_{s\restriction 1}, \ldots, U_s, U \rangle) : U \in T_x\}.$$

Observe that for each neighborhood U of x, $\sigma(\langle U_{s\restriction 1}, \ldots, U_s, U\rangle) \in X_s \cap U$, so $x \in \overline{X_s}$. As X is Fréchet, let

$$A_s = \{\sigma(\langle U_{s\restriction 1}, \ldots, U_s, U_{s^\frown\langle n\rangle}\rangle) : n < \omega\} \subseteq X_s$$

satisfy $x \in \overline{A_s}$.

For any $f \in \omega^\omega$ and $n < \omega$, observe that $x \in \overline{A_{f\restriction n}}$; therefore there is $x_{f\restriction n} \in A_{f\restriction n}$ where $x_{f\restriction n} \to x$.

Finally, pick $f \in \omega^\omega$ such that

$$x_{f\restriction n} = \sigma(\langle U_{f\restriction 1}, \ldots, U_{f\restriction (n+1)}\rangle)$$

and observe that σ is defeated by the counterattack $\langle U_{f\restriction 1}, U_{f\restriction 2}, \ldots\rangle$ as $x_{f\restriction n} \to x$.

$(d) \to (c)$: Suppose there are subsets A_n of X and a point x with $x \in \overline{A_n}$ for each $n \in \omega$, but there is no sequence (x_n) with $x_n \in A_n$ for each $n \in \omega$ and $x_n \to x$. If U_n is the latest move of player O in $G_{O,P}(X, x)$, then player P may choose $p_n \in U_n \cap A_n$. Then $p_n \not\to x$, so this outlines a winning strategy for player P in $G_{O,P}(X, x)$. Thus X is not a w-space.

$(c) \to (e)$: Trivial.

$(e) \to (f) \to (g)$: Trivial. ■

Proof of Example 348

For each countable subset C of $T \cup \{\infty\}$, it follows that $C \subseteq T_\alpha \cup \{\infty\}$ for some $\alpha < \omega_1$ where T_α is everything below the αth level of T.

Since T is an open first-countable subset of $T \cup \{\infty\}$, we need only show ∞ has a countable base when $\infty \in C$. Note $\{\{C \setminus t^\downarrow : t \in T_{\alpha+1}\}$ is countable. Each $C \setminus t^\downarrow$ is a neighborhood of ∞. We claim we have a subbase. A subbasic open set of the entire space is $C \setminus s^\downarrow$ for $s \in T$. If $s \in T_{\alpha+1}$ we are done. Otherwise, note $s \restriction \alpha \in T_{\alpha+1}$ and $C \setminus s^\downarrow = C \setminus (s \restriction \alpha)^\downarrow$.

We now will show that $T \cup \{\infty\}$ is countably tight. Consider $A \subseteq T$, and note that for each $t \in \overline{A} \cap T$, $t \in \overline{A \cap t^\downarrow}$. If A contains an infinite antichain B, then choose a countably infinite subset $B' \subseteq B$ and note $\infty \in \overline{B'}$. If A contains a infinite subset B of comparable elements with no upper bound in T, then B is countable and $\infty \in \overline{B}$.

Otherwise, consider the minimal elements of A. There must be finitely many since A lacks an infinite antichain. Then for each minimal element, consider the branches containing that element and a distinct element from other such branches; again there must be finitely many since A lacks an infinite antichain. Finally, since A lacks an infinite subset of comparable elements, there must be an element of T in each such branch which is above every element of A in that branch. Thus $\infty \notin \overline{A}$.

Finally, let σ be a strategy for O in $G_{O,P}(T \cup \{\infty\}, \infty)$. For each open neighborhood U of ∞, let $F(U) \in [T]^{<\omega}$ satisfy $T \setminus F(U)^{\downarrow} \subseteq U$.

Well order the limit levels $L(T)$ of T with $\prec$ such that $|s| < |t| \to s \prec t$. Since levels of T are countable, it follows that $\langle T, \prec \rangle$ is order-isomorphic to ω_1. Let

$$C = \{t \in L(t) : q \in \{s : s \prec t\}^{<\omega} \to F(\sigma(q)) \in [\{s : s \prec t\}]^{<\omega}\}$$

which is closed and unbounded. Choose $t_n \in C$ with $|t_n| < |t_{n+1}|$ converging to $t \in C$, then note that playing $t \restriction |t_n|$ during round n is a legal counterattack to σ which converges to t, not ∞, so σ is not a winning strategy. ■

Proof of Theorem 349

For each $n < \omega$ let $c_n = a_n \setminus (\bigcup_{m \leqslant n} \omega \setminus b_m)$. Put $c = \bigcup_{n<\omega} c_n$ and fix $n < \omega$. Note $a_n \subset^* c_n$ since $|a_n \cap (\bigcup_{m \leqslant n} \omega \setminus b_m)| < \omega$, so $a_n \subset^* c$. If $m < n$ we have $c_m \subset^* a_m \subset^* b_n$, and if $m \geqslant n$ we have $c_m \subset b_n$. Thus $|c \setminus b_n| \leqslant |\bigcup_{m \leqslant n} c_m \setminus b_n| < \omega$, so $c \subset^* b_n$. ■

Proof of Example 350

We show X satisfies property (c) in Theorem 96. Suppose $x \in \overline{A}_n$ for each $n \in \omega$. If $x \in \omega$ then $x \in A_n$ for each $n \in \omega$, so we may set $x_n = x$ for each $n \in \omega$. Now suppose $x = \infty$. For each $n \in \omega$ there exists $\alpha_n < \omega_1$ such that $|A_n \cap a_{\alpha_n}| = \omega$. Let $\alpha = \sup_{n \in \omega} \alpha_n$. Then $|A_n \cap a_\alpha| = \omega$ for each $n \in \omega$. Let $x_0 \in A_0 \cap a_\alpha$. Assuming x_k has been defined, choose $x_{k+1} \in A_{k+1} \cap a_\alpha \setminus \{x_j : j \leqslant k\}$. Since a_α converges to ∞ (every open nbhd of ∞ contains all but finitely many points of a_α), $x_n \to x$.

By a previous theorem, X is a W-space implies X is first-countable. We will show X is not first-countable by contradiction. Suppose X is first-countable, and let $U_0 \supset U_1 \supset U_2 \supset \ldots$ be a decreasing open base

at ∞. For each $\alpha \in \omega_1$, there is $n_\alpha \in \omega$ such that $b_\alpha \supset U_{n_\alpha} \setminus \{\infty\}$. So, there is a $k \in \omega$ such that $A = \{\alpha : n_\alpha = k\}$ is uncountable, and $U_k \setminus \{\infty\} \subset b_\alpha$ for each $\alpha \in A$. Therefore, $U_k \setminus \{\infty\} \subset^* b_\alpha$ for each $\alpha \in \omega_1$. Since every a_α converges to ∞, $a_\alpha \subset^* U_k \setminus \{\infty\}$ for each $\alpha \in \omega_1$. So $a_\alpha \subset^* U_k \setminus \{\infty\} \subset^* b_\alpha$ for all $\alpha \in \omega_1$, a contradiction. We conclude X is not first-countable, and therefore not a W-space. ■